just imagine
NEC MULTIMEDIA

NEC

SEMICONDUCTORS SELECTION GUIDE



Microcomputer 1

IC Memory 2

Semi-Custom IC 3

Particular Purpose 4

General Purpose Linear IC 5

Transistor/Diode/Thyristor 6

GaAs Device/
Silicon Microwave Semiconductor 7

Optical Device 8

Packages 9

Index 10

October 1998

www.datasheet4u.com

The export of these products from Japan is regulated by the Japanese government. The export of some or all of these products may be prohibited without governmental license. To export or re-export some or all of these products from a country other than Japan may also be prohibited without a license from that country. Please call an NEC sales representative.

MS-DOS, Microsoft and Windows are either registered trademarks or trademarks of Microsoft Corporation in the United States and/or other countries.
PC/AT and PC DOS are trademarks of IBM Corporation.
Pentium is a trademark of Intel Corporation.
Z80 is a trademark of ZILOG.
Direct Rumbus, RIMM are the trademarks of Rambus Inc.
EEPROM, *emIC-17K*, FIP, IEBus, OCMOS FET, *SIMPLEHOST*, V Series, V20, V20HL, V25, V25+, V30, V30HL, V30MX, V33A, V40, V40HL, V50, V50HL, V53A, V55PI, V800 Series, V810, V810 family, V830 family, V850 family, V810, V821, V830, V831, V832, V851, V852, V853, V854, V850/SA1, V850/SB1, V850/SB2, V850/SB3, V850E/MS1, VR Series, VR4100, VR4102, VR4110, VR4111, VR4300, VR4305, VR4310, VR5000, VR10000, D²BGA, OPENCAD are the trademarks of NEC Corporation.

The information in this document is based on documents issued in July, 1998 at the latest. The information is subject to change without notice. For actual design-in, refer to the latest publications of data sheet, etc., for the most up-to-date specifications of the device.

No part of this document may be copied or reproduced in any form or by any means without the prior written consent of NEC Corporation. NEC Corporation assumes no responsibility for any errors which may appear in this document.
NEC Corporation does not assume any liability for infringement of patents, copyrights or other intellectual property rights of third parties by or arising from use of a device described herein or any other liability arising from use of such device. No license, either express, implied or otherwise, is granted under any patents, copyrights or other intellectual property rights of NEC Corporation or others.
While NEC Corporation has been making continuous effort to enhance the reliability of its semiconductor devices, the possibility of defects cannot be eliminated entirely. To minimize risks of damage or injury to persons or property arising from a defect in an NEC semiconductor device, customers must incorporate sufficient safety measures in its design, such as redundancy, fire-containment, and anti-failure features.
NEC devices are classified into the following three quality grades:
"Standard", "Special", and "Specific". The Specific quality grade applies only to devices developed based on a customer designated “quality assurance program“ for a specific application. The recommended applications of a device depend on its quality grade, as indicated below. Customers must check the quality grade of each device before using it in a particular application.

- Standard: Computers, office equipment, communications equipment, test and measurement equipment, audio and visual equipment, home electronic appliances, machine tools, personal electronic equipment and industrial robots
- Special: Transportation equipment (automobiles, trains, ships, etc.), traffic control systems, anti-disaster systems, anti-crime systems, safety equipment and medical equipment (not specifically designed for life support)
- Specific: Aircrafts, aerospace equipment, submersible repeaters, nuclear reactor control systems, life support systems or medical equipment for life support, etc.

The quality grade of NEC devices is "Standard" unless otherwise specified in NEC's Data Sheets or Data Books. If customers intend to use NEC devices for applications other than those specified for Standard quality grade, they should contact an NEC sales representative in advance.
Anti-radioactive design is not implemented in this product.

The μPD178123, 178124, 178F124, 178133, 178134, 178F134, 754144, 754244, 754264, 75F4264, 789144, 789146, 789154, 789156, 78F9156, 789196AY, 789197AY, 78F9197AY, 789216AY, 789217AY, 78F9217AY, 780973, 78F0974, 780949, 78F0949, 78243, 78244 is manufactured and sold in accordance with the license agreement with CP8 Transac pertinent to the patent of microcomputers with on-chip EEPROM.

Note that this product can not be used for the IC card (SMART CARD).

Caution: The I^2C bus interface circuit is incorporated in the μPD703008Y, 70F3008Y, 703015Y, 703017Y, 70F3017Y, 703033Y, 70F3033Y, 70F3035Y, 70F3037Y, 17001, 17P001, 17003A, 17005, 17P005, 17010, 17P010, μPD17709 subseries, 17719 subseries, 178003 subseries, 178018A subseries, 178098 subseries, 178078 subseries, 178048 subseries, 789197AY subseries, 789217AY subseries, 78018FY subseries, 780018AY subseries, 780024Y subseries, 780034Y subseries, 780058Y subseries, 78054Y subseries, 78058FY subseries, 78064Y subseries, 78078Y subseries, 780308Y subseries, 780701Y subseries, 780833Y subseries, μPD78070AY, μPD784038Y subseries, 784216Y subseries, 784218Y subseries, 784225Y subseries, 784928Y subseries, μPC1884, μPD61882, μPC1851B, μPC1854A, μPC1853, μPC1857A, μPD6221 and μPD6222.

Those who use the I^2C bus interface can be granted the license below by giving prior notification before ordering the custom code.

Purchase of NEC I^2C components conveys a license under the Philips I^2C Patent Rights to use these components in an I^2C system, provided that the system conforms to the I^2C Standard Specification as defined by Philips.

[MEMO]

Table of Contents

1. Microcomputer 1

4-Bit Single Chip Microcontroller 2
- 75XL Series 2
- 17K Series 14
- μPD6133, 6604, 63 Series 28

8/16-Bit Single Chip Microcontroller 30
- 78K Series 30

V Series™ 103
- 16-Bit V series 103
- V800 Series™ 104
- VR Series™ 111

2. IC Memory 113

Dynamic RAM 114
Dynamic RAM Module 120
Static RAM 124
Mask ROM 128
COMBO Memory 129
Flash MEMORY 130
MCP (Flash Memory + SRAM) 131
Other 132
- Synchronous Graphics RAM 132
- Field/Line Buffer 132

3. Semi-Custom IC 133

Gate Array 134
- CMOS Gate Array 134
- BiCMOS Gate Array 138

Cell-Based IC 139
Embedded Array 145
Analog Master 146
Mixed Signal ASIC 148

4. Particular Purpose IC 155

DSP 156
CCD Linear Image Sensor 157
Communication IC 159
Mass Storage IC 163
Global Positioning System IC 164
Display IC 164
Motor Driver 167
Power MOS IC for Voltage Regulator 167
Power Switch for USB Application 168
Image IC 168
MPEG IC 168
Audio IC 169
TV IC 171
Multisync Monitor IC 173
Digital Image IC 174
On-Screen Character Display IC 175
Video Camera IC 176
Remote Control IC 176
Clock IC 176
Rotary Encoder IC 177
A/D Converter IC 177
D/A Converter IC 177
Line Driver Receiver IC 178

5. General Purpose Linear IC 179

Operational Amplifier 180
Comparator 181
Fixed Output Voltage, 3-Terminal Regulator 181
Variable Output Voltage Regulator 182
Regulator with System Reset 182
Regulator with ON/OFF Function 183
High Precision Reference Voltage 183
Switching Regulator Control Circuit 183
Functional Block 184

6. Transistor/Diode/Thyristor 185

Transistor 186
- Quick Reference by Package 186
- Quick Reference Table by Function/Application 196

Field Effect Transistor 202
- Small Signal FET 202
- Power MOS FET 205

Transistor with Internal Resistor 215
Transistor for Array 220
Transistor Array 221
Zener Diode 224
Noise Clipping Diode 226
Thyristor 227

7. GaAs Device/Silicon Microwave Semiconductor 229

GaAs Device 230
- GaAs MMIC 230
- GaAs MCM 231
- Field Effect Transistor 232

Silicon Microwave Semiconductor 234
- Si MMIC 234
- Transistor, FET 241

8. Optical Device 245

Fiber Optic Communication/Light Beam Application Devices 246
- Laser Diodes 247
- Detectors 248

Photo Detectors 250
- PIN Photo-diode 250
- Light Receiving IC 250

Photocoupler 251
OCMOS FET™ 253
POF Link 254
Infra-Red Link Unit 254

9. Packages 255

Surface Mount IC Packages 256
Through Hole IC Packages 272
Surface Mount Discrete Device Packages 276
Through Hole Discrete Device Packages 278

10. Index 279

Microcomputer

1

4-Bit Single Chip Microcontroller 2

- 75XL Series 2
- 17K Series 14
- μPD6133, 6604, 63 Series 28

8/16-Bit Single Chip Microcontroller 30

- 78K Series 30

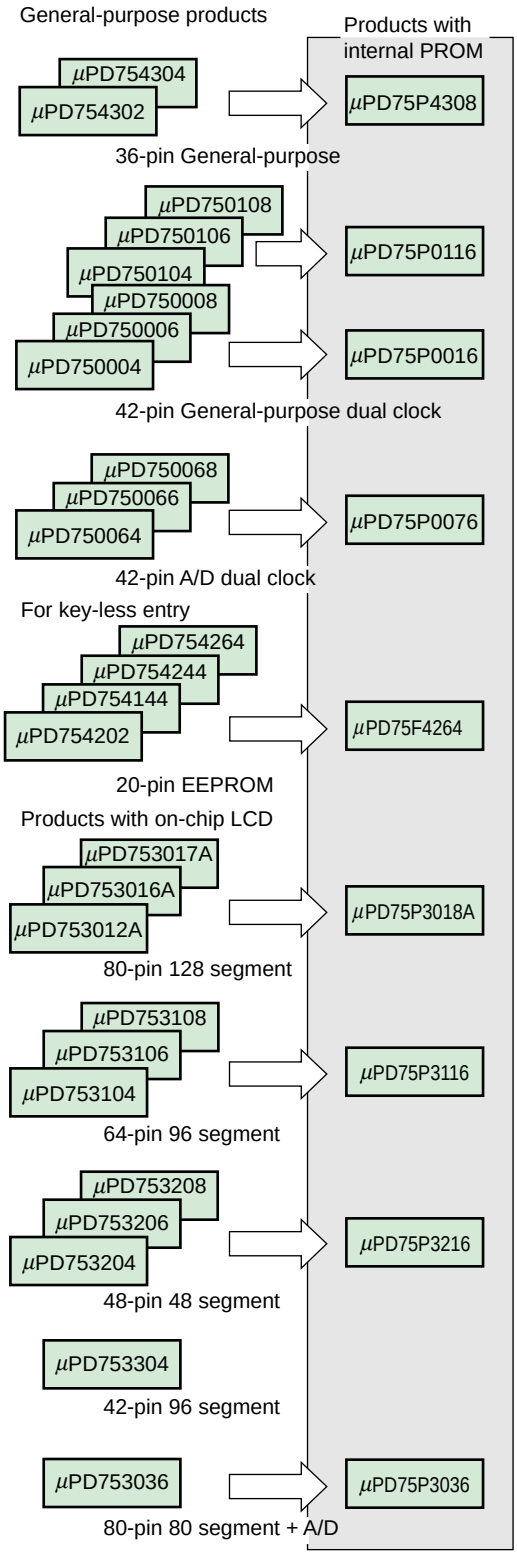
V Series™ 103

- 16-Bit V Series 103
- V800 Series™ 104
- V_R Series™ 111

4-Bit Single Chip Microcontroller

75XL Series

■ 75XL Series Product Evolution



★: Under development

4-Bit Single Chip Microcontroller

■ 75XL Series Products (1/10)

Part number			μPD750004	μPD750006	μPD750008	μPD75P0016
ROM (bytes)			4096 (Mask ROM)	6144 (Mask ROM)	8192 (Mask ROM)	16384 (One-time PROM)
RAM (× 4 bits)			512			
General-purpose register			(4-bit × 8) × 4 banks or (8-bit × 4) × 4 banks			
Instruction cycle			• 0.95 μs, 1.91 μs, 3.81 μs, 15.3 μs (with 4.19 MHz main system clock) • 0.67 μs, 1.33 μs, 2.67 μs, 10.7 μs (with 6.0 MHz main system clock) • 122 μs (with 32.768 kHz subsystem clock)			
Minimum instruction execution time			• 0.67 μs (V_{DD} = 2.7 to 5.5 V) • 0.95 μs (V_{DD} = 2.2 to 5.5 V)			
I/O port	CMOS input	34	8	On-chip pull-up resistors are connectable by software: 7		
	CMOS I/O		18	Can drive LED On-chip pull-up resistors are connectable by software		
	N-ch open-drain I/O		8	Can drive LED, 13 V withstand voltage Pull-up resistors can be incorporated by mask option		Can drive LED 13 V withstand voltage No on-chip pull-up resistors
Timer		4 channels		• 8-bit timer/event counter × 1 channel • Basic interval timer/watchdog timer × 1 channel • Clock timer × 1 channel • 8-bit timer × 1 channel		
Serial interface			Any of three modes can be used: • 3-wire serial I/O mode: MSB/LSB switchable •2-wire serial I/O mode •SBI mode			
Vectored interrupt			External: 3, Internal: 4			
Test input			External: 1, Internal: 1			
Clock output (PCL)			• Φ, 524 kHz, 262 kHz, 65.5 kHz (with 4.19 MHz main system clock) • Φ, 750 kHz, 375 kHz, 93.8 kHz (with 6.0 MHz main system clock)			
Buzzer output (BUZ)			• 2 kHz, 4 kHz, 32 kHz (with 4.19 MHz main system clock, or 32.768 kHz subsystem clock) • 2.93 kHz, 5.86 kHz, 46.9 kHz (with 6.0 MHz main system clock)			
Instruction set			• Bit data set/reset/test/boolean operations • 4-/8-bit data transfer/operations/increment and decrement/comparison			
Power supply voltage			V_{DD} = 2.2 to 5.5 V (V_{DD} = 1.8 to 5.5 V with external clock)			
Package			• 42-pin plastic shrink DIP (600 mil) • 44-pin plastic QFP (10 × 10 mm)			
Applications			VCRs, CD players, cordless telephones, radios, cameras, home electric appliances, etc.			

4-Bit Single Chip Microcontroller

■ 75XL Series Products (2/10)

Part number		**μPD750104**		**μPD750106**	**μPD750108**	**μPD75P0116**
ROM (bytes)		4096 (Mask ROM)		6144 (Mask ROM)	8192 (Mask ROM)	16384 (One-time PROM)
RAM (× 4 bits)		512				
General-purpose register		(4-bit × 8) × 4 banks or (8-bit × 4) × 4 banks				
Instruction cycle		• 2 μs, 4 μs, 8 μs, 32 μs (with RC 2.0 MHz main system clock) • 4 μs, 8 μs, 16 μs, 64 μs (with RC 1.0 MHz main system clock) • 122 μs (with 32.768 kHz subsystem clock)				
Minimum instruction execution time		2 μs (V_{DD} = 1.8 to 5.5 V)				
I/O port	CMOS input	34	8	On-chip pull-up resistors are connectable by software: 7		
	CMOS I/O		18	Can drive LED On-chip pull-up resistors are connectable by software		
	N-ch open-drain I/O		8	Can drive LED, 13 V withstand voltage Pull-up resistors can be incorporated by mask option		Can drive LED 13 V withstand voltage No on-chip pull-up resistors
Timer		4 channels		• 8-bit timer/event counter × 1 channel • Basic interval timer/watchdog timer × 1 channel • Clock timer × 1 channel • 8-bit timer × 1 channel		
Serial interface		Any of three modes can be used: • 3-wire serial I/O mode: MSB/LSB switchable • 2-wire serial I/O mode • SBI mode				
Vectored interrupt		External: 3, Internal: 4				
Test input		External: 1, Internal: 1				
Clock output (PCL)		Φ, 125 kHz, 62.5 kHz, 15.6 kHz (with 1.0 MHz main system clock)				
Buzzer output (BUZ)		• 2 kHz, 4 kHz, 32 kHz (with 32.768 kHz subsystem clock) • 0.49 kHz, 0.98 kHz, 7.81 kHz (with 1.0 MHz main system clock)				
Instruction set		• Bit data set/reset/test/boolean operations • 4-/8-bit data transfer/operations/increment and decrement/comparison				
Power supply voltage		V_{DD} = 1.8 to 5.5 V				
Package		• 42-pin plastic shrink DIP (600 mil) • 44-pin plastic QFP (10 × 10 mm)				
Applications		VCRs, CD players, cordless telephones, radios, cameras, home electric appliances, etc.				

4-Bit Single Chip Microcontroller

■ 75XL Series Products (3/10)

Part number				μPD750064	μPD750066	μPD750068	μPD75P0076
ROM (bytes)				4096 (Mask ROM)	6144 (Mask ROM)	8192 (Mask ROM)	16384 (One-time PROM)
RAM (× 4 bits)				512			
General-purpose register				(4-bit × 8) × 4 banks or (8-bit × 4) × 4 banks			
Instruction cycle				• 0.95 μs, 1.91 μs, 3.81 μs, 15.3 μs (with 4.19 MHz main system clock) • 0.67 μs, 1.33 μs, 2.67 μs, 10.7 μs (with 6.0 MHz main system clock) • 122 μs (with 32.768 kHz subsystem clock)			
Minimum instruction execution time				• 0.67 μs (V_{DD} = 2.7 to 5.5 V) • 0.95 μs (V_{DD} = 1.8 to 5.5 V)			
I/O port	CMOS input	32	12	On-chip pull-up resistors are connectable by software: 7 Also used for analog input pins: 4			
	CMOS I/O		12	Can drive LED On-chip pull-up resistors are connectable by software Also used for analog input pins: 4			
	N-ch open-drain I/O		8	Can drive LED, 13 V withstand voltage Pull-up resistors can be incorporated by mask option			Can drive LED 13 V withstand voltage No on-chip pull-up resistors
Timer		4 channels		• 8-bit timer/event counter: 2 channels (Usable as 16-bit timer/event counter) • Basic interval timer/watchdog timer: 1 channel • Clock timer: 1 channel			
Serial interface				Any of two modes can be used: • 3-wire serial I/O mode (MSB/LSB switchable) • 2-wire serial I/O mode			
A/D converter				8-bit resolution × 8 channels			
Vectored interrupt				External: 3, Internal: 4			
Test input				External: 1, Internal: 1			
Clock output (PCL)				• Φ, 1.05 MHz, 262 kHz, 65.5 kHz (with 4.19 MHz main system clock) • Φ, 1.5 MHz, 375 kHz, 93.8 kHz (with 6.0 MHz main system clock)			
Buzzer output (BUZ)				• 2 kHz, 4 kHz, 32 kHz (with 4.19 MHz main system clock, or 32.768 kHz subsystem clock) • 2.93 kHz, 5.86 kHz, 46.9 kHz (with 6.0 MHz main system clock)			
Instruction set				• Bit data set/reset/test/boolean operations • 4-/8-bit data transfer/oprations/increment and decrement/comparison			
Power supply voltage				V_{DD} = 1.8 to 5.5 V			
Package				• 42-pin plastic shrink DIP (600 mil, 1.778-mm pitch) • 42-pin plastic shrink SOP (375 mil, 0.8-mm pitch)			
Applications				Cordless telephones, AV equipment, home electric appliances, OA equipment, health equipment, meters, gas tables, etc.			

4-Bit Single Chip Microcontroller

■ 75XL Series Products (4/10)

Part number		**μPD754302**		**μPD754304**	**μPD75P4308**
ROM (bytes)		2048 (Mask ROM)		4096 (Mask ROM)	8192 (One-time PROM)
RAM (× 4 bits)		256			
General-purpose register		(4-bit × 8) × 4 banks or (8-bit × 4) × 4 banks			
Instruction cycle		• 0.95 μs, 1.91 μs, 3.81 μs, 15.3 μs (with 4.19 MHz main system clock) • 0.67 μs, 1.33 μs, 2.67 μs, 10.7 μs (with 6.0 MHz main system clock)			
Minimum instruction execution time		• 0.67 μs (V_{DD} = 2.7 to 5.5 V) • 0.95 μs (V_{DD} = 1.8 to 5.5 V)			
I/O port	CMOS input	30	8	On-chip pull-up resistors are connectable by software: 7	
	CMOS I/O		18	Can drive LED On-chip pull-up resistors are connectable by software	
	N-ch open-drain I/O		4	Can drive LED, 13 V withstand voltage Pull-up resistors can be incorporated by mask option	Can drive LED 13 V withstand voltage No on-chip pull-up resistors
Timer		3 channels		8-bit timer/event counter × 2 channels (usable as 16-bit timer/event counter) Basic interval timer/watchdog timer × 1 channel	
Serial interface		Either of two modes can be used: • 3-wire serial I/O mode: MSB/LSB switchable • 2-wire serial I/O mode			
Vectored interrupt		External: 3, Internal: 4			
Test input		External: 1			
Clock output (PCL)		• Φ, 524 kHz, 262 kHz, 65.5 kHz (with 4.19 MHz main system clock) • Φ, 750 kHz, 375 kHz, 93.8 kHz (with 6.0 MHz main system clock)			
Instruction set		• Bit data set/reset/test/boolean operations • 4-/8-bit data transfer/operations/increment and decrement/comparison			
Power supply voltage		V_{DD} = 1.8 to 5.5 V			
Package		• 36-pin plastic shrink SOP (300 mil, 0.8-mm pitch)			
Applications		Cordless telephones, TVs, VCRs, audio equipment, home electric appliances, OA equipment, etc.			

4-Bit Single Chip Microcontroller

■ 75XL Series Products (5/10)

Part number		μPD754202		μPD754144	μPD754244	μPD754264	μPD75F4264★
ROM (bytes)		2048 (Mask ROM)		4096 (Mask ROM)			4096 (Flash memory)
RAM (× 4 bits)		128					
EEPROM™ (bytes)		—		16		32	
General-purpose register		(4-bit × 8) × 4 banks or (8-bit × 4) × 4 banks					
Instruction cycle		• 0.95 μs, 1.91 μs, 3.81 μs, 15.3 μs (with 4.19 MHz System clock) • 0.67 μs, 1.33 μs, 2.67 μs, 10.7 μs (with 6.0 MHz System clock)		• 4 μs, 8 μs, 16 μs, 64 μs (with 1 MHz system clock)	• 0.95 μs, 1.91 μs, 3.81 μs, 15.3 μs (with 4.19 MHz system clock) • 0.67 μs, 1.33 μs, 2.67 μs, 10.7 μs (with 6.0 MHz system clock)		
Minimum instruction execution time		• 0.67 μs (V_{DD} = 2.7 to 6.0 V) • 0.95 μs (V_{DD} = 1.8 to 6.0 V)		2 μs (V_{DD} = 1.8 to 6.0 V)	• 0.67 μs (V_{DD} = 2.7 to 6.0 V) • 0.95 μs (V_{DD} = 1.8 to 6.0 V)		
I/O port	CMOS input	12	4	Pull-up resistors can be incorporated by mask option			No on-chip pull-up resistors
	CMOS I/O		8	Can drive LED On-chip pull-up resistors are connectable by software			
Systrem clock oscillator		Crystal/ceramic oscillator		RC oscillator (external resistors and capacitors)	Crystal/ceramic oscillator		
Timer		4 channels		• 8-bit timer counter: 3 channels (Usable as 16-bit timer counter or remote controller signal carrier generator) • Basic interval timer/watchdog timer: 1 channel			
Programmable threshold port		—		4-bit resolution × 2		—	
A/D converter		—				8-bit resolution × 2	
Others		Key return reset function					
Vectored interrupt		External: 1, internal: 5					
Test input		External: 1					
Instruction set		• Bit data set/reset/test/boolean operations • 4-/8-bit data transfer/operations/increment and decrement/comparison					
Power supply voltage		V_{DD} = 1.8 to 6.0 V					
Package		• 20-pin plastic SOP (300 mil, 1.27-mm pitch) • 20-pin plastic shrink SOP (300 mil, 0.65-mm pitch)				20-pin plastic SOP (300 mil)	
Applications		Keyless entries, data carriers					

★: Under development

4-Bit Single Chip Microcontroller

■ 75XL Series Products (6/10)

Part number		μPD753012A		μPD753016A	μPD753017A	μPD75P3018A
ROM (bytes)		12288 (Mask ROM)		16384 (Mask ROM)	24576 (Mask ROM)	32768 (One-time PROM)
RAM (× 4 bits)		1024				
General-purpose register		(4-bit × 8) × 4 banks or (8-bit × 4) × 4 banks				
Instruction cycle		• 0.95 μs, 1.91 μs, 3.81 μs, 15.3 μs (with 4.19 MHz main system clock) • 0.67 μs, 1.33 μs, 2.67 μs, 10.7 μs (with 6.0 MHz main system clock) • 122 μs (with 32.768 kHz subsystem clock)				
Minimum instruction execution time		• 0.67 μs (V_{DD} = 2.7 to 5.5 V) • 0.95 μs (V_{DD} = 1.8 to 5.5 V)				
I/O port	CMOS input	40	8	On-chip pull-up resistors are connectable by software: 7		
	CMOS I/O		16	Can drive LED On-chip pull-up resistors are connectable by software		
	CMOS I/O		8	Also used for segment pins		
	N-ch open-drain I/O		8	Can drive LED, 13 V withstand voltage Pull-up resistors can be incorporated by mask option		Can drive LED 13 V withstand voltage No on-chip pull-up resistors
LCD controller/driver		• Number of segments selectable: 24/28/32 segments (switchable to CMOS ports in multiples of 4: 8 maximum) • Selectable display mode: Static, 1/2 duty (1/2 bias), 1/3 duty (1/2 bias), 1/3 duty (1/3 bias), 1/4 duty (1/3 bias)				
		Voltage divider resistors for LCD drive can be incorporated by mask option				No on-chip voltage divider resistors for LCD drive
Timer		5 channels		• 8-bit timer/event counter: 3 channels (usable as 16-bit timer/event counter or remote-controller signal carrier generator) • Basic interval timer/watchdog timer × 1 channel • Clock timer × 1 channel		
Serial interface		Any of three modes can be used: • 3-wire serial I/O mode: MSB/LSB switchable • 2-wire serial I/O mode • SBI mode				
Vectored interrupt		External: 3, Internal: 5				
Test input		External: 1, Internal: 1				
Clock output (PCL)		• Φ, 524 kHz, 262 kHz, 65.5 kHz (with 4.19 MHz main system clock) • Φ, 750 kHz, 375 kHz, 93.8 kHz (with 6.0 MHz main system clock)				
Buzzer output (BUZ)		• 2 kHz, 4 kHz, 32 kHz (with 4.19 MHz main system clock, or with 32.768 kHz subsystem clock) • 2.93 kHz, 5.86 kHz, 46.9 kHz (with 6.0 MHz main system clock)				
Instruction set		• Bit data set/reset/test/boolean operations • 4-/8-bit data transfer/operations/increment and decrement/comparison				
Power supply voltage		V_{DD} = 1.8 to 5.5 V				
Package		• 80-pin plastic QFP (14 × 14 mm) • 80-pin plastic TQFP (fine pitch) (12 × 12 mm)				
Applications		Camcorders, CD players, telephones, cameras, pagers, remote controllers, etc.				

4-Bit Single Chip Microcontroller

■ 75XL Series Products (7/10)

Part number		μPD753104		μPD753106	μPD753108	μPD75P3116
ROM (bytes)		4096 (Mask ROM)		6144 (Mask ROM)	8192 (Mask ROM)	16384 (One-time PROM)
RAM (× 4 bits)		512				
General-purpose register		(4-bit × 8) × 4 banks or (8-bit × 4) × 4 banks				
Instruction cycle		• 0.95 μs, 1.91 μs, 3.81 μs, 15.3 μs (with 4.19 MHz main system clock) • 0.67 μs, 1.33 μs, 2.67 μs, 10.7 μs (with 6.0 MHz main system clock) • 122 μs (with 32.768 kHz subsystem clock)				
Minimum instruction execution time		• 0.67 μs (V_{DD} = 2.7 to 5.5 V) • 0.95 μs (V_{DD} = 1.8 to 5.5 V)				
I/O port	CMOS input	32	8	On-chip pull-up resistors are connectable by software: 7		
	CMOS I/O		12	Can drive LED On-chip pull-up resistors are connectable by software		
	CMOS I/O		8	Also used for segment pins		
	N-ch open-drain I/O		4	Can drive LED, 13 V withstand voltage Pull-up resistors can be incorporated by mask option		Can drive LED 13 V withstand voltage No on-chip pull-up resistors
LCD controller/driver		• Number of segments selectable: 16/20/24 segments (switchable to CMOS ports in multiples of 4: 8 maximum) • Selectable display mode: Static, 1/2 duty (1/2 bias), 1/3 duty (1/2 bias), 1/3 duty (1/3 bias), 1/4 duty (1/3 bias)				
		Voltage divider resistors for LCD drive can be incorporated by mask option				No on-chip voltage divider resistors for LCD drive
Timer		5 channels		• 8-bit timer/event counter: 3 channels (usable as 16-bit timer/event counter or remote-controller signal carrier generator) • Basic interval timer/watchdog timer × 1 channel • Clock timer × 1 channel		
Serial interface		Any of three modes can be used: • 3-wire serial I/O mode: MSB/LSB switchable • 2-wire serial I/O mode • SBI mode				
Vectored interrupt		External: 3, Internal: 5				
Test input		External: 1, Internal: 1				
Clock output (PCL)		• Φ, 524 kHz, 262 kHz, 65.5 kHz (with 4.19 MHz main system clock) • Φ, 750 kHz, 375 kHz, 93.8 kHz (with 6.0 MHz main system clock)				
Buzzer output (BUZ)		• 2 kHz, 4 kHz, 32 kHz (with 4.19 MHz main system clock, or with 32.768 kHz subsystem clock) • 2.93 kHz, 5.86 kHz, 46.9 kHz (with 6.0 MHz main system clock)				
Instruction set		• Bit data set/reset/test/boolean operations • 4-/8-bit data transfer/operations/increment and decrement/comparison				
Power supply voltage		V_{DD} = 1.8 to 5.5 V				
Package		• 64-pin plastic QFP (14 × 14 mm) • 64-pin plastic TQFP (12 × 12 mm)				
Applications		Telephones, cameras, sphygmomanometers, gas meters, electronic scales, remote controllers, etc.				

4-Bit Single Chip Microcontroller

■ 75XL Series Products (8/10)

Part number		**μPD753204**		**μPD753206**	**μPD753208**	**μPD75P3216**
ROM (bytes)		4096 (Mask ROM)		6144 (Mask ROM)	8192 (Mask ROM)	16384 (One-time PROM)
RAM (× 4 bits)		512				
General-purpose register		(4-bit × 8) × 4 banks or (8-bit × 4) × 4 banks				
Instruction cycle		• 0.95 μs, 1.91 μs, 3.81 μs, 15.3 μs (with 4.19 MHz main system clock) • 0.67 μs, 1.33 μs, 2.67 μs, 10.7 μs (with 6.0 MHz main system clock)				
Minimum instruction execution time		• 0.67 μs (V_{DD} = 2.7 to 5.5 V) • 0.95 μs (V_{DD} = 1.8 to 5.5 V)				
I/O port	CMOS input	30	6	On-chip pull-up resistors are connectable by software: 5		
	CMOS I/O		12	Can drive LED On-chip pull-up resistors are connectable by software		
	CMOS I/O		8	Also used for segment pins On-chip pull-up resistors are connectable by software		
	N-ch open-drain I/O		4	Can drive LED, 13 V withstand voltage Pull-up resistors can be incorporated by mask option		Can drive LED 13 V withstand voltage No on-chip pull-up resistors
LCD controller/driver		• Number of segments selectable: 4/8/12 segments • Selectable display mode: Static, 1/2 duty (1/2 bias), 1/3 duty (1/2 bias), 1/3 duty (1/3 bias), 1/4 duty (1/3 bias)				
		Voltage divider resistors for LCD drive can be incorporated by mask option				No on-chip voltage divider resistors for LCD drive
Timer		5 channels		• 8-bit timer/event counter × 1 channel • Basic interval timer/watchdog timer × 1 channel • Clock timer × 1 channel • 8-bit timer × 2 channels (usable as 16-bit timer or remote controller signal carrier generator)		
Serial interface		Any of three modes can be used: • 3-wire serial I/O mode: MSB/LSB switchable • 2-wire serial I/O mode • SBI mode				
Vectored interrupt		External: 2, Internal: 5				
Test input		External: 1, Internal: 1				
Clock output (PCL)		• Φ, 524 kHz, 262 kHz, 65.5 kHz (with 4.19 MHz main system clock) • Φ, 750 kHz, 375 kHz, 93.8 kHz (with 6.0 MHz main system clock)				
Buzzer output (BUZ)		• 2 kHz, 4 kHz, 32 kHz (with 4.19 MHz main system clock) • 2.93 kHz, 5.86 kHz, 46.9 kHz (with 6.0 MHz main system clock)				
Instruction set		• Bit data set/reset/test/boolean operations • 4-/8-bit data transfer/operations/increment and decrement/comparison				
Power supply voltage		V_{DD} = 1.8 to 5.5 V				
Package		48-pin plastic shrink SOP (375 mil, 0.65-mm pitch)				
Applications		Remote controllers, cameras, sphygmomanometers, CD radio cassette players, gas meters, etc.				

4-Bit Single Chip Microcontroller

■ 75XL Series Products (9/10)

Part number				μPD753036	μPD75P3036
ROM (bytes)				16384 (Mask ROM)	16384 (One-time PROM)
RAM (× 4 bits)				768	
General-purpose register				(4-bit × 8) × 4 banks or (8-bit × 4) × 4 banks	
Instruction cycle				• 0.95 μs, 1.91 μs, 3.81 μs, 15.3 μs (with 4.19 MHz main system clock) • 0.67 μs, 1.33 μs, 2.67 μs, 10.7 μs (with 6.0 MHz main system clock) • 122 μs (with 32.768 kHz subsystem clock)	
Minimum instruction execution time				• 0.67 μs (V_{DD} = 2.7 to 5.5 V) • 0.95 μs (V_{DD} = 1.8 to 5.5 V)	
I/O port	CMOS input	44	8	On-chip pull-up resistors are connectable by software: 7	
	CMOS I/O		20	Can drive LED On-chip pull-up resistors are connectable by software	
	Bit-port output		8	Also used for segment pins	
	N-ch open-drain I/O		8	Can drive LED 13 V withstand voltage Pull-up resistors can be incorporated by mask option	Can drive LED, 13 V withstand voltage No on-chip pull-up resistors
LCD controller/driver				• Number of segments selectable: 12/16/20 segments (switchable to bit-port output in multiples of 4: 8 maximum) • Selectable display mode: Static, 1/2 duty (1/2 bias), 1/3 duty (1/2 bias), 1/3 duty (1/3 bias), 1/4 duty (1/3 bias)	
				Voltage divider resistors for LCD drive can be incorporated by mask option	No on-chip voltage divider resistors for LCD drive
Timer				5 channels	• 8-bit timer/event counter: 3 channels (Usable as 16-bit timer/event counter or remote controller signal carrier generator) • Basic interval timer/watchdog timer: 1 channel • Clock timer: 1 channel
Serial interface				Any of three modes can be used: • 3-wire serial I/O mode (MSB/LSB switchable) • 2-wire serial I/O mode • SBI mode	
A/D converter				8-bit resolution × 8 channels	
Vectored interrupt				External: 3, Internal: 5	
Test input				External: 1, Internal: 1	
Clock output (PCL)				• Φ, 524 kHz, 262 kHz, 65.5 kHz (with 4.19 MHz main system clock) • Φ, 750 kHz, 375 kHz, 93.8 kHz (with 6.0 MHz main system clock)	
Buzzer output (BUZ)				• 2 kHz, 4 kHz, 32 kHz (with 4.19 MHz main system clock, or 32.768 kHz subsystem clock) • 2.93 kHz, 5.86 kHz, 46.9 kHz (with 6.0 MHz main system clock)	
Instruction set				• Bit data set/reset/test/boolean operations • 4-/8-bit data transfer/operations/increment and decrement/comparison	
Power supply voltage				V_{DD} = 1.8 to 5.5 V	
Package				• 80-pin plastic QFP (14 × 14 mm) • 80-pin plastic TQFP (fine pitch) (12 × 12 mm)	
Applications				Cordless telephones, AV equipment, home electric appliances, OA equipment, health equipment, meters, gas tables, etc.	

4-Bit Single Chip Microcontroller

■ 75XL Series Products (10/10)

Part number		**μPD753304**		
ROM (bytes)		4096 (Mask ROM)		
RAM (× 4 bits)		256		
General-purpose register		(4-bit × 8) × 4 banks or (8-bit × 4) × 4 banks		
Instruction cycle		1.1 μs, 2.2 μs, 4.4 μs, 17.8 μs (with 3.6 MHz main system clock, RC oscillator)		
		85.1 μs (with 47 kHz subsystem clock, RC oscillator)		
Minimum instruction execution time		• 0.67 μs (V_{DD} = 4.5 to 5.5 V) • 0.8 μs (V_{DD} = 2.5 to 5.5 V)		
I/O port	CMOS I/O	12	8	On-chip pull-up resistors are connectable by software: 4
	CMOS I/O		4	Also used for segment pins
LCD controller/driver		• Number of segments selectable: 20/24 segments (switchable to CMOS I/O ports in multiples of 4 : 4) • Selectable display mode: Static, 1/2 duty (1/2 bias), 1/3 duty (1/2 bias), 1/3 duty (1/3 bias), 1/4 duty (1/3 bias)		
		On-chip voltage divider resistors for LCD drive		
Timer		3 channels		• 8-bit timer: 1 channel • Basic interval timer/watchdog timer: 1 channel • Watch timer: 1 channel
Vectored interrupt		External: 1, Internal: 2		
Test input		Internal: 1		
Clock output (PCL)		Φ, 3.6 MHz, 450 kHz, 225 kHz (with 3.6 MHz main system clock)		
Buzzer output (BUZ)		• 2.94 kHz, 5.88 kHz, 47 kHz (with 47 kHz subsystem clock) • 1.76 kHz, 3.52 kHz, 28.13 kHz (with 3.6 MHz main system clock)		
Instruction set		• Bit data set/reset/test/boolean operations • 4-/8-bit data transfer/operations/increment and decrement/comparison		
Power supply voltage		V_{DD} = 2.5 to 5.5 V		
Operating ambient temperature		T_A = –10 to +60°C		
Package		• 42-pin pellet/wafer • 42-pin ceramic shrink DIP (600 mil) (ES only)		
Applications		Small-size LCD displays, etc.		

4-Bit Single Chip Microcontroller

■ 75XL Series Development Tools

Target device		Hardware			Software	
Device name	Package	In-circuit emulator	Emulation probe	Program adapter	Relocatable assembler/ device file	IE controller
μPD75000× **μPD75010×**	42-pin plastic SDIP	IE-75001-R + IE-75300-R-EM	EP-75008CU-R	PA-75P008CU	μS××××RA75X + μS××××DF750008	μS××××IE75X[Note]
	44-pin plastic QFP		EP-75008GB-R	PA-75P0016GB		
μPD75430×	36-pin plastic SSOP		EP-754304GS-R + EV-9500GS-36	PA-75P4308GS	μS××××RA75X + μS××××DF754304	
μPD75006×	42-pin plastic SDIP		EP-750068CU-R	PA-75P008CU PA-75P0076CU	μS××××RA75X + μS××××DF750068	
	42-pin plastic SSOP		EP-750068GT-R	PA-75P0076CU		
μPD754144 **μPD754244**	20-pin plastic SOP		EP-754144GS-R	—	μS××××RA75X + μS××××DF754244	
	20-pin plastic SSOP					
μPD7530×× **μPD7530××A**	80-pin plastic QFP 80-pin plastic TQFP		EP-753017GC-R EP-753017GK-R	PA-75P3016BGC PA-75P3016BGK	μS××××RA75X + μS××××DF753017	
μPD7531××	64-pin plastic QFP 64-pin plastic TQFP		EP-753108GC-R EP-753108GK-R	PA-75P3116GC PA-75P3116GK	μS××××RA75X + μS××××DF753108	
μPD753036	80-pin plastic QFP 80-pin plastic TQFP		EP-75336GC-R EP-75336GK-R	PA-75P328GC PA-75P336GK	μS××××RA75X + μS××××DF753036	
μPD7532××	48-pin plastic SSOP		EP-753208GT-R	PA-75P3216GT	μS××××RA75X + μS××××DF753208	
μPD753304	42-pin ceramic SDIP		EP-753304DU-R	—	μS××××RA75X + μS××××DF753304	

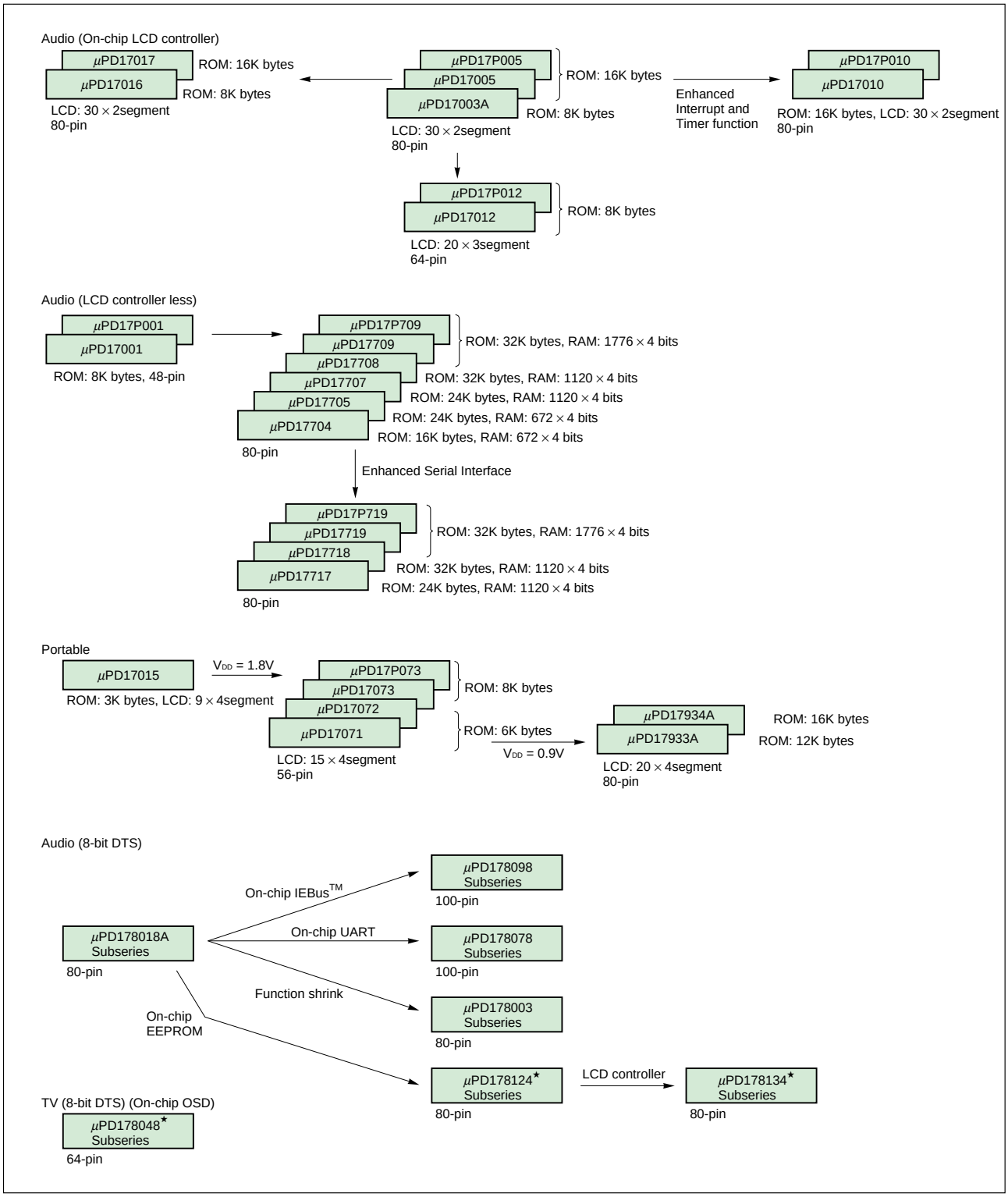
Note The assembler and controller can run on PC-9800 (MS-DOS™) and IBM PC/AT™ compatible (PC DOS™) machines. Those parts indicated by ×××× depend on the host machine.

4-Bit Single Chip Microcontroller

17K Series

■ 17K Series Product Evolution (1/2)

DTS

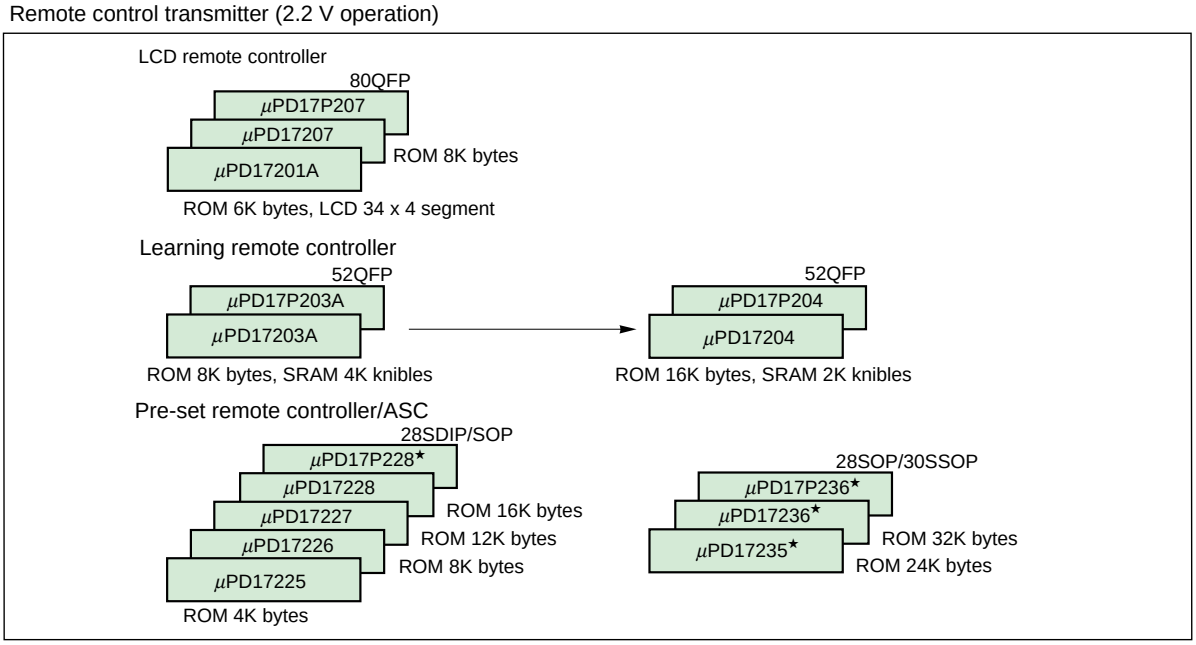


★: Under development

4-Bit Single Chip Microcontroller

17K Series

■ 17K Series Product Evolution (2/2)



★: Under development

4-Bit Single Chip Microcontroller

■ 17K Series (DTS) Products (1/4)

Part number	**μPD17001**	**μPD17003A**	**μPD17005**	**μPD17010**	**μPD17016**	**μPD17017**	**μPD17012**
Main application	High-end tuner, automotive radio						
ROM size	3836 × 16 bits	3836 × 16 bits	7932 × 16 bits	7932 × 16 bits	3836 × 16 bits	7932 × 16 bits	4096 × 16 bits
RAM size	224 × 4 bits	320 × 4 bits	432 × 4 bits	432 × 4 bits	320 × 4 bits	432 × 4 bits	316 × 4 bits
Display	—	On-chip LCD driver Segment signal: 30 Common signal: 2					On-chip LCD driver Segment signal: 20 Common signal: 3
Input port	8	8					
Output port	12	9 (+30)			9 (+8)		8 (+ 20)
I/O port	12	16					14
D/A converter	4	4			2		2
A/D converter	6	6			6		2
Modulo timer	—	—					1
Power supply voltage	+5 V ±10%						
Crystal	4.5 MHz						
PLL reference frequency	1, 1.25, 2.5, 3, 5, 6.25, 9, 10, 12.5, 25, 50, 100 kHz						
Applicable prescaler	On-chip (130MHz)	On-chip (150 MHz)					On-chip (250 MHz)
Package	48-pin plastic QFP	80-pin plastic QFP					64-pin plastic QFP
One-time PROM product	μPD17P001	μPD17P005		μPD17P010	μPD17P005		μPD17P012
Assembler	RA17K						
C compiler	*emIC-17K* ™						
Device file	AS17001	AS17003	AS17005	AS17010	AS17016		AS17012
SIMPLEHOST ™	ID17K						
In-circuit emulator	IE-17K or IE-17K-ET						
	SE-17001	SE-17010					SE-17012
	EP-17001GH	EP-17003GF					EP-17202GF

4-Bit Single Chip Microcontroller

■ 17K Series (DTS) Products (2/4)

Part number	μPD17015	μPD17071	μPD17072	μPD17073	μPD17933A	μPD17934A
Main application	Portable radio					
ROM size	1528 × 16 bits	3072 × 16 bits		4096 × 16 bits	6144 × 16 bits	8192 × 16 bits
RAM size	97 × 4 bits	174 × 4 bits			448 × 4 bits	
Display	On-chip LCD driver Segment signal: 9 Common signal: 4	On-chip LCD driver Segment signal: 15 Common signal: 4			On-chip LCD driver Segment signal: 20 Common signal: 4	
Input port	3	4			11	
Output port	7	9			6	
I/O port	2	8			20	
A/D converter	—		2		3	
Power supply voltage	1.7 to 2.6 V (when PLL is stopped) 1.8 to 3.6 V (when PLL is operating)	1.8 to 3.6 V			0.9 to 1.8 V	
Crystal	75 kHz					
PLL reference frequency	1, 3, 5, 12.5 kHz	1, 3, 5, 6.25, 12.5, 25 kHz				
Applicable prescaler	On-chip (220 MHz)	On-chip (230 MHz)				
Package	38-pin plastic shrink SOP	56-pin plastic QFP	56-pin plastic QFP 64-pin plastic TQFP		80-pin plastic TQFP	
One-time PROM product	—	μPD17P073			—	
Assembler	RA17K					
C compiler	*emIC-17K*					
Device file	AS17015	AS17072			AS17934	
SIMPLEHOST	ID17K					
In-circuit emulator	IE-17K or IE-17K-ET					
	SE-17015	SE-17072			SE-17934	
	EP-17K38GT	EP-17K56GB	EP-17K56GB, EP-17K64GB		EP-17K80GK	

4-Bit Single Chip Microcontroller

■ 17K Series (DTS) Products (3/4)

Subseries name	**μPD17709 Subseries**				
Part number	**μPD17704**	**μPD17705**	**μPD17707**	**μPD17708**	**μPD17709**
Main application	High-end tuner, automotive radio				
ROM size	8192 × 16 bits	12288 × 16 bits		16384 × 16 bits	
RAM size	672 × 4 bits		1120 × 4 bits		1776 × 4 bits
Input port	12				
Output port	4				
I/O port	46				
D/A converter	3				
A/D converter	6				
Modulo timer	4 (Used also PWM: 1)				
Power supply voltage	+5 V ±10%				
Crystal	4.5 MHz				
PLL reference frequency	1, 1.25, 2.5, 3, 5, 6.25, 9, 10, 12.5, 18, 20, 25, 50 kHz				
Applicable prescaler	On-chip (130 MHz)				
Package	80-pin plastic QFP				
One-time PROM product	μPD17P709				
Assembler	RA17K				
C compiler	*emIC-17K*				
Device file	AS17707				
SIMPLEHOST	ID17K				
In-circuit emulator	IE-17K or IE-17K-ET				
	SE-17709				
	EP-17K80GC				

4-Bit Single Chip Microcontroller

■ 17K Series (DTS) Products (4/4)

Subseries name	**μPD17719 Subseries**		
Part number	**μPD17717**	**μPD17718**	**μPD17719**
Main application	High-end tuner, automotive radio		
ROM size	12288 × 16 bits	16384 × 16 bits	
RAM size	1120 × 4 bits		1776 × 4 bits
Input port	12		
Output port	4		
I/O port	46		
D/A converter	3		
A/D converter	6		
Serial interface	2-wire/3-wire/SBI/I²C bus (selectable): 1 channel 3-wire/UART (selectable): 1 channel		
Modulo timer	4 (Used also PWM: 1)		
Power supply voltage	+5 V ±10%		
Crystal	4.5 MHz		
PLL reference frequency	1, 1.25, 2.5, 3, 5, 6.25, 9, 10, 12.5, 18, 20, 25, 50 kHz		
Applicable prescaler	On-chip (130 MHz)		
Package	80-pin plastic QFP		
One-time PROM product	μPD17P719		
Assembler	RA17K		
C compiler	*emIC-17K*		
Device file	AS17707		
SIMPLEHOST	ID17K		
In-circuit emulator	IE-17K or IE-17K-ET		
	SE-17709		
	EP-17K80GC		

4-Bit Single Chip Microcontroller

■ 8-bit DTS Series Products (1/6)

Subseries name	**μPD178003 Subseries**		**μPD178018A Subseries**			
Part number	**μPD178002**	**μPD178003**	**μPD178004A**	**μPD178006A**	**μPD178016A**	**μPD178018A**
Main application	High-end tuner, automotive radio					
ROM size	16K bytes	24K bytes	32K bytes	48K bytes		60K bytes
RAM size	1K bytes				3K bytes	
Input port	1					
Output port	3					
I/O port	58					
D/A converter	–		3			
A/D converter	3		6			
Modulo timer	3 (Used also PWM: 1)					
Power supply voltage	+5 V ±10%					
Crystal	4.5 MHz					
PLL reference frequency	1, 3, 9, 10, 12.5, 25, 50 kHz		1, 1.25, 2.5, 3, 5, 6.25, 9, 10, 12.5, 18, 20, 25, 50 kHz			
Applicable prescaler	On-chip (160 MHz)					
Package	80-pin plastic QFP					
One-time PROM product	μPD178P018A					
Assembler	RA78K0					
C compiler	CC78K0					
Real-time OS	RX78K0, MX78K0					
Integrated debugger	ID78K0 or ID78K0-NS					
	DF178018					
In-circuit emulator	IE-78001-R-A or IE-78K0-NS					
	With IE-78001-R-A: IE-178018-R-EM, with IE-78K0-NS: IE-178018-NS-EM1					
	With IE-78001-R-A: EP-78230GC-R, with IE-78K0-NS: NP-80GC *					

*: This is a product of Naito Densei Machidaseisakusho Co., LTD.

4-Bit Single Chip Microcontroller

■ 8-bit DTS Series Products (2/6)

Subseries name	**μPD178124 Subseries**			
Part number	**μPD178023★**	**μPD178024★**	**μPD178123★**	**μPD178124★**
Main application	High-end tuner, automotive radio			
ROM size	24K bytes	32K bytes	24K bytes	32K bytes
RAM size	0.5K bytes	1K bytes	0.5K bytes	1K bytes
EEPROM	–		128 bytes	
Input port	6			
Output port	3			
I/O port	53			
Serial interface	I^2C bus: 1 channel 3-wire: 1 channel UART: 1 channel			
A/D converter	6			
Modulo timer	8 bits × 2 ch 16 bits × 1 ch (only μPD178F124)			
Power supply voltage	4.5 to 5.5 V (When PLL or A/D converter is operating) 3.5 to 5.5 V (When only CPU is operating)			
Crystal	4.5, 6.3 MHz			
PLL reference frequency	1, 3, 9, 10, 12.5, 25, 30, 50 kHz			
Applicable prescaler	On-chip (160 MHz)			
Package	80-pin plastic QFP (14 × 14 mm, 0.65 mm pitch) 80-pin plastic QFP (14 × 20 mm, 0.8 mm pitch)			
Flash memory product	μPD178F124★			
Assembler	RA78K0			
C compiler	CC78K0			
Real-time OS	RX78K0, MX78K0			
Integrated debugger	ID78K0 or ID78K0-NS			
	DF178134★			
In-circuit emulator	IE-78001-R-A or IE-78K0-NS			
	With IE-78001-R-A: IE-178134-NS-EM1★ + IE-78K0-R-EX1 With IE-78K0-NS: IE-178134-NS-EM1★			
	With IE-78001-R-A: EP-78230GC-R, EP-78130GF-R With IE-78K0-NS: NP-80GC*, NP-80GF*			

*: This is a product of Naito Densei Machidaseisakusho Co., LTD.

★: Under development

4-Bit Single Chip Microcontroller

■ 8-bit DTS Series Products (3/6)

Subseries name	**μPD178134 Subseries**			
Part number	**μPD178033★**	**μPD178034★**	**μPD178133★**	**μPD178134★**
Main application	High-end tuner, automotive radio			
ROM size	24K bytes	32K bytes	24K bytes	32K bytes
RAM size	0.5K bytes	1K bytes	0.5K bytes	1K bytes
EEPROM	–		128 bytes	
Input port	6			
Output port	3			
I/O port	27			
Display	On-chip LCD driver Segment signal: MAX. 20 Common: MAX. 4			
Serial interface	I^2C bus: 1 channel 3-wire: 1 channel UART: 1 channel			
A/D converter	6			
Modulo timer	8 bits × 2 ch 16 bits × 1 ch (only μPD178F134)			
Power supply voltage	4.5 to 5.5 V (When PLL or A/D converter is operating) 3.5 to 5.5 V (When only CPU is operating)			
Crystal	4.5, 6.3 MHz			
PLL reference frequency	1, 3, 9, 10, 12.5, 25, 30, 50 kHz			
Applicable prescaler	On-chip (160 MHz)			
Package	80-pin plastic QFP (14 × 14 mm, 0.65 mm pitch) 80-pin plastic QFP (14 × 20 mm, 0.8 mm pitch)			
Flash memory product	μPD178F134★			
Assembler	RA78K0			
C compiler	CC78K0			
Real-time OS	RX78K0, MX78K0			
Integrated debugger	ID78K0 or ID78K0-NS			
	DF178134★			
In-circuit emulator	IE-78001-R-A or IE-78K0-NS			
	With IE-78001-R-A: IE-178134-NS-EM1★ + IE-78K0-R-EX1 With IE-78K0-NS: IE-178134-NS-EM1★			
	With IE-78001-R-A: EP-78230GC-R, EP-78130GF-R With IE-78K0-NS: NP-80GC*, NP-80GF*			

*: This is a product of Naito Densei Machidaseisakusho Co., LTD.

★: Under development

4-Bit Single Chip Microcontroller

■ 8-bit DTS Series Products (4/6)

Subseries name	**μPD178098 Subseries**	
Part number	**μPD178096**	**μPD178098**
Main application	High-end tuner, automotive radio	
ROM size	48K bytes	60K bytes
RAM size	2048 bytes	3072 bytes
Input port	8	
Output port	8	
I/O port	65	
Serial interface	IEBus (mode 1 only): 1 channel 3-wire/2-wire/SBI/I²C bus (selectable): 1 channel 3-wire: 1 channel 3-wire (with function for automatic transmission and reception): 1 channel	
D/A converter	–	
A/D converter	8	
Modulo timer	3 (Used also PWM: 1)	
Power supply voltage	+5 V ±10%	
Crystal	6.3 MHz	
PLL reference frequency	1, 3, 9, 10, 12.5, 25, 50 kHz	
Applicable prescaler	On-chip (160 MHz)	
Package	100-pin plastic QFP	
Flash memory product	μPD178F098	
Assembler	RA78K0	
C compiler	CC78K0	
Real-time OS	RX78K0, MX78K0	
Integrated debugger	ID78K0 or ID78K0-NS	
	DF178098	
In-circuit emulator	IE-78001-R-A or IE-78K0-NS	
	With IE-78001-R-A: IE-178098-NS-EM1 + IE-78K0-R-EX1 With IE-78K0-NS: IE-178098-NS-EM1	
	With IE-78001-R-A: EP-78064GF-R, with IE-78K0-NS: NP-100GF*	

*: This is a product of Naito Densei Machidaseisakusho Co., LTD.

4-Bit Single Chip Microcontroller

■ 8-bit DTS Series Products (5/6)

Subseries name	**μPD178078 Subseries**	
Part number	**μPD178076**	**μPD178078**
Main application	High-end tuner, automotive radio	
ROM size	48K bytes	60K bytes
RAM size	2K bytes	3K bytes
Input port	8	
Output port	8 (N-ch open drain)	
I/O port	64 (N-ch open drain: 3)	
Serial interface	3-wire/2-wire/SBI/I^2C bus (selectable): 1 channel 3-wire: 1 channel 3-wire (with function for automatic transmission and reception): 1 channel UART: 1 channel	
A/D converter	8-bit resolution × 8 channels	
Power supply voltage	+5 V ±10%	
Crystal	6.3 MHz, 4.5 MHz	
PLL reference frequency	1, 3, 9, 10, 12.5, 25, 50 kHz	
Applicable prescaler	On-chip (160 MHz)	
Package	100-pin plastic QFP	
Flash memory product	μPD178F098	
Assembler	RA78K0	
C compiler	CC78K0	
Real-time OS	RX78K0, MX78K0	
Integrated debugger	ID78K0 or ID78K0-NS	
	DF178098	
In-circuit emulator	IE-78001-R-A or IE-78K0-NS	
	With IE-78001-R-A: IE-178098-NS-EM1 + IE-78K0-R-EX1 With IE-78K0-NS: IE-178098-NS-EM1	
	With IE-78001-R-A: EP-78064GF-R, with IE-78K0-NS: NP-100GF*	

*: This is a product of Naito Densei Machidaseisakusho Co., LTD.

4-Bit Single Chip Microcontroller

■ 8-bit DTS Series Products (6/6)

Subseries name	**μPD178048 Subseries**	
Part number	**μPD178046★**	**μPD178048★**
Main application	TV sets, CATV	
ROM size	48K bytes	60K bytes
RAM size	1K bytes	
Display	On-chip OSD Kind of character: 255 Display character: 12 lines × 24 columns Character type: 12 × 18 dot Character color: 8	
Input port	6	
Output port	10 (N-ch open drain: 5)	
I/O port	37 (N-ch open drain: 4)	
Serial interface	I^2C bus: 2 channels 3-wire: 1 channel	
PWM	14-bit × 1 8-bit × 4	
A/D converter	8-bit resolution × 4 channels	
Power supply voltage	+5 V ±10%	
Crystal	5.0 MHz	
Package	64-pin plastic shrink DIP	
Flash memory product	μPD178F048★	
Assembler	RA78K0	
C compiler	CC78K0	
Real-time OS	RX78K0, MX78K0	
Integrated debugger	ID78K0 or ID78K0-NS	
	DF178048	
In-circuit emulator	IE-78001-R-A or IE-78K0-NS	
	With IE-78001-R-A: IE-178048-NS-EM1 + IE-78K0-R-EX1 With IE-78K0-NS: IE-178048-NS-EM1	
	With IE-78001-R-A: EP-78240CW-R, with IE-78K0-NS: NP-64CW*	

*: This is a product of Naito Densei Machidaseisakusho Co., LTD.

★: Under development

4-Bit Single Chip Microcontroller

■ 17K Series (Remote-Controller) Products (1/2)

Part number	**μPD17201A**	**μPD17207**	**μPD17203A**	**μPD17204**
Main application	Infrared remote controller with LCD display		Infrared remote controller for learning	
ROM size	3072 × 16 bits	4096 × 16 bits	4096 × 16 bits	7936 × 16 bits
RAM size	336 × 4 bits		336 × 4 bits	
Static RAM	—	—	4096 × 4 bits	2048 × 4 bits
LCD controller/driver	136 segments maximum		None	
Carrier generator for infrared remote-controller signal	On-chip		On-chip	
Infrared remote-controller signal receiver preamplifier	None		On-chip	
I/O port	19		28	
External interrupt	1		1	
Analog input	A/D converter: 4 channels (8-bit resolution)		—	
Timer	2 channels		4 channel	
Serial interface	1 channel		1 channel	
Watchdog timer	On-chip ($\overline{\text{WDOUT}}$ output)		On-chip ($\overline{\text{WDOUT}}$ output)	
Dual clock	On-chip		On-chip	
Instruction execution time	4 μs at 4 MHz		4 μs at 4 MHz	
Power supply voltage	2.2 to 5.5 V		2.2 to 5.5 V	
Package	80-pin plastic QFP		52-pin plastic QFP	
One-time PROM product	μPD17P207	μPD17P207	μPD17P203A	μPD17P204
Assembler	RA17K			
C compiler	*emIC-17K*			
Device file	AS17201	AS17207	AS17203	AS17204
SIMPLEHOST	ID17K			
In-circuit emulator	IE-17K or IE-17K-ET			
	SE-17207		SE-17204	
	EP-17201GF		EP-17203GC	

4-Bit Single Chip Microcontroller

■ 17K Series (Remote-Controller) Products (2/2)

Part number	**μPD17225**	**μPD17226**	**μPD17227**	**μPD17228**
Main application	Pre-set remote controller, toy, portable system			
ROM size	2048 × 16 bits	4096 × 16 bits	6144 × 16 bits	8192 × 16 bits
RAM size	111 × 4 bits		223 × 4 bits	
Carrier generator for infrared remote-controller signal	On-chip			
I/O port	CMOS input: 9 N-ch open-drain output: 8 CMOS I/O: 4			
External interrupt	1			
Timer	2 channels			
Watchdog timer	On-chip ($\overline{\text{WDOUT}}$ output)			
Low voltage detection circuit	On-chip ($\overline{\text{WDOUT}}$ output)			
P0C/P0D standby operation	The output level existing immediately before setting standby mode is maintained.			
Instruction execution time	High speed mode: 2 μs (at 8 MHz) Normal mode: 4 μs (at 8 MHz)			
Power supply voltage	2.2 to 3.6 V (4 μs, 2.0 to 3.6 V)			
Package	28-pin plastic SDIP, 28-pin plastic SOP			
One-time PROM product	μPD17P228★			
Assembler	RA17K			
C compiler	*emIC-17K*			
Device file	AS17225			
SIMPLEHOST	ID17K			
In-circuit emulator	IE-17K or IE-17K-ET			
	SE-17225			
	EP-17K28CT, EP-17K28GT			

★: Under development

4-Bit Single Chip Microcontroller

μPD6133, 6604, 63 Series

■ μPD6133 Series (Remote Controller) (1/3)

Part number	μPD6133	μPD6134	μPD61P34B
Main application	AV, household electric appliances		
ROM size	512 × 10 bits	1002 × 10 bits	
	Mask ROM		One time PROM
RAM size	32 × 4 bits		
Number of supported keys	32 (standard), 48 (when using key expansion pin), 96 (when expanded by diode)		
Operation clock frequency (f_x)	300 kHz to 1 MHz (ceramic oscillator)		
Instruction execution time	16 μs (at f_x = 500 kHz)		
Modulation carrier frequency	f_x, $f_x/2$, $f_x/8$, $f_x/12$, $f_x/16$, $f_x/24$, no carrier (high level)		
Timer	9-bit programmable timer		
POC circuit	Mask option		On-chip
Power supply voltage (V_{DD})	1.8 to 3.6 V, 2.2 to 3.6 V (when POC circuit is incorporated)		2.2 to 3.6 V
Package	20-pin plastic SOP, 20-pin plastic shrink SOP★		
Assembler	AS6133		
Emulator	EB-6133		

★: Under development

■ μPD6604 Series (Remote Controller) (2/3)

Part number	μPD6604	μPD66P04B
Main application	AV, household electric appliances	
ROM size	1002 × 10 bits	
	Mask ROM	One time PROM
RAM size	32 × 4 bits	
Number of supported keys	32 (standard), 48 (when using key expansion pin), 96 (when expanded by diode)	
Operation clock frequency (f_x)	300 kHz to 1 MHz (RC oscillator)	
Instruction execution time	16 μs (at f_x = 500 kHz)	
Modulation carrier frequency	f_x, $f_x/2$, $f_x/8$, $f_x/12$, $f_x/16$, $f_x/24$, no carrier (high level)	
Timer	9-bit programmable timer	
POC circuit	Mask option	On-chip
Power supply voltage (V_{DD})	1.8 to 3.6 V, 2.2 to 3.6 V (when POC circuit is incorporated)	2.2 to 3.6 V
Package	20-pin plastic SOP, 20-pin plastic shrink SOP (Only μPD66P04B is under development)	
Assembler	AS6133	
Emulator	EB-6133	

4-Bit Single Chip Microcontroller

■ μPD63 Series (Remote Controller) (3/3)

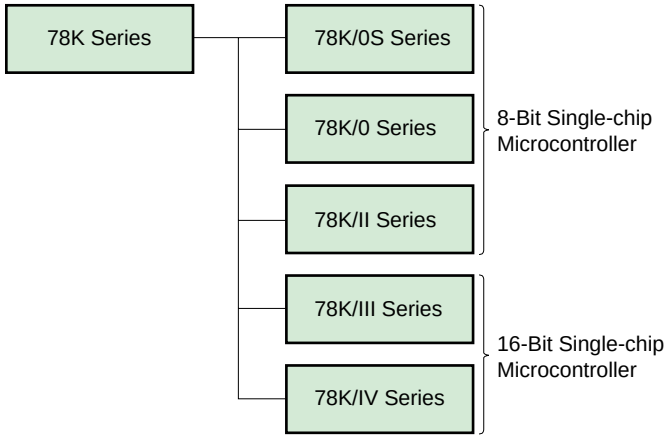
Part number	μPD63	μPD63A	μPD64	μPD6P4B
Main application	AV, household electric appliances			
ROM size	512 × 10 bits	768 × 10 bits	1002 × 10 bits	
	Mask ROM			One time PROM
RAM size	32 × 4 bits			
Number of supported keys	32 (standard), 48 (when using key expansion pin), 96 (when expanded by diode)			
Operation clock frequency (f_X)	2.4 MHz to 8 MHz (ceramic oscillator)			
Instruction execution time	16 μs (at f_X = 4 MHz)			
Modulation carrier frequency	f_X/8, f_X/16, f_X/64, f_X/96, f_X/128, f_X/192, no carrier (high level)			
Timer	9-bit programmable timer			
POC circuit	Mask option			On-chip
Power supply voltage (V_{DD})	1.8 to 3.6 V, 2.2 to 3.6 V (when POC circuit is incorporated)			2.2 to 3.6 V
Package	20-pin plastic SOP, 20-pin plastic shrink SOP★			
Assembler	AS6133			
Emulator	EB-6133			

★: Under development

8/16-Bit Single Chip Microcontroller

78K Series

■ 78K Series Configuration

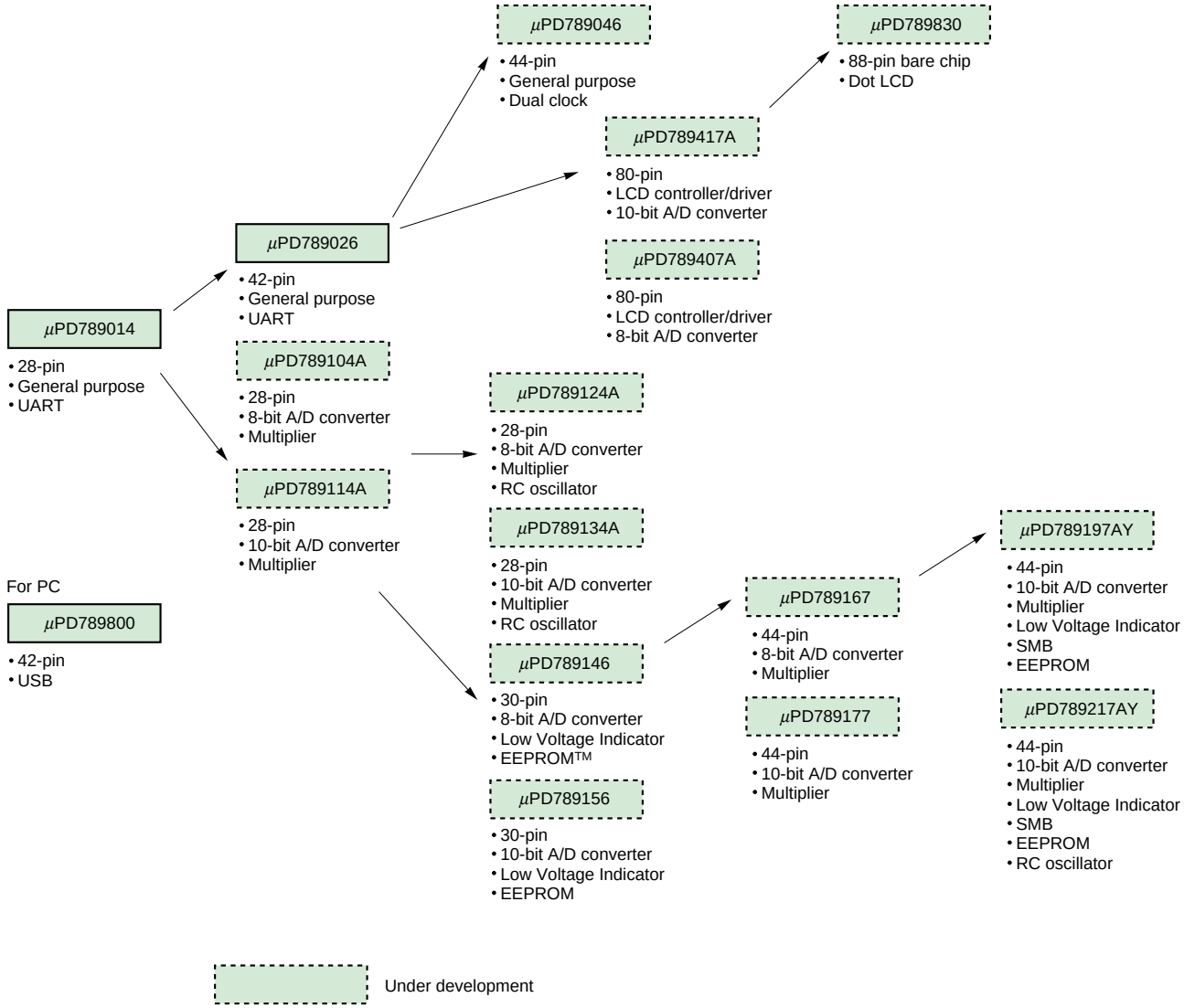


8/16-Bit Single Chip Microcontroller

■ 78K Series Product Evolution (1/6)

Subseries names are shown inside frames.

78K/0S series



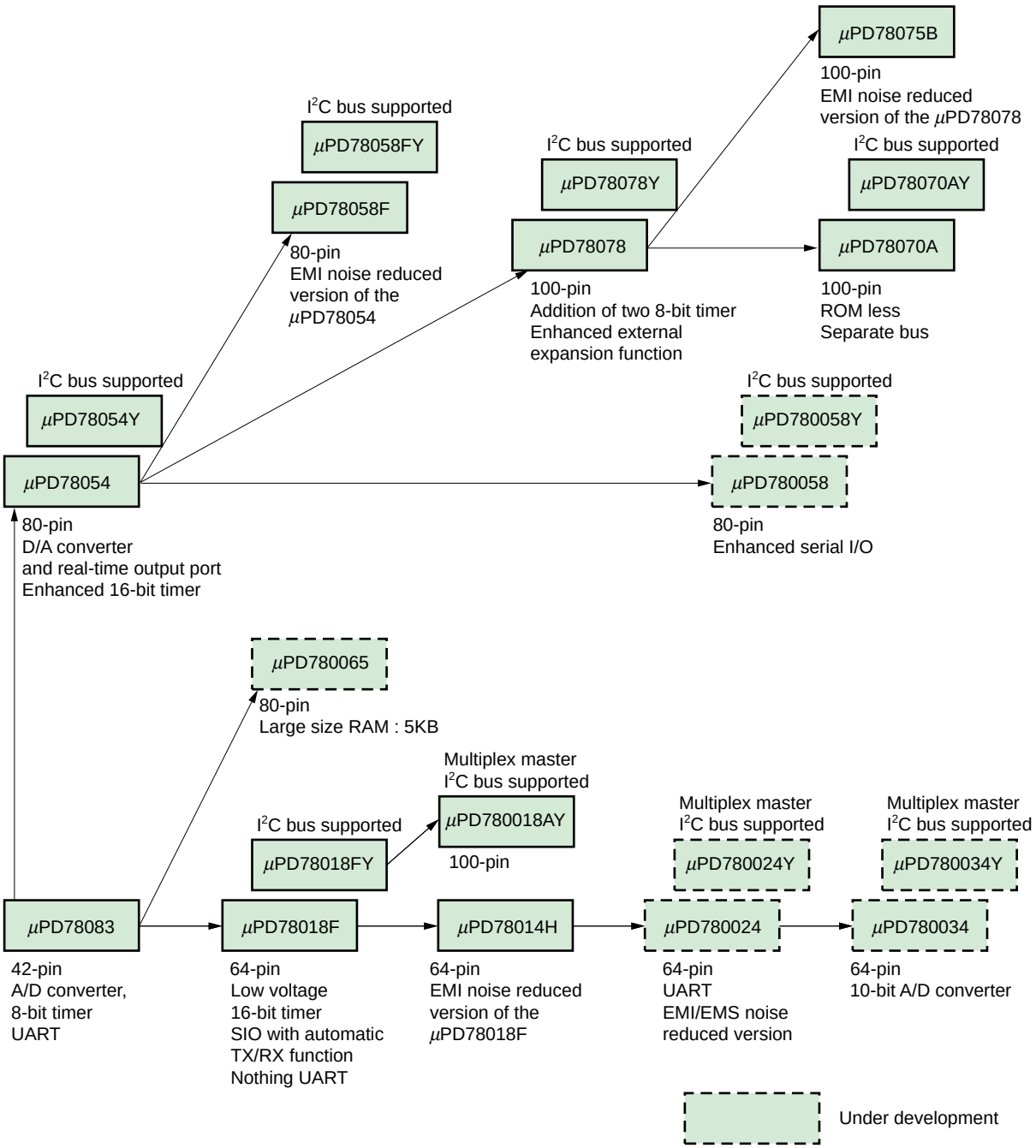
8/16-Bit Single Chip Microcontroller

■ 78K Series Product Evolution (2/6)

Subseries names are shown inside frames.

78K/0 series (1/3)

Controller series



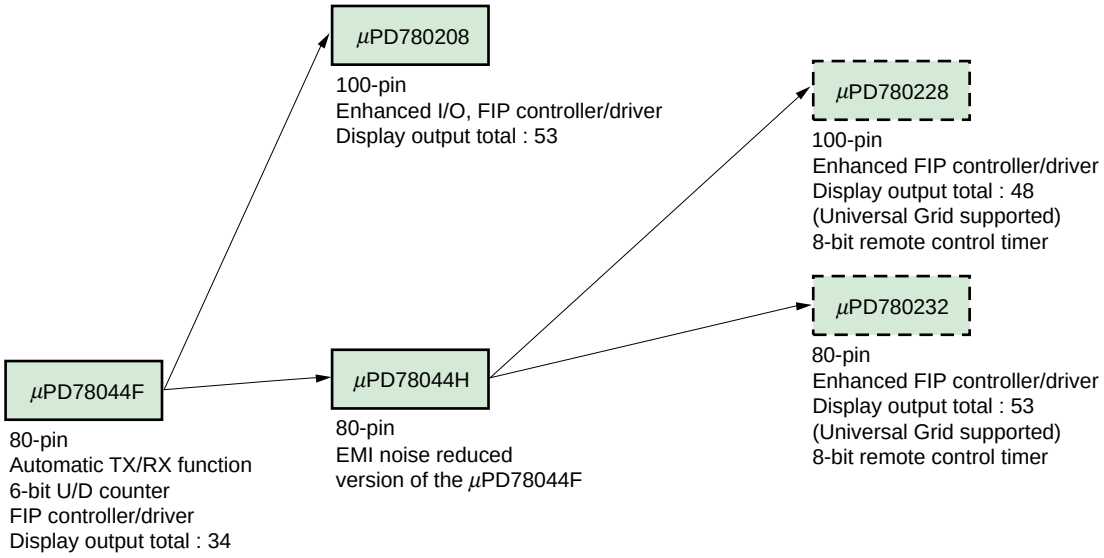
8/16-Bit Single Chip Microcontroller

■ 78K Series Product Evolution (3/6)

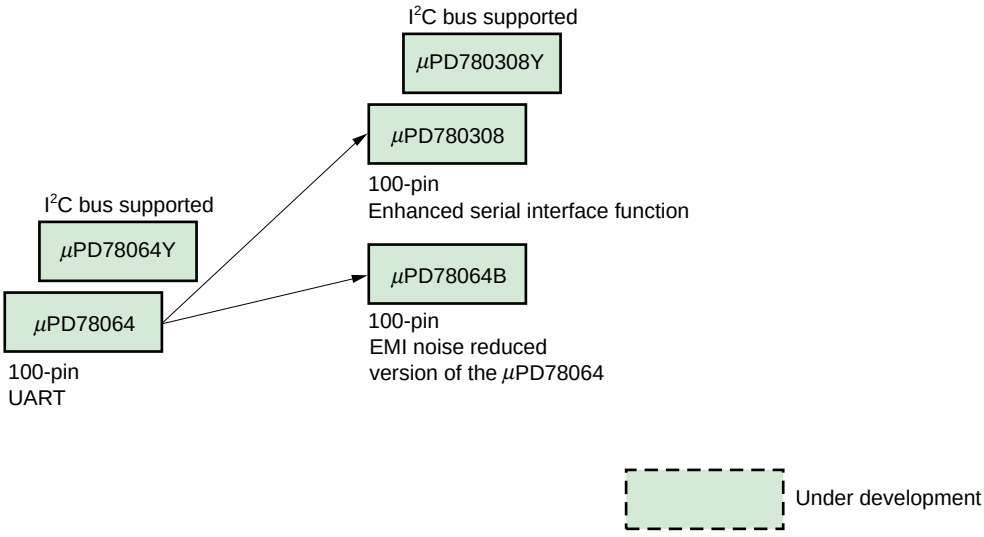
Subseries names are shown inside frames.

78K/0 series (2/3)

FIP™ drive series



LCD drive series



8/16-Bit Single Chip Microcontroller

■ 78K Series Product Evolution (4/6)

Subseries names are shown inside frames.

78K/0 series (3/3)

Inverter controller series

μPD780988
64-pin
10-bit A/D converter

Super low power consumption series

μPD780955
80-pin
Consumption current : 3.4 μA typical (operation)
1.1 μA typical (HALT)

Dashboard controller series

μPD780973
80-pin
Meter controller/driver

Bus controller series

μPD78098B
80-pin
IEBus controller

μPD780701Y
80-pin
(μPD780701Y : DCAN controller
μPD780702Y : IEBus controller
μPD78F0701Y : DCAN/IEBus controller)

μPD780833Y
80-pin
J1850 (CLASS2) controller

μPD780948
100-pin
DCAN controller

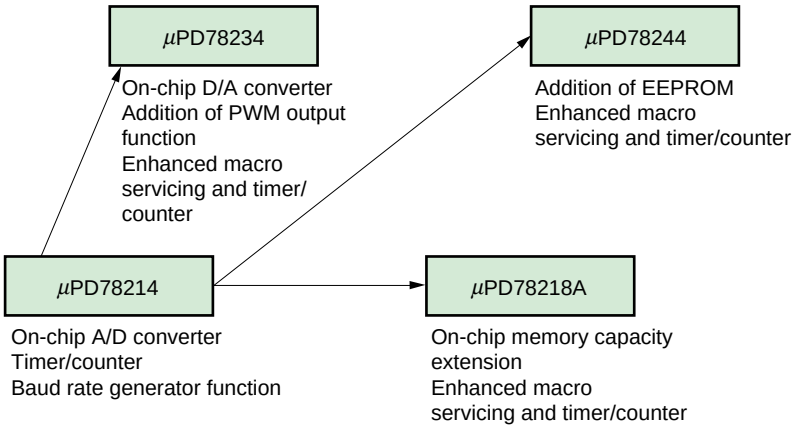
Under development

8/16-Bit Single Chip Microcontroller

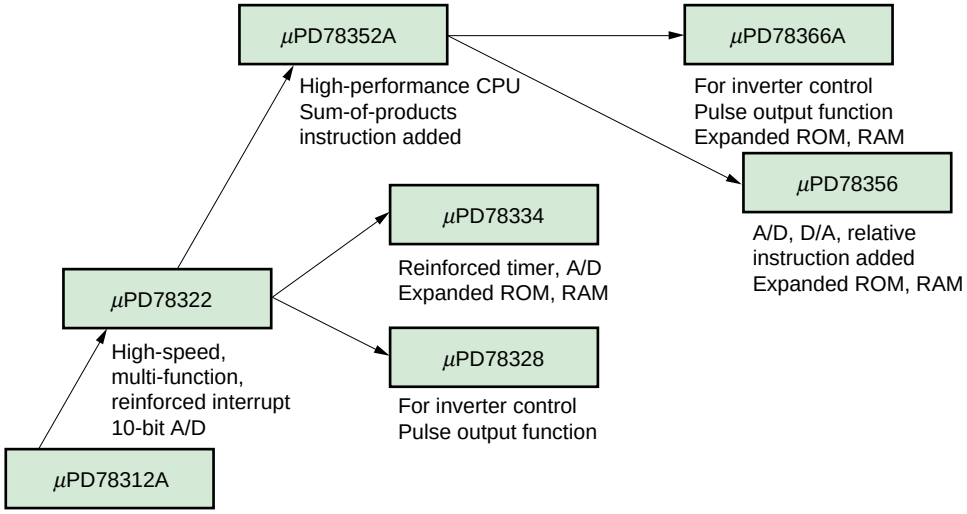
■ 78K Series Product Evolution (5/6)

Subseries names are shown inside frames.

78K/II series



78K/III series

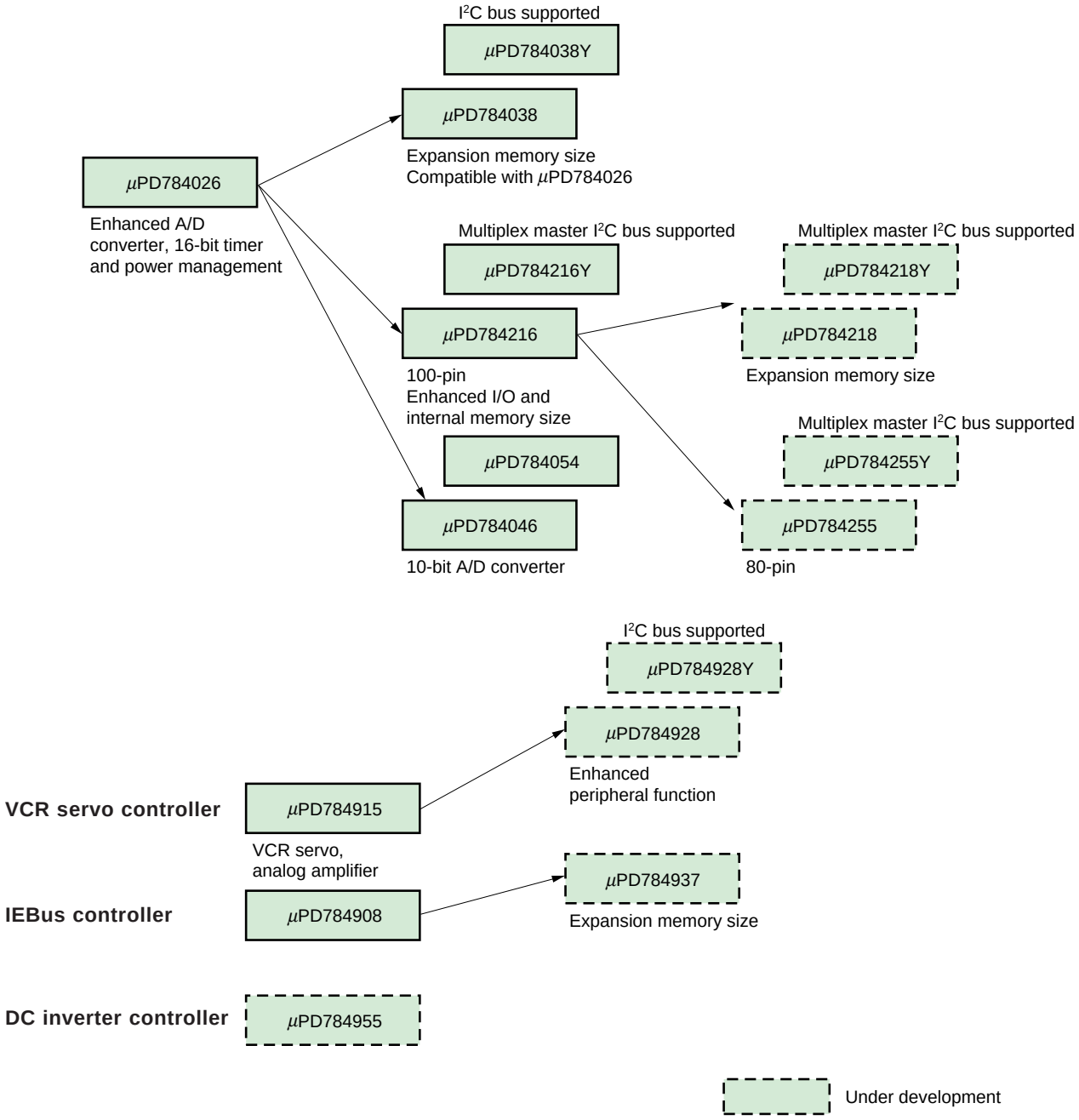


8/16-Bit Single Chip Microcontroller

■ 78K Series Product Evolution (6/6)

Subseries names are shown inside frames.

78K/IV series



8/16-Bit Single Chip Microcontroller

78K/0S Series

■ 78K/0S Series Products (1/12)

Subseries name			μPD789014 Subseries		
Part number			μPD789011	μPD789012	μPD78P9014
Number of basic instructions			48		
Minimum instruction execution time			0.4 μs (5.0 MHz)		
Instruction cycle			Selectable from 0.4 μs and 1.6 μs (5.0 MHz)		
Internal memory	ROM	Size (bytes)	2K	4K	8K (PROM)
	RAM	Size (bytes)	128		256
Address space			64K bytes		
Interrupt function		External	3		
		Internal	7		
I/O pin	CMOS I/O		22		
Timer			8-bit timer/event counter × 2 channels Watchdog timer × 1 channel		
Timer output			2		
Serial interface			3-wire/UART mode (selectable): 1 channel		
Standby function			STOP/HALT		
Operating power supply voltage range			V_{DD} = 1.8 to 5.5 V		
Package			• 28-pin plastic shrink DIP (400 mil) • 28-pin plastic SOP (375 mil)		
Assembler			RA78K0S		
C compiler			CC78K0S		
Real-time OS			MX78K0S		
Device file			DF789014		
In-circuit emulator			IE-78K0S-NS + IE-789014-NS-EM1		
			NP-28CT ∗ NP-28GT ∗		
Applications			Remote control, game machines, etc.		

∗: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

8/16-Bit Single Chip Microcontroller

■ 78K/0S Series Products (2/12)

Subseries name			μPD789026 Subseries				
Part number			μPD789022★	μPD789024★	μPD789025	μPD789026	μPD78F9026A★
Number of basic instructions			48				
Minimum instruction execution time			0.4 μs (5.0 MHz)				
Instruction cycle			Selectable from 0.4 μs and 1.6 μs (5.0 MHz)				
Internal memory	ROM	Size (bytes)	4K	8K	12K	16K	16K (Flash memory)
	RAM	Size (bytes)	256		512		
Address space			64K bytes				
Interrupt function		External	4				
		Internal	6				
I/O pin	CMOS I/O		34				
Timer			16-bit timer counter × 1 channel 8-bit timer/event counter × 1 channel Watchdog timer × 1 channel				
Timer output			2				
Serial interface			3-wire/UART mode (selectable): 1 channel				
Standby function			STOP/HALT				
Operating power supply voltage range			V_{DD} = 1.8 to 5.5 V				
Package			• 42-pin plastic shrink DIP (600 mil) • 44-pin plastic QFP (10 × 10 mm)				
Assembler			RA78K0S				
C compiler			CC78K0S				
Real-time OS			MX78K0S				
Device file			DF789026				
In-circuit emulator			IE-78K0S-NS + IE-789026-NS-EM1				
			NP-44GB ＊ NP-42CU ＊				
Applications			Compact household appliances, car accessories, air-conditioners, games, etc.				

＊: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0S Series Products (3/12)

Subseries name			**μPD789046 Subseries**	
Part number			**μPD789046★**	**μPD78F9046★**
Number of basic instructions			48	
Minimum instruction execution time			0.4 μs	
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs and 1.6 μs (5.0 MHz)	
	When subsystem clock is selected		122 μs (32.768 kHz)	
Internal memory	ROM	Size (bytes)	16K	16K (Flash memory)
	RAM	Size (bytes)	512	
Address space			64K bytes	
Interrupt function		External	4	
		Internal	8	
I/O pin	CMOS I/O		34	
Timer			16-bit timer counter × 1 channel 8-bit timer/event counter × 1 channel Watch timer × 1 channel Watchdog timer × 1 channel	
Timer output			2	
Serial interface			3-wire/UART mode (selectable): 1 channel	
Standby function			STOP/HALT	
Operating power supply voltage range			V_{DD} = 1.8 to 5.5 V	
Package			44-pin plastic QFP (10 × 10 mm)	
Assembler			RA78K0S	
C compiler			CC78K0S	
Real-time OS			MX78K0S	
Device file			DF789046	
In-circuit emulator			IE-78K0S-NS + IE-789046-NS-EM1	
			NP-44GB *	
Applications			Cordless phones, etc.	

*: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0S Series Products (4/12)

Subseries name			μPD789104A Subseries			μPD789114A Subseries			
Part number			μPD789101A★	μPD789102A★	μPD789104A★	μPD789111A★	μPD789112A★	μPD789114A★	μPD78F9116A★
Number of basic instructions			48						
Minimum instruction execution time			0.4 μs (5.0 MHz)						
Instruction cycle			Selectable from 0.4 μs and 1.6 μs (5.0 MHz)						
System clock			Ceramic oscillator						
Internal memory	ROM	Size (bytes)	2K	4K	8K	2K	4K	8K	16K (Flash memory)
	RAM	Size (bytes)	256						
Address space			64K bytes						
Interrupt function		External	3						
		Internal	7						
Multiplier			8 × 8 = 16 bits						
I/O pin	CMOS input		4						
	CMOS I/O		12						
	N-ch open-drain I/O		4						
Timer			16-bit timer counter × 1 channel 8-bit timer/event counter × 1 channel Watchdog timer × 1 channel						
Timer output			2						
Serial interface			3-wire/UART mode (selectable): 1 channel						
A/D converter			8-bit resolution × 4 channels			10-bit resolution × 4 channels			
Standby function			STOP/HALT						
Power supply voltage range			V_{DD} = 1.8 to 5.5 V						
Package			• 28-pin plastic shrink DIP (400 mil) • 30-pin plastic shrink SOP (300 mil)						
Assembler			RA78K0S						
C compiler			CC78K0S						
Real-time OS			MX78K0S						
Device file			DF789136						
In-circuit emulator			IE-78K0S-NS + IE-789136-NS-EM1						
			NP-28CT ∗ NP-36GS ∗★ + NGS-30 ∗★						
Applications			Cleaners, washing machines, refrigerators, etc.						

∗: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0S Series Products (5/12)

Subseries name			μPD789124A Subseries			μPD789134A Subseries			
Part number			μPD789121A★	μPD789122A★	μPD789124A★	μPD789131A★	μPD789132A★	μPD789134A★	μPD78F9136A★
Number of basic instructions			48						
Minimum instruction execution time			0.5 μs (4.0 MHz)						
Instruction cycle			Selectable from 0.5 μs and 2.0 μs (4.0 MHz)						
System clock			RC oscillator						
Internal memory	ROM	Size (bytes)	2K	4K	8K	2K	4K	8K	16K (Flash memory)
	RAM	Size (bytes)	256						
Address space			64K bytes						
Interrupt function		External	3						
		Internal	7						
Multiplier			8 × 8 = 16 bits						
I/O pin	CMOS input		4						
	CMOS I/O		12						
	N-ch open-drain I/O		4						
Timer			16-bit timer counter × 1 channel 8-bit timer/event counter × 1 channel Watchdog timer × 1 channel						
Timer output			2						
Serial interface			3-wire serial I/O mode/UART mode (selectable): 1 channel						
A/D converter			8-bit resolution × 4 channels			10-bit resolution × 4 channels			
Standby function			STOP/HALT						
Power supply voltage range			V_{DD} = 1.8 to 5.5 V						
Package			• 28-pin plastic shrink DIP (400 mil) • 30-pin plastic shrink SOP (300 mil)						
Assembler			RA78K0S						
C compiler			CC78K0S						
Real-time OS			MX78K0S						
Device file			DF789136★						
In-circuit emulator			IE-78K0S-NS + IE-789136-NS-EM1★						
			NP-28CT ∗ NP-36GS ∗★ + NGS-30 ∗★						
Applications			Battery chargers, etc.						

∗: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0S Series Products (6/12)

Subseries name			μPD789146 Subseries		μPD789156 Subseries		
Part number			μPD789144★	μPD789146★	μPD789154★	μPD789156★	μPD78F9156★
Number of basic instructions			48				
Minimum instruction execution time			0.4 μs (5.0 MHz)				
Instruction cycle			Selectable from 0.4 μs and 1.6 μs (5.0 MHz)				
Internal memory	ROM	Size (bytes)	8K	16K	8K	16K	16K (Flash memory)
	RAM	Size (bytes)	256				
	EEPROM	Size (bytes)	256				
Address space			64K bytes				
Interrupt function		External	3				
		Internal	9				
Multiplier			8 × 8 =16 bits				
I/O pin	CMOS input		4				
	CMOS I/O		12				
	N-ch open-drain I/O		4				
Timer			16-bit timer counter × 1 channel 8-bit timer/event counter × 1 channel Watchdog timer × 1 channel				
Timer output			2				
Serial interface			3-wire/UART mode (selectable): 1 channel				
A/D converter			8-bit resolution × 4 channels		10-bit resolution × 4 channels		
Low voltage indicator			Indicate a level of supply voltage				
Standby function			STOP/HALT				
Power supply voltage range			V_{DD} = 1.8 to 5.5 V				
Package			30-pin plastic shrink SOP (300 mil)				
Assembler			RA78K0S				
C compiler			CC78K0S				
Real-time OS			MX78K0S				
Device file			DF789156★				
In-circuit emulator			IE-78K0S-NS + IE-789156-NS-EM1★				
			NP-36GS *★ + NGS-30*★				
Applications			Side air bags, etc.				

*: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0S Series Products (7/12)

Subseries name			μPD789167 Subseries		μPD789177 Subseries		
Part number			μPD789166★	μPD789167★	μPD789176★	μPD789177★	μPD78F9177★
Number of basic instructions			48				
Minimum instruction execution time			0.4 μs (5.0 MHz)				
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs and 1.6 μs (5.0 MHz)				
	When subsystem clock is selected		122 μs (32.768 kHz)				
System clock			Ceramic oscillator				
Internal memory	ROM	Size (bytes)	16K	24K	16K	24K	24K (Flash memory)
	RAM	Size (bytes)	512				
Address space			64K bytes				
Interrupt function		External	4				
		Internal	11				
Multiplier			8 × 8 =16 bits				
I/O pin	CMOS input		8				
	CMOS I/O		17				
	N-ch open-drain I/O		6				
Timer			16-bit timer counter × 1 channel 8-bit timer counter × 1 channel 8-bit timer/event counter × 2 channels Watch timer × 1 channel Watchdog timer × 1 channel				
Timer output			4				
Serial interface			3-wire/UART mode (selectable): 1 channel				
A/D converter			8-bit resolution × 8 channels		10-bit resolution × 8 channels		
Standby function			STOP/HALT				
Power supply voltage range			V_{DD} = 1.8 to 5.5 V				
Package			44-pin plastic QFP (10 × 10 mm)				
Assembler			RA78K0S				
C compiler			CC78K0S				
Real-time OS			MX78K0S				
Device file			DF789197★				
In-circuit emulator			IE-78K0S-NS + IE-789197-NS-EM1★				
			NP-44GB ＊				
Applications			Power windows, keyless entries, battery management units, side air bags, etc.				

＊: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0S Series Products (8/12)

Subseries name			μPD789197AY Subseries		
Part number			μPD789196AY★	μPD789197AY★	μPD78F9197AY★
Number of basic instructions			48		
Minimum instruction execution time			0.4 μs (5.0 MHz)		
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs and 1.6 μs (5.0 MHz)		
	When subsystem clock is selected		122 μs (32.768 kHz)		
System clock			Ceramic oscillator		
Internal memory	ROM	Size (bytes)	16K	24K	24K (Flash memory)
	RAM	Size (bytes)	512		
	EEPROM	Size (bytes)	128		
Address space			64K bytes		
Interrupt function		External	4		
		Internal	15		
Multiplier			8 × 8 =16 bits		
I/O pin	CMOS input		8		
	CMOS I/O		17		
	N-ch open-drain I/O		6		
Timer			16-bit timer counter × 1 channel 8-bit timer counter × 1 channel 8-bit timer/event counter × 2 channels Watch timer × 1 channel Watchdog timer × 1 channel		
Timer output			4		
Serial interface			3-wire/UART mode (selectable): 1 channel SMB: 1 channel		
A/D converter			10-bit resolution × 8 channels		
Low voltage indicator			Indicate a level of supply voltage		
Standby function			STOP/HALT		
Power supply voltage range			V_{DD} = 1.8 to 5.5 V		
Package			• 44-pin plastic QFP (10 × 10 mm) • 48-pin plastic TQFP (fine pitch) (7 × 7 mm)		
Assembler			RA78K0S		
C compiler			CC78K0S		
Real-time OS			MX78K0S		
Device file			DF789197★		
In-circuit emulator			IE-78K0S-NS + IE-789197-NS-EM1★		
			NP-44GB ＊ NP-48GA ＊		
Applications			Battery management units, etc.		

＊: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0S Series Products (9/12)

Subseries name			μPD789217AY Subseries		
Part number			μPD789216AY★	μPD789217AY★	μPD78F9217AY★
Number of basic instructions			48		
Minimum instruction execution time			0.4 μs (5.0 MHz)		
Instruction cycle	When main system clock is selected		Selectable from 0.5 μs and 2.0 μs (4.0 MHz)		
	When subsystem clock is selected		122 μs (32.768 kHz)		
System clock			RC oscillator		
Internal memory	ROM	Size (bytes)	16K	24K	32K
	RAM	Size (bytes)	512		
	EEPROM	Size (bytes)	128		
Address space			64K bytes		
Interrupt function		External	4		
		Internal	15		
Multiplier			8 × 8 =16 bits		
I/O pin	CMOS input		8		
	CMOS I/O		17		
	N-ch open-drain I/O		6		
Timer			16-bit timer counter × 1 channel 8-bit timer counter × 1 channel 8-bit timer/event counter × 2 channels Watch timer × 1 channel Watchdog timer × 1 channel		
Timer output			4		
Serial interface			3-wire/UART mode (selectable): 1 channel SMB: 1 channel		
A/D converter			10-bit resolution × 8 channels		
Low voltage indicator			Indicate a level of supply voltage		
Standby function			STOP/HALT		
Power supply voltage range			V_{DD} = 1.8 to 5.5 V		
Package			• 44-pin plastic QFP (10 × 10 mm) • 48-pin plastic TQFP (fine pitch) (7 × 7 mm)		
Assembler			RA78K0S		
C compiler			CC78K0S		
Real-time OS			MX78K0S		
Device file			DF789197★		
In-circuit emulator			IE-78K0S-NS + IE-789197-NS-EM1★		
			NP-44GB ＊ NP-48GA ＊		
Applications			Battery management units, etc.		

＊: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0S Series Products (10/12)

Subseries name			μPD789407A Subseries			μPD789417A Subseries			
Part number			μPD789405A★	μPD789406A★	μPD789407A★	μPD789415A★	μPD789416A★	μPD789417A★	μPD78F9418A★
Number of basic instructions			48						
Minimum instruction execution time			0.4 μs (5.0 MHz)						
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs and 1.6 μs (5.0 MHz)						
	When subsystem clock is selected		122 μs (32.768 kHz)						
Internal memory	ROM	Size (bytes)	12K	16K	24K	12K	16K	24K	32K (Flash memory)
	RAM	Size (bytes)	512						
	LCD data RAM	Size (bytes)	28						
Address space			64K bytes						
Interrupt function		External	5						
		Internal	12						
I/O pin	CMOS input		7						
	CMOS I/O		32						
	N-ch open-drain I/O		4						
Timer			16-bit timer counter × 1 channel 8-bit timer counter × 1 channel 8-bit timer/event counter × 2 channels Watch timer × 1 channel Watchdog timer × 1 channel						
Timer output			2						
Serial interface			3-wire/UART mode (selectable): 1 channel						
A/D converter			8-bit resolution × 7 channels			10-bit resolution × 7 channels			
LCD controller/driver			Number of segment signal outputs: 28 maximum Number of common signal outputs: 4 maximum Bias: 1/2 or 1/3 (switchable)						
Standby function			STOP/HALT						
Power supply voltage range			V_{DD} = 1.8 to 5.5 V						
Package			• 80-pin plastic QFP (14 × 14 mm) • 80-pin plastic TQFP (fine pitch) (12 × 12 mm)						
Assembler			RA78K0S						
C compiler			CC78K0S						
Real-time OS			MX78K0S						
Device file			DF789418						
In-circuit emulator			IE-78K0S-NS + IE-789418-NS-EM1						
			NP-80GC ＊ NP-80GK ＊						
Applications			APS compact cameras, etc.						

＊: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0S Series Products (11/12)

Subseries name			**μPD789800 Subseries**	
Part number			**μPD789800**	**μPD78F9801★**
Number of basic instructions			48	
Minimum instruction execution time			0.33 μs (6.0 MHz)	
Instruction cycle			Selectable from 0.33 μs and 1.33 μs (6.0 MHz)	
Internal memory	ROM	Size (bytes)	8K	16K (Flash memory)
	RAM	Size (bytes)	256	
Address space			64K bytes	
Interrupt function		External	2	
		Internal	10	
I/O pin	CMOS I/O		31 (18 ports of the above CMOS I/O ports can be switched to N-ch open-drain I/O ports)	
Timer			8-bit timer × 1 channel 8-bit timer/event counter × 1 channel Watchdog timer × 1 channel	
Serial interface			USB (Universal Serial Bus): 1 channel 3-wire: 1 channel	
Regulator			Incorporated (V_{REG} = 3.3 ±0.3 V)	
Standby function			STOP/HALT	
Power supply voltage range			V_{DD} = 4.0 to 5.5 V	
Operating ambient temperature			T_A = –40 to +85°C (When the USB is not operating) T_A = 0 to 70°C (When the USB is operating)	
Package			• 42-pin plastic shrink DIP (600 mil) • 44-pin plastic QFP (10 × 10 mm)	
Assembler			RA78K0S	
C compiler			CC78K0S	
Real-time OS			MX78K0S	
Device file			DF789801	
In-circuit emulator			IE-78K0S-NS + IE-789801-NS-EM1	
			NP-42CU ＊ NP-44GB ＊	
Applications			USB keyboards, etc.	

＊: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0S Series Products (12/12)

Subseries name			**μPD789830 Subseries**	
Part number			**μPD789830★**	**μPD78F9831★**
Number of basic instructions			48	
Minimum instruction execution time			0.4 μs	
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs and 1.6 μs (5.0 MHz)	
	When subsystem clock is selected		122 μs (32.768 kHz)	
Internal memory	ROM	Size (bytes)	24K	48K (Flash memory)
	RAM	Size (bytes)	1024	2048
	LCD data RAM	Size (bytes)	80	
Address space			64K bytes	
Interrupt function		External	3	5
		Internal	11	
I/O pin	CMOS I/O		30	38
Timer			16-bit timer counter × 1 channel 8-bit timer counter × 1 channel Watch timer × 1 channel Watchdog timer × 1 channel	
Timer output			2	
Serial interface			UART: 1 channel	
LCD controller/driver			Segment signal output: 40 maximum Common signal output: 16 maximum	
Standby function			STOP/HALT	
Operating power supply voltage range			V_{DD} = 2.7 to 5.5 V	
Delivery form			88-pin bare chip	100-pin plastic LQFP (14 × 14 mm)
Assembler			RA78K0S	
C compiler			CC78K0S	
Real-time OS			MX78K0S	
Device file			DF789831	
In-circuit emulator			IE-78K0S-NS + IE-789831-NS-EM1	
			NP-100GC *	
Applications			Card readers, etc.	

*: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

78K/0 Series

■ 78K/0 Series Products (1/32)

Subseries name			μPD78018F Subseries							
Part number			μPD78011F	μPD78012F	μPD78013F	μPD78014F	μPD78015F	μPD78016F	μPD78018F	μPD78P018F
Number of basic instructions			63							
Minimum instruction execution time			0.4 μs (10.0 MHz)							
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, and 6.4 μs (10.0 MHz)							
	When subsystem clock is selected		122 μs (32.768 kHz)							
Internal memory	ROM	Size (bytes)	8K	16K	24K	32K	40K	48K	60K	60K (PROM)
	RAM	Size (bytes)	544		1056		1568		2080	
Address space			64K bytes							
Interrupt function		External	4							
		Internal	10							
I/O pin	CMOS input		2							
	CMOS I/O		47							
	N-ch open-drain I/O		4							
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 2 channels Clock timer × 1 channel Watchdog timer × 1 channel							
Timer output			3 (PWM output: 1)							
Serial interface			3-wire/SBI/2-wire mode (selectable): 1 channel 3-wire (with function for automatic transmission and reception of up to 32 bytes): 1 channel							
A/D converter			8-bit resolution × 8 channels							
Others			Clock output function Buzzer output function							
Standby function			STOP/HALT							
Operating power supply voltage range			1.8 to 5.5 V							
Package			• 64-pin plastic shrink DIP (750 mil) • 64-pin plastic QFP (14 × 14 mm, 0.8-mm pitch) • 64-pin plastic QFP (12 × 12 mm, 0.65-mm pitch)							∗1
Assembler			RA78K0							
C compiler			CC78K0							
Real-time OS			MX78K0	RX78K0, MX78K0						
Integrated debugger			ID78K0 or ID78K0-NS							
			DF78014							
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)							
			With IE-78001-R-A: IE-78014-R-EM-A, with IE-78K0-NS: IE-78018-NS-EM1							
			With IE-78001-R-A: EP-78240CW-R, EP-78240GC-R, EP-78012GK-R With IE-78K0-NS:NP-64CW ∗2, NP-64GC ∗2, NP-64GK ∗2							
Applications			VCRs, TVs, audio equipments, etc.							

∗1: • 64-pin plastic shrink DIP (750 mil)
• 64-pin plastic QFP (14 × 14 mm, 0.8-mm pitch)
• 64-pin plastic QFP (12 × 12 mm, 0.65-mm pitch)
• 64-pin ceramic shrink DIP with window (750 mil)
• 64-pin ceramic WQFN (14 × 14 mm)

∗2: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (2/32)

Subseries name			μPD78018FY Subseries							
Part number			μPD78011FY	μPD78012FY	μPD78013FY	μPD78014FY	μPD78015FY	μPD78016FY	μPD78018FY	μPD78P018FY
Number of basic instructions			63							
Minimum instruction execution time			0.4 μs (10.0 MHz)							
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, and 6.4 μs (10.0 MHz)							
	When subsystem clock is selected		122 μs (32.768 kHz)							
Internal memory	ROM	Size (bytes)	8K	16K	24K	32K	40K	48K	60K	60K (PROM)
	RAM	Size (bytes)	544		1056		1568		2080	
Address space			64K bytes							
Interrupt function		External	4							
		Internal	10							
I/O pin	CMOS input		2							
	CMOS I/O		47							
	N-ch open-drain I/O		4							
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 2 channels Clock timer × 1 channel Watchdog timer × 1 channel							
Timer output			3 (PWM output: 1)							
Serial interface			3-wire/2-wire/I²C bus mode (selectable): 1 channel 3-wire (with function for automatic transmission and reception of up to 32 bytes): 1 channel							
A/D converter			8-bit resolution × 8 channels							
Others			Clock output function Buzzer output function							
Standby function			STOP/HALT							
Operating power supply voltage range			1.8 to 5.5 V							
Package			• 64-pin plastic shrink DIP (750 mil) • 64-pin plastic QFP (14 × 14 mm, 0.8-mm pitch) • 64-pin plastic QFP (12 × 12 mm, 0.65-mm pitch) (except μPD78013FY, 78015FY, 78016FY, 78018FY)							∗1
Assembler			RA78K0							
C compiler			CC78K0							
Real-time OS			MX78K0		RX78K0, MX78K0					
Integrated debugger			ID78K0 or ID78K0-NS							
			DF78014							
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)							
			With IE-78001-R-A: IE-78014-R-EM-A, with IE-78K0-NS: IE-78018-NS-EM1							
			With IE-78001-R-A:EP-78240CW-R, EP-78240GC-R, EP-78012GK-R With IE-78K0-NS: NP-64CW *2, NP-64GC *2, NP-64GK *2							
Applications			VCRs, TVs, audio equipments, etc.							

∗1: • 64-pin plastic shrink DIP (750 mil)
• 64-pin plastic QFP (14 × 14 mm, 0.8-mm pitch)
• 64-pin plastic QFP (12 × 12 mm, 0.65-mm pitch)
• 64-pin ceramic shrink DIP with window (750 mil)
• 64-pin ceramic WQFN (14 × 14 mm)

∗2: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (3/32)

Subseries name			μPD78014H Subseries			
Part number			μPD78011H	μPD78012H	μPD78013H	μPD78014H
Number of basic instructions			63			
Minimum instruction execution time			0.4 μs (10.0 MHz)			
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, and 6.4 μs (10.0 MHz)			
	When subsystem clock is selected		122 μs (32.768 kHz)			
Internal memory	ROM	Size (bytes)	8K	16K	24K	32K
	RAM	Size (bytes)	544		1056	
Address space			64K bytes			
Interrupt function		External	4			
		Internal	10			
I/O pin	CMOS input		2			
	CMOS I/O		47			
	N-ch open-drain I/O		4			
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 2 channels Clock timer × 1 channel Watchdog timer × 1 channel			
Timer output			3 (PWM output: 1)			
Serial interface			3-wire/SBI/2-wire mode (selectable): 1 channel 3-wire (with function for automatic transmission and reception of up to 32 bytes): 1 channel			
A/D converter			8-bit resolution × 8 channels			
Others			Clock output function Buzzer output function			
Standby function			STOP/HALT			
Operating power supply voltage range			1.8 to 5.5 V			
Package			• 64-pin plastic shrink DIP (750 mil) • 64-pin plastic QFP (14 × 14 mm, 0.8-mm pitch) • 64-pin plastic QFP (12 × 12 mm, 0.65-mm pitch)			
Assembler			RA78K0			
C compiler			CC78K0			
Real-time OS			MX78K0	RX78K0, MX78K0		
Integrated debugger			ID78K0 or ID78K0-NS			
			DF78014			
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)			
			With IE-78001-R-A: IE-78014-R-EM-A, IE-78K0-NS: IE-78018-NS-EM1			
			With IE-78001-R-A: EP-78240CW-R, EP-78240GC-R, EP-78012GK-R With IE-78K0-NS:NP-64CW *, NP-64GC *, NP-64GK *			
Applications			Cellular phones, VCRs, audio equipments, cameras, home appliances, etc.			

*: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (4/32)

Subseries name			μPD780024 Subseries μPD780034 Subseries				
Part number			μPD780021★ μPD780031★	μPD780022★ μPD780032★	μPD780023★ μPD780033★	μPD780024★ μPD780034★	μPD78F0034★
Number of basic instructions			63				
Minimum instruction execution time			0.24 μs (8.38 MHz)				
Instruction cycle	When main system clock is selected		Selectable from 0.24 μs, 0.48 μs, 0.95 μs, 1.91 μs, and 3.81 μs (8.38 MHz)				
	When subsystem clock is selected		122 μs (32.768 kHz)				
Internal memory	ROM	Size (bytes)	8K	16K	24K	32K	32K (Flash memory)
	RAM	Size (bytes)	512		1024		
Address space			64K bytes				
Interrupt function		External	5				
		Internal	15				
I/O pin	CMOS input		8				
	CMOS I/O		39				
	N-ch open-drain I/O		4				
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 2 channels Clock timer × 1 channel Watchdog timer × 1 channel				
Timer output			3 (PWM output: 2)				
Serial interface			3-wire: 2 channels UART: 1 channel				
A/D converter			8-bit resolution × 8 channels (μPD780021, 780022, 780023, 780024) 10-bit resolution × 8 channels (μPD780031, 780032, 780033, 780034, 78F0034)				
Others			Clock output function Buzzer output function				
Standby function			STOP/HALT				
Operating power supply voltage range			1.8 to 5.5 V				
Package			• 64-pin plastic shirink DIP (750 mil) • 64-pin plastic QFP (14 × 14 mm, 0.8-mm pitch) • 64-pin plastic QFP (12 × 12 mm, 0.65-mm pitch)				
Assembler			RA78K0				
C compiler			CC78K0				
Real-time OS			RX78K0, MX78K0				
Integrated debugger			ID78K0 or ID78K0-NS				
			DF780024, DF780034				
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)				
			With IE-78001-R-A: IE-78K0-SL-P01, IE-780034-SL-EM4 With IE-78K0-NS: IE-780034-NS-EM1				
			With IE-78001-R-A: EP-78240CW-R, EP-78240GC-R, EP-78012GK-R With IE-78K0-NS: NP-64CW *, NP-64GC *, NP-64GK *				
Applications			VCRs, TVs, audio equipments, etc.				

*: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (5/32)

Subseries name			**μPD780024Y Subseries μPD780034Y Subseries**				
Part number			**μPD780021Y★ μPD780031Y★**	**μPD780022Y★ μPD780032Y★**	**μPD780023Y★ μPD780033Y★**	**μPD780024Y★ μPD780034Y★**	**μPD78F0034Y★**
Number of basic instructions			63				
Minimum instruction execution time			0.24 μs (8.38 MHz)				
Instruction cycle	When main system clock is selected		Selectable from 0.24 μs, 0.48 μs, 0.95 μs, 1.91 μs, and 3.81 μs (8.38 MHz)				
	When subsystem clock is selected		122 μs (32.768 kHz)				
Internal memory	ROM	Size (bytes)	8K	16K	24K	32K	32K (Flash memory)
	RAM	Size (bytes)	512		1024		
Address space			64K bytes				
Interrupt function		External	5				
		Internal	15				
I/O pin	CMOS input		8				
	CMOS I/O		39				
	N-ch open-drain I/O		4				
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 2 channels Clock timer × 1 channel Watchdog timer × 1 channel				
Timer output			3 (PWM output: 2)				
Serial interface			3-wire: 1 channel UART: 1 channel I^2C bus (supporting multi-master): 1 channel				
A/D converter			8-bit resolution × 8 channels (μPD780021Y, 780022Y, 780023Y, 780024Y) 10-bit resolution × 8 channels (μPD780031Y, 780032Y, 780033Y, 780034Y, 78F0034Y)				
Others			Clock output function Buzzer output function				
Standby function			STOP/HALT				
Operating power supply voltage range			1.8 to 5.5 V				
Package			• 64-pin plastic shirink DIP (750 mil) • 64-pin plastic QFP (14 × 14 mm, 0.8-mm pitch) • 64-pin plastic QFP (12 × 12 mm, 0.65-mm pitch)				
Assembler			RA78K0				
C compiler			CC78K0				
Real-time OS			RX78K0, MX78K0				
Integrated debugger			ID78K0 or ID78K0-NS				
			DF780024, DF780034				
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)				
			With IE-78001-R-A: IE-78K0-SL-P01, IE-780034-SL-EM4 With IE-78K0-NS: IE-780034-NS-EM1				
			With IE-78001-R-A: EP-78240CW-R, EP-78240GC-R, EP-78012GK-R With IE-78K0-NS: NP-64CW *, NP-64GC *, NP-64GK *				
Applications			VCRs, TVs, audio equipments, etc.				

*: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (6/32)

Subseries name			**μPD780065 Subseries**	
Part number			**μPD780065★**	**μPD78F0066★**
Number of basic instructions			63	
Minimum instruction execution time			0.24 μs (8.38 MHz)	
Instruction cycle	When main system clock is selected		Selectable from 0.24 μs, 0.48 μs, 0.96 μs, 1.92 μs, 3.84 μs and 7.68 μs (8.38 MHz)	
	When subsystem clock is selected		122 μs (32.768 kHz)	
Internal memory	ROM	Size (bytes)	40K	48K (Flash memory)
	RAM	Size (bytes)	5K	
Address space			64K bytes	
Interrupt function		External	4	
		Internal	13	
I/O pin	CMOS I/O		53	
Timer			16-bit timer/event counter × 1 channel 8-bit timer × 2 channels Watch timer × 1 channel Watchdog timer × 1 channel	
Timer output			3 (PWM output: 2)	
Serial interface			3-wire/2-wire mode (selectable): 1 channel 3-wire (with function for automatic transmission and reception of up to 32 bytes): 1 channel UART: 1 channel	
A/D converter			8-bit resolution × 8 channels	
Others			Clock output function Buzzer output function	
Standby function			STOP/HALT	
Operating power supply voltage range			V_{DD} = 2.7 to 5.5 V	
Package			80-pin plastic QFP (14 × 14 mm, 0.65-mm pitch)	
Assembler			RA78K0	
C compiler			CC78K0	
Real-time OS			RX78K0, MX78K0	
Integrated debugger			ID78K0-NS	
			DF780065	
In-circuit emulator			IE-78001-R-A or IE-78K0-NS	
			IE-78K0-NS-P01 + IE-780066-NS-EM4	
			With IE-78001-R-A: EP-78230GC-R + IE-78K0-R-EX1 With IE-78K0-NS: NP-80GC ＊	
Applications			Control of audio systems compatible with CD text, etc.	

＊: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (7/32)

Subseries name			**μPD780018AY Subseries**		
Part number			**μPD780016AY**	**μPD780018AY**	**μPD78P0018AY★**
Number of basic instructions			63		
Minimum instruction execution time			0.4 μs (5.0 MHz)		
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, and 6.4 μs (5.0 MHz)		
	When subsystem clock is selected		122 μs (32.768 kHz)		
Internal memory	ROM	Size (bytes)	48K	60K	60K (PROM)
	RAM	Size (bytes)	2080		
Address space			64K bytes		
Interrupt function		External	7		
		Internal	13		
I/O pin	CMOS input		9		
	CMOS I/O		79		
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 4 channels Clock timer × 1 channel Watchdog timer × 1 channel		
Timer output			5 (PWM output: 3)		
Serial interface			3-wire (with function for automatic transmission and reception of up to 32 bytes): 1 channel 3-wire (with time-division transmission function): 1 channel I^2C bus (supporting multi-master): 1 channel		
A/D converter			8-bit resolution × 8 channels		
Others			Clock output function Buzzer output function		
Standby function			Main STOP/Sub STOP/HALT		
Operating power supply voltage range			2.7 to 5.5 V		
Package			• 100-pin plastic QFP (14 × 20 mm, 0.65-mm pitch)		∗1
Assembler			RA78K0		
C compiler			CC78K0		
Real-time OS			RX78K0, MX78K0		
Integrated debugger			ID78K0 or ID78K0-NS		
			DF780018		
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)		
			With IE-78001-R-A: IE-780018-R-EM, with IE-78K0-NS: IE-780018-NS-EM1		
			With IE-78001-R-A: EP-78064GF-R, with IE-78K0-NS: NP-100GF ∗2		
Applications			Mini-component stereo systems, car stereos, etc.		

∗1: • 100-pin plastic QFP (14 × 20 mm, 0.65-mm pitch)
• 100-pin ceramic WQFN (14 × 20 mm)

∗2: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (8/32)

Subseries name			μPD78044F Subseries				
Part number			μPD78042F	μPD78043F	μPD78044F	μPD78045F	μPD78P048A
Number of basic instructions			63				
Minimum instruction execution time			0.4 μs (5.0 MHz)				
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, and 6.4 μs (5.0 MHz)				
	When subsystem clock is selected		122 μs (32.768 kHz)				
Internal memory	ROM	Size (bytes)	16K	24K	32K	40K	60K (PROM)
	RAM	Size (bytes)	624		1136		
Interrupt function		External	4				
		Internal	12				
I/O pin	CMOS input		2				
	CMOS I/O		27				
	N-ch open-drain I/O		5				
	P-ch open-drain I/O		16				
	P-ch open-drain output		18				
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 2 channels Clock timer × 1 channel Watchdog timer × 1 channel 6-bit up/down counter × 1 channel				
Timer output			3 (PWM output: 1)				
Serial interface			3-wire/SBI/2-wire mode (selectable): 1 channel 3-wire (with function for automatic transmission and reception of up to 64 bytes): 1 channel				
A/D converter			8-bit resolution × 8 channels				
FIP controller/driver			Number of segments: 9 to 24 Number of digits: 2 to 16				
Others			Clock output function Buzzer output function				
Standby function			STOP/HALT				
Operating power supply voltage range			2.7 to 5.5 V				
Package			• 80-pin plastic QFP (14 × 20 mm, 0.8-mm pitch)				∗1
Assembler			RA78K0				
C compiler			CC78K0				
Real-time OS			MX78K0		RX78K0, MX78K0		
Integrated debugger			ID78K0 or ID78K0-NS				
			DF78044				
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)				
			With IE-78001-R-A: IE-78044-R-EM, with IE-78K0-NS: IE-78048-NS-EM1				
			With IE-78001-R-A: EP-78130GF-R, with IE-78K0-NS: NP-80GF ∗2				
Applications			VCRs, audio equipments, microwave ovens, etc.				

∗1: • 80-pin plastic QFP (14 × 20 mm, 0.8-mm pitch)
• 80-pin ceramic WQFN (14 × 20 mm)
∗2: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (9/32)

Subseries name			μPD78044H Subseries		
Part number			**μPD78044H**	**μPD78045H**	**μPD78046H**
Number of basic instructions			63		
Minimum instruction execution time			0.4 μs (5.0 MHz)		
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, and 6.4 μs (5.0 MHz)		
	When subsystem clock is selected		122 μs (32.768 kHz)		
Internal memory	ROM	Size (bytes)	32K	40K	48K
	RAM	Size (bytes)	1136		
Interrupt function		External	4		
		Internal	10		
I/O pin	CMOS input		2		
	CMOS I/O		19		
	N-ch open-drain I/O		13		
	P-ch open-drain I/O		16		
	P-ch open-drain output		18		
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 2 channels Clock timer × 1 channel Watchdog timer × 1 channel		
Timer output			3 (PWM output: 1)		
Serial interface			3-wire: 1 channel		
A/D converter			8-bit resolution × 8 channels		
FIP controller/driver			Number of segments: 9 to 24 Number of digits: 2 to 16		
Others			Clock output function Buzzer output function		
Standby function			STOP/HALT		
Operating power supply voltage range			2.7 to 5.5 V		
Package			• 80-pin plastic QFP (14 × 20 mm, 0.8-mm pitch)		∗1
Assembler			RA78K0		
C compiler			CC78K0		
Real-time OS			RX78K0, MX78K0		
Integrated debugger			ID78K0 or ID78K0-NS		
			DF78044		
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS(with ID78K0-NS)		
			With IE-78001-R-A: IE-78044-R-EM, with IE-78K0-NS: IE-78048-NS-EM1		
			With IE-78001-R-A: EP-78130GF-R, with IE-78K0-NS: NP-80GF ∗2		
Applications			Mini-component stereo systems, microwave ovens, ECRs, etc.		

∗1: • 80-pin plastic QFP (14 × 20 mm, 0.8-mm pitch)
• 80-pin ceramic WQFN (14 × 20 mm)
∗2: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (10/32)

Subseries name			μPD780208 Subseries				
Part number			**μPD780204**	**μPD780205**	**μPD780206**	**μPD780208**	**μPD78P0208**
Number of basic instructions			63				
Minimum instruction execution time			0.4 μs (5.0 MHz)				
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, and 6.4 μs (5.0 MHz)				
	When subsystem clock is selected		122 μs (32.768 kHz)				
Internal memory	ROM	Size (bytes)	32K	40K	48K	60K	60K (PROM)
	RAM	Size (bytes)	1168		2192		
Interrupt function		External	4				
		Internal	12				
I/O pin	CMOS input		2				
	CMOS I/O		27				
	N-ch open-drain I/O		5				
	P-ch open-drain I/O		24				
	P-ch open-drain output		16				
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 2 channels Clock timer × 1 channel Watchdog timer × 1 channel				
Timer output			3 (PWM output: 1)				
Serial interface			3-wire/SBI/2-wire mode (selectable): 1 channel 3-wire (with function for automatic transmission and reception of up to 64 bytes): 1 channel				
A/D converter			8-bit resolution × 8 channels				
FIP controller/driver			Number of segments: 9 to 40 Number of digits: 2 to 16				
Others			Clock output function Buzzer output function				
Standby function			STOP/HALT				
Operating power supply voltage range			2.7 to 5.5 V				
Package			• 100-pin plastic QFP (14 × 20 mm, 0.65-mm pitch)				∗1
Assembler			RA78K0				
C compiler			CC78K0				
Real-time OS			RX78K0, MX78K0				
Integrated debugger			ID78K0 or ID78K0-NS				
			DF780208				
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)				
			With IE-78001-R-A: IE-780208-R-EM, with IE-78K0-NS: IE-780208-NS-EM1				
			With IE-78001-R-A: EP-78064GF-R, with IE-78K0-NS: NP-100GF ∗2				
Applications			VCRs, audio equipments, microwave ovens, etc.				

∗1: • 100-pin plastic QFP (14 × 20 mm, 0.65-mm pitch)
• 100-pin ceramic WQFN (14 × 20 mm)
∗2: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (11/32)

Subseries name			**μPD780228 Subseries**	
Part number			**μPD780226**	**μPD780228**
Number of basic instructions			63	
Minimum instruction execution time			0.4 μs (5.0 MHz)	
Instruction cycle			Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, and 6.4 μs (5.0 MHz)	
Internal memory	ROM	Size (bytes)	48K	60K
	RAM	Size (bytes)	2192	
Interrupt function		External	4	
		Internal	8	
I/O pin	CMOS input		8	
	CMOS I/O		16	
	N-ch open-drain I/O		16	
	P-ch open-drain I/O		24	
	P-ch open-drain output		8	
Timer			8-bit remote control timer × 1 channel 8-bit PWM timer × 2 channels Watchdog timer × 1 channel	
Serial interface			3-wire: 1 channel	
A/D converter			8-bit resolution × 8 channels	
FIP controller/driver			Number of Display outputs: 48	
Standby function			STOP/HALT	
Operating power supply voltage range			4.5 to 5.5 V	
Package			• 100-pin plastic QFP (14 × 20 mm, 0.65-mm pitch)	
Assembler			RA78K0	
C compiler			CC78K0	
Real-time OS			RX78K0, MX78K0	
Integrated debugger			ID78K0	
			DF780228	
In-circuit emulator			IE-78001-R-A	
			IE-78K0-SL-P01, IE-780228-SL-EM4	
			EP-78064GF-R	
Applications			Combined mini-component stereo systems, separate mini-component stereo systems, tuners, cassette decks, CD players, audio amplifiers, and car audio systems	

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (12/32)

Subseries name			**μPD780232 Subseries**	
Part number			**μPD780232★**	**μPD78F0233★**
Number of basic instructions			63	
Minimum instruction execution time			0.4 μs (5.0 MHz)	
Instruction cycle			Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs and 6.4 μs (5.0 MHz)	
Internal memory	ROM	Size (bytes)	16K	24K (Flash memory)
	RAM	Size (bytes)	768	
Interrupt function		External	2	
		Internal	12	
I/O pin	CMOS I/O		11	
	P-ch open-drain I/O		29	
	P-ch open-drain output		24	
Timer			8-bit remote control timer × 1 channel 8-bit timer × 2 channels Watchdog timer × 1 channel	
Timer outupt			3 (PWM output: 2)	
Serial interface			2-wire: 1 channel 3-wire (with function for automatic transmission and reception of up to 32 bytes): 1 channel	
A/D converter			8-bit resolution × 4 channels	
FIP controller/driver			Display outputs: 53	
Standby function			STOP/HALT	
Operating power supply voltage range			V_{DD} = 4.5 to 5.5 V	
Package			80-pin plastic QFP (14 × 14 mm, 0.65-mm pitch)	
Assembler			RA78K0	
C compiler			CC78K0	
Real-time OS			RX78K0, MX78K0	
Integrated debugger			ID78K0 or ID78K0-NS	
			DF780234★	
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)	
			IE-78K0-NS-P01 + IE-780233-NS-EM4★	
			With IE-78001-R-A: EP-78230GC-R + IE-78K0-R-EX1 With IE-78K0-NS: NP-80GC ＊	
Applications			Combined mini-component stereo systems, separate mini-component stereo systems, tuners, cassette tape decks, compact disc/mini disc players, audio amplifiers, etc.	

＊: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (13/32)

Subseries name			μPD78054 Subseries							
Part number			μPD78052	μPD78053	μPD78054	μPD78P054	μPD78055	μPD78056	μPD78058	μPD78P058
Number of basic instructions			63							
Minimum instruction execution time			0.4 μs (5.0 MHz)							
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, 6.4 μs, and 12.8 μs (5.0 MHz)							
	When subsystem clock is selected		122 μs (32.768 kHz)							
Internal memory	ROM	Size (bytes)	16K	24K	32K	32K (PROM)	40K	48K	60K	60K (PROM)
	RAM	Size (bytes)	544	1056					2080	
Address space			64K bytes							
Interrupt function		External	7							
		Internal	15							
I/O pin	CMOS input		2							
	CMOS I/O		63							
	N-ch open-drain I/O		4							
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 2 channels Clock timer × 1 channel Watchdog timer × 1 channel							
Timer output			3 (PWM output: 1)							
Serial interface			3-wire/SBI/2-wire mode (selectable): 1 channel 3-wire (with function for automatic transmission and reception of up to 32 bytes): 1 channel 3-wire/UART mode (selectable): 1 channel							
A/D converter			8-bit resolution × 8 channels							
D/A converter			8-bit resolution × 2 channels							
Others			Clock output function Buzzer output function							
Standby function			STOP/HALT							
Operating power supply voltage range			2.0 to 6.0 V							
Package			• 80-pin plastic QFP (14 × 14 mm, 0.65-mm pitch) • 80-pin plastic QFP (12 × 12 mm, 0.5-mm pitch)			∗1	• 80-pin plastic QFP (14 × 14 mm)		∗2	∗3
Assembler			RA78K0							
C compiler			CC78K0							
Real-time OS			MX78K0	RX78K0, MX78K0						
Integrated debugger			ID78K0 or ID78K0-NS							
			DF78054							
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)							
			With IE-78001-R-A: IE-780308-R-EM, with IE-78K0-NS: IE-780308-NS-EM1							
			With IE-78001-R-A: EP-78230GC-R, EP-78054GK-R With IE-78K0-NS: NP-80GC ∗4, NP-80GK ∗4							
Applications			Cellular phones, pagers, printers, AV equipments, air conditioners, cameras, PPCs, fuzzy home appliances, vending machines, etc.							

∗1: • 80-pin plastic QFP (14 × 14 mm, 0.65-mm pitch)
• 80-pin plastic QFP (12 × 12 mm, 0.5-mm pitch)
• 80-pin ceramic WQFN (14 × 14 mm)
∗2: • 80-pin plastic QFP (14 × 14 mm)
• 80-pin plastic QFP (12 × 12 mm)
∗3: • 80-pin plastic QFP (14 × 14 mm)
• 80-pin ceramic WQFN (14 × 14 mm)
∗4: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (14/32)

Subseries name			μPD78054Y Subseries						
Part number			μPD78052Y	μPD78053Y	μPD78054Y	μPD78055Y	μPD78056Y	μPD78058Y	μPD78P058Y
Number of basic instructions			63						
Minimum instruction execution time			0.4 μs (5.0 MHz)						
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, 6.4 μs, and 12.8 μs (5.0 MHz)						
	When subsystem clock is selected		122 μs (32.768 kHz)						
Internal memory	ROM	Size (bytes)	16K	24K	32K	40K	48K	60K	60K (PROM)
	RAM	Size (bytes)	544	1056				2080	
Address space			64K bytes						
Interrupt function		External	7						
		Internal	15						
I/O pin	CMOS input		2						
	CMOS I/O		63						
	N-ch open-drain I/O		4						
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 2 channels Clock timer × 1 channel Watchdog timer × 1 channel						
Timer output			3 (PWM output: 1)						
Serial interface			3-wire/I²C bus/2-wire mode (selectable): 1 channel 3-wire (with function for automatic transmission and reception of up to 32 bytes): 1 channel 3-wire/UART mode (selectable): 1 channel						
A/D converter			8-bit resolution × 8 channels						
D/A converter			8-bit resolution × 2 channels						
Others			Clock output function Buzzer output function						
Standby function			STOP/HALT						
Operating power supply voltage range			2.0 to 6.0 V						
Package			• 80-pin plastic QFP (14 × 14 mm, 0.65-mm pitch)			• 80-pin plastic QFP (14 × 14 mm)		∗1	∗2
Assembler			RA78K0						
C compiler			CC78K0						
Real-time OS			MX78K0	RX78K0, MX78K0					
Integrated debugger			ID78K0 or ID78K0-NS						
			DF78054						
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)						
			With IE-78001-R-A: IE-780308-R-EM, with IE-78K0-NS: IE-780308-NS-EM1						
			With IE-78001-R-A: EP-78230GC-R, EP-78054GK-R With IE-78K0-NS: NP-80GC *3, NP-80GK *3						
Applications			Cellular phones, pagers, printers, AV equipments, air conditioners, cameras, PPCs, fuzzy home appliances, vending machines, etc.						

∗1: • 80-pin plastic QFP (14 × 14 mm)
• 80-pin plastic QFP (12 × 12 mm)
∗2: • 80-pin plastic QFP (14 × 14 mm)
• 80-pin ceramic WQFN (14 × 14 mm)
∗3: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (15/32)

Subseries name			μPD78058F Subseries		
Part number			μPD78056F	μPD78058F	μPD78P058F
Number of basic instructions			63		
Minimum instruction execution time			0.4 μs (5.0 MHz)		
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, 6.4 μs, and 12.8 μs (5.0 MHz)		
	When subsystem clock is selected		122 μs (32.768 kHz)		
Internal memory	ROM	Size (bytes)	48K	60K	60K (PROM)
	RAM	Size (bytes)	1056	2080	
Address space			64K bytes		
Interrupt function		External	7		
		Internal	15		
I/O pin	CMOS input		2		
	CMOS I/O		63		
	N-ch open-drain I/O		4		
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 2 channels Clock timer × 1 channel Watchdog timer × 1 channel		
Timer output			3 (PWM output: 1)		
Serial interface			3-wire/SBI/2-wire mode (selectable): 1 channel 3-wire (with function for automatic transmission and reception of up to 32 bytes): 1 channel 3-wire/UART mode (selectable): 1 channel		
A/D converter			8-bit resolution × 8 channels		
D/A converter			8-bit resolution × 2 channels		
Others			Clock output function Buzzer output function		
Standby function			STOP/HALT		
Operating power supply voltage range			2.7 to 6.0 V		
Package			• 80-pin plastic QFP (14 × 14 mm, 0.65-mm pitch)	• 80-pin plastic QFP (14 × 14 mm, 0.65-mm pitch) • 80-pin plastic QFP (12 × 12 mm, 0.5-mm pitch)	• 80-pin plastic QFP (14 × 14 mm)
Assembler			RA78K0		
C compiler			CC78K0		
Real-time OS			RX78K0, MX78K0		
Integrated debugger			ID78K0 or ID78K0-NS		
			DF78054		
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)		
			With IE-78001-R-A: IE-780308-R-EM, with IE-78K0-NS: IE-780308-NS-EM1		
			With IE-78001-R-A: EP-78230GC-R, EP-78054GK-R With IE-78K0-NS: NP-80GC *, NP-80GK *		
Applications			Cellular phones, AV equipments, car stereo, cameras, PPCs, etc.		

*: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (16/32)

Subseries name			μPD78058FY Subseries		
Part number			μPD78056FY	μPD78058FY	μPD78P058FY
Number of basic instructions			63		
Minimum instruction execution time			0.4 μs (5.0 MHz)		
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, 6.4 μs, and 12.8 μs (5.0 MHz)		
	When subsystem clock is selected		122 μs (32.768 kHz)		
Internal memory	ROM	Size (bytes)	48K	60K	60K (PROM)
	RAM	Size (bytes)	1056	2080	
Address space			64K bytes		
Interrupt function		External	7		
		Internal	15		
I/O pin	CMOS input		2		
	CMOS I/O		63		
	N-ch open-drain I/O		4		
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 2 channels Clock timer × 1 channel Watchdog timer × 1 channel		
Timer output			3 (PWM output: 1)		
Serial interface			3-wire/I²C bus/2-wire mode (selectable): 1 channel 3-wire (with function for automatic transmission and reception of up to 32 bytes): 1 channel 3-wire/UART mode (selectable): 1 channel		
A/D converter			8-bit resolution × 8 channels		
D/A converter			8-bit resolution × 2 channels		
Others			Clock output function Buzzer output function		
Standby function			STOP/HALT		
Operating power supply voltage range			2.7 to 6.0 V		
Package			• 80-pin plastic QFP (14 × 14 mm, 0.65-mm pitch)	• 80-pin plastic QFP (14 × 14 mm, 0.65-mm pitch) • 80-pin plastic QFP (12 × 12 mm, 0.5-mm pitch)	• 80-pin plastic QFP (14 × 14 mm)
Assembler			RA78K0		
C compiler			CC78K0		
Real-time OS			RX78K0, MX78K0		
Integrated debugger			ID78K0 or ID78K0-NS		
			DF78054		
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)		
			With IE-78001-R-A: IE-780308-R-EM, with IE-78K0-NS: IE-780308-NS-EM1		
			With IE-78001-R-A: EP-78230GC-R, EP-78054GK-R With IE-78K0-NS: NP-80GC *, NP-80GK *		
Applications			Cellular phones, AV equipments, car stereo, cameras, PPCs, etc.		

*: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (17/32)

Subseries name			μPD780058 Subseries					
Part number			μPD780053★	μPD780054★	μPD780055★	μPD780056★	μPD780058★	μPD78F0058★
Number of basic instructions			63					
Minimum instruction execution time			0.4 μs (5.0 MHz)					
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, 6.4 μs, and 12.8 μs (5.0 MHz)					
	When subsystem clock is selected		122 μs (32.768 kHz)					
Internal memory	ROM	Size (bytes)	24K	32K	40K	48K	60K	60K (Flash memory)
	RAM	Size (bytes)	1056				2080	
Address space			64K bytes					
Interrupt function		External	7					
		Internal	15					
I/O pin	CMOS input		2					
	CMOS I/O		62					
	N-ch open-drain I/O		4					
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 2 channels Clock timer × 1 channel Watchdog timer × 1 channel					
Timer output			3 (PWM output: 1)					
Serial interface			3-wire/SBI/2-wire mode (selectable): 1 channel 3-wire (with function for automatic transmission and reception of up to 32 bytes): 1 channel 3-wire/UART mode (with time-division transmission function) (selectable): 1 channel					
A/D converter			8-bit resolution × 8 channels					
D/A converter			8-bit resolution × 2 channels					
Others			Clock output function Buzzer output function					
Standby function			STOP/HALT					
Operating power supply voltage range			1.8 to 5.5 V					
Package			• 80-pin plastic QFP (14 × 14 mm, 0.65-mm pitch) • 80-pin plastic TQFP (12 × 12 mm, 0.5-mm pitch)					
Assembler			RA78K0					
C compiler			CC78K0					
Real-time OS			RX78K0, MX78K0					
Integrated debugger			ID78K0 or ID78K0-NS					
			DF780058★					
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)					
			With IE-78001-R-A: IE-780308-R-EM, with IE-78K0-NS: IE-780308-NS-EM1					
			With IE-78001-R-A: EP-78230GC-R With IE-78K0-NS: NP-80GC ＊, NP-80GK ＊					
Applications			Cellular phones, pagers, printers, AV equipments, air conditioners, cameras, PPCs, fuzzy home appliances, vending machines, etc.					

＊: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (18/32)

Subseries name			μPD780058Y Subseries					
Part number			μPD780053Y★	μPD780054Y★	μPD780055Y★	μPD780056Y★	μPD780058Y★	μPD78F0058Y★
Number of basic instructions			63					
Minimum instruction execution time			0.4 μs (5.0 MHz)					
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, 6.4 μs, and 12.8 μs (5.0 MHz)					
	When subsystem clock is selected		122 μs (32.768 kHz)					
Internal memory	ROM	Size (bytes)	24K	32K	40K	48K	60K	60K (Flash memory)
	RAM	Size (bytes)	1056				2080	
Address space			64K bytes					
Interrupt function		External	7					
		Internal	15					
I/O pin	CMOS input		2					
	CMOS I/O		62					
	N-ch open-drain I/O		4					
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 2 channels Clock timer × 1 channel Watchdog timer × 1 channel					
Timer output			3 (PWM output: 1)					
Serial interface			3-wire/I²C bus/2-wire mode (selectable): 1 channel 3-wire (with function for automatic transmission and reception of up to 32 bytes): 1 channel 3-wire/UART mode (with time-division transmission function) (selectable): 1 channel					
A/D converter			8-bit resolution × 8 channels					
D/A converter			8-bit resolution × 2 channels					
Others			Clock output function Buzzer output function					
Standby function			STOP/HALT					
Operating power supply voltage range			1.8 to 5.5 V					
Package			• 80-pin plastic QFP (14 × 14 mm, 0.65-mm pitch) • 80-pin plastic TQFP (12 × 12 mm, 0.5-mm pitch)					
Assembler			RA78K0					
C compiler			CC78K0					
Real-time OS			RX78K0, MX78K0					
Integrated debugger			ID78K0 or ID78K0-NS					
			DF780058★					
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)					
			With IE-78001-R-A: IE-780308-R-EM, with IE-78K0-NS: IE-780308-NS-EM1					
			With IE-78001-R-A: EP-78230GC-R, EP-78054GK-R With IE-78K0-NS: NP-80GC *, NP-80GK *					
Applications			Cellular phones, pagers, printers, AV equipments, air conditioners, cameras, PPCs, fuzzy home appliances, vending machines, etc.					

*: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (19/32)

Subseries name			μPD78064 Subseries				μPD78064B Subseries	
Part number			μPD78062	μPD78063	μPD78064	μPD78P064	μPD78064B	μPD78P064B
Number of basic instructions			63					
Minimum instruction execution time			0.4 μs (5.0 MHz)					
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, 6.4 μs, and 12.8 μs (5.0 MHz)					
	When subsystem clock is selected		122 μs (32.768 kHz)					
Internal memory	ROM	Size (bytes)	16K	24K	32K	32K (PROM)	32K	32K (PROM)
	RAM	Size (bytes)	532	1044				
Interrupt function		External	6					
		Internal	14					
I/O pin	CMOS input		2					
	CMOS I/O		55					
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 2 channels Clock timer × 1 channel Watchdog timer × 1 channel					
Timer output			3 (PWM output: 1)					
Serial interface			3-wire/SBI/2-wire mode (selectable): 1 channel 3-wire/UART mode (selectable): 1 channel					
A/D converter			8-bit resolution × 8 channels					
LCD controller/driver			Number of segment signal outputs: 40 maximum Number of common signal outputs: 4 maximum Bias: 1/2 or 1/3 (switchable)					
Others			Clock output function Buzzer output function					
Standby function			STOP/HALT					
Operating power supply voltage range			2.0 to 6.0 V					
Package			• 100-pin plastic QFP (14 × 20 mm, 0.65-mm pitch) • 100-pin plastic QFP (14 × 14 mm, 0.5-mm pitch)					
Assembler			RA78K0					
C compiler			CC78K0					
Real-time OS			MX78K0	RX78K0, MX78K0				
Integrated debugger			ID78K0 or ID78K0-NS					
			DF78064					
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)					
			With IE-78001-R-A: IE-780308-R-EM, with IE-78K0-NS: IE-780308-NS-EM1					
			With IE-78001-R-A: EP-78064GF-R, EP-78064GC-R With IE-78K0-NS: NP-100GF *, NP-100GC *					
Applications			Cellular phones, CD players, cameras, etc.					

*: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (20/32)

Subseries name			μPD78064Y Subseries		
Part number			μPD78062Y	μPD78063Y	μPD78064Y
Number of basic instructions			63		
Minimum instruction execution time			0.4 μs (5.0 MHz)		
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, 6.4 μs, and 12.8 μs (5.0 MHz)		
	When subsystem clock is selected		122 μs (32.768 kHz)		
Internal memory	ROM	Size (bytes)	16K	24K	32K
	RAM	Size (bytes)	532	1044	
Interrupt function		External	6		
		Internal	14		
I/O pin	CMOS input		2		
	CMOS I/O		55		
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 2 channels Clock timer × 1 channel Watchdog timer × 1 channel		
Timer output			3 (PWM output: 1)		
Serial interface			3-wire/2-wire/I^2C bus mode (selectable): 1 channel 3-wire/UART mode (selectable): 1 channel		
A/D converter			8-bit resolution × 8 channels		
LCD controller/driver			Number of segment signal outputs: 40 maximum Number of common signal outputs: 4 maximum Bias: 1/2 or 1/3 (switchable)		
Others			Clock output function Buzzer output function		
Standby function			STOP/HALT		
Operating power supply voltage range			2.0 to 6.0 V		
Package			• 100-pin plastic QFP (14 × 20 mm, 0.65-mm pitch) • 100-pin plastic QFP (14 × 14 mm, 0.5-mm pitch)		
Assembler			RA78K0		
C compiler			CC78K0		
Real-time OS			MX78K0	RX78K0, MX78K0	
Integrated debugger			ID78K0 or ID78K0-NS		
			DF78064		
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)		
			With IE-78001-R-A: IE-780308-R-EM, with IE-78K0-NS: IE-780308-NS-EM1		
			With IE-78001-R-A: EP-78064GF-R, EP-78064GC-R With IE-78K0-NS: NP-100GF *, NP-100GC *		
Applications			Cellular phones, CD players, cameras, etc.		

*: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (21/32)

Subseries name			μPD780308 Subseries			μPD780308Y Subseries		
Part number			μPD780306	μPD780308	μPD78P0308★	μPD780306Y	μPD780308Y	μPD78P0308Y★
Number of basic instructions			63					
Minimum instruction execution time			0.4 μs (5.0 MHz)					
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, 6.4 μs, and 12.8 μs (5.0 MHz)					
	When subsystem clock is selected		122 μs (32.768 kHz)					
Internal memory	ROM	Size (bytes)	48K	60K		48K	60K	60K (PROM)
	RAM	Size (bytes)	2068					
Address space			64K bytes					
Interrupt function		External	6					
		Internal	15					
I/O pin	CMOS input		2					
	CMOS I/O		55					
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 2 channels Clock timer × 1 channel Watchdog timer × 1 channel					
Timer output			3 (PWM output: 1)					
Serial interface			3-wire/SBI/2-wire mode (selectable): 1 channel 3-wire/UART mode (selectable): 1 channel 3-wire (with time-division transmission function): 1 channel			3-wire/I²C bus/2-wire mode (selectable): 1 channel 3-wire/UART mode (selectable): 1 channel 3-wire (with time-division transmission function): 1 channel		
A/D converter			8-bit resolution × 8 channels					
LCD controller/driver			Number of segment signal outputs: 40 maximum Number of common signal outputs: 4 maximum Bias: 1/2 or 1/3 (switchable)					
Others			Clock output function Buzzer output function					
Standby function			STOP/HALT					
Operating power supply voltage range			2.0 to 5.5 V					
Package			• 100-pin plastic QFP (14 × 20 mm, 0.65-mm pitch) • 100-pin plastic LQFP (14 × 14 mm, 0.5-mm pitch)		∗1	• 100-pin plastic QFP (14 × 20 mm, 0.65-mm pitch) • 100-pin plastic LQFP (14 × 14 mm, 0.5-mm pitch)		∗1
Assembler			RA78K0					
C compiler			CC78K0					
Real-time OS			RX78K0, MX78K0					
Integrated debugger			ID78K0 or ID78K0-NS					
			DF780308 (Part number: μS××××DF78064)					
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)					
			With IE-78001-R-A: IE-780308-R-EM, with IE-78K0-NS: IE-780308-NS-EM1					
			With IE-78001-R-A: EP-78064GF-R, EP-78064GC-R With IE-78K0-NS: NP-100GF ∗2, NP-100GC ∗2					
Applications			Cellular phones, pagers, CD players, meters, etc.					

∗1: • 100-pin plastic QFP (14 × 20 mm, 0.65-mm pitch)
• 100-pin plastic QFP (14 × 14 mm, 0.5-mm pitch)
• 100-pin ceramic WQFN (14 × 20 mm)

∗2: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (22/32)

Subseries name			μPD78075B Subseries		μPD78078 Subseries		
Part number			μPD78074B	μPD78075B	μPD78076	μPD78078	μPD78P078
Number of basic instructions			63				
Minimum instruction execution time			0.4 μs (5.0 MHz)				
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, 6.4 μs, and 12.8 μs (5.0 MHz)				
	When subsystem clock is selected		122 μs (32.768 kHz)				
Internal memory	ROM	Size (bytes)	32K	40K	48K	60K	60K (PROM)
	RAM	Size (bytes)	1056		2080		
Address space			64K bytes				
Interrupt function		External	7				
		Internal	17				
I/O pin	CMOS input		2				
	CMOS I/O		78				
	N-ch open-drain I/O		8				
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 4 channels Clock timer × 1 channel Watchdog timer × 1 channel				
Timer output			3 (PWM output: 1)				
Serial interface			3-wire/SBI/2-wire mode (selectable): 1 channel 3-wire (with function for automatic transmission and reception of up to 32 bytes): 1 channel 3-wire/UART mode (selectable): 1 channel				
A/D converter			8-bit resolution × 8 channels				
D/A converter			8-bit resolution × 2 channels				
Others			Clock output function Buzzer output function				
Standby function			STOP/HALT				
Operating power supply voltage range			1.8 to 5.5 V				
Package			• 100-pin plastic QFP (14 × 20 mm, 0.65-mm pitch) • 100-pin plastic QFP (14 × 14 mm, 0.5-mm pitch)				∗1
Assembler			RA78K0				
C compiler			CC78K0				
Real-time OS			RX78K0, MX78K0				
Integrated debugger			ID78K0 or ID78K0-NS				
			DF78078				
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)				
			With IE-78001-R-A: IE-78078-R-EM, with IE-78K0-NS: IE-78078-NS-EM1				
			With IE-78001-R-A: EP-78064GF-R, EP-78064GC-R With IE-78K0-NS: NP-100GF ∗2, NP-100GC ∗2				
Applications			Cellular phones, cordless phones, audio equipments, VCRs, printers, etc.				

∗1: • 100-pin plastic QFP (14 × 20 mm, 0.65-mm pitch)
• 100-pin plastic QFP (14 × 14 mm, 0.5-mm pitch)
• 100-pin ceramic WQFN (14 × 20 mm)

∗2: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (23/32)

Subseries name			μPD78078Y Subseries		
Part number			μPD78076Y	μPD78078Y	μPD78P078Y
Number of basic instructions			63		
Minimum instruction execution time			0.4 μs (5.0 MHz)		
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, 6.4 μs, and 12.8 μs (5.0 MHz)		
	When subsystem clock is selected		122 μs (32.768 kHz)		
Internal memory	ROM	Size (bytes)	48K	60K	60K (PROM)
	RAM	Size (bytes)	2080		
Address space			64K bytes		
Interrupt function		External	7		
		Internal	17		
I/O pin	CMOS input		2		
	CMOS I/O		78		
	N-ch open-drain I/O		8		
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 4 channels Clock timer × 1 channel Watchdog timer × 1 channel		
Timer output			3 (PWM output: 1)		
Serial interface			3-wire/2-wire/I²C bus mode (selectable): 1 channel 3-wire (with function for automatic transmission and reception of up to 32 bytes): 1 channel 3-wire/UART mode (selectable): 1 channel		
A/D converter			8-bit resolution × 8 channels		
D/A converter			8-bit resolution × 2 channels		
Others			Clock output function Buzzer output function		
Standby function			STOP/HALT		
Operating power supply voltage range			1.8 to 5.5 V		
Package			• 100-pin plastic QFP (14 × 20 mm, 0.65-mm pitch) • 100-pin plastic QFP (14 × 14 mm, 0.5-mm pitch)		∗1
Assembler			RA78K0		
C compiler			CC78K0		
Real-time OS			RX78K0, MX78K0		
Integrated debugger			ID78K0 or ID78K0-NS		
			DF78078		
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)		
			With IE-78001-R-A: IE-78078-R-EM, with IE-78K0-NS: IE-78078-NS-EM1		
			With IE-78001-R-A: EP-78064GF-R, EP-78064GC-R With IE-78K0-NS: NP-100GF ∗2, NP-100GC ∗2		
Applications			VCRs, TVs, audio equipments, etc.		

∗1: • 100-pin plastic QFP (14 × 20 mm, 0.65-mm pitch)
• 100-pin plastic QFP (14 × 14 mm, 0.5-mm pitch)
• 100-pin ceramic WQFN (14 × 20 mm)

∗2: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (24/32)

Part number			μPD78070A	μPD78070AY
Number of basic instructions			63	
Minimum instruction execution time			0.4 μs (5.0 MHz)	
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, 6.4 μs, and 12.8 μs (5.0 MHz)	
	When subsystem clock is selected		122 μs (32.768 kHz)	
Internal memory	ROM	Size (bytes)	—	
	RAM	Size (bytes)	1056	
Address space			64K bytes	
Interrupt function		External	7	
		Internal	17	
I/O pin	CMOS input		2	
	CMOS I/O		51	
	N-ch open-drain I/O		8	
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 4 channels Clock timer × 1 channel Watchdog timer × 1 channel	
Timer output			3 (PWM output: 1)	
Serial interface			∗1	∗2
A/D converter			8-bit resolution × 8 channels	
D/A converter			8-bit resolution × 2 channels	
Others			Clock output function Buzzer output function	
Standby function			STOP/HALT	
Operating power supply voltage range			2.7 to 5.5 V	
Package			• 100-pin plastic QFP (14 × 20 mm, 0.65-mm pitch) • 100-pin plastic QFP (14 × 14 mm, 0.5-mm pitch)	
Assembler			RA78K0	
C compiler			CC78K0	
Real-time OS			RX78K0, MX78K0	
Integrated debugger			ID78K0 or ID78K0-NS	
			DF78078	
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)	
			With IE-78001-R-A: IE-78078-R-EM, with IE-78K0-NS: IE-78078-NS-EM1	
			With IE-78001-R-A: EP-78064GF-R, EP-78064GC-R With IE-78K0-NS: NP-100GF *3, NP-100GC *3	
Applications			CD-ROMs, printers, PPCs, etc.	

∗1: • 3-wire/SBI/2-wire mode (selectable): 1 channel
• 3-wire (with function for automatic transmission and reception of up to 32 bytes): 1 channel
• 3-wire/UART mode (selectable): 1 channel

∗2: • 3-wire/2-wire/I^2C bus mode selectable: 1 channel
• 3-wire (with function for automatic transmission and reception of up to 32 bytes): 1 channel
• 3-wire/UART mode (selectable): 1 channel

∗3: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (25/32)

Subseries name			**μPD78083 Subseries**		
Part number			**μPD78081**	**μPD78082**	**μPD78P083**
Number of basic instructions			63		
Minimum instruction execution time			0.4 μs (5.0 MHz)		
Instruction cycle	When main system clock is selected		Selectable from 0.4 μs, 0.8 μs, 1.6 μs, 3.2 μs, 6.4 μs and 12.8 μs (5.0 MHz)		
Internal memory	ROM	Size (bytes)	8K	16K	24K (PROM)
	RAM	Size (bytes)	256	384	512
Address space			64K bytes		
Interrupt function		External	3		
		Internal	10		
I/O pin	CMOS input		1		
	CMOS I/O		32		
Timer			8-bit timer/event counter × 2 channels Watchdog timer × 1 channel		
Timer output			2 (PWM output: 2)		
Serial interface			3-wire/UART mode (selectable): 1 channel		
A/D converter			8-bit resolution × 8 channels		
Others			Clock output function Buzzer output function		
Standby function			STOP/HALT		
Operating power supply voltage range			1.8 to 5.5 V		
Package			• 42-pin plastic shrink DIP (600 mil) • 44-pin plastic QFP (10 × 10 mm, 0.8-mm pitch)		• 42-pin plastic shrink DIP (600 mil) • 44-pin plastic QFP (10 × 10 mm, 0.8-mm pitch) • 42-pin ceramic shrink DIP with window (600 mil)
Assembler			RA78K0		
C compiler			CC78K0		
Real-time OS			MX78K0		
Integrated debugger			ID78K0 or ID78K0-NS		
			DF78083		
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)		
			With IE-78001-R-A: IE-78078-R-EM, with IE-78K0-NS: IE-78078-NS-EM1		
			With IE-78001-R-A: EP-78083CU-R, EP-78083GB-R With IE-78K0-NS: NP-42CU *, NP-44GB *		
Applications			Airbags, OA equipments, display equipments, air conditioners, keyless entry, etc.		

*: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (26/32)

Subseries name			μPD78098B Subseries			
Part number			μPD78095B	μPD78096B	μPD78098B	μPD78P098B
Number of basic instructions			63			
Minimum instruction execution time			0.4 μs (5.0 MHz)			
Instruction cycle	When main system clock is selected		Selectable from 0.5 μs, 1.0 μs, 2.0 μs, 4.0 μs, 8.0 μs, and 16.0 μs (6.0 MHz)			
	When subsystem clock is selected		122 μs (32.768 kHz)			
Internal memory	ROM	Size (bytes)	40K	48K	60K	60K (PROM)
	RAM	Size (bytes)	1056		3104	
Address space			64K bytes			
Interrupt function		External	7			
		Internal	16			
I/O pin	CMOS input		2			
	CMOS I/O		63			
	N-ch open-drain I/O		4			
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 2 channels Clock timer × 1 channel Watchdog timer × 1 channel			
Timer output			3 (PWM output: 1)			
Serial interface			3-wire/SBI/2-wire mode (selectable): 1 channel 3-wire (with function for automatic transmission and reception of up to 32 bytes): 1 channel 3-wire/UART mode (selectable): 1 channel			
A/D converter			8-bit resolution × 8 channels			
D/A converter			8-bit resolution × 2 channels			
IEBus controller			Effective transmission speed: 26 kbps maximum			
Others			Clock output function Buzzer output function			
Standby function			STOP/HALT			
Operating power supply voltage range			2.7 to 6.0 V			
Package			• 80-pin plastic QFP (14 × 14 mm, 0.65-mm pitch)			*
Assembler			RA78K0			
C compiler			CC78K0			
Real-time OS			RX78K0, MX78K0			
Integrated debugger			ID78K0			
			DF78098			
In-circuit emulator			IE-78001-R-A			
			IE-780908-R-EM			
			EP-78230GC-R			
Applications			Car audios, CD changers, etc.			

*: • 80-pin plastic QFP (14 × 14 mm, 0.65-mm pitch)
• 80-pin ceramic WQFN (14 × 14 mm)

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (27/32)

Subseries name			μPD780988 Subseries				
Part number			μPD780983★	μPD780984★	μPD780986★	μPD780988★	μPD78F0988★
Number of basic instructions			63				
Minimum instruction execution time			0.24 μs (8.38 MHz)				
Instruction cycle			Selectable from 0.24 μs, 0.48 μs, 0.95 μs, 1.9 μs and 3.8 μs (8.38 MHz)				
Internal memory	ROM	Size (bytes)	24K	32K	48K	60K	60K (Flash memory)
	RAM	Size (bytes)	1024		2048		
Address space			64K bytes				
Interrupt function		External	8				
		Internal	16				
I/O pin	CMOS input		8				
	CMOS I/O		39				
Timer			10-bit timer for inverter control × 1 channel 16-bit timer/event counter × 2 channels 8-bit timer/event counter × 3 channels Watchdog timer × 1 channel				
Timer output			11 (PWM output: 5, inverter control output: 6)				
Serial interface			3-wire: 1 channel UART: 2 channels				
A/D converter			10-bit resolution × 8 channels				
Others			Real-time output port				
Standby function			STOP/HALT				
Operating power supply voltage range			V_{DD} = 4.0 to 5.5 V				
Package			64-pin plastic shrink DIP (750 mil) 64-pin plastic QFP (14 × 14 mm, 0.8 mm pitch)				
Assembler			RA78K0				
C compiler			CC78K0				
Real-time OS			RX78K0, MX78K0				
Integrated debugger			ID78K0 or ID78K0-NS				
			DF780988				
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)				
			IE-78K0-NS-P01 + IE-780988-NS-EM4				
			With IE-78001-R-A: EP-78240CW-R + IE-78K0-R-EX1, EP-78240GC-R + IE-78K0-R-EX1 With IE-78K0-NS: NP-64CW *, NP-64GC *				
Applications			Motor control for inverter-type air conditioners, washing machines, etc.				

*: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (28/32)

Part number			**μPD780955★**
Number of basic instructions			62
Minimum instruction execution time			52 μs (38.4 kHz)
Internal memory	ROM	Size (bytes)	40K
	RAM	Size (bytes)	1024
Address space			64K bytes
Interrupt function		External	7
		Internal	10
I/O pin	CMOS I/O		47
	N-ch open drain output		3
Timer			16-bit counter × 1 channel 8-bit timer/event counter × 1 channel 8-bit timer × 5 channels Watchdog timer × 1 channel
Timer output			1
Serial interface			UART: 2 channels
A/D converter			8-bit resolution × 1 channel
Others			Quality grade: Special (μPD780955(A)) only (for high reliability electronic equipment)
Standby function			HALT
Operating power supply voltage range			V_{DD} = 2.2 to 3.5 V
Package			80-pin plastic QFP (14 × 20 mm, 0.8 mm pitch)
Assembler			RA78K0
C compiler			CC78K0
Real-time OS			RX78K0, MX78K0
Integrated debugger			ID78K0
			DF780955
In-circuit emulator			IE-78001-R-A or IE-78K0-NS
			IE-78K0-SL-P01 + IE-780955-SL-EM4
			EP-80GF-SL
Applications			Gas meters, etc.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (29/32)

Subseries name			μPD780973 Subseries	
Part number			μPD780973★	μPD78F0974★
Number of basic instructions			63	
Minimum instruction execution time			0.24 μs (8.38 MHz)	
Instruction cycle			Selectable from 0.24 μs, 0.48 μs, 0.95 μs, 1.91 μs, and 3.81 μs (8.38 MHz)	
Internal memory	ROM	Size (bytes)	24K	32K (Flash memory)
	RAM	Size (bytes)	778	1034
	EEPROM	Size (bytes)	256	
Address space			64K bytes	
Interrupt function		External	3	
		Internal	17	
I/O pin	CMOS input		5	
	CMOS Output		18	
	CMOS I/O		35	
Timer			16-bit timer/event counter × 1 channel 8-bit timer/event counter × 3 channels Clock timer × 1 channel Watchdog timer × 1 channel	
Timer output			2 (PWM output: 2)	
Serial interface			3-wire: 1 channel UART: 1 channel	
A/D converter			8-bit resolution × 5 channels	
LCD controller/driver			Number of segment signal outputs: 20 maximum Number of common signal outputs: 4 maximum Bias: 1/3	
Sound generator			Buzzer output signal with composed amplitude	
Meter controller/driver			4 meter control (Cross-coil or stepper motor)	
Others			Clock output function	
Standby function			STOP/HALT	
Operating power supply voltage range			4.5 to 5.5 V	
Package			• 80-pin plastic QFP (14 × 20 mm, 0.8-mm pitch)	
Assembler			RA78K0	
C compiler			CC78K0	
Real-time OS			RX78K0, MX78K0	
Integrated debugger			ID78K0 or ID78K0-NS	
			DF780974	
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)	
			With IE-78001-R-A: IE-780974-SL-EM1 With IE-78K0-NS: IE-780974-NS-EM1	
			With IE-78001-R-A: EP-80GF-SL, with IE-78K0-NS: NP-80GF *	
Applications			Vehicle meters control (dashboard mounted)	

*: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (30/32)

Subseries name			μPD780948 Subseries			
Part number			μPD780948★	μPD780949★	μPD78F0948★	μPD78F0949★
Number of basic instructions			63			
Minimum instruction execution time			0.25 μs (8.0 MHz)			
Instruction cycle	When main system clock is selected		Selectable from 0.25 μs, 0.5 μs, 1 μs, 2 μs and 4 μs (8.0 MHz)			
	When subsystem clock is selected		122 μs (32.768 kHz)			
Internal memory	ROM	Size (bytes)	60K		60K (Flash memory)	
	Internal fast RAM	Size (bytes)	1024			
	Internal expansion RAM	Size (bytes)	1024			
	EEPROM	Size (bytes)	–	256	–	256
Address space			64K bytes			
Interrupt function		External	5			
		Internal	24			
I/O pin	CMOS input		8			
	CMOS I/O		71			
Timer			16-bit timer/event counter × 1 channel 16-bit timer × 1 channel 8-bit timer/event counter × 2 channels Watch timer × 1 channel Watchdog timer × 1 channel			
Timer output			4 (16-bit PWM output: 1, 8-bit PWM output: 2)			
Serial interface			3-wire: 1 channel 2-wire: 1 channel UART: 1 channel			
CAN controller			Conformed on CAN protocol Ver. 2.0 (part B), transmission speed: 500 kbps maximum			
A/D converter			8-bit resolution × 8 channels			
LCD controller/driver			Segment signal output: 40 maximum Common signal output: 4 maximum Bias: 1/3 bias			
Power fail detect function			Battery voltage detect function			
Sound generator			Buzzer output function (with composed amplitude)			
Standby function			STOP/HALT			
Operating power supply voltage range			V_{DD} = 4.0 to 5.5 V			
Package			100-pin plastic QFP (14 × 20 mm, 0.65 mm pitch)			
Assembler			RA78K0			
C compiler			CC78K0			
Real-time OS			RX78K0, MX78K0			
Integrated debugger			ID78K0			
			DF780948			
In-circuit emulator			IE-78001-R-A			
			IE-780948-SL-EM1 + IE780948-SL-EM4			
			EP-100GF-SL			
Applications			Control of automotive dashboard and electrical equipment in car body, etc.			

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (31/32)

Subseries name			μPD780701Y Subseries		
Part number			μPD780701Y★	μPD780702Y★	μPD78F0701Y★
Number of basic instructions			63		
Minimum instruction execution time			0.32 μs (6.29 MHz)		
Instruction cycle			Selectable from 0.32 μs, 0.64 μs, 1.27 μs, 2.54 μs, and 5.09 μs (6.29 MHz)		
Internal memory	ROM	Size (bytes)	60K		60K (Flash memory)
	RAM	Size (bytes)	3552		
Address space			64K bytes		
Interrupt function		External	8		
		Internal	21		
I/O pin	CMOS I/O		56		
	TTL input/CMOS output		8		
	N-ch open-drain I/O		3		
Timer			16-bit timer/event counter × 2 channels 8-bit timer/event counter × 3 channels Watch timer × 1 channel Watchdog timer × 1 channel		
Timer output			5 (PWM output: 3)		
Serial interface			3-wire: 2 channels UART: 1 channel I^2C bus (supporting multi-master): 1 channel		
A/D converter			8-bit resolution × 16 channels		
Power fail detect function			battery voltage detect function		
Others			Clock output function Buzzer output function		
Bus controller			DCAN controller	IEBus controller	DCAN controller or IEBus controller
Standby function			STOP/HALT		
Power supply voltage range			V_{DD} = 4.5 to 5.5 V		
Package			80-pin plastic QFP (14 × 14 mm)		
Assembler			RA78K0		
C compiler			CC78K0		
Real-time OS			RX78K0, MX78K0		
Integrated debugger			ID78K0 or ID78K0-NS		
			DF780701★		
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)		
			IE-780701-NS-EM1★		
			With IE-78001-R-A: EP-78230GC-R + IE-78K0-R-EX1 With IE-78K0-NS: NP-80GC ＊		
Applications			Car audio systems, etc.		

＊: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/0 Series Products (32/32)

Subseries name			μPD780833Y Subseries		
Part number			μPD780831Y★	μPD780833Y★	μPD78F0833Y★
Number of basic instructions			63		
Minimum instruction execution time			0.48 μs (4.19 MHz)		
Instruction cycle			Selectable from 0.48 μs, 0.96 μs, 1.92 μs, 3.84 μs, and 7.68 μs (4.19 MHz)		
Internal memory	ROM	Size (bytes)	60K		60K (Flash memory)
	RAM	Size (bytes)	3104		
Address space			64K bytes		
Interrupt function		External	9		
		Internal	20		
I/O pin	CMOS I/O		54		
	TTL input/CMOS output		8		
	N-ch open-drain I/O		3		
Timer			16-bit timer/event counter × 2 channels Watch timer × 1 channel Watchdog timer × 1 channel	16-bit timer/event counter × 2 channels 8-bit timer/event counter × 3 channels Watch timer × 1 channel Watchdog timer × 1 channel	
Timer output			5 (PWM output: 3)		
Serial interface			3-wire: 2 channels UART: 1 channel I^2C bus (supporting multi-master): 1 channel		
A/D converter			8-bit resolution × 8 channels × 2		
Others			Clock output function		
Bus controller			J1850 (CLASS2) bus interface		
Standby function			STOP/HALT		
Power supply voltage range			V_{DD} = 4.5 to 5.5 V		
Package			80-pin plastic QFP (14 × 14 mm)		
Assembler			RA78K0		
C compiler			CC78K0		
Real-time OS			RX78K0, MX78K0		
Integrated debugger			ID78K0 or ID78K0-NS		
			DF780831★		
In-circuit emulator			IE-78001-R-A (with ID78K0), IE-78K0-NS (with ID78K0-NS)		
			IE-780831-NS-EM4★ + IE-78K0-NS-P02★		
			With IE-78001-R-A: EP-78230GC-R + IE-78K0-R-EX1 With IE-78K0-NS: NP-80GC *		
Applications			Car audio systems, etc.		

*: This is a product of Naito Densei Machidaseisakusho Co., Ltd.

★: Under development

8/16-Bit Single Chip Microcontroller

78K/II Series

■ **78K/II Series Products (1/2)**

Series name			μPD78214 Series				μPD78218A Series			μPD78244 Series	
Part number			μPD78212	μPD78213	μPD78214	μPD78P214	μPD78217A	μPD78218A	μPD78P218A	μPD78243	μPD78244
Number of instructions			65								
Minimum instruction cycle			333 ns/12 MHz	500 ns/12 MHz	333 ns/12 MHz		500 ns/12 MHz	333 ns/12 MHz		500 ns/12 MHz	333 ns/12 MHz
Memory space			Program memory: 64K bytes, Data memory: 1M bytes								
Internal memory		ROM	8K bytes	ROM-less	16K bytes	16K bytes (PROM)	ROM-less	32K bytes	32K bytes (PROM)	ROM-less EEPROM: 512 bytes	16K bytes EEPROM: 512 bytes
		RAM	384 bytes	512 bytes			1024 bytes			512 bytes	
Interrupt			2-level programmable priority, vectored/macro service								
	External source		7								
	Internal source		12							14	
	Number of interrupts supporting the use of a macro service		15								
	Type of macro service		3 types, 6 kinds				3 types, 11 kinds				
I/O pin		Input	14								
		Output	12								
		I/O	28	10	28		10	28		10	28
Timer/counter			16-bit timer/counter × 1 channel 8-bit timer/counter × 3 channels								
Serial interface			UART (with dedicated on-chip baud rate generator): 1 channel CSI (SBI capability): 1 channel								
A/D converter			8-bit resolution × 8 channels								
Comparator			None								
D/A converter			None								
PWM output			None								
Standby function			HALT/STOP mode								
Pseudo-SRAM refresh function			Provided								
Feature			• General-purpose type with real-time output ports for easy stepper motor control				• Upwardly compatible with μPD78214 Series			• General-purpose type with real-time output ports for easy stepper motor control • Internal EEPROM	
Package			• 64-pin plastic shrink DIP (750 mil) • 68-pin plastic QFJ • 64-pin plastic QFP (14 × 14 mm) • 74-pin plastic QFP (20 × 20 mm) • 64-pin plastic QUIP • 64-pin ceramic shrink DIP with window (for μPD78P214 only)				• 64-pin plastic QFP (14 × 14 mm) • 64-pin plastic shrink DIP (750 mil) • 64-pin ceramic shrink DIP with window (for μPD78P218A only)			• 64-pin plastic QFP (14 × 14 mm) • 64-pin plastic shrink DIP (750 mil)	
Assembler			RA78K2								
C compiler			CC78K2								
Real-time OS			—				RX78K2			—	
Screen debugger			SD78K2								
			DF78210							DF78240	
In-circuit emulator			IE-78240-R-A								
Application			OA equipment such as printer, PPC, Fax, etc.								

8/16-Bit Single Chip Microcontroller

■ 78K/II Series Products (2/2)

Series name			**μPD78234 Series**				
Part number			**μPD78233**	**μPD78234**	**μPD78237**	**μPD78238**	**μPD78P238**
Number of instructions			65				
Minimum instruction cycle			500 ns/12 MHz	333 ns/12 MHz	500 ns/12 MHz	333 ns/12 MHz	
Memory space			Program memory: 64K bytes, Data memory: 1M bytes				
Internal memory		ROM	ROM-less	16K bytes	ROM-less	32K bytes	32K/16K bytes (PROM) *
		RAM	640 bytes		1024 bytes		1024/640 bytes *
Interrupt			2-level programmable priority, vectored/macro service				
	External source		7				
	Internal source		12				
	Number of interrupts supporting the use of a macro service		15				
	Type of macro service		3 types, 11 kinds				
I/O pin		Input	16				
		Output	12				
		I/O	18	36	18	36	
Timer/counter			16-bit timer/counter × 1 channel 8-bit timer/counter × 3 channels				
Serial interface			UART (with dedicated on-chip baud rate generator): 1 channel CSI (SBI capability): 1 channel				
A/D converter			8-bit resolution × 8 channels				
Comparator			None				
D/A converter			8-bit resolution × 2 channels				
PWM output			12-bit resolution × 2 channels				
Standby function			HALT/STOP mode				
Pseudo-SRAM refresh function			Provided				
Feature			• Enhanced analog output function • Large internal memory				
Package			• 84-pin plastic QFJ • 80-pin plastic QFP (14 × 14 mm) • 94-pin ceramic WQFN (for μPD78P238 only)				
Assembler			RA78K2				
C compiler			CC78K2				
Real-time OS			—		RX78K2		
Screen debugger			SD78K2				
			DF78230				
In-circuit emulator			IE-78230-R-A				
Applications			OA equipment such as LBP, PPC, Fax, etc.				

*: Set by software

8/16-Bit Single Chip Microcontroller

78K/III Series

■ **78K/III Series Products (1/4)**

Series name		μPD78312A Series			μPD78322 Series					
Part number		μPD78310A	μPD78312A	μPD78P312A	μPD78320	μPD78322	μPD78P322	μPD78323	μPD78324	μPD78P324
Number of instructions		96			111					
Minimum instruction cycle		500 ns/12 MHz			250 ns/16MHz					
Internal memory	ROM size	—	8K bytes	8K bytes (PROM)	—	16K bytes	16K bytes (PROM)	—	32K bytes	32K bytes (PROM)
	RAM size	256 bytes			640 bytes			1024 bytes		
Interrupt function	External source	4			8					
	Internal source	14			13					
		• 8-level programmable priority • Macro service (high-speed data transfer) function • Context switching function			• 3-level programmable priority • Macro service (high-speed data transfer/operation) function: 9 types • Context switching function					
I/O pin	Input	8			16					
	Output	—								
	I/O	24	40		21	39		21	39	
Analog input		8-bit A/D converter × 4 channels			10-bit resolution × 8 channels					
Timer/counter		16-bit up/down counter × 2 channels 16-bit interval timer × 2 channels			18/16-bit free running timer × 1 channel 16-bit timer/event counter × 1 channel					
Serial interface		UART (with dedicated on-chip baud rate generator): 1 channel (Synchronous)			UART (with dedicated on-chip baud rate generator): 1 channel Synchronous (SBI capability): 1 channel					
Standby function		HALT/STOP mode								
Feature		• On-chip up/down counter for DC servo			• Real-time pulse unit outputting multi-purpose programmable pulse signals					
Package		• 64-pin plastic SDIP • 64-pin plastic QUIP • 64-pin plastic QFP • 68-pin plastic QFJ		∗1	• 68-pin plastic QFJ • 74-pin plastic QFP • 80-pin plastic QFP		∗2	• 68-pin plastic QFJ • 74-pin plastic QFP		∗3
Assembler		RA78K3								
C compiler		CC78K3								
Real-time OS		—			RX78K3					
In-circuit emulator		IE-78310A-R			IE-78327-R					
Applications		Control equipment for DC servo motors, stepping motors, etc.								

∗1: • 64-pin plastic SDIP
• 64-pin plastic QUIP
• 64-pin plastic QFP
• 68-pin plastic QFJ
• 64-pin ceramic SDIP with window
• 64-pin ceramic QUIP with window

∗2: • 68-pin plastic QFJ
• 74-pin plastic QFP
• 80-pin plastic QFP
• 68-pin ceramic WQFN
• 74-pin ceramic WQFN
• 80-pin ceramic WQFN

∗3: • 68-pin plastic QFJ
• 74-pin plastic QFP
• 68-pin ceramic WQFN
• 74-pin ceramic WQFN

8/16-Bit Single Chip Microcontroller

■ 78K/III Series Products (2/4)

Series name		μPD78328 Series			μPD78334 Series			μPD78350 Series			
Part number		μPD78327	μPD78328	μPD78P328	μPD78330	μPD78334	μPD78P334	μPD78350	μPD78350A	μPD78352A	μPD78P352
Number of instructions		111						113			
Minimum instruction cycle		250 ns/16 MHz						160 ns/25 MHz	125 ns/32 MHz		
Internal memory	ROM size	—	16K bytes	16K bytes (PROM)	—	32K bytes	32K bytes (PROM)	—		32K bytes	32K bytes (PROM)
	RAM size	512 bytes			1024 bytes			640 bytes			
Interrupt function	External source	4			8			5			
	Internal source	11			14			4			
		• 3-level programmable priority • Macro service (high-speed data transfer/operation):10 types • Context switching function			• 3-level programmable priority • Macro service (high-speed data transfer/operation) function: 8 types • Context switching function			• 4-level programmable priority • Macro service (high-speed data transfer/operation) function • Context switching function			
I/O pin	Input	11			24			6			
	Output	—			—			—			
	I/O	23	41		28	46		24	44		
Analog input		10-bit resolution × 8 channels			10-bit resolution × 16 channels			—			
Timer/counter		16-bit timer × 2 channels 16-bit timer/event counter × 1			18/16-bit free running timer × 1 channel 16-bit timer/counter × 3 channels			16-bit timer × 3 channels			
Serial interface		UART (with dedicated on-chip baud rate generator): 1 channel Synchronous (SBI capability): 1 channel						—			
Standby function		HALT/STOP mode									
Feature		6-channel PWM output function for inverter control			Real-time pulse unit supporting output of versatile programmable pulse signals			• High-speed sum-of-products operation function • Suitable for ASIC control			
Package		• 64-pin plastic QFP • 64-pin plastic SDIP	*		• 84-pin plastic QFJ • 94-pin plastic QFP		• 84-pin plastic QFJ • 94-pin plastic QFP • 84-pin ceramic WQFN • 94-pin ceamic WQFN	• 64-pin plastic QFP	• 64-pin plastic QFP • 64-pin ceramic WQFN		
Assembler		RA78K3						RA78K3			
C compiler		CC78K3						CC78K3			
Real-time OS		RX78K3						—			
Integrated debugger		—						ID78K3			
In-circuit emulator		IE-78327-R			IE-78330-R			IE-78350-R+IE-78350-R-EM1 (with IE controller) IE-784000-R+IE-78350-R-EM-A+IE-78350-R-EM1 (with integrated debugger)			
Applications		Inverter air-conditioners			FA field, etc.			HDDs, FDDs, etc.			

*: • 64-pin plastic QFP
• 64-pin plastic SDIP
• 64-pin ceramic SDIP with window

8/16-Bit Single Chip Microcontroller

■ 78K/III Series Products (3/4)

Series name		μPD78356 Series		
Part number		**μPD78355**	**μPD78356**	**μPD78P356**
Number of instructions		115		
Minimum instruction cycle		125 ns/32 MHz		
Internal memory	ROM size	—	48K bytes	48K bytes(PROM)
	RAM size	2048 bytes		
Interrupt function	External source	5		
	Internal source	25		
		4-level programmable priority Macro service (high-speed data transfer/operation) function Context switching function		
I/O pin	Input	8		
	Output	—		
	I/O	49	68	
A/D converter		10-bit resolution × 8 channels		
D/A converter		8-bit resolution × 2 channels		
Timer/counter		16-bit timer × 5 channels/10-bit timer × 1 channel		
Serial interface		UART (with dedicated on-chip baud rate generator): 1 channel Synchronous (SBI capability): 1 channel Synchronous (with pin switching function): 1 channel		
Standby function		HALT/STOP mode		
Feature		• Very high-speed on-chip A/D converter (Conversion time: 2 μs) • Small QFP package (0.5-mm pitch, 1.45-mm thick)		
Package		100-pin plastic QFP 120-pin plastic QFP		100-pin plastic QFP 120-pin plastic QFP 120-pin ceramic WQFN
Assembler		RA78K3		
C compiler		CC78K3		
Integrated debugger		ID78K3		
In-circuit emulator		IE-78350-R+IE-78355-R-EM1 (with IE controller) IE-784000-R+IE-78350-R-EM-A+ IE-78355-R-EM1 (with integrated debugger)		
Applications		HDDs, video camcorders, single-lens reflex cameras, etc.		

8/16-Bit Single Chip Microcontroller

■ 78K/III Series Products (4/4)

Series name		**μPD78366A Series**							
Part number		**μPD78361A**	**μPD78362A**	**μPD78P364A**	**μPD78365A**	**μPD78363A**	**μPD78366A**	**μPD78368A**	**μPD78P368A**
Number of instructions		115							
Minimum instruction cycle		125 ns/8 MHz (with on-chip PLL circuit)							
Internal memory	ROM size	32K bytes	24K bytes	48K bytes (PROM)	—	24K bytes	32K bytes	48K bytes	48K bytes(PROM)
	RAM size	2K bytes	768 bytes	2K bytes		768 byte	2K bytes		
Interrupt function	External source	6							
	Internal source	14							
		4-level programmable priority Macro service (high-speed data transfer/operation) function Context switching function							
I/O pin	Input	14 (Used also for analog input: 8)							
	Output	—							
	I/O	38			31	49			
A/D converter		10-bit resolution × 8 channels							
D/A converter		—							
Timer/counter		16-bit timer × 5 channels							
Serial interface		UART (with dedicated on-chip baud rate generator): 1 channel Synchronous (SBI capability): 1 channel							
Standby function		HALT/STOP mode							
Feature		• High-speed timer with 62.5-ns resolution • Real-time pulse unit suitable for inverter control (with on-chip dead time timer, capture with noise elimination function, etc.)							
Package		64-pin plastic SDIP			80-pin plastic QFP				80-pin plastic QFP 80-pin ceramic WQFN
Assembler		RA78K3							
C compiler		CC78K3							
Integrated debugger		ID78K3							
In-circuit emulator		IE-78350-R+IE-78365-R-EM1 (with IE controller) IE-784000-R+IE-78350-R-EM-A+IE-78365-R-EM1 (with integrated debugger)							
Applications		Inverter air-conditioners, robots, machine tools, etc.							

8/16-Bit Single Chip Microcontroller

78K/IV Series

■ **78K/IV Series Products (1/16)**

Subseries name		**μPD784026 Subseries**				
Part number		**μPD784020**	**μPD784021**	**μPD784025**	**μPD784026**	**μPD78P4026**
Number of instructions		113				
Minimum instruction cycle		160 ns (25 MHz)				
Memory expansion function		Total of 1M bytes for both programs and data				
Internal memory	ROM size (bytes)	—		48K	64K	64K (PROM)
	RAM size (bytes)	512	2048			
Interrupt function		4-level programmable priority, macro service, vectored interrupt, context switching function				
	External source	7				
	Internal source	18				
	Number of interrupts supporting the use of a macro service	19				
	Type of macro service	4 types, 21 kinds				
I/O pin	Input	8				
	Output	4		0		
	I/O	34		56		
Timer/counter		16-bit timer/counter × 3 channels 16-bit timer × 1 channel				
Serial interface		UART/IOE (with on-chip baud rate generator): 2 channels CSI (SBI capability): 1 channel				
A/D converter		8-bit resolution × 8 channels				
D/A converter		8-bit resolution × 2 channels				
PWM output		12-bit resolution × 2 channels				
Standby function		HALT/IDLE/STOP mode				
Pseudo-SRAM refresh function		Provided				
Feature		• Large internal memory • Wide operating voltage range (2.7 V to 5.5 V) • System clock dividing function (1/2, 1/4, 1/8, 1/16)				
Package		80-pin plastic QFP	80-pin plastic QFP 80-pin plastic TQFP	80-pin plastic QFP		80-pin plastic QFP 80-pin ceramic WQFN
Assembler		RA78K4				
C compiler		CC78K4				
Real-time OS		RX78K4, MX78K4				
System performance analyzer		AZ78K4				
Integrated debugger		ID78K4				
Device file		DF784026				
In-circuit emulator		IE-784000-R+IE-784000-R-EM+IE-784038-NS-EM1+IE-78K4-R-EX2 (with ID78K4) IE-78K4-NS+IE-784038-NS-EM1 (with ID78K4-NS)				
Applications		PPCs, LBPs, autofocus cameras, etc.				

8/16-Bit Single Chip Microcontroller

■ 78K/IV Series Products (2/16)

Subseries name		**μPD784038 Subseries**					
Part number		**μPD784031**	**μPD784035**	**μPD784036**	**μPD784037**	**μPD784038**	**μPD78P4038**
Number of instructions		113					
Minimum instruction cycle		125 ns (32 MHz)					
Memory expansion function		Total of 1M bytes for both programs and data					
Internal memory	ROM size (bytes)	–	48K	64K	96K	128K	128K (PROM)
	RAM size (bytes)	2K			3.5K	4.25K	
Interrupt function		4-level programmable priority, macro service, vectored interrupt, context switching function					
	External source	7					
	Internal source	18					
	Number of interrupts supporting the use of a macro service	19					
	Type of macro service	4 types, 21 kinds					
I/O pin	Input	8					
	Output	4	0				
	I/O	34	56				
Timer/counter		16-bit timer/counter × 3 channels 16-bit timer × 1 channel					
Serial interface		UART/IOE (with on-chip baud rate generator): 2 channels CSI (3-wire/2wire mode): 1 channel					
A/D converter		8-bit resolution × 8 channels					
D/A converter		8-bit resolution × 2 channels					
PWM output		12-bit resolution × 2 channels					
Standby function		HALT/IDLE/STOP mode					
Pseudo-SRAM refresh function		Provided					
Feature		• Large internal memory • Wide operating voltage range (2.7 V to 5.5 V) • System clock dividing function (1/2, 1/4, 1/8, 1/16)					
Package		80-pin plastic QFP 80-pin plastic TQFP					80-pin plastic QFP 80-pin plastic TQFP 80-pin ceramic WQFN
Assembler		RA78K4					
C compiler		CC78K4					
Real-time OS		RX78K4, MX78K4					
System performance analyzer		AZ78K4					
Integrated debugger		ID78K4 or ID78K4-NS					
Device file		DF784038					
In-circuit emulator		IE-784000-R+IE-784000-R-EM+IE-784038-NS-EM1+IE-78K4-R-EX2 (with ID78K4) IE-78K4-NS+IE-784038-NS-EM1 (with ID78K4-NS)					
Applications		PPCs, LBPs, autofocus cameras, etc.					

8/16-Bit Single Chip Microcontroller

■ 78K/IV Series Products (3/16)

Subseries name		μPD784038Y Subseries					
Part number		**μPD784031Y**	**μPD784035Y**	**μPD784036Y**	**μPD784037Y**	**μPD784038Y**	**μPD78P4038Y**
Number of instructions		113					
Minimum instruction cycle		125 ns (32 MHz)					
Memory expansion function		Total of 1M bytes for both programs and data					
Internal memory	ROM size (bytes)	–	48K	64K	96K	128K	128K (PROM)
	RAM size (bytes)	2K			3.5K	4.25K	
Interrupt function		4-level programmable priority, macro service, vectored interrupt, context switching function					
	External source	8					
	Internal source	19					
	Number of interrupts supporting the use of a macro service	20					
	Type of macro service	4 types, 21 kinds					
I/O pin	Input	8					
	Output	4	0				
	I/O	34	56				
Timer/counter		16-bit timer/counter × 3 channels 16-bit timer × 1 channel					
Serial interface		UART/IOE (with on-chip baud rate generator): 2 channels CSI (3-wire/2wire/I^2C bus mode): 1 channel					
A/D converter		8-bit resolution × 8 channels					
D/A converter		8-bit resolution × 2 channels					
PWM output		12-bit resolution × 2 channels					
Standby function		HALT/IDLE/STOP mode					
Pseudo-SRAM refresh function		Provided					
Feature		• Large internal memory • Wide operating voltage range (2.7 V to 5.5 V) • System clock dividing function (1/2, 1/4, 1/8, 1/16)					
Package		80-pin plastic QFP 80-pin plastic TQFP					80-pin plastic QFP 80-pin plastic TQFP 80-pin ceramic WQFN
Assembler		RA78K4					
C compiler		CC78K4					
Real-time OS		RX78K4, MX78K4					
System performance analyzer		AZ78K4					
Integrated debugger		ID78K4 or ID78K4-NS					
Device file		DF784038					
In-circuit emulator		IE-784000-R+IE-784000-R-EM+IE-784038-NS-EM1+IE-78K4-R-EX2 (with ID78K4) IE-78K4-NS+IE-784038-NS-EM1 (with ID78K4-NS)					
Applications		Cellular phones, cordless phones, AV equipment, etc.					

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/IV Series Products (4/16)

Subseries name		μPD784046 Subseries			μPD784054 Subseries
Part number		**μPD784044**	**μPD784046**	**μPD78F4046**	**μPD784054**
Number of instructions		113			
Minimum instruction cycle		125 ns (32 MHz)			
Memory expansion function		Total of 1M bytes for both programs and data			
Internal memory	ROM size (bytes)	32K	64K	64K (flash memory)	32K
	RAM size (bytes)	1024	2048	2048	1024
Interrupt function		4-level programmable priority, macro service, vectored interrupt, context switching function			
	External source	8			
	Internal source	23			19
	Number of interrupts supporting the use of a macro service	25			21
	Type of macro service	7 types, 27 kinds			7 types, 23 kinds
I/O pin	Input	17			
	Output	0			
	I/O	48			47
Timer/counter		16-bit timer × 3 channels 16-bit timer/counter × 2 channels			16-bit timer × 3
Serial interface		UART/CSI × 2 channels			
A/D converter		10-bit resolution × 16 channels			
Standby function		HALT/IDLE/STOP mode			HALT/IDLE/STOP/standby invalidation mode
Feature		On-chip timer unit optimum for real-time control			
Package		80-pin plastic QFP			
Assembler		RA78K4			
C compiler		CC78K4			
Real-time OS		RX78K4, MX78K4			
System performance analyzer		AZ78K4			
Integrated debugger		ID78K4 or ID78K4-NS			
Device file		DF784046			
In-circuit emulator		IE-784000-R+IE-784000-R-EM+IE-784046-NS-EM1+IE-78K4-R-EX2 (with ID78K4) IE-78K4-NS+IE-784046-NS-EM1 (with ID78K4-NS)			
Applications		Hot water supplier, vending machines, etc.			

8/16-Bit Single Chip Microcontroller

■ 78K/IV Series Products (5/16)

Subseries name		μPD784216 Subseries			
Part number		μPD784214	μPD784215	μPD784216	μPD78F4216★
Number of instructions		113			
Minimum instruction cycle		160 ns (Main system clock 12.5 MHz)/61 μs (Subsystem clock 32.768 kHz)			
Memory expansion function		Total of 1M bytes for both programs and data			
Internal memory	ROM size (bytes)	96K	128K	128K	128K (Flash memory)
	RAM size (bytes)	3584	5120	8192	
Interrupt function		4-level programmable priority, macro service, vectored interrupt, context switching function			
	External source	9			
	Internal source	20			
	Number of interrupts supporting the use of a macro service	27			
	Type of macro service	4 types, 21kinds			
I/O pin	Input	8			
	I/O	72			
	N-ch open-drain I/O	6			
Timer/counter		16-bit timer/counter × 1 channel 8-bit timer/counter × 6 channels			
Real-time output port		4-bit × 2 ports or 8-bit × 1 port			
Serial interface		UART/IOE (with dedicated on-chip baud rate generator): 2 channels CSI (3-wire mode): 1 channel			
A/D converter		8-bit resolution × 8 channels			
D/A converter		8-bit resolution × 2 channels			
Clock output		Select of f_{XX}, $f_{XX}/2$, $f_{XX}/2^2$, $f_{XX}/2^3$, $f_{XX}/2^4$, $f_{XX}/2^5$, $f_{XX}/2^6$, $f_{XX}/2^7$, f_{XT}			
Buzzer output		Select of $f_{XX}/2^{10}$, $f_{XX}/2^{11}$, $f_{XX}/2^{12}$, $f_{XX}/2^{13}$			
Clock timer		1 channel			
Standby function		HALT/STOP/IDLE mode HALT/IDLE mode at Low power consumption mode (CPU operation using subsystem clock)			
Feature		• Large internal memory • Wide operation voltage range (1.8 V to 5.5 V) • Using peripheral functions of 78K/0 series			
Package		100-pin plastic QFP (14 × 14 mm) 100-pin plastic QFP (14 × 20 mm)			
Assembler		RA78K4			
C compiler		CC78K4			
Real-time OS		RX78K4, MX78K4			
System performance analyzer		AZ78K4			
Integrated debugger		ID78K4 or ID78K4-NS			
Device file		DF784218★			
In-circuit emulator		IE-784000-R+IE-784000-R-EM+IE-784225-NS-EM1+IE-78K4-R-EX3 (with ID78K4) IE-78K4-NS+IE-784225-NS-EM1 (with ID78K4-NS)			
Applications		Cellular phones, PHS, cordless phones, CD-ROMs, AV equipment, etc.			

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/IV Series Products (6/16)

Subseries name		μPD784216Y Subseries			
Part number		μPD784214Y	μPD784215Y	μPD784216Y	μPD78F4216Y★
Number of instructions		113			
Minimum instruction cycle		160 ns (Main system clock 12.5 MHz)/61 μs (Subsystem clock 32.768 kHz)			
Memory expansion function		Total of 1M bytes for both programs and data			
Internal memory	ROM size (bytes)	96K	128K	128K	128K (Flash memory)
	RAM size (bytes)	3584	5120	8192	
Interrupt function		4-level programmable priority, macro service, vectored interrupt, context switching function			
	External source	9			
	Internal source	20			
	Number of interrupts supporting the use of a macro service	27			
	Type of macro service	4 types, 21kinds			
I/O pin	Input	8			
	I/O	72			
	N-ch open-drain I/O	6			
Timer/counter		16-bit timer/counter × 1 channel 8-bit timer/counter × 6 channels			
Real-time output port		4-bit × 2 ports or 8-bit × 1 port			
Serial interface		UART/IOE (with dedicated on-chip baud rate generator): 2 channels CSI (3-wire/I^2C bus supporting multi-master mode): 1 channel			
A/D converter		8-bit resolution × 8 channels			
D/A converter		8-bit resolution × 2 channels			
Clock output		Select of f_{XX}, $f_{XX}/2$, $f_{XX}/2^2$, $f_{XX}/2^3$, $f_{XX}/2^4$, $f_{XX}/2^5$, $f_{XX}/2^6$, $f_{XX}/2^7$, f_{XT}			
Buzzer output		Select of $f_{XX}/2^{10}$, $f_{XX}/2^{11}$, $f_{XX}/2^{12}$, $f_{XX}/2^{13}$			
Clock timer		1 channel			
Standby function		HALT/STOP/IDLE mode HALT/IDLE mode at Low power consumption mode (CPU operation using subsystem clock)			
Feature		• Large internal memory • Wide operation voltage range (1.8 V to 5.5 V) • Using peripheral functions of 78K/0 series			
Package		100-pin plastic QFP (14 × 14 mm) 100-pin plastic QFP (14 × 20 mm)			
Assembler		RA78K4			
C compiler		CC78K4			
Real-time OS		RX78K4, MX78K4			
System performance analyzer		AZ78K4			
Integrated debugger		ID78K4 or ID78K4-NS			
Device file		DF784218★			
In-circuit emulator		IE-784000-R+IE-784000-R-EM+IE-784225-NS-EM1+IE-78K4-EX3 (with ID78K4) IE-78K4-NS+IE-784225-NS-EM1 (with ID78K4-NS)			
Applications		Cellular phones, PHS, cordless phones, CD-ROMs, AV equipment, etc.			

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/IV Series Products (7/16)

Subseries name		μPD784218 Subseries		
Part number		μPD784217★	μPD784218★	μPD78F4218★
Number of instructions		113		
Minimum instruction cycle		160 ns (Main system clock 12.5 MHz)/61 μs (Subsystem clock 32.768 kHz)		
Memory expansion function		Total of 1M bytes for both programs and data		
Internal memory	ROM size (bytes)	192K	256K	256K (Flash memory)
	RAM size (bytes)	12800		
Interrupt function		4-level programmable priority, macro service, vectored interrupt, context switching function		
	External source	9		
	Internal source	20		
	Number of interrupts supporting the use of a macro service	27		
	Type of macro service	4 types, 21 kinds		
I/O pin	Input	8		
	I/O	72		
	N-ch open-drain I/O	6		
Timer/counter		16-bit timer/counter × 1 channel 8-bit timer/counter × 6 channels		
Real-time output port		4-bit × 2 ports or 8-bit × 1 port		
Serial interface		UART/IOE (with dedicated on-chip baud rate generator): 2 channels CSI (3-wire mode): 1 channel		
A/D converter		8-bit resolution × 8 channels		
D/A converter		8-bit resolution × 2 channels		
Clock output		Select of f_{XX}, $f_{XX}/2$, $f_{XX}/2^2$, $f_{XX}/2^3$, $f_{XX}/2^4$, $f_{XX}/2^5$, $f_{XX}/2^6$, $f_{XX}/2^7$, f_{XT}		
Buzzer output		Select of $f_{XX}/2^{10}$, $f_{XX}/2^{11}$, $f_{XX}/2^{12}$, $f_{XX}/2^{13}$		
Clock timer		1 channel		
Standby function		HALT/STOP/IDLE mode HALT/IDLE mode at Low power consumption mode (CPU operation using subsystem clock)		
Feature		• Large internal memory • Wide operation voltage range (1.8 to 5.5 V) • Using peripheral functions of 78K/0 series		
Package		100-pin plastic QFP (14 × 20 mm, 0.65-mm pitch) 100-pin plastic QFP (14 × 14 mm, 0.5-mm pitch)		
Assembler		RA78K4		
C compiler		CC78K4		
Real-time OS		RX78K4, MX78K4		
System performance analyzer		AZ78K4		
Integrated debugger		ID78K4 or ID78K4-NS		
Device file		DF784218★		
In-circuit emulator		IE-784000-R+IE-784000-R-EM+IE-784225-NS-EM1+IE-78K4-R-EX3 (with ID78K4) IE-78K4-NS+IE-784225-NS-EM1 (with ID78K4-NS)		
Applications		Cellular phones, PHS, cordless phones, CD-ROMs, AV equipment, etc.		

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/IV Series Products (8/16)

Subseries name		μPD784218Y Subseries		
Part number		μPD784217Y★	μPD784218Y★	μPD78F4218Y★
Number of instructions		113		
Minimum instruction cycle		160 ns (Main system clock 12.5 MHz)/61 μs (Subsystem clock 32.768 kHz)		
Memory expansion function		Total of 1M bytes for both programs and data		
Internal memory	ROM size (bytes)	192K	256K	256K (Flash memory)
	RAM size (bytes)	12800		
Interrupt function		4-level programmable priority, macro service, vectored interrupt, context switching function		
	External source	9		
	Internal source	20		
	Number of interrupts supporting the use of a macro service	27		
	Type of macro service	4 types, 21 kinds		
I/O pin	Input	8		
	I/O	72		
	N-ch open-drain I/O	6		
Timer/counter		16-bit timer/counter × 1 channel 8-bit timer/counter × 6 channels		
Real-time output port		4-bit × 2 ports or 8-bit × 1 port		
Serial interface		UART/IOE (with dedicated on-chip baud rate generator): 2 channels CSI (3-wire/I^2C bus supporting multi-master mode): 1 channel		
A/D converter		8-bit resolution × 8 channels		
D/A converter		8-bit resolution × 2 channels		
Clock output		Select of f_{XX}, $f_{XX}/2$, $f_{XX}/2^2$, $f_{XX}/2^3$, $f_{XX}/2^4$, $f_{XX}/2^5$, $f_{XX}/2^6$, $f_{XX}/2^7$, f_{XT}		
Buzzer output		Select of $f_{XX}/2^{10}$, $f_{XX}/2^{11}$, $f_{XX}/2^{12}$, $f_{XX}/2^{13}$		
Clock timer		1 channel		
Standby function		HALT/STOP/IDLE mode HALT/IDLE mode at Low power consumption mode (CPU operation using subsystem clock)		
Feature		• Large internal memory • Wide operation voltage range (1.8 to 5.5 V) • Using peripheral functions of 78K/0 series		
Package		100-pin plastic QFP (14 × 20 mm, 0.65-mm pitch) 100-pin plastic LQFP (14 × 14 mm, 0.5-mm pitch)		
Assembler		RA78K4		
C compiler		CC78K4		
Real-time OS		RX78K4, MX78K4		
System performance analyzer		AZ78K4		
Integrated debugger		ID78K4 or ID78K4-NS		
Device file		DF784218★		
In-circuit emulator		IE-784000-R+IE-784000-R-EM+IE-784225-NS-EM1+IE-78K4-R-EX3 (with ID78K4) IE-78K4-NS+IE-784225-NS-EM1 (with ID78K4-NS)		
Applications		Cellular phones, PHS, cordless phones, CD-ROMs, AV equipment, etc.		

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/IV Series Products (9/16)

Subseries name		μPD784225 Subseries		
Part number		μPD784224★	μPD784225★	μPD78F4225★
Number of instructions		113		
Minimum instruction cycle		160 ns (Main system clock 12.5 MHz)/61 μs (Subsystem clock 32.768 kHz)		
Memory expansion function		Total of 1M bytes for both programs and data		
Internal memory	ROM size (bytes)	96K	128K	128K (Flash memory)
	RAM size (bytes)	3584	4352	
Interrupt function		4-level programmable priority, macro service, vectored interrupt, context switching function		
	External source	7		
	Internal source	18		
	Number of interrupts supporting the use of a macro service	23		
	Type of macro service	4 types, 21 kinds		
I/O pin	Input	8		
	I/O	59		
Timer/counter		16-bit timer/counter × 1 channel 8-bit timer/counter × 4 channels		
Real-time output port		4-bit × 2 ports or 8-bit × 1 port		
Serial interface		UART/IOE (with dedicated on-chip baud rate generator): 2 channels CSI (3-wire mode): 1 channel		
A/D converter		8-bit resolution × 8 channels		
D/A converter		8-bit resolution × 2 channels		
Clock output		Select of f_{XX}, $f_{XX}/2$, $f_{XX}/2^2$, $f_{XX}/2^3$, $f_{XX}/2^4$, $f_{XX}/2^5$, $f_{XX}/2^6$, $f_{XX}/2^7$, f_{XT}		
Buzzer output		Select of $f_{XX}/2^{10}$, $f_{XX}/2^{11}$, $f_{XX}/2^{12}$, $f_{XX}/2^{13}$		
Clock timer		1 channel		
Standby function		HALT/STOP/IDLE mode HALT/IDLE mode at Low power consumption mode (CPU operation using subsystem clock)		
Feature		• Large internal memory • Wide operation voltage range (1.8 to 5.5 V) • Using peripheral functions of 78K/0 series		
Package		80-pin plastic QFP (14 × 14 mm) 80-pin plastic TQFP (12 × 12 mm)		
Assembler		RA78K4		
C compiler		CC78K4		
Real-time OS		RX78K4, MX78K4		
System performance analyzer		AZ78K4		
Integrated debugger		ID78K4 or ID78K4-NS		
Device file		DF784225★		
In-circuit emulator		IE-784000-R+IE-784000-R-EM+IE-784225-NS-EM1+IE-78K4-R-EX2 (with ID78K4) IE-78K4-NS+IE-784225-NS-EM1 (with ID78K4-NS)		
Applications		Car audio, portable audio, telephones, etc.		

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/IV Series Products (10/16)

Subseries name		μPD784225Y Subseries		
Part number		μPD784224Y★	μPD784225Y★	μPD78F4225Y★
Number of instructions		113		
Minimum instruction cycle		160 ns (Main system clock 12.5 MHz)/61 μs (Subsystem clock 32.768 kHz)		
Memory expansion function		Total of 1M bytes for both programs and data		
Internal memory	ROM size (bytes)	96K	128K	128K (Flash memory)
	RAM size (bytes)	3584	4352	
Interrupt function		4-level programmable priority, macro service, vectored interrupt, context switching function		
	External source	7		
	Internal source	18		
	Number of interrupts supporting the use of a macro service	23		
	Type of macro service	4 types, 21 kinds		
I/O pin	Input	8		
	I/O	59		
Timer/counter		16-bit timer/counter × 1 channel 8-bit timer/counter × 4 channels		
Real-time output port		4-bit × 2 ports or 8-bit × 1 port		
Serial interface		UART/IOE (with dedicated on-chip baud rate generator): 2 channels CSI (3-wire mode): 1 channel		
A/D converter		8-bit resolution × 8 channels		
D/A converter		8-bit resolution × 2 channels		
Clock output		Select of f_{XX}, $f_{XX}/2$, $f_{XX}/2^2$, $f_{XX}/2^3$, $f_{XX}/2^4$, $f_{XX}/2^5$, $f_{XX}/2^6$, $f_{XX}/2^7$, f_{XT}		
Buzzer output		Select of $f_{XX}/2^{10}$, $f_{XX}/2^{11}$, $f_{XX}/2^{12}$, $f_{XX}/2^{13}$		
Clock timer		1 channel		
Standby function		HALT/STOP/IDLE mode HALT/IDLE mode at Low power consumption mode (CPU operation using subsystem clock)		
Feature		• Large internal memory • Wide operation voltage range (1.8 to 5.5 V) • Using peripheral functions of 78K/0 series		
Package		80-pin plastic QFP (14 × 14 mm) 80-pin plastic TQFP (12 × 12 mm)		
Assembler		RA78K4		
C compiler		CC78K4		
Real-time OS		RX78K4, MX78K4		
System performance analyzer		AZ78K4		
Integrated debugger		ID78K4 or ID78K4-NS		
Device file		DF784225★		
In-circuit emulator		IE-784000-R+IE-784000-R-EM+IE-784225-NS-EM1+IE-78K4-R-EX2 (with ID78K4) IE-78K4-NS+IE-784225-NS-EM1 (with ID78K4-NS)		
Applications		Car audio, portable audio, telephones, etc.		

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/IV Series Products (11/16)

Subseries name		μPD784915 Subseries				
Part number		μPD784915, 784915A	μPD784915B	μPD784916A	μPD784916B	μPD78P4916
Number of instructions		113				
Minimum instruction cycle		250 ns (Internal clock: 8 MHz)				
Internal memory	ROM size (bytes)	48K		62K		62K (PROM)
	RAM size (bytes)	1280				2048
Interrupt function		4 levels (programmable), vectored interrupt, macro service, vectored interrupt, context switching function				
	External source	9 (including NMI)				
	Internal source	20 (including software interrupts)				
I/O pin	Input	8				
	Output	0				
	I/O	46				
Real-time output port		11 (including pseudo V sync, head amplifier switch, chroma-rotation outputs)				
Super timer unit	Timer/counter	22-bit FRC counter × 1 channel 16-bit timer × 6 channels (including 3 channels for general-purpose timers) 8-bit counter × 2 channels 5-bit UDC counter × 1 channel				
	Special circuit for VCR	V sync separator circuit, H sync separator circuit VISS detection circuit, wide aspect detection circuit Field identification circuit Head amplifier switch/chroma-rotation output circuit				
	PWM output	16-bit resolution × 3 channels (Carrier frequency: 62.5 kHz) 8-bit resolution × 3 channels (Carrier frequency: 62.5 kHz)				
Serial interface		Synchronous serial interface (3-wire): 2 channels (including 1 channel supporting BUSY/STRB control)				
A/D converter		8-bit resolution × 12 channels (Conversion time: 10 μs)				
Analog VCR circuits		CTL amplifier, RECCTL driver (with rewrite capability) CFG amplifier DFG amplifier, DPG comparator, DPFG separator circuit (3-value separator circuit) Reel FG comparator (2 channels) Csync comparator				
Standby function		HALT/STOP mode HALT mode for low power consumption				
Clock function		0.5-second measurement, low voltage operation (V_{DD} = 2.7 V) allowed				
Package		100-pin plastic QFP (14 × 20 mm)				
Assembler		RA78K4				
C compiler		CC78K4				
Integrated debugger		ID78K4 or ID78K4-NS				
Device file		DF784915				
In-circuit emulator		IE-784000-R+IE-784000-R-EM+IE-784928-NS-EM1+IE-78K4-R-EX3 (with ID78K4) IE-78K4-NS+IE-784928-NS-EM1 (with ID78K4-NS)				
Applications		VCRs, camcorder, integrated TV-VCR machines, etc.				

8/16-Bit Single Chip Microcontroller

■ 78K/IV Series Products (12/16)

Subseries name		μPD784928 Subseries		
Part number		μPD784927★	μPD784928★	μPD78F4928★
Number of instructions		113		
Minimum instruction cycle		250 ns (Internal clock: 8 MHz)		
Internal memory	ROM size (bytes)	96K	128K	128K (Flash memory)
	RAM size (bytes)	2048	3584	
Interrupt function		4 levels (programmable), vectored interrupt, macro service, context switching function		
	External source	9 (including NMI)		
	Internal source	22 (including software interrupts)		
I/O pin	Input	20		
	Output	0		
	I/O	54		
Real-time output port		11 (including pseudo V sync, head amplifier switch, chroma-rotation outputs)		
Super timer unit	Timer/counter	22-bit FRC counter × 1 channel 16-bit timer × 6 channels (including 3 channels for general-purpose timers) 8-bit counter × 2 channels 5-bit UDC counter × 1 channel		
	Special circuit for VCR	V sync separator circuit, H sync separator circuit VISS detection circuit, wide aspect detection circuit Field identification circuit Head amplifier switch/chroma-rotation output circuit		
	PWM output	16-bit resolution × 3 channels (Carrier frequency: 62.5 kHz) 8-bit resolution × 3 channels (Carrier frequency: 62.5 kHz)		
Serial interface		Synchronous serial interface (3-wire): 2 channels (including 1 channel supporting BUSY/STRB control)		
A/D converter		8-bit resolution × 12 channels (Conversion time: 10 μs)		
Analog VCR circuits		CTL amplifier, RECCTL driver (with rewrite capability) CFG amplifier DFG amplifier, DPG amplifier, DPFG separator circuit (3-value separator circuit) Reel FG comparator (2 channels) Csync comparator		
Standby function		HALT/STOP mode Low power consumption mode/HALT mode for low power consumption		
Clock function		0.5-second measurement, low voltage operation (V_{DD} = 2.7 V) allowed		
Package		100-pin plastic QFP (14 × 20 mm)		
Assembler		RA78K4		
C compiler		CC78K4		
Integrated debugger		ID78K4 or ID78K4-NS		
Device file		DF784928		
In-circuit emulator		IE-784000-R+IE-784000-R-EM+IE-784928-NS-EM1+IE-78K4-EX3 (with ID78K4) IE-78K4-NS+IE-784928-NS-EM1 (with ID78K4-NS)		
Applications		VCRs, camcorder, integrated TV-VCR machines, etc.		

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/IV Series Products (13/16)

Subseries name		μPD784928Y Subseries		
Part number		μPD784927Y★	μPD784928Y★	μPD78F4928Y★
Number of instructions		113		
Minimum instruction cycle		250 ns (Internal clock: 8 MHz)		
Internal memory	ROM size (bytes)	96K	128K	128K (Flash memory)
	RAM size (bytes)	2048	3584	
Interrupt function		4 levels (programmable), vectored interrupt, macro service, context switching function		
	External source	9 (including NMI)		
	Internal source	23 (including software interrupts)		
I/O pin	Input	20		
	Output	0		
	I/O	54		
Real-time output port		11 (including pseudo V sync, head amplifier switch, chroma-rotation outputs)		
Super timer unit	Timer/counter	22-bit FRC counter × 1 channel 16-bit timer × 6 channels (including 3 channels for general-purpose timers) 8-bit counter × 2 channels 5-bit UDC counter × 1 channel		
	Special circuit for VCR	V sync separator circuit, H sync separator circuit VISS detection circuit, wide aspect detection circuit Field identification circuit Head amplifier switch/chroma-rotation output circuit		
	PWM output	16-bit resolution × 3 channels (Carrier frequency: 62.5 kHz) 8-bit resolution × 3 channels (Carrier frequency: 62.5 kHz)		
Serial interface		Synchronous serial interface (3-wire): 2 channels (including 1 channel supporting BUSY/STRB control) I^2C bus interface: 1 channel		
A/D converter		8-bit resolution × 12 channels (Conversion time: 10 μs)		
Analog VCR circuits		CTL amplifier, RECCTL driver (with rewrite capability) CFG amplifier DFG amplifier, DPG amplifier, DPFG separator circuit (3-value separator circuit) Reel FG comparator (2 channels) Csync comparator		
Standby function		HALT/STOP mode Low power consumption mode/HALT mode for low power consumption		
Clock function		0.5-second measurement, low voltage operation (V_{DD} = 2.7 V) allowed		
Package		100-pin plastic QFP (14 × 20 mm)		
Assembler		RA78K4		
C compiler		CC78K4		
Integrated debugger		ID78K4 or ID78K4-NS		
Device file		DF784928		
In-circuit emulator		IE-784000-R+IE-784000-R-EM+IE-784928-NS-EM1+IE-78K4-R-EX3 (with ID78K4) IE-78K4-NS+IE-784928-NS-EM1 (with ID78K4-NS)		
Applications		VCRs, camcorder, integrated TV-VCR machines, etc.		

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/IV Series Products (14/16)

Subseries name		μPD784908 Subseries		
Part number		μPD784907	μPD784908	μPD78P4908
Number of instructions		113		
Minimum instruction cycle		160 ns (12.58 MHz)		
Memory expansion function		Total of 1M bytes for both programs and data		
Internal memory	ROM size (bytes)	96K	128K	128K (PROM)
	RAM size (bytes)	3584	4352	
Interrupt function		4-level programmable priority, macro service, vectored interrupt, context switching function		
	External source	7		
	Internal source	20		
I/O pin	Input	8		
	I/O	72		
	LED direct drive output	24		
	Transistor direct drive	8		
	N-ch open-drain I/O	4		
Timer/counter		16-bit timer/counter × 4 channels		
Real-time output port		4-bit × 2 ports or 8-bit × 1 port		
IEBus controller		Internal		
Serial interface		UART/IOE (with dedicated on-chip baud rete generator): 2 channels CSI (3-wire): 2 channels		
A/D converter		8-bit resolution × 8 channels		
PWM output		12-bit resolution × 2 channels		
Clock output		Select of f_{XX}, $f_{XX}/2$, $f_{XX}/2^2$, $f_{XX}/2^3$, $f_{XX}/2^4$		
Watchdog timer		1 channel		
Standby function		HALT/STOP/IDLE mode		
Package		100-pin plastic QFP (14 × 20 mm)		
Assembler		RA78K4		
C compiler		CC78K4		
Real-time OS		RX78K4, MX78K4		
System simulator		SM78K4		
Integrated debugger		ID78K4 or ID78K4-NS		
Device file		DF784908		
In-circuit emulator		IE-784000-R+IE-784000-R-EM+IE-784908-NS-EM1+IE-78K4-R-EX3 (with ID78K4) IE-78K4-NS+IE-784908-NS-EM1 (with ID78K4-NS)		
Applications		Car audio, etc.		

8/16-Bit Single Chip Microcontroller

■ 78K/IV Series Products (15/16)

Subseries name		μPD784937 Subseries			
Part number		μPD784935★	μPD784936★	μPD784937★	μPD78F4937★
Number of instructions		113			
Minimum instruction cycle		160 ns (12.58 MHz)			
Memory expansion function		Total of 1M bytes for both programs and data			
Internal memory	ROM size (bytes)	96K	128K	192K	192K (Flash memory)
	RAM size (bytes)	5120	6656	8192	
Interrupt function		4-level programmable priority, macro service, vectored interrupt, context switching function			
	External source	7			
	Internal source	20			
I/O pin	Input	8			
	I/O	72			
	LED direct drive output	24			
	Transistor direct drive	8			
	N-ch open-drain I/O	4			
Timer/counter		16-bit timer/counter × 4 channels			
Real-time output port		4-bit × 2 ports or 8-bit × 1 port			
IEBus controller		Internal			
Serial interface		UART/IOE (with dedicated on-chip baud rete generator): 2 channels CSI (3-wire): 2 channels			
A/D converter		8-bit resolution × 8 channels			
PWM output		12-bit resolution × 2 channels			
Clock output		Select of f_{XX}, $f_{XX}/2$, $f_{XX}/2^2$, $f_{XX}/2^3$, $f_{XX}/2^4$			
Watchdog timer		1 channel			
Standby function		HALT/STOP/IDLE mode			
Package		100-pin plastic QFP (14 × 20 mm, 0.65-mm pitch) 100-pin plastic LQFP (14 × 14 mm, 0.5-mm pitch)			
Assembler		RA78K4			
C compiler		CC78K4			
Real-time OS		RX78K4, MX78K4			
System simulator		SM78K4			
Integrated debugger		ID78K4 or ID78K4-NS			
Device file		DF784937★			
In-circuit emulator		IE-784000-R+IE-784000-R-EM+IE-784937-NS-EM1★+IE78K4-R-EX3 (with ID78K4) IE-78K4-NS+IE-784937-NS-EM1★ (with ID78K4-NS)			
Applications		Car audio, etc.			

★: Under development

8/16-Bit Single Chip Microcontroller

■ 78K/IV Series Products (16/16)

Subseries name		μPD784955 Subseries		
Part number		μPD784953★	μPD784955★	μPD78F4956★
Number of instructions		113		
Minimum instruction cycle		160 ns (Internal 12.5 MHz)		
Internal memory	ROM size (bytes)	24K	48K	64K (Flash memory)
	RAM size (bytes)	768	2048	
Interrupt function		4-level programmable priority, macro service, vectored interrupt, context switching function		
	External source	8 (Also internal source: 2)		
	Internal source	22		
I/O pin	Input	8		
	I/O	59		
	Pin with pull-up register	59		
	LED direct drive output	32		
Timer/counter		16-bit timer/counter × 6 channels 8-bit timer/counter × 2 channels		
Serial interface		Synchronous serial interface (3-wire): 1 channel UART: 1 channel		
A/D converter		8-bit resolution × 8 channels		
Watchdog timer		1 channel		
Standby function		HALT/STOP/IDLE mode		
Package		80-pin plastic QFP (14 × 14 mm, 1.4 mm thick)		
Assembler		RA78K4		
C compiler		CC78K4		
Real-time OS		RX78K4, MX78K4		
System simulator		SM78K4		
Integrated debugger		ID78K4 or ID78K4-NS		
Device file		DF784956★		
In-circuit emulator		IE-784000-R+IE-784000-R-EM+IE-784956-NS-EM1+IE-78K4-R-EX2 (with ID78K4) IE-78K4-NS+IE-784956-NS-EM1 (with ID78K4-NS)		
Applications		Motor control for inverter air-conditioners, etc.		

★: Under development

V Series™

16-bit V Series

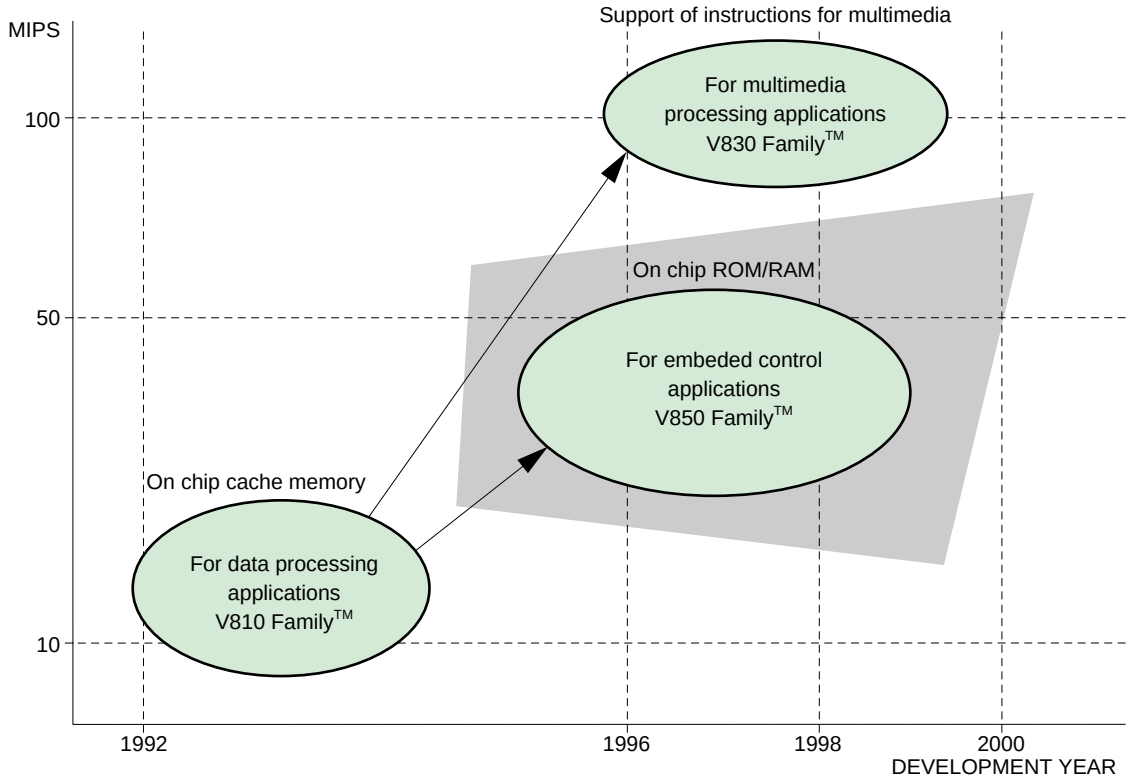
■ 16-bit V Series Products

Part number	Alias	Features	Bus width (Bits)		Real memory space (Bytes)	Operating frequency (MHz)	Package
			Internal	External			
μPD70108H	V20HL™	V20™/V30™ compatible, high speed low voltage operation • Upward software and pin compatible, including interfacing with peripheral controller • Capable of operating at 16 MHz (at 5 V) • Capable of operating at 3 V (at 8 MHz) • Clock can be stopped • Standby function	16	8	1M	10 12.5 16	• 40-pin DIP • 44-pin QFJ • 52-pin QFP
μPD70116H	V30HL™		16	16			
μPD70208H	V40HL™	V40™/V50™ upper compatible, high speed low voltage operation • Capable of operating at 20 MHz (at 5 V) • Capable of operating at 3 V (at 10 MHz) V20HL/V30HL CPU plus the following peripheral functions: • 4-ch DMA controller • 3-ch 16-bit timer/counter • Interrupt controller • Programmable clock generator • DRAM refresh controller • Programmable wait generator (Internal peripheral controllers are subset functions of μPD7105X series) • 2 kinds of standby function	16	8	1M	10 12.5 16 20	• 68-pin QFJ • 80-pin QFP • 80-pin TQFP
μPD70216H	V50HL™		16	16			
μPD70236A	V53A™	V20/V30 compatible high-performance + enhanced peripheral functions of V40/V50 • Incorporates same peripheral functions as those of V40HL and V50HL • Performance of approximately 4 times that of V30 (10 MHz) • Delivers 5.28 VAX MIPS performance in 20 MHz operation • Support of 16M-byte address space • Internal dynamic bus sizing function • 3 kinds of power consumption reduction control functions • Capable of operating at 3 V and 10 MHz	16	8/16 (Selectable)	16M	10 12.5 16 20	• 132-pin PGA • 120-pin QFP (0.8 or 0.5 mm pin pitch) • 120-pin TQFP (0.4 mm pin pitch)
μPD70320	V25™	Software compatible with V20/V30 Following additional peripheral features including RAM: • 2-ch DMA controller • 2-ch serial controller • 2-ch 16-bit timer/counter • Interrupt controller • 24 general purpose I/O port pins • 8 general purpose register banks • Macro service function	16	8	1M	5 8	• 84-pin QFJ • 94-pin QFP
μPD70325	V25+™	Speed up V25 and achieves high-speed DMA transfer • High-speed DMA transfer of 5.3M bytes/second • Capable of operating at 10 MHz (at 5 V)	16	8	1M	8 10	• 84-pin QFJ • 94-pin QFP
μPD70433	V55PI™	Software compatible with V20/V30/V25 and on-chip following various periheral features for mechanical control • 16 general purpose register banks (512-byte internal RAM) • Multi-mode interrupt controller (Vectored/Register bank switching/Macro service) • Real time output ports for controlling stepping moters • 8-bit A/D converter with 4 inputs • 2 DMA controllers with intelligent modes • 5 16-bit timer/counters • 2 asynchronous/synchronous serial controllers • 8-bit parallel interface • Watchdog timer • MH/MR codec instructions	16	8/16 (Selectable)	16M	12.5 16	• 132-pin PGA • 120-pin QFP (0.8 or 0.5 mm pin pitch)

V Series

V800 Series™

■ V800 Series Road Map



V Series

■ V800 Series Products (1/6)

Part number		μPD70732	μPD70741	μPD705100	μPD705101	μPD705102★
Alias		V810™	V821™	V830™	V831™	V832™
Performance		18 MIPS/25 MHz	16 MIPS/25 MHz	118 MIPS/100 MHz	118 MIPS/100 MHz	170 MIPS/143 MHz
Operating frequency		25 MHz	25 MHz	100 MHz	100 MHz	143 MHz
Operating voltage		2.2 to 5.5 V	4.5 to 5.5 V	3.0 to 3.6 V	3.0 to 3.6 V	2.4 to 2.7 V 3.0 to 3.6 V
Power consumption (max.)		500 mW (5 V/25 MHz)	500 mW (5 V/25 MHz)	400 mW (3.3 V/100 MHz)	550 mW (3.3 V/100 MHz)	380 mW (2.5 V, 3.3 V/ 143 MHz)
Address bus		32-bit	24-bit	32-bit	24-bit	
Data bus		32-bit	16-bit	16-/32-bit	16-/32-bit	
Address space		4G bytes	Internal: 4 G bytes External: 64M bytes (maximum)	4G bytes	Internal: 4 G bytes External: 128M bytes (maximum)	
Cache		Instruction cache: 1K bytes		Instruction cache: 4K bytes, Data cache: 4K bytes		
Internal RAM		—		Instruction RAM: 4K bytes, Data RAM: 4K bytes		
DSP		—		32 bits		
Peripheral I/O	Timer	—	2 channels	—	2 channels	
	Interrupt	—	External: 9, Internal: 11	—	External: 9, Internal: 11	
	Serial I/O	—	2 channels	—	2 channels	
	DMA	—	2 channels	—	4 channels	
	Memory interface	—	Fast page mode DRAM, SRAM, I/O, Page-ROM	—	EDO DRAM, SRAM, I/O, Page-ROM	SDRAM, SRAM, I/O, Page-ROM
Others		—	Watchdog timer	—	Debug controller	
Package		120-pin plastic QFP 120-pin plastic TQFP	100-pin plastic QFP	144-pin plastic LQFP	160-pin plastic LQFP	
Applications		Control equipment that incorporates microcomputers, including facsimile machines, digital PPCs, word processors, image processing devices, and real-time control devices	Printers, facsimiles, word processors, game machines, car navigation, PDAs, etc.	Car navigation, high-performance TVs, color facsimile machines, PDAs, PHS (Personal Handyphone System), etc.	Internet/intra-net systems, car navigation, color facsimile machines, high-performance TVs, etc.	Internet/intra-net systems, car navigation, color facsimile machines, digital still cameras, etc.

★: Under development

V Series

■ V800 Series Products (2/6)

Part number		μPD703100★	μPD703101★	μPD703102★	μPD70F3102★
Alias		V850E/MS1™			
CPU core		V850E			
CPU performance (Dhrystone)		—	43 MIPS (33 MHz)		
On-chip ROM (bytes)		—	96K	128K	128K (Flash memory)
On-chip RAM (bytes)		4K			
External bus interface	Address bus	24-bit			
	Data bus	16-bit			
	Programmable wait	0 to 7			
Interrupt controller		External: 25, Internal: 47			
DSP	32 × 32 → 64	0.025 to 0.05 μs (40 MHz)	0.03 to 0.06 μs (33 MHz)		
	32 × 32 + 32 → 32	0.075 μs (40 MHz)	0.09 μs (33 MHz)		
	16 × 16 → 32	—			
	16 × 16 + 32 → 32	—			
Timer/counter (RPU)		16-bit × 8 channels			
Serial interface (SIO)	CSI	2 channels			
	CSI/I²C	—			
	CSI/UART	2 channels			
	UART	—			
	Dedicated baud rate generator	3 channels			
A/D converter		10-bit resolution × 8 channels			
DMA controller		4 channels			
Real-time output port		—			
Port	Input/output	114			
	Input	9			
Other peripheral function		Memory access control function (Direct connectable various memory)			
Power save function		HALT/IDLE/STOP			
Operating frequency		2 to 40 MHz (Internal: 3.3 V, External: 5 V or 3.3 V)	2 to 33 MHz (Internal: 3.3 V, External: 5 V or 3.3 V)		
Operating voltage		Internal unit: 3.3 V, A/D converter: 5 V or 3.3 V, External pin: 5 V or 3.3 V			
Power consumption (TYP.)		570 mW (Internal: 3.3 V, External: 5 V, 40 MHz)	470 mW (Internal: 3.3 V, External: 5 V, 33 MHz)		
Package		144-pin plastic LQFP (20 × 20 mm) 157-pin plastic FBGA (14 × 14 mm)			
Real-time OS		RX850 (Reduced version) RX850 Pro (Advanced version)			
C compiler		CA850			
Debugger		ID850			
Simulator		SM850			
System performance analyzer		AZ850			
In-circuit emulator		IE-703100-MC			
Applications		Printers, facsimiles, PPCs, digital still cameras, video printers, DVD systems, etc.			

★: Under development

V Series

■ V800 Series Products (3/6)

Part number		μPD703015/ μPD703015Y★	μPD703017★/ μPD703017Y★	μPD70F3017★/ μPD70F3017Y★
Alias		V850/SA1™		
CPU core		V850		
CPU performance (Dhrystone)		19 MIPS (17 MHz)		
On-chip ROM (bytes)		128K	256K	256K (Flash memory)
On-chip RAM (bytes)		4K	8K	8K
External bus interface	Address bus	22-bit		
	Data bus	16-bit		
	Programmable wait	0 to 3		
Interrupt controller		External: 8, Internal: 23		
DSP	32 × 32 → 64	—		
	32 × 32 + 32 → 32	—		
	16 × 16 → 32	0.059 to 0.118 μs (17 MHz)		
	16 × 16 + 32 → 32	0.177 μs (17 MHz)		
Timer/counter (RPU)		16-bit × 2 channels 8-bit × 4 channels (Cascade connection supported)		
Serial interface (SIO)	CSI	2 channels/1 channel	2 channels/1 channel	2 channels/1 channel
	CSI/I²C	0 channels/1 channel	0 channels/1 channel	0 channels/1 channel
	CSI/UART	1 channel		
	UART	1 channel		
	Dedicated baud rate generator	2 channels (UART only)		
A/D converter		10-bit resolution × 12 channels		
DMA controller		3 channels (internal RAM ←→ internal peripheral I/O)		
Real-time output port		8-bit × 1 or 4-bit × 2		
Port	Input/output	72		
	Input	13		
Other peripheral function		Watch timer: 1 channel Watchdog timer: 1 channel		
Power save function		HALT/IDLE/STOP		
Operating frequency		1 to 17 MHz (main clock, 3 V) 32.768 kHz (subclok)		
Operating voltage		2.7 to 3.6 V		
Power consumption (TYP.)		30 mW (main clock: 10 MHz, 3 V)		
Package		100-pin plastic LQFP (14 × 14 mm) 121-pin plastic FBGA (12 × 12 mm)		
Real-time OS		RX850 (Reduced version) RX850 Pro (Advanced version)		
C compiler		CA850		
Debugger		ID850		
Simulator		SM850		
System performance analyzer		AZ850		
In-circuit emulator		IE-703017-MC-EM1 + IE-703002-MC		
Applications		Portable equipment such as camcorders, VCRs, PHS (Personal Handyphone System), digital still cameras, cellular phones, portable MDs, etc.		

★: Under development

V Series

■ V800 Series Products (4/6)

Part number		μPD703033★/ μPD703033Y★	μPD70F3033★/ μPD70F3033Y★	μPD70F3035★/ μPD70F3035Y★	μPD70F3037★/ μPD70F3037Y★	μPD703008★/ μPD703008Y★	μPD70F3008★/ μPD70F3008Y★
Alias		V850/SB1™		V850/SB2™	V850/SB3™	V854™	
CPU core		V850					
CPU performance (Dhrystone)		22 MIPS (20 MHz)				38 MIPS (33 MHz)	
On-chip ROM (bytes)		256K	256K (Flash memory)			128K	128K (Flash memory)
On-chip RAM (bytes)		16K				4K	
External bus interface	Address bus	22-bit				24-bit	
	Data bus	16-bit					
	Programmable wait	0 to 3					
Interrupt controller		External: 8, Internal: 30 (31 for Y products)		External: 8, Internal: 32 (33 for Y products)	External: 8, Internal: 31 (32 for Y products)	External: 22, Internal: 31	
DSP	32 × 32 → 64	—					
	32 × 32 + 32 → 32	—					
	16 × 16 → 32	0.050 to 0.100 μs (20 MHz)				0.03 to 0.06 μs (33 MHz)	
	16 × 16 + 32 → 32	0.150 μs (20 MHz)				0.09 μs (33 MHz)	
Timer/counter (RPU)		16-bit × 2 channels 8-bit × 6 channels (Cascade connection supported)				24-bit × 2 channels 16-bit × 6 channels	
Serial interface (SIO)	CSI	3 channels/1 channel	3 channels/1 channel	3 channels/1 channel	3 channels/1 channel	3 channels/2 channels	3 channels/2 channels
	CSI/I²C	0 channels/2 channels	0 channels/2 channels	0 channels/2 channels	0 channels/2 channels	0 channels/1 channel	0 channels/1 channel
	CSI/UART	2 channels				1 channel	
	UART	—				—	
	Dedicated baud rate generator	3 channels				4 channels	
A/D converter		10-bit resolution × 12 channels				8-bit resolution × 16 channels	
DMA controller		6 channels (internal RAM ←→ internal peripheral I/O)				—	
Real-time output port		8-bit × 1 or 4-bit × 2				8	
Port	Input/output	71				96	
	Input	12				16	
Other peripheral function		Watch timer: 1 channel Watchdog timer: 1 channel		Watch timer: 1 channel Watchdog timer: 1 channel IEBus (Simple version)	Watch timer: 1 channel Watchdog timer: 1 channel J1850	PWM: 4 channels (12 to 16-bit resolution)	
Power save function		HALT/IDLE/STOP					
Operating frequency		2 to 20 MHz (main clock) 32.768 kHz (subclock)				2 to 33 MHz (3 V)	
Operating voltage		4.5 to 5.5 V				2.7 to 3.6 V	
Power consumption (TYP.)		40 mW (target value) (main clock: 5 V, 10 MHz)				120 mW (3 V, 33 MHz)	
Package		100-pin plastic LQFP (14 × 14 mm) 100-pin plastic QFP (14 × 20 mm)				144-pin plastic LQFP (20 × 20 mm)	
Real-time OS		RX850 (Reduced version) RX850 Pro (Advanced version)					
C compiler		CA850					
Debugger		ID850					
Simulator		SM850					
System performance analyzer		AZ850					
In-circuit emulator		IE-703037-MC-EM1★ + IE-703002-MC				IE-703008-MC-EM1 + IE-703002-MC	
Applications		AV equipment such as car audio systems, TVs, and VCRs, and other applications				Low-voltage-operation equipment including camcorders, VCRs, cellular phones, PDAs, and digital still cameras	

★: Under development

V Series

■ V800 Series Products (5/6)

Part number		μPD703003/ μPD703003A★	μPD703004A★	μPD70F3003	μPD70F3003A★	μPD703025A★	μPD70F3025A★
Alias		V853™					
CPU core		V850					
CPU performance (Dhrystone)		38 MIPS (33 MHz)		29 MIPS (25 MHz)	38 MIPS (33 MHz)		
On-chip ROM (bytes)		128K	96K	128K (Flash memory)		256K	256K (Flash memory)
On-chip RAM (bytes)		4K				8K	
External bus interface	Address bus	20-bit					
	Data bus	16-bit					
	Programmable wait	0 to 3					
Interrupt controller		External: 17, Internal: 32					
DSP	32 × 32 → 64	—					
	32 × 32 + 32 → 32	—					
	16 × 16 → 32	0.03 to 0.06 μs (33 MHz)					
	16 × 16 + 32 → 32	0.09 μs (33 MHz)					
Timer/counter (RPU)		16 bit × 5 channels					
Serial interface (SIO)	CSI	2 channels					
	CSI/I²C	—					
	CSI/UART	2 channels					
	UART	—					
	Dedicated baud rate generator	3 channels					
A/D converter		10-bit resolution × 8 channels					
DMA controller		—					
Real-time output port		—					
Port	Input/output	67					
	Input	8					
Other peripheral function		PWM: 4 ch (8/9/10/12-bit resolution) D/A converter: 2 channels					
Power save function		HALT/IDLE/STOP					
Operating frequency		5 to 33 MHz (5 V)		5 to 25 MHz (5 V)	5 to 33 MHz (5 V)		
Operating voltage		4.5 to 5.5 V		4.75 to 5.25 V	4.5 to 5.5 V		
Power consumption (TYP.)		450 mW/365 mW (5 V, 33 MHz)	365 mW (5 V, 33 MHz)	385 mW (5 V, 25 MHz)	425 mW (5 V, 33 MHz)	450 mW (5 V, 33 MHz)	480 mW (5 V, 33 MHz)
Package		100-pin plastic QFP (14 × 14 mm)					
Real-time OS		RX850 (Reduced version) RX850 Pro (Advanced version)					
C compiler		CA850					
Debugger		ID850					
Simulator		SM850					
System performance analyzer		AZ850					
In-circuit emulator		IE-703003-MC-EM1 + IE-703002-MC					
Applications		HDDs, PPCs, VCRs, motor control, robots, LBPs, printers, NC machine tools, single-lens reflex cameras, digital still cameras, etc.					

★: Under development

V Series

■ V800 Series Products (6/6)

Part number		μPD703002	μPD70P3002	μPD703001	μPD703000	μPD70P3000
Alias		V852™		V851™		
CPU core		V850				
CPU performance (Dhrystone)		29 MIPS (25 MHz)		38 MIPS (33 MHz)		
On-chip ROM (bytes)		90K	90K (PROM)	—	32K	32K (PROM)
On-chip RAM (bytes)		3K		1K		
External bus interface	Address bus	24-bit				
	Data bus	16-bit				
	Programmable wait	0 to 3				
Interrupt controller		External: 9, Internal: 16		External: 9, Internal: 10		
DSP	32 × 32 → 64	—				
	32 × 32 + 32 → 32	—				
	16 × 16 → 32	0.04 to 0.08 μs (25 MHz)		0.03 to 0.06 μs (33 MHz)		
	16 × 16 + 32 → 32	0.12 μs (25 MHz)		0.09 μs (33 MHz)		
Timer/counter (RPU)		16-bit × 2 channels				
Serial interface (SIO)	CSI	3 channels		1 channel		
	CSI/I²C	—				
	CSI/UART	—				
	UART	1 channel				
	Dedicated baud rate generator	2 channels		1 channel		
A/D converter		—				
DMA controller		—				
Real-time output port		—				
Port	Input/output	67				
	Input	1				
Other peripheral function		—				
Power save function		HALT/IDLE/STOP				
Operating frequency		3 to 25 MHz (5 V)		3 to 33 MHz (5 V) 3 to 12 MHz (3 V)		
Operating voltage		4.5 to 5.5 V		3.0 to 5.5 V		
Power consumption (TYP.)		250 mW (5 V, 25 MHz)	320 mW (5 V, 25 MHz)	270 mW (5 V, 33 MHz)		330 mW (5 V, 33 MHz)
Package		100-pin plastic QFP (14 × 14 mm)				
Real-time OS		RX850 (Reduced version) RX850 Pro (Advanced version)				
C compiler		CA850				
Debugger		ID850				
Simulator		SM850				
System performance analyzer		AZ850				
In-circuit emulator		IE-703002-MC				
Applications		HDDs, PPCs, VCRs, motor control, robots, LBPs, printers, NC machine tools, digital phones, engine control, etc.				

V Series

VR Series™

■ VR Series Products (1/2)

Part number	μPD30102	μPD30111	μPD30200	
Alias	VR4102™	VR4111™	VR4300™	VR4305™
Internal frequency	49 MHz	80 MHz 100 MHz	100 MHz 133 MHz	80 MHz
Operating voltage	3.3 V	External: 3.3 V Internal: 2.5 V	3.3 V	
Cache size	Instruction: 4K bytes Data: 1K bytes	Instruction: 16K bytes Data: 8K bytes	Instruction: 16K bytes Data: 8K bytes	
Performance	59 MIPS	130 MIPS	177 MIPS 80 SPECint92 60 SPECfp92	106 MIPS 48 SPECint92 36 SPECfp92
Features	CPU core: VR4100™ On-chip peripheral function ideal for PDAs High-speed processing by 49 MHz operation and 32-bit memory bus Low power consumption design	CPU core: VR4110™ On-chip peripheral function ideal for PDAs High-speed processing by 100 MHz operation and 0.25 μm process rule Low power consumption design	Software compatible with VR4000™ Series 32-bit external bus High-speed floating-point operation	
Package	216-pin plastic LQFP 224-pin plastic FBGA	224-pin plastic FBGA	120-pin plastic QFP	
Applications	PDAs such as handheld PCs and palm-size PCs		Embedded equipment for LBP, STB, and game machines	

■ VR Series Products (2/2)

Part number	μPD30210★	μPD30500	μPD30700	
Alias	VR4310™	VR5000™	VR10000™	
Internal frequency	167 MHz	200 MHz 250 MHz	200 MHz	250 MHz
Operating voltage	3.3 V	3.3 V	3.3 V	2.6 V
Cache size	Instruction: 16K bytes Data: 8K bytes	Instruction: 32K bytes Data: 32K bytes	Instruction: 32K bytes Data: 32K bytes	
Performance	222 MIPS 100 SPECint92 75 SPECfp92	391 MIPS 6.8 SPECint95 6.8 SPECfp95	9.0 SPECint95 19.0 SPECfp95	14 SPECint95 23 SPECfp95
Features	Software compatible with VR4300 32-bit external bus High-speed floating-point operation	Two-way superscalar On-chip large primary cache On-chip secondary cache interface	Five-way superscalar On-chip FPU and MMU Multiprocessor On-chip secondary cache interface	
Package	120-pin plastic QFP	272-pin plastic BGA 223-pin ceramic PGA	599-pin ceramic LGA	
Applications	Embedded equipment for LBP, STB, and game machines	High-end embedded equipment for color LBPs, network routers, etc	Advanced WSs, servers	

★: Under development

[MEMO]

IC Memory

Dynamic RAM 114

Dynamic RAM Module 120

Static RAM 124

Mask ROM 128

COMBO Memory 129

Flash Memory 130

MCP (Flash Memory + SRAM) 131

Other 132

- Synchronous Graphics RAM 132
- Field/Line Buffer 132

Dynamic RAM

■ Synchronous DRAM: SDR (Single Data Rate) (1/4)

Density (bits)	Organization (words × bits × banks)	Part number	Class	/CAS latency	Clock frequency (MHz)	Clock access time MAX. (ns)	Maximum supply current (mA)				Supply voltage (V)	Refresh cycle (cycles /ms)	Interface level	Package	Remark
							Active		Refresh						
							Normal (BL = 1)	Burst	Auto	Self					
256M★	16M×4×4	**μPD45256441**	A80	3	125	6	115	115	180	2	3.3±0.3	8K/64	LVTTL	G5: 54-pin TSOP(II) (400 mil)	PC100 Compliant
				2	100		110	100							
			A10	3	100		115								
				2	77	7	110	80							
			A10B	3	100		95	95	150						—
				2	67	8	90	75							
			A80L	3	125	6	115	115	180	1					PC100 Compliant
				2	100		110	100							
			A10L	3	100		115								
				2	77	7	110	80							
			A10BL	3	100		95	95	150						—
				2	67	8	90	75							
	8M×8×4	**μPD45256841**	A80	3	125	6	125	130	180	2					PC100 Compliant
				2	100		120	110							
			A10	3	100		125	115							
				2	77	7	120	90							
			A10B	3	100		105	105	150						—
				2	67	8	100	80							
			A80L	3	125	6	125	130	180	1					PC100 Compliant
				2	100		120	110							
			A10L	3	100		125	115							
				2	77	7	120	90							
			A10BL	3	100		105	105	150						—
				2	67	8	100	80							
	4M×16×4	**μPD45256163**	A80	3	125	6	115	115	180	2					PC100 Compliant
				2	100		110	100							
			A10	3	100		115	100							
				2	77	7	110	80							
			A10B	3	100		95	95	150						—
				2	67	8	90	75							
			A80L	3	125	6	115	115	180	1					PC100 Compliant
				2	100		110	100							
			A10L	3	100		115								
				2	77	7	110	80							
			A10BL	3	100		95	95	150						—
				2	67	8	90	75							

BL: Burst Length, ★: Under development

Dynamic RAM

■ Synchronous DRAM: SDR (Single Data Rate) (2/4)

Density (bits)	Organization (words × bits × banks)	Part number	Class	/CAS latency	Clock frequency (MHz)	Clock access time MAX. (ns)	Maximum supply current (mA) Active Normal (BL = 1)	Active Burst	Refresh Auto	Refresh Self	Supply voltage (V)	Refresh cycle (cycles /ms)	Interface level	Package	Remark
128M	8M×4×4	**μPD45128441**	A80★	3	125	6	115	115	270	2	3.3±0.3	4K/64	LVTTL	G5: 54-pin TSOP(II) (400 mil)	PC100 Compliant
				2	100		110	100	260						
			A10★	3	100		100		250						
				2	77	7	95	80	240						
			A10B	3	100			95	230						—
				2	67	8	90	75	220						
			A80L★	3	125	6	115	115	270	0.8					PC100 Compliant
				2	100		110	100	260						
			A10L★	3	100		100		250						
				2	77	7	95	80	240						
			A10BL★	3	100			95	230						—
				2	67	8	90	75	220						
	4M×8×4	**μPD45128841**	A80★	3	125	6	125	135	270	2					PC100 Compliant
				2	100		120	115	260						
			A10★	3	100		110		250						
				2	77	7	105	90	240						
			A10B	3	100			105	230						—
				2	67	8	100	80	220						
			A80L★	3	125	6	125	135	270	0.8					PC100 Compliant
				2	100		120	115	260						
			A10L★	3	100		110		250						
				2	77	7	105	90	240						
			A10BL★	3	100			105	230						—
				2	67	8	100	80	220						
	2M×16×4	**μPD45128163**	A80★	3	125	6	150	195	270	2					PC100 Compliant
				2	100		135	165	260						
			A10★	3	100		130		250						
				2	77	7	120	130	240						
			A10B	3	100			150	230						—
				2	67	8	110	120	220						
			A80L★	3	125	6	150	195	270	0.8					PC100 Compliant
				2	100		135	165	260						
			A10L★	3	100		130		250						
				2	77	7	120	130	240						
			A10BL★	3	100			150	230						—
				2	67	8	110	120	220						

BL: Burst Length, ★: Under development

Dynamic RAM

■ Synchronous DRAM: SDR (Single Data Rate) (3/4)

Density (bits)	Organization (words × bits × banks)	Part number	Class	/CAS latency	Clock frequency (MHz)	Clock access time MAX. (ns)	Maximum supply current (mA) Active Normal (BL = 1)	Active Burst	Refresh Auto	Refresh Self	Supply voltage (V)	Refresh cycle (cycles /ms)	Interface level	Package	Remark
64M	4M×4×4	**μPD4564441**	A80	3	125	6	80	105	135	1	3.3±0.3	4K/64	LVTTL	G5: 54-pin TSOP(II) (400 mil)	PC100 Compliant
				2	83		75	90	130						
			A10	3	100		70		135						
				2	77	7	65	70	130						
			A10B	3	100		70	90	115						—
				2	67	8	60	65	105						
			A80L	3	125	6	80	105	135	0.4					PC100 Compliant
				2	83		75	90	130						
			A10L	3	100		70		135						
				2	77	7	65	70	130						
			A10BL	3	100		70	90	115						—
				2	67	8	60	65	105						
	2M×8×4	**μPD4564841**	A80	3	125	6	85	125	135	1					PC100 Compliant
				2	83		80	105	130						
			A10	3	100		75		135						
				2	77	7	70	80	130						
			A10B	3	100		75	105	115						—
				2	67	8	65	70	105						
			A80L	3	125	6	85	125	135	0.4					PC100 Compliant
				2	83		80	105	130						
			A10L	3	100		75		135						
				2	77	7	70	80	130						
			A10BL	3	100		75	105	115						—
				2	67	8	65	70	105						
	1M×16×4	**μPD4564163**	A80	3	125	6	115	195	135	1					PC100 Compliant
				2	83		90	165	130						
			A10	3	100				135						
				2	77	7	80	130	130						
			A10B	3	100		90	165	115						—
				2	67	8	70	110	105						
			A80L	3	125	6	115	195	135	0.4					PC100 Compliant
				2	83		90	165	130						
			A10L	3	100				135						
				2	77	7	80	130	130						
			A10BL	3	100		90	165	115						—
				2	67	8	70	110	105						
	512K×32×4	**μPD4564323**	A80★	3	125	6	180	230	205	1				G5: 86-pin TSOP(II) (400 mil)	PC100 Compliant
				2	100		170	180	200						
			A10★	3	100		160	200	205						
				2	77	7	150	160	200						
			A10B	3	100		160	200	175						—
				2	67	8	150	140	170						
			A80L★	3	125	6	180	230	205	0.4					PC100 Compliant
				2	100		170	180	200						
			A10L★	3	100		160	200	205						
				2	77	7	150	160	200						
			A10BL★	3	100		160	200	175						—
				2	67	8	150	140	170						

BL: Burst Length, ★: Under development

Dynamic RAM

■ Synchronous DRAM: SDR (Single Data Rate) (4/4)

Density (bits)	Organization (words × bits × banks)	Part number	Class	/CAS latency	Clock frequency (MHz)	Clock access time MAX. (ns)	Maximum supply current (mA) Active Normal (BL = 1)	Active Burst	Refresh Auto	Refresh Self	Supply voltage (V)	Refresh cycle (cycles /ms)	Interface level	Package	Remark
16M	2M×4×2	**μPD4516421A**	A80	3	120	6	110	110	90	1	3.3±0.3	2K/32	LVTTL	G5: 44-pin TSOP(II) (400 mil)	PC100 Compliant
				2	100		100	95							
			A10	3	100		110	90							
				2	77	8	100	75							
			A10B	3	100	7	90	90							—
				2	77	8	85	75							
			A80L	3	120	6	110	110		0.25		2K/64			PC100 Compliant
				2	100		100	95							
			A10L	3	100		110	90							
				2	77	8	100	75							
			A10BL	3	100	7	90	90							—
				2	77	8	85	75							
	1M×8×2	**μPD4516821A**	A80	3	120	6	115	120		1		2K/32			PC100 Compliant
				2	100		105	105							
			A10	3	100		115	100							
				2	77	8	105	85							
			A10B	3	100	7	95	100							—
				2	77	8	90	85							
			A80L	3	120	6	115	120		0.25		2K/64			PC100 Compliant
				2	100		105	105							
			A10L	3	100		115	100							
				2	77	8	105	85							
			A10BL	3	100	7	95	100							—
				2	77	8	90	85							
	512K×16×2	**μPD4516161A**	A80	3	120	6	120	125		1		2K/32		G5: 50-pin TSOP(II) (400 mil)	PC100 Compliant
				2	100		110	110							
			A10	3	100		120	105							
				2	77	8	110	90							
			A10B	3	100	7	100	105							—
				2	77	8	95	90							
			A80L	3	120	6	120	125		0.25		2K/64			PC100 Compliant
				2	100		110	110							
			A10L	3	100		120	105							
				2	77	8	110	90							
			A10BL	3	100	7	100	105							—
				2	77	8	95	90							
4M	128K×16×2	**μPD4504161**	A10	3	100		90	140	85	2		1K/16			
				2	67	10									
			A12	3	83	9		120							
				2	67	11		100							

BL: Burst Length, ★: Under development

Dynamic RAM

■ Virtual Channel Synchronous DRAM:SDR (Single Data Rate)

Density (bits)	Organization (words × bits × banks)	Part number	Class	Read latency	Clock frequency (MHz)	Clock access time MAX. (ns)	Maximum supply current (mA)				Supply voltage (V)	Refresh cycle (cycles /ms)	Interface level	Package	Remark
							Active		Refresh						
							Normal (BL = 1)	Burst	Auto	Self					
64M	8M×4×2	**μPD4565421**	A70★	2	143	5.5	90	60	145	1.0	3.3±0.3	4K/64	LVTTL	G5: 54-pin TSOP(II) (400 mil)	
			A80★		125	6	85	55	135						
			A10		100	6	80	45	115						
	4M×8×2	**μPD4565821**	A70★		143	5.5	90	70	145						
			A80★		125	6	85	60	135						
			A10		100	6	80	50	115						
	2M×16×2	**μPD4565161**	A70★		143	5.5	90	100	145						
			A80★		125	6	85	90	135						
			A10		100	6	80	75	115						

BL: Burst Length, ★: Under development

■ Synchronous DRAM: DDR (Double Data Rate)★

Density (bits)	Organization (words × bits × banks)	Part number	Class	/CAS latency	Clock frequency (MHz)	Maximum supply current (mA)				Supply voltage (V)		Refresh cycle (cycles /ms)	Interface level	Package	Remark
						Active		Refresh							
						Normal (BL = 1)	Burst	Auto	Self	V_{CC}	V_{CCQ}				
128M	8M×4×4	**μPD45D128442**	C10	2.5	125	TBD	TBD	TBD	2	2.5±0.2	2.5±0.2	4K/64	SSTL_2	G5: 66-pin TSOP(II) (400 mil)	
				2	100										
			C12	2.5	100										
				2	83										
	4M×8×4	**μPD45D128842**	C10	2.5	125										
				2	100										
			C12	2.5	100										
				2	83										
	2M×16×4	**μPD45D128164**	C10	2.5	125										
				2	100										
			C12	2.5	100										
				2	83										

BL: Burst Length, ★: Under development

Dynamic RAM

■ Direct Rambus™ DRAM★

Density (bits)	Organization (words × bits × banks)	Part number	Clock frequency		Access time		Maximum supply current (mA)			Supply voltage (V)	Refresh cycle (cycles /ms)	Interface level	Package	Remark
								Refresh						
			Class	1/t_{CYCLE} (MHz)	Class	t_{RAC} (ns)	Active	Auto	Self					
72M	256K×18×16	**μPD488385**	C80	400	40	40	TBD	TBD	TBD	2.5±0.13	8K/32	RSL	FB-BF1: 74-pin D^2BGA^{TM}	
					45	45								
					50	50								
			C60	300	60	60								

★: Under development

Dynamic RAM Module

■ 168-pin SDRAM DIMM (Dual In-line Memory Module): Unbuffered Type

Density (bytes)	Organization (words × bits)	Bank organization	Part number	Class	/CAS latency	Clock frequency (MHz)	Supply voltage (V)	Refresh cycle (cycles /ms)	Interface level	Mounted devices	Package height (mm/inch)	Remark
256M	32M×72	2	MC-4532CC726F★	A80	3	125	3.3±0.3	4K/64	LVTTL	μPD45128841G5 ×18 pcs.	34.93/1.375	PC100 Compliant
					2	100						
				A10	3	100						
					2	77						
	32M×64	2	MC-4532CD646F★	A80	3	125				μPD45128841G5 ×16 pcs.		
					2	100						
				A10	3	100						
					2	77						
128M	16M×72	2	MC-4516CC726F	A80	3	125				μPD4564841G5 ×18 pcs.		
					2	100						
				A10	3	100						
					2	77						
		2	MC-4516CC725FA	A10B	3	100					33.02/1.30	—
					2	67						
	16M×64	2	MC-4516CD646F	A80	3	125				μPD4564841G5 ×16 pcs.	34.93/1.375	PC100 Compliant
					2	100						
				A10	3	100						
					2	77						
		2	MC-4516CD645FA	A10B	3	100				μPD4564841G5 ×16 pcs.	33.02/1.30	—
					2	67						
64M	8M×72	1	MC-458CA726F	A80	3	125				μPD4564841G5 ×9 pcs.	34.93/1.375	PC100 Compliant
					2	100						
				A10	3	100						
					2	77						
		1	MC-458CA725FA	A10B	3	100					31.75/1.25	—
					2	67						
	8M×64	1	MC-458CB646F	A80	3	125				μPD4564841G5 ×8 pcs.	34.93/1.375	PC100 Compliant
					2	100						
				A10	3	100						
					2	77						
		1	MC-458CB645FA	A10B	3	100					31.75/1.25	—
					2	67						
32M	4M×72	1	MC-454CA726F	A80	3	125		4K/64		μPD4564163G5 ×5 pcs.	34.93/1.375	PC100 Compliant
					2	100						
				A10	3	100						
					2	77						
	4M×64	1	MC-454CB646F	A80	3	125		4K/64		μPD4564163G5 ×4 pcs.	34.93/1.375	
					2	100						
				A10	3	100						
					2	77						
		1	MC-454CB645FA	A10B	3	100					29.21/1.15	—
					2	67						

★: Under development

Dynamic RAM Module

■ 168-pin SDRAM DIMM (Dual In-line Memory Module): Registered Type

Density (bytes)	Organization (words × bits)	Bank organization	Part number	Class	/CAS latency	Clock frequency (MHz)	Supply voltage (V)	Refresh cycle (cycles /ms)	Interface level	Mounted devices	Package height (mm/inch)	Remark
256M	32M×72	1	MC-4532DA726F★	A80	3	125	3.3±0.3	4K/64	LVTTL	μPD45128441G5 ×18 pcs.	43.18/1.70	PC100 Compliant
					2	100						
				A10	3	100						
					2	77						
128M	16M×72	1	MC-4516DA726F	A80	3	125				μPD4564441G5 ×18 pcs.		
					2	100						
				A10	3	100						
					2	77						
64M	8M×72	1	MC-458DA726F	A80	3	125				μPD4546841G5 ×9 pcs.	38.1/1.50	
					2	100						
				A10	3	100						
					2	77						
32M	4M×72	1	MC-454DA726F	A80	3	125				μPD4564163G5 ×5 pcs.		
					2	100						
				A10	3	100						
					2	77						

★: Under development

Dynamic RAM Module

■ 144-pin SDRAM SO DIMM (Small Outline Dual In-line Memory Module)

Density (bytes)	Organization (words × bits)	Bank Organization	Part number	class	/CAS latency	Clock frequency (MHz)	Self refresh current (mA)	Supply voltage (V)	Refresh cycle (cycles/ms)	Interface level	Mounted devices	Package height (mm/inch)	Remark
128M	16M×64	1	**MC-4516CB64S**	A80★	3	125	16	3.3±0.3	4K/64	LVTTL	μPD45128841G5 ×8 pcs.	26.67/1.05	
					2	100							
				A10★	3	100							
					2	70							
				A10B	3	100							
					2	67							
				A10BL★	3	100	6.4						
					2	67							
		2	**MC-4516CD64S**	A80★	3	125	16				μPD45128163G5 ×8 pcs.		
					2	100							
				A10★	3	100							
					2	77							
				A10B	3	100							
					2	67							
				A10BL★	3	100	6.4						
					2	67							
64M	8M×64	1	**MC-458CB64SA**	A80	3	100	8				μPD4564841G5 ×8 pcs.		
					2	67							
				A10	3	83							
					2	55							
				A10B	3	100							
					2	67							
				A10BL	3	100	3.2						
					2	67							
		2	**MC-458CD64S**	A80	3	125	8				μPD4564163G5 ×8 pcs.		
					2	100							
				A10	3	100							
					2	77							
				A10B	3	100							
					2	67							
				A10BL	3	100	3.2						
					2	67							
32M	4M×64	1	**MC-454CB64S**	A80	3	125	4				μPD4564163G5 ×4 pcs.	25.4/1.00	
					2	100							
				A10	3	100							
					2	77							
				A10B	3	100							
					2	67							
				A10BL	3	100	2						
					2	67							

★: Under development

Dynamic RAM Module

■ 200-pin SDRAM DIMM (Dual In-line Memory Module): Registered Type

Density (bytes)	Organization (words × bits)	Bank organization	Part number	Class	/CAS latency	Clock frequency (MHz)	Supply voltage (V)	Refresh cycle (cycles /ms)	Interface level	Mounted devices	Package height (mm/inch)	Remark
128M	16M × 72	1	**MC-4516DA72F**	A80	3	100	3.3±0.3	4K/64	LVTTL	μPD4564441G5 × 18 pcs.	38.1/1.50	
					2	100						
				A10	3	100						
					2	77						
				A10B	3	100						
					2	67						
	8M × 72	1	**MC-458DA72F**	A10	3	100				μPD4564841G5 × 9 pcs.		
					2	67						
				A12	3	83						
					2	55						
				A10B	3	100						
					2	67						

■ 184-pin Direct Rambus™ RIMM™★

Density (bytes)	Organization (words × bits)	Bank organization	Part number	Clock frequency		Access time		Supply voltage (V)	Interface level	Mounted devices	Package height (mm/inch)	Remark
				class	$1/t_{CYCLE}$ (MHz)	class	t_{RAC} (ns)					
128M	64M × 18	16 × 16	**MC-48R128AAG181F**	C80	400	40	40	2.5±0.13	RSL	μPD488385FB × 16 pcs.	TBD	
						45	45					
						50	50					
				C60	600	60	60					
64M	32M × 18	16 × 8	**MC-48R64AAG181F**	C80	400	40	40			μPD488385FB × 8 pcs.		
						45	45					
						50	50					
				C60	600	60	60					

★: Under development

Static RAM

■ Low power SRAM (Temperature range T_A = 0 to 70°C)

Density (bits)	Organization (words × bits)	Part number	Supply voltage (V)	Version	Class	Access time MAX. (ns)	Maximum supply current			Package
							Active (mA)	Standby (μA)	Data retention (μA)	
1M	128K × 8	**μPD431000A**	4.5 to 5.5	L	70	70	70	100	50	CZ: 32-pin DIP (600 mil)
					85	85				GW: 32-pin SOP (525 mil)
				LL	70	70		20	10	CZ: 32-pin DIP (600 mil)
					85	85				GW: 32-pin SOP (525 mil)
										GU: 32-pin TSOP(I) (8 × 13.4 mm)
										GZ: 32-pin TSOP(I) (8 × 20 mm)
			3.0 to 5.5	A	10	100	35	13		GW: 32-pin SOP (525 mil)
						70 (V_{CC}=4.5 to 5.5)	70	20		GU: 32-pin TSOP(I) (8 × 13.4 mm)
					12	120	35	13		GZ: 32-pin TSOP(I) (8 × 20 mm)
						70 (V_{CC}=4.5 to 5.5)	70	20		
			2.7 to 5.5	B	10	100	30	11		
						70 (V_{CC}=4.5 to 5.5)	70	20		
					12	120	30	11		
						70 (V_{CC}=4.5 to 5.5)	70	20		
					15	150	30	11		
						70 (V_{CC}=4.5 to 5.5)	70	20		
256K	32K × 8	**μPD43257B**	4.5 to 5.5	L	70	70	45	50	20	CZ: 28-pin DIP (600 mil)
					85	85				GU: 28-pin SOP (450 mil)
				LL	70	70		15	7	
					85	85				
		μPD43256B		L	70	70		50	20	
					85	85				
				LL	70	70		15	7	CZ: 28-pin DIP (600 mil)
					85	85				GU: 28-pin SOP (450 mil)
										GW: 28-pin TSOP(I) (8 × 13.4 mm)
			3.0 to 5.5	A	85	85				GU: 28-pin SOP (450 mil)
					10	100				GW: 28-pin TSOP(I) (8 × 13.4 mm)
						85 (V_{CC}=4.5 to 5.5)				
					12	120				
						85 (V_{CC}=4.5 to 5.5)				
			2.7 to 5.5	B	10	100	20	10		
						85 (V_{CC}=4.5 to 5.5)	45	15		
					12	120	20	10		
						85 (V_{CC}=4.5 to 5.5)	45	15		
					15	150	20	10		
						85 (V_{CC}=4.5 to 5.5)	45	15		

Static RAM

■ Low power SRAM (Extended Temperature T_A = –25 to +85°C)

Density (bits)	Organization (words × bits)	Part number	Supply voltage (V)	Version	Class	Access time MAX. (ns)	Maximum supply current			Package
							Active (mA)	Standby (μA)	Data retention (μA)	
2M	256K × 8	**μPD442000L-X**	2.7 to 3.6	B	85	85	35	2	2	GU: 32-pin TSOP(I) (8 × 13.4 mm) GZ: 32-pin TSOP(I) (8 × 20 mm) F1: 36-pin FPBGA (10.5 × 6.5 mm)
					10	100				
			2.2 to 3.6	C						
					12	120				
					15	150				
			1.8 to 3.6	D						
					20	200				
1M	128K × 8	**μPD441000L-X**	2.7 to 3.6	B	70	70	25			GU: 32-pin TSOP(I) (8 × 13.4 mm) GZ: 32-pin TSOP(I) (8 × 20 mm) F1: 36-pin FPBGA (6 × 6 mm)
					85	85				
					10	100				
			2.2 to 3.6	C	10					
					12	120				
			1.8 to 3.6	D	12					
					15	150				
		μPD431000A-X	4.5 to 5.5	—	70	70	70	50	20	GU: 32-pin TSOP(I) (8 × 1.4 mm) GZ: 32-pin TSOP(I) (8 × 20 mm)
					85	85				
					10	100				
			3.0 to 5.5	A	10		35	26		
						70 (V_{CC}=4.5 to 5.5)	70	50		
					12	120	35	26		
						70 (V_{CC}=4.5 to 5.5)	70	50		
			2.7 to 5.5	B	10	100	30	22		
						70 (V_{CC}=4.5 to 5.5)	70	50		
					12	120	30	22		
						70 (V_{CC}=4.5 to 5.5)	70	50		
					15	150	30	22		
						70 (V_{CC}=4.5 to 5.5)	70	50		
256K	32K × 8	**μPD43256B-X**	4.5 to 5.5	—	70	70	45			GW: 28-pin TSOP(I) (8 × 13.4 mm)
					85	85				
					10	100	40			
			3.0 to 5.5	A	85	85	45			
					10	100				
						85 (V_{CC}=4.5 to 5.5)				
					12	120	40			
						85 (V_{CC}=4.5 to 5.5)				
			2.7 to 5.5	B	10	100	25	25		
						85 (V_{CC}=4.5 to 5.5)	40	50		
					12	120	25	25		
						85 (V_{CC}=4.5 to 5.5)	40	50		
					15	150	25	25		
						85 (V_{CC}=4.5 to 5.5)	40	50		

Static RAM

■ CMOS Fast SRAM

Density (bits)	Organization (words × bits)	Part number	Class	Access time MAX. (ns)	Maximum supply current			Supply voltage (V)	Package	Remark
					Active (mA)	Standby (mA)	Power down (mA)			
4M	4M × 1	**μPD434001A**	15	15	170	10	–	5.0±0.5	LE: 32-pin SOJ (400 mil)	
			17	17	160					
			20	20	140					
		μPD434001AL	A15	15	150	5		3.3±0.3	LE: 32-pin SOJ (400 mil) G5: 32-pin TSOP (II) (400 mil)	
			A17	17	140					
			A20	20	130					
	1M × 4	**μPD434004A**	15	15	170	10		5.0±0.5	LE: 32-pin SOJ (400 mil)	
			17	17	160					
			20	20	150					
		μPD434004AL	A15	15		5		3.3±0.3		
			A17	17	140					
			A20	20	130					
	512K × 8	**μPD434008A**	12	12	200	10		5.0±0.5	LE: 36-pin SOJ (400 mil)	
			15	15	190					
			17	17	180					
			20	20	170					
		μPD434008AL	A15	15		5		3.3±0.3		
			A17	17	160					
			A20	20	150					
	256K × 16	**μPD434016A**	12	12	230	10		5.0±0.5	LE: 44-pin SOJ (400 mil) G5: 44-pin TSOP (II) (400 mil)	
			15	15	220					
			17	17	210					
			20	20	200					
		μPD434016AL	A15	15		5		3.3±0.3		
			A17	17	190					
			A20	20	180					

Static RAM

■ CMOS Synchronous Fast SRAM

Density (bits)	Organization (words × bits)	Part number	Class	Clock frequency (MHz)	Clock access time MAX. (ns)	Maximum supply current			Supply voltage (V)		Package	Remark
						Active (mA)	Standby (mA)	Power down (mA)	Chip	I/O		
2M	64K × 36	**μPD432836L**	A40	250	3.9	350	3.5	3.5	3.3±0.2	2.0	GF: 100-pin LQFP	Pipeline operation
			A44	225	4.34							
			A50	200	4.9							
			A55	183	5.3	335						
			A57	175		325						
			A60	167		290						
	64K × 32	**μPD432232L**	A5	100	5.0	180	2	2	3.3+0.3/–0.2	3.3 or 2.5		
			A7	83	7.0							
			A8	66	8.0							
1M	32K × 32	**μPD431231L**	A7	83	7.0	130	5	1		2.5		
			A8	66	8.0							
		μPD431632L	A6	150	4.6	220	20	20	3.3±0.165	3.3 or 2.5		
			A7	133	5.0	200						
		μPD431532L	A8	100	8.5		5	5				Flow Through operation
			A9		9.0							
			A10	83	10.0							
			A12	66	12.0							
	32K × 36	**μPD431636L**	A6	150	4.6	220	20	20				Pipeline operation
			A7	133	5.0	200						
		μPD431536L	A8	100	8.5		5	5				Flow Through operation
			A9		9.0							
			A10	83	10.0							
			A12	66	12.0							

■ Bi-CMOS Synchronous Fast SRAM

Density (bits)	Organization (words × bits)	Part number	Class	Clock frequency (MHz)	Clock access time MAX. (ns)	Maximum supply current			Supply voltage (V)	Interface level	Package	Remark
						Active (mA)	Standby (mA)	Power down (mA)				
4M	256K × 18	**μPD464318L**	A5	200	2.6	600	–	–	3.3±0.15	HSTL	S1: 119-pin BGA	Single clock RR mode
			A6	175	3.0							
			A65	143	3.5							
		μPD464518L	A8	125	8.0	450	150			LVTTL		Single clock RL mode
	128K × 36	**μPD464336L**	A5	200	2.6	700	–			HSTL		Single clock RR mode
			A6	175	3.0							
			A65	143	3.5							
		μPD464536L	A8	125	8.0	550	150			LVTTL		Single clock RL mode

Mask ROM

■ Mask ROM

Density (bits)	Organization (words × bits)	Part number	Access time MAX. (ns)		Maximum supply current		Supply voltage (V)	Package
			Normal	Page acccess	Active (mA)	Standby (μA)		
128M	16M × 8 or 8M × 16	**μPD23C128000L**	100	–	45	30	3.3±0.3	GY: 48-pin TSOP (I) (12 × 18 mm)
			120		40	30	3.0±0.3	
		μPD23C128040L	100	30	60	30	3.3±0.3	GY: 48-pin TSOP (I) (12 × 18 mm)
			120	40	55	30	3.0±0.3	
64M	8M × 8 or 4M × 16	**μPD23C64000**	120	–	100	100	5.0±0.5	GX: 44-pin SOP (600 mil) GY: 48-pin TSOP (I) (12 × 18)
	4M × 16	**μPD23C64020**	120		70	–	5.0±0.5	CZ: 42-pin DIP (600 mil)
	8M × 8 or 4M × 16	**μPD23C64000AL**	100		100	30	3.3±0.3	GX: 44-pin SOP (600 mil)
			120		120	30	3.0±0.3	GY: 48-pin TSOP (I) (12 × 18 mm)
		μPD23C64040AL	100	30	60	30	3.3±0.3	GX: 44-pin SOP (600 mil)
			120	40	55	30	3.0±0.3	GY: 48-pin TSOP (I) (12 × 18 mm)
32M	4M × 8 or 2M × 16	**μPD23C32000A**	120	–	70	100	5.0±0.5	GX: 44-pin SOP (600 mil) GY: 48-pin TSOP (I) (12 × 18 mm) G5: 44-pin TSOP (II) (400 mil)
	2M × 16	**μPD23C32020A**	120		70	100	5.0±0.5	CZ: 42-pin DIP (600 mil)
	4M × 8 or 2M × 16	**μPD23C32000L**	120		40	30	3.3±0.3	GX: 44-pin SOP (600 mil)
			140		35	30	3.0±0.3	GY: 48-pin TSOP (I) (12 × 18 mm) G5: 44-pin TSOP (II) (400 mil)
		μPD23C32040L	140	40	60	30	3.3±0.3	GX: 44-pin SOP (600 mil) GY: 48-pin TSOP (I) (12 × 18 mm) G5: 44-pin TSOP (II) (400 mil)
	2M × 16 or 1M × 32	**μPD23C32082L**	100	30	80	30	3.3±0.3	G7: 70-pin SSOP (500 mil)
		μPD23C32202L Synchronous operation	66MHz (/CAS latency = 5)	–	50	100	3.3±0.3	G5: 86-pin TSOP (II) (400 mil)
24M	3M × 8 or 1.5M × 16	**μPD23C24000**	120	–	70	100	5.0±0.5	GX: 44-pin SOP (600 mil)
	1.5M × 16	**μPD23C24020**	120		70	100	5.0±0.5	CZ: 42-pin DIP (600 mil)
16M	2M × 8 or 1M × 16	**μPD23C16000AL**	100	–	45	30	3.3±0.3	GX: 44-pin SOP (600 mil)
			120		40		3.0±0.3	GY: 48-pin TSOP (I) (12 × 18 mm)
		μPD23C16040AL	100	30	60	30	3.3±0.3	G5: 44-pin TSOP (II) (400 mil)
			120	40	55		3.0±0.3	
8M	1M × 8 or 512K × 16	**μPD23C8000X**	120	–	50	100	5.0±0.5	CZ: 42-pin DIP (600 mil) GX: 44-pin SOP (600 mil) GY: 48-pin TSOP (I) (12 × 18 mm) G5: 44-pin TSOP (II) (400 mil)
	1M × 8	**μPD23C8001EJ**	120		50	100	5.0±0.5	CZ: 32-pin DIP (600 mil) GW: 32-pin SOP (525 mil)
	1M × 8 or 512K × 16	**μPD23C8000LX**	170		25	25	3.3±0.3	CZ: 42-pin DIP (600 mil)
			200		20	20	3.0±0.3	GX: 44-pin SOP (600 mil) GY: 48-pin TSOP (I) (12 × 18 mm) G5: 44-pin TSOP (II) (400 mil)

COMBO Memory

■ COMBO Memory

Part number	Function	Density (bits) Organization (words × bits)	Supply Voltage (V)	Access time MAX. (ns)	Maximum supply current			Package
					Active (mA)	Standby (μA)	Data Retention (μA)	
μPD26401	SRAM	1M (128K × 8)	1.8 to 2.2	500	3.5	6.5	10	GU: 32-pin TSOP (I) (8 × 13.4 mm)
	+	+	2.2 to 2.7	375	5.5	8.0		
	ROM	2M (256K × 8)	2.7 to 3.3	250	7.5	10.0		
μPD26411	SRAM	1M (128K × 8)	1.8 to 2.2	500	4.0	6.5		GZ: 32-pin TSOP (I) (8 × 20 mm)
	+	+	2.2 to 2.7	375	6.0	8.0		
	ROM	2M (256K × 8)	2.7 to 3.3	250	10.0	10.0		

Flash MEMORY

■ Flash Memory

Density (bits)	Organization (words × bits)	Part number	Access time MAX. (ns)	Maximum supply current		Supply voltage (V)	Boot code Sector architecture	Package
				Active (mA)	Standby (μA)			
8M	1M × 8	**μPD29F008AL-X★**	90	12	5	2.7 to 3.6	top or bottom	GZ: 40-pin TSOP (I) (10 × 20 mm)
			120					
			120			2.2 to 2.7		
			150					
	1M × 8 or 1M × 16	**μPD29F800AL-X★**	90			2.7 to 3.6		GZ: 48-pin TSOP (I) (12 × 20 mm) GX: 44-pin TSOP (II) (600 mil)
			120					
			120			2.2 to 2.7		
			150					

★: Under development

MCP (Flash Memory + SRAM)

■ MCP (Flash Memory and SRAM)

Part number	Function	Density (bits) Organization (words × bits)	Access time MAX. (ns)	Maximum supply current			Supply Voltage (V)	Boot code Sector architecture	Package
				Active (mA)	Standby (μA)	Data Retention (μA)			
MC-22100	Flash Memory	16M (2M × 8)	100	35	5	–	2.7 to 3.6	Top	F1-DE: 48-pin BGA (10 × 14 mm)
	SRAM	2M (256K × 8)			2	2		–	
MC-22101	Flash Memory	16M (2M × 8)			5	–		Bottom	
	SRAM	2M (256K × 8)			2	2		–	
MC-22102	Flash Memory	16M (1M × 16)			5	–		Top	
	SRAM	2M (256K × 8)			2	2		–	
MC-22103	Flash Memory	16M (1M × 16)			5	–		Bottom	
	SRAM	2M (256K × 8)			2	2		–	
MC-22104	Flash Memory	16M (2M × 8)			5	–		Top	
	SRAM	1M (128K × 8)			26	20		–	
MC-22105	Flash Memory	16M (2M × 8)			5	–		Bottom	
	SRAM	1M (128K × 8)			26	20		–	
MC-22106	Flash Memory	16M (1M × 16)			5	–		Top	
	SRAM	1M (128K × 8)			26	20		–	
MC-22107	Flash Memory	16M (1M × 16)			5	–		Bottom	
	SRAM	1M (128K × 8)			26	20		–	
MC-22000	Flash Memory	8M (1M × 8)			5	–		Top	F1-DB: 48-pin BGA (10 × 11 mm)
	SRAM	2M (256K × 8)			2	2		–	
MC-22001	Flash Memory	8M (1M × 8)			5	–		Bottom	
	SRAM	2M (256K × 8)			2	2		–	
MC-22002	Flash Memory	8M (1M × 8 or 512K × 16)			5	–		Top	
	SRAM	2M (256K × 8)			2	2		–	
MC-22003	Flash Memory	8M (1M × 8 or 512K × 16)			5	–		Bottom	
	SRAM	2M (256K × 8)			2	2		–	
MC-22004	Flash Memory	8M (1M × 8)			5	–		Top	
	SRAM	1M (128K × 8)			26	20		–	
MC-22005	Flash Memory	8M (1M × 8)			5	–		Bottom	
	SRAM	1M (128K × 8)			26	20		–	
MC-22006	Flash Memory	8M (1M × 8 or 512K × 16)			5	–		Top	
	SRAM	1M (128K × 8)			26	20		–	
MC-22007	Flash Memory	8M (1M × 8 or 512K × 16)			5	–		Bottom	
	SRAM	1M (128K × 8)			26	20		–	

Other

■ Synchronous Graphics RAM

Density (bits)	Organization (words × bits × banks)	Part number	Class	/CAS latency	Clock frequency (MHz)	Clock access time MAX. (ns)	Maximum supply current (mA)				Supply voltage (V)	Refresh cycle (cycles/ms)	Interface level	Package	Remark
							Active		Refresh						
							Normal (BL=1)	Burst	Auto	Self					
16M	256K × 32 × 2	**μPD4811650 (Rev.E)**	A70R	3	143	6	230	345	190	2	3.5±0.1	2K/32	LVTTL	GF: 100-pin TQFP (14 × 20 mm)	
			A80	3	125	6.5	220	300	180		3.3±0.3				
				2	83	9		220							
			A10	3	100	7.5	205	250	170						
				2	77	10		200							
			A12	3	83	8	180	210	150						
				2	67	11		170							
8M	128K × 32 × 2	**μPD481850 (Rev.L)**	A10	3	100	9	105	200	85	6		1K/16		GF: 100-pin QFP (14 × 20 mm)	
				2	66	12		155							
			A12	3	83	11	90	170	80						
				2	55	15		130							

BL: Burst Length

■ Field/Line Buffer

Density (bits)	Organization (words × bits)	Part number	Read/Write cycle time MIN. (ns)	Data hold period (ms)	Maximum supply current active (mA)	Supply voltage (V)	Package	Remarks
2M	256 × 8	**μPD42280**	30/30 60/60	–	90 60	5.0±0.5	• 28-pin SOP	Field buffer FIFO structure
80K	10096 × 8 5048 × 16	**μPD485506**	25/25 35/35	–	140	5.0±0.5	• 44-pin TSOP II	FAX, PPC Line buffer FIFO structure
40K	5048 × 8	**μPD485505**	25/25 35/35	–	80		• 24-pin SOP	

Semi-Custom IC

Gate Array 134

- CMOS Gate Array 134
- BiCMOS Gate Array 138

Cell-Based IC 139

Embedded Array 145

Analog Master 146

Mixed Signal ASIC 148

Gate Array

CMOS Gate Array

■ CMOS-6 family (channelless type)

Part number		μPD65658	μPD65664	μPD65672	μPD65676
Inte-gration	Internal gates	42240	72576	119232	177408
	Usable gate number	31680	54432	89424	133056
	Pad number *	236	304	384	464
Delay time	Internal gate	0.5 ns (fan-outs: 2, wiring length: 2 mm)/0.4 ns (power gate, fan-outs: 2, wiring length: 2 mm)			
	Input buffer	1.0 ns			
	Output buffer	2.0 ns (C_L = 15 pF)			
Output drive capability		I_{OL} = 4.5, 9.0, 13.5, 18.0, 24.0 mA			
Supply voltage		5 V			

∗: V_{DD}, GND terminals included. Number of pins which can be actually used depends on package.

■ CMOS-6A family (channelless type)

Part number		μPD65630	μPD65636	μPD65640	μPD65646	μPD65650	μPD65654	μPD65656	μPD65662
Inte-gration	Internal gates	5376	8000	11520	16240	21120	30720	40480	70272
	Usable gate number	3225	4800	6912	9744	12672	18432	24288	42163
	Pad number *	100	116	136	156	176	208	220	288
Delay time	Internal gate	0.5 ns (fan-outs: 2, wiring length: 2 mm)/0.4 ns (power gate, fan-outs: 2, wiring length: 2 mm)							
	Input buffer	1.0 ns							
	Output buffer	2.0 ns (C_L = 15 pF)							
Output drive capability		I_{OL} = 4.5, 9.0, 13.5, 18.0, 24.0 mA							
Supply voltage		5 V							

∗: V_{DD}, GND terminals included. Number of pins which can be actually used depends on package.

■ CMOS-6S family (channelless type)

Part number		μPD65611	μPD65621	μPD65625	μPD65643	μPD65653
Inte-gration	Internal gates	1836	3484	5312	13416	25488
	Usable gate number	1101	2090	3187	8049	15292
	Pad number *	80	100	120	176	236
Delay time	Internal gate	0.5 ns (fan-outs: 2, wiring length: 2 mm)/0.4 ns (power gate, fan-outs: 2, wiring length: 2 mm)				
	Input buffer	1.0 ns				
	Output buffer	2.0 ns (C_L = 15 pF)				
Output drive capability		I_{OL} = 4, 5, 9, 13.5, 18, 24 mA				
Supply voltage		5 V ±10 %				

∗: V_{DD}, GND terminals included. Number of pins which can be actually used depends on package.

Gate Array

CMOS Gate Array

■ CMOS-8 family (channelless type) 2-Metal layer (1/2)

Part number		μPD65800	μPD65801	μPD65802	μPD65803
Inte-gration	Internal gates	11712	21504	32000	42688
	Usable gate number	7027	12902	19200	25612
	Pad number *	172	228	268	308
Delay time	Internal gate	0.29 ns (fan-outs: 2, wiring length: 2 mm)/0.20 ns (power gate, fan-outs: 2, wiring length: 2 mm)			
	Input buffer	0.36 ns			
	Output buffer	1.73 ns (C_L = 15 pF)			
Output drive capability		I_{OL} = 3, 6, 9, 12, 18, 24 mA			
Supply voltage		5 V			

*: V_{DD}, GND terminals included. Number of pins which can be actually used depends on package.

■ CMOS-8 family (channelless type) 3-Metal layer (2/2)

Part number		μPD65804	μPD65806	μPD65808	μPD65810	μPD65811	μPD65812	μPD65813
Inte-gration	Internal gates	42688	58752	82432	103680	138776	176720	233280
	Usable gate number	32016	44064	61824	77760	104082	132540	174960
	Pad number *	308	356	420	468	524	588	676
Delay time	Internal gate	0.29 ns (fan-outs: 2, wiring length: 2 mm)/0.20 ns (power gate, fan-outs: 2, wiring length: 2 mm)						
	Input buffer	0.36 ns						
	Output buffer	1.73 ns (C_L = 15 pF)						
Output drive capability		I_{OL} = 3, 6, 9, 12, 18, 24 mA						
Supply voltage		5 V						

*: V_{DD}, GND terminals included. Number of pins which can be actually used depends on package.

■ CMOS-N5 family (channelless type) (1/2)

Part number		μPD65880	μPD65881	μPD65882	μPD65883	μPD65884
Inte-gration	Internal gates	3456	5880	13952	25344	33864
	Usable gate number	2073	3528	8371	15206	20318
	Pad number *	72	88	120	160	184
Delay time	Internal gate	0.29 ns (fan-outs: 2, standard wiring length)/0.20 ns (power gate, fan-outs: 2, standard wiring length)				
	Input buffer	0.36 ns				
	Output buffer	1.73 ns (C_L = 15 pF)				
Output drive capability		I_{OL} = 3, 6, 9, 12 mA				
Supply voltage		5 V ±10% (CMOS level)				

*: V_{DD}, GND terminals included. Number of pins which can be actually used depends on package.

■ CMOS-N5 family (channelless type) (2/2)

Part number		μPD65885	μPD65887	μPD65889	μPD65890	μPD65893
Inte-gration	Internal gates	40768	56496	76000	99528	123384
	Usable gate number	24460	33897	45600	59716	74030
	Pad number *	244	284	324	372	412
Delay time	Internal gate	0.29 ns (fan-outs: 2, standard wiring length)/0.20 ns (power gate, fan-outs: 2, standard wiring length)				
	Input buffer	0.36 ns				
	Output buffer	1.73 ns (C_L = 15 pF)				
Output drive capability		I_{OL} = 3, 6, 9, 12 mA				
Supply voltage		5 V ±10% (CMOS level)				

*: V_{DD}, GND terminals included. Number of pins which can be actually used depends on package.

Gate Array

CMOS Gate Array

■ CMOS-8L family (channelless type) 2-Metal layer (1/3)

Part number		μPD65837	μPD65839	μPD65840	μPD65841	μPD65842	μPD65843	μPD65845
Inte-gration	Internal gates	4480	7904	10912	20832	30192	40592	52528
	Usable gate number	2912	5137	6547	12499	18115	24355	31516
	Pad number *	80	104	164	204	212	244	276
Delay time	Internal gate	0.13 ns (fan-outs: 1, wiring length: 0 mm), 0.33 ns (fan-outs: 2, wiring length: 2 mm)/ 0.21 ns (power gate, fan-outs: 2, wiring length: 2 mm)						
	Input buffer	0.71 ns						
	Output buffer	1.44 ns (FO01, C_L = 15 pF)						
Output drive capability		I_{OL} = 3, 6, 9, 12, 18, 24 mA						
Supply voltage		3.3 V ±0.3 V						

∗: V_{DD}, GND terminals included. Number of pins which can be actually used depends on package.

■ CMOS-8L family (channelless type) 2-Metal layer (2/3)

Part number		μPD65846	μPD65848	μPD65849	μPD65850	μPD65851
Inte-gration	Internal gates	61904	81984	102272	120768	148256
	Usable gate number	37142	49190	61363	72460	88953
	Pad number *	300	340	380	412	452
Delay time	Internal gate	0.13 ns (fan-outs: 1, wiring length: 0 mm), 0.33 ns (fan-outs: 2, wiring length: 2 mm)/ 0.21 ns (power gate, fan-outs: 2, wiring length: 2 mm)				
	Input buffer	0.71 ns				
	Output buffer	1.44 ns (FO01, C_L = 15 pF)				
Output drive capability		I_{OL} = 3, 6, 9, 12, 18, 24 mA				
Supply voltage		3.3 V ±0.3 V				

∗: V_{DD}, GND terminals included. Number of pins which can be actually used depends on package.

■ CMOS-8L family (channelless type) 2-Metal layer (3/3)

Part number		μPD65852	μPD65853	μPD65855	μPD65858
Inte-gration	Internal gates	202752	255744	342000	488720
	Usable gate number	121651	153446	205200	293232
	Pad number *	524	588	676	804
Delay time	Internal gate	0.13 ns (fan-outs: 1, wiring length: 0 mm), 0.33 ns (fan-outs: 2, wiring length: 2 mm)/ 0.21 ns (power gate, fan-outs: 2, wiring length: 2 mm)			
	Input buffer	0.71 ns			
	Output buffer	1.44 ns (FO01, C_L = 15 pF)			
Output drive capability		I_{OL} = 3, 6, 9, 12, 18, 24 mA			
Supply voltage		3.3 V ±0.3 V			

∗: V_{DD}, GND terminals included. Number of pins which can be actually used depends on package.

■ CMOS-8L family (channelless type) 3-Metal layer (1/3)

Part number		μPD65860	μPD65861	μPD65862	μPD65863	μPD65865
Inte-gration	Internal gates	10912	20832	30192	40592	52528
	Usable gate number	8184	15624	22644	30444	39396
	Pad number *	164	204	212	244	276
Delay time	Internal gate	0.13 ns (fan-outs: 1, wiring length: 0 mm), 0.33 ns (fan-outs: 2, wiring length: 2 mm)/ 0.21 ns (power gate, fan-outs: 2, wiring length: 2 mm)				
	Input buffer	0.71 ns				
	Output buffer	1.44 ns (FO01, C_L = 15 pF)				
Output drive capability		I_{OL} = 1, 2, 3, 6, 9, 12, 18, 24 mA				
Supply voltage		3.3 V ±0.3 V				

∗: V_{DD}, GND terminals included. Number of pins which can be actually used depends on package.

Gate Array

CMOS Gate Array

■ CMOS-8L family (channelless type) 3-Metal layer (2/3)

Part number		μPD65866	μPD65868	μPD65869	μPD65870	μPD65871
Inte-gration	Internal gates	61904	81984	102272	120768	148256
	Usable gate number	46428	61488	76704	90576	111192
	Pad number *	300	340	380	412	452
Delay time	Internal gate	0.13 ns (fan-outs: 1, wiring length: 0 mm), 0.33 ns (fan-outs: 2, wiring length: 2 mm)/ 0.21 ns (power gate, fan-outs: 2, wiring length: 2 mm)				
	Input buffer	0.71 ns				
	Output buffer	1.44 ns (FO01, C_L = 15 pF)				
Output drive capability		I_{OL} = 1, 2, 3, 6, 9, 12, 18, 24 mA				
Supply voltage		3.3 V ±0.3 V				

∗: V_{DD}, GND terminals included. Number of pins which can be actually used depends on package.

■ CMOS-8L family (channelless type) 3-Metal layer (3/3)

Part number		μPD65872	μPD65873	μPD65875	μPD65878
Inte-gration	Internal gates	202752	255744	342000	488720
	Usable gate number	152064	191808	256500	366540
	Pad number *	524	588	676	804
Delay time	Internal gate	0.13 ns (fan-outs: 1, wiring length: 0 mm), 0.33 ns (fan-outs: 2, wiring length: 2 mm)/ 0.21 ns (power gate, fan-outs: 2, wiring length: 2 mm)			
	Input buffer	0.71 ns			
	Output buffer	1.44 ns (FO01, C_L = 15 pF)			
Output drive capability		I_{OL} = 1, 2, 3, 6, 9, 12, 18, 24 mA			
Supply voltage		3.3 V ±0.3 V			

∗: V_{DD}, GND terminals included. Number of pins which can be actually used depends on package.

■ CMOS-9HD family (channelless type) (1/2)

[3-Metal layer]

Part number		μPD65943	μPD65944	μPD65945	μPD65946	μPD65948
Inte-gration	Internal gates	75740	100602	128338	202630	312684
	Usable gate counts	53018	70421	89836	141841	218878
	Pad number *	172	196	216	268	324
Delay time	Internal gates	131 ps (fan-out = 1, standard wiring length)				
	Input buffer	229 ps				
	Output buffer	1396 ps (I_{OL} = 9 mA, C_L = 15 pF)				
Output drive capacity		I_{OL} = 1, 2, 3, 6, 9, 12, 18, 24 mA				
Supply voltage		3.3 V ±0.3 V				

∗: V_{DD}, GND terminals included. Number of pins which can be actually used depends on package.

■ CMOS-9HD family (channelless type) (2/2)

[3-Metal layer]

Part number		μPD65949	μPD65951	μPD65954	μPD65956	μPD65958
Inte-gration	Internal gates	437136	585390	835664	1096452	1615646
	Usable gate counts	305995	351234	501398	657871	969387
	Pad number *	380	436	516	588	708
Delay time	Internal gates	131 ps (fan-out = 1, standard wiring length)				
	Input buffer	229 ps				
	Output buffer	1396 ps (I_{OL} = 9 mA, C_L = 15 pF)				
Output drive capacity		I_{OL} = 1, 2, 3, 6, 9, 12, 18, 24 mA				
Supply voltage		3.3 V ±0.3 V				

∗: V_{DD}, GND terminals included. Number of pins which can be actually used depends on package.

Gate Array

BiCMOS Gate Array

■ QB-8 family (1/2)

Part number		μPD67821	μPD67822	μPD67823	μPD67824	μPD67825	μPD67826
Inte-gration	Raw gate counts *1	32832	44352	56800	69520	93184	123808
	Usable gate counts *2	22982	31046	39760	48664	65229	86666
	Signal pin counts	156	180	204	224	260	300
Delay time	Internal gate	107 ps (fan-outs: 1, wiring length: 0 mm) 191 ps (fan-outs: 2, wiring length: 2 mm)					
	Power gate	142 ps (fan-outs: 2, wiring length: 2 mm)					
	Input buffer	289 ps (fan-outs: 2, wiring length: 2 mm)					
	Output buffer	1.64 ns (C_L = 15 pF)					
Output drive capability		I_{OL} = 1, 2, 3, 6, 9, 12, 18, 24 mA					
Supply voltage		3.3 V ±0.3 V					

*1: 1 gate = 2-input NAND gate.
*2: usable gate: 70%

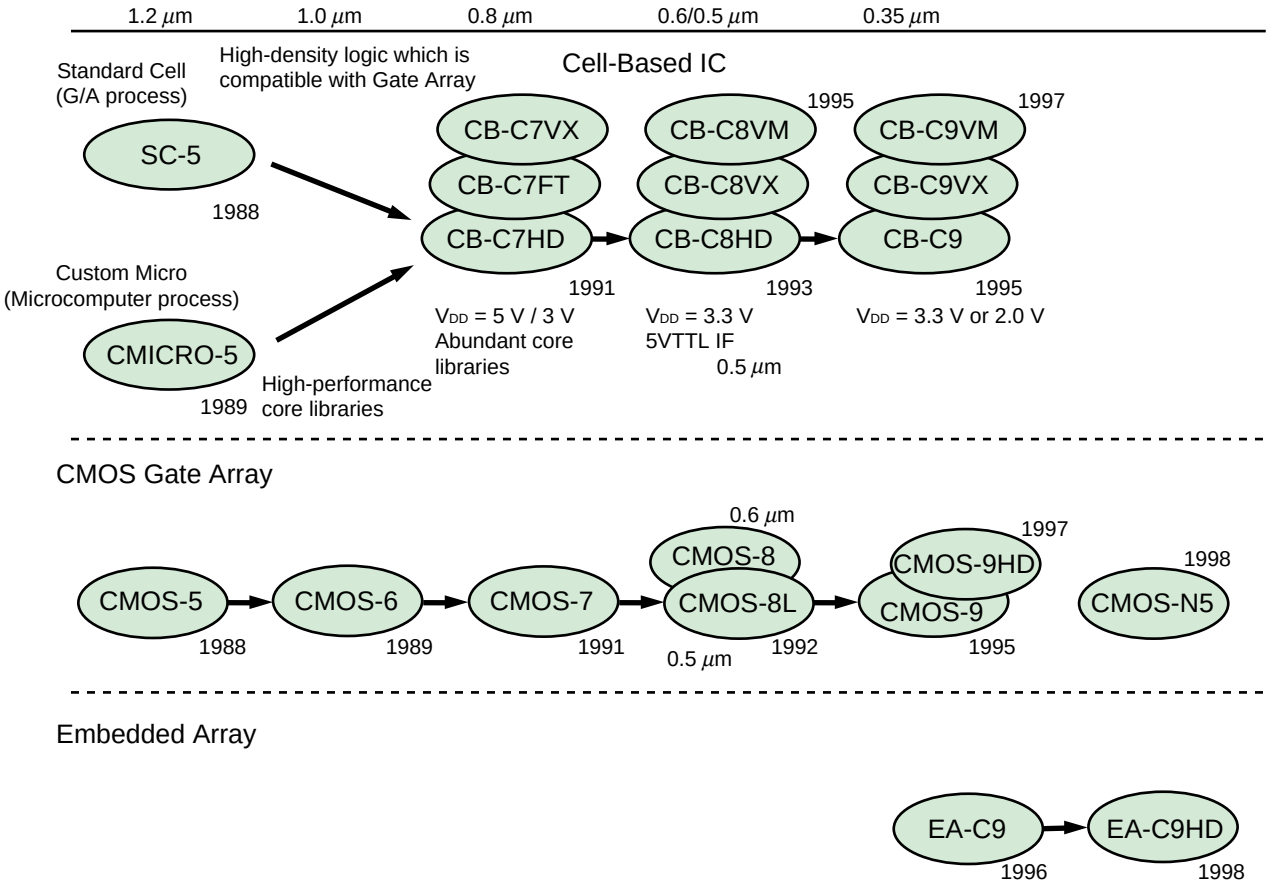
■ QB-8 family (2/2)

Part number		μPD67827	μPD67828	μPD67829	μPD67830	μPD67831
Inte-gration	Raw gate counts *1	136752	167280	234320	292896	381840
	Usable gate counts *2	95726	117096	164024	205027	267288
	Signal pin counts	316	348	412	460	524
Delay time	Internal gate	107 ps (fan-outs: 1, wiring length: 0 mm) 191 ps (fan-outs: 2, wiring length: 2 mm)				
	Power gate	142 ps (fan-outs: 2, wiring length: 2 mm)				
	Input buffer	289 ps (fan-outs: 2, wiring length: 2 mm)				
	Output buffer	1.64 ns (C_L = 15 pF)				
Output drive capability		I_{OL} = 1, 2, 3, 6, 9, 12, 18, 24 mA				
Supply voltage		3.3 V ±0.3 V				

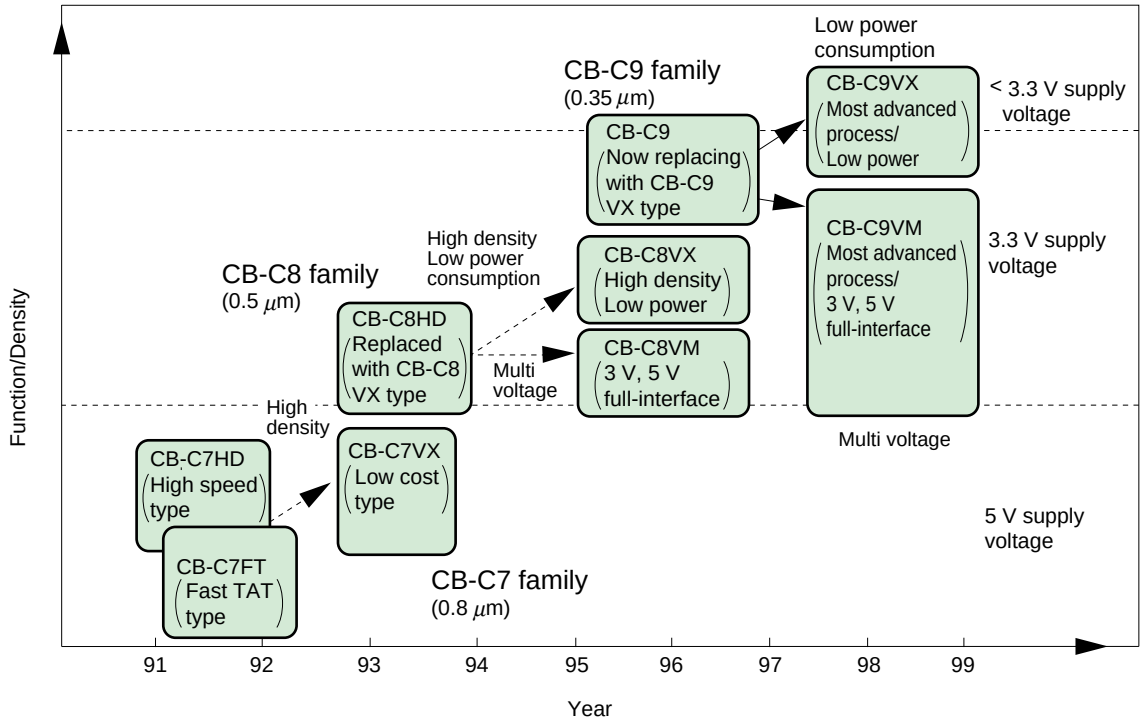
*1: 1 gate = 2-input NAND gate.
*2: usable gate: 70%

Cell-Based IC

■ NEC ASIC Product Development

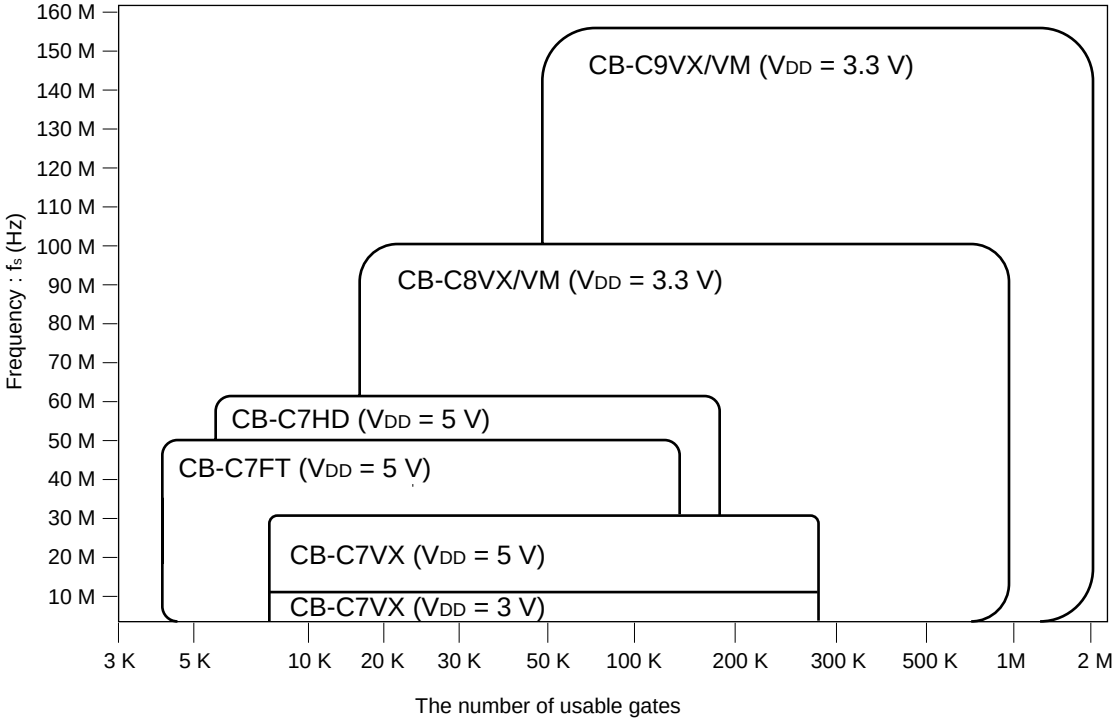


■ Cell-Based IC Product Development

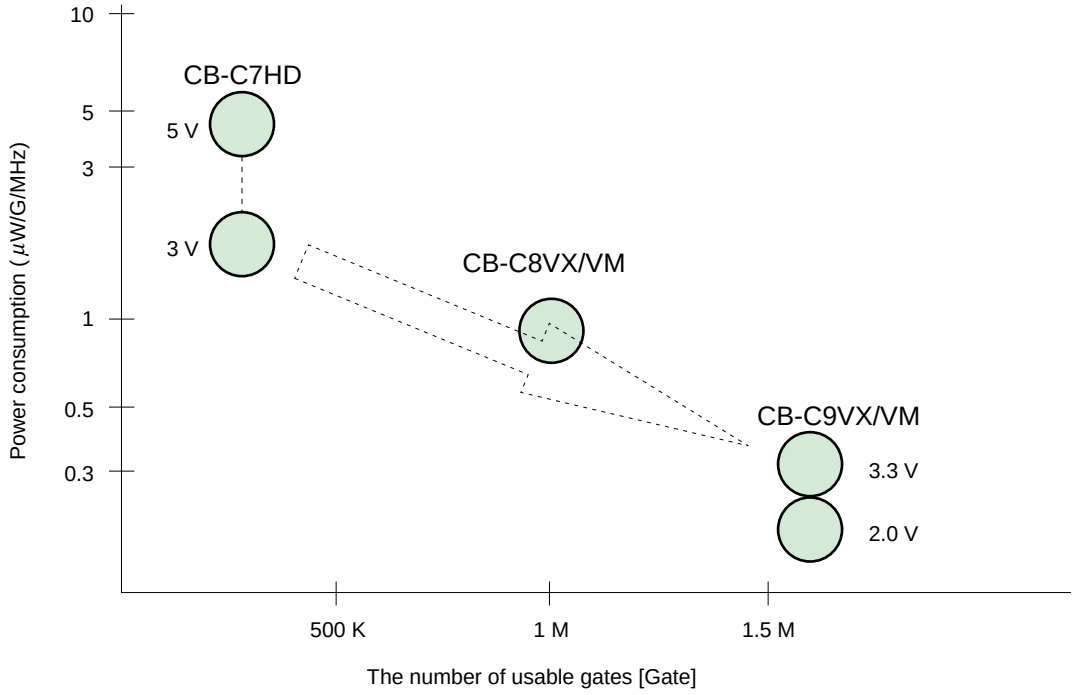


Cell-Based IC

■ Application of Cell-Based IC



■ Power Consumption of Cell-Based IC



Cell-Based IC

■ 0.8 μm CMOS Cell-Based IC: CB-C7 family FT type/HD type/VX type

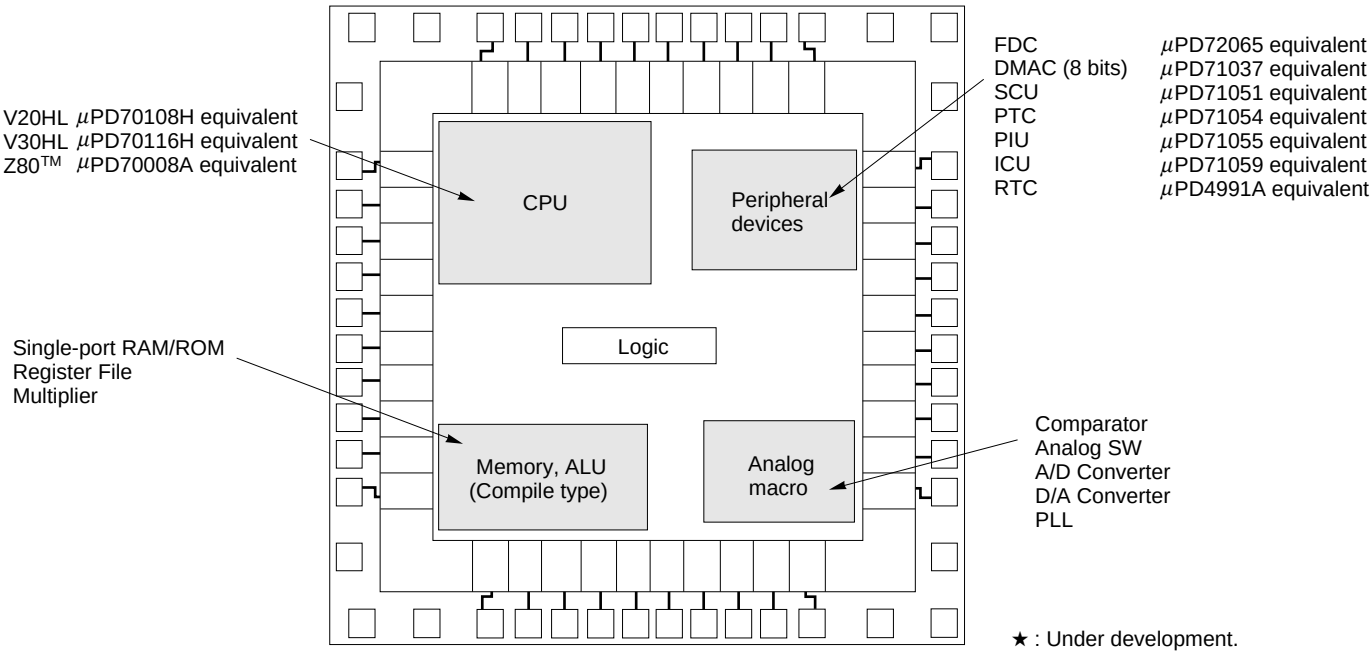
- FT type : Fast TAT embedded type.
- HD type: High speed and high density type.
- VX type : Middle speed and super high density type.

- Product overview

Family name			CB-C7 FT type / CB-C7 HD type / CB-C7 VX type			
Part number			**μPD93000**	**μPD94000**	**μPD95100**	**μPD96100**
Derivative product *1			No	Yes	No	Yes
Supply voltage			5.0 V ±10%, 3 V ±10%			
Interface level			CMOS (5 V I/F), TTL compatible, CMOS (3 V I/F)			
Technology			0.8 μm Si gate CMOS process, Al 2-layer / 3-layer wiring			
Integration level *2			FT: 167 K gates (MAX.) / HD: 223 K gates (MAX.) / VX: 273 K gates (MAX.)			
Maximum I/O pads			440			
Delay time	Internal gate		FT, HD: 0.18 ns (@5 V) / 0.43 ns (@3 V) [F/O = 1, L = 0 mm] 0.33 ns *3 / 0.44 ns (@5 V) / 0.67 ns *3 / 0.90 ns (@3 V) [F/O = 2, L = 2 mm] VX : 0.40 ns (@5 V) / 0.73 ns (@3 V) [F/O = 1, L = 0 mm] 0.90 ns *3 / 2.10 ns (@5 V) / 1.54 ns *3 / 3.42 ns (@3 V) [F/O = 2, L = 2 mm]			
	Input buffer		0.60 ns (@5 V) / 0.76 ns (@3 V) [F/O = 2, L = 2 mm] 1.25 ns (@5 V) / 2.18 ns (@3 V) [F/O = 2, L = 2 mm]			
	Output buffer		FT, HD: 1.6 ns (@5 V, I_{OL} = 8 mA buffer) / 3.0 ns (@3 V, 4.4 mA buffer) [C_L = 15 pF] VX : 1.7 ns (@5 V, I_{OL} = 8 mA buffer) / 3.0 ns (@3 V, 4.4 mA buffer) [C_L = 15 pF]			
Consumed power (internal gate)			[V_{DD} = 5.0 V] [V_{DD} = 3.0 V] FT, HD: 3.90 μW/MHz/gate (@5 V) 1.41 μW/MHz/gate (@3 V) VX : 3.21 μW/MHz/gate (@5 V) 1.17 μW/MHz/gate (@3 V)			
Main Library	Macro	CPU	—		V20HL, V30HL, V30MX™ *4, μPD70008(Z80)	
		Peripheral	—		FDC, DMAC, 7105x, etc.	
		Memory	RAM	ROM, RAM	RAM	ROM, RAM
		Analog	A/D converter, D/A converter, Analog switch, A-PLL, etc.			
		Others	Register file, Multiplier			
	I/O buffer		Input buffer : 3 V CMOS, 5 V TTL/CMOS interface buffer Output buffer : I_{OL} = 4, 8, 12, 24, 48 mA (@5 V) I_{OL} = 2.2, 4.4, 6.6 mA (@3 V)			

∗1: ROM cord are variable.
∗2: Number of usable gates when not mounting hard macros (2-input NAND Gate conversion)
∗3: power gate
∗4: V30MX is a CPU core developed specially for Cell-Based ICs, and has software compatibility with V20HL/V30HL.

Cell-Based IC



★ : Under development.

Cell-Based IC

■ CB-C8 family VX/VM type

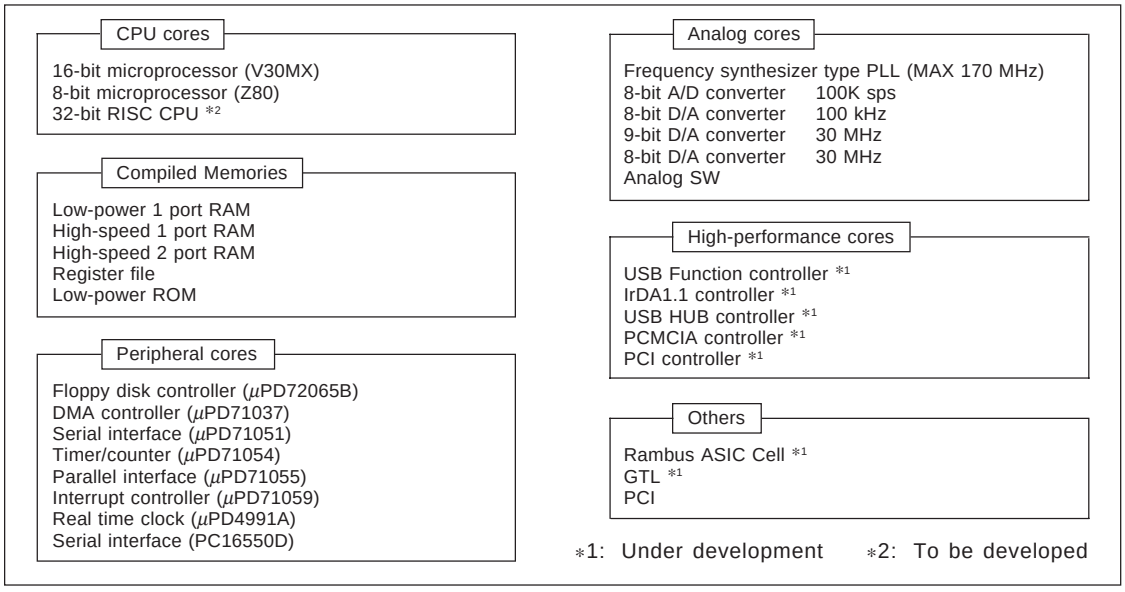
- 0.5 μm Cell-Based IC of high density and low power consumption
  - · Maximum number of usable gates is 1.0M gates.
  - · Power consumption is 20% lower than CB-C8 HD type.
- Multi voltage (VM type): The interface with a device with 5 V/3 V mixed power supply is possible.
- Equipped with high-performance cores

- Product overview

Family name		CB-C8 VX type	CB-C8 VM type
Part number		**μPD97000**	**μPD99000**
Derivative product *1		Yes	
Supply voltage		3.3 V ±0.3 V	Internal: 3.3 V ±0.3 V I/O : 5 V ±10%, 3.3 V ±0.3 V
Interface level		LVTTL level (5 V tolerant buffer included)	5 V CMOS/TTL level LVTTL level (5 V tolerant buffer included)
Technology		0.5 μm Si gate CMOS process, Al 2-layer / 3-layer wiring	
Integration level *2		1000K gates (MAX.)	
Delay time	Internal gate	0.16 ns *3 / 0.34 ns	[F/O = 2, L = 1 mm]
	Input buffer	3.3 V I/F : 0.26 ns 5 V Tolerant : 0.33 ns 5 V Full Swing: 0.72 ns	[F/O = 2, L = 1 mm] [F/O = 2, L = 1 mm] [F/O = 2, L = 1 mm]
	Output buffer	3.3 V I/F : 1.24 ns (I_{OL} = 9 mA) 5 V Tolerant : 1.88 ns (I_{OL} = 9 mA) 5 V Full Swing: 2.30 ns (I_{OL} = 9 mA)	[C_L = 15 pF] [C_L = 15 pF] [C_L = 15 pF]
Consumed power (internal gate)		1.04 μW/MHz/gate 0.75 μW/MHz/gate	[Lxxx block] [Yxxx block]
Output drive capability		I_{OL} = 3, 6, 9, 12, 18, 24 mA I_{OL} = 1, 2, 3, 6, 9 mA	[3.0 V I/F] [5 V Tolerant]
		—	I_{OL} = 1, 2, 3, 6, 9, 12, 18 mA [5 V Full Swing]

∗1: ROM cord are variable.
∗2: Number of usable gates when not mounting hard macros (2-input NAND Gate conversion)
∗3: power gate

- Core overview



PC16550D is a product name of National Semiconductor Corp.

Cell-Based IC

■ CB-C9 family

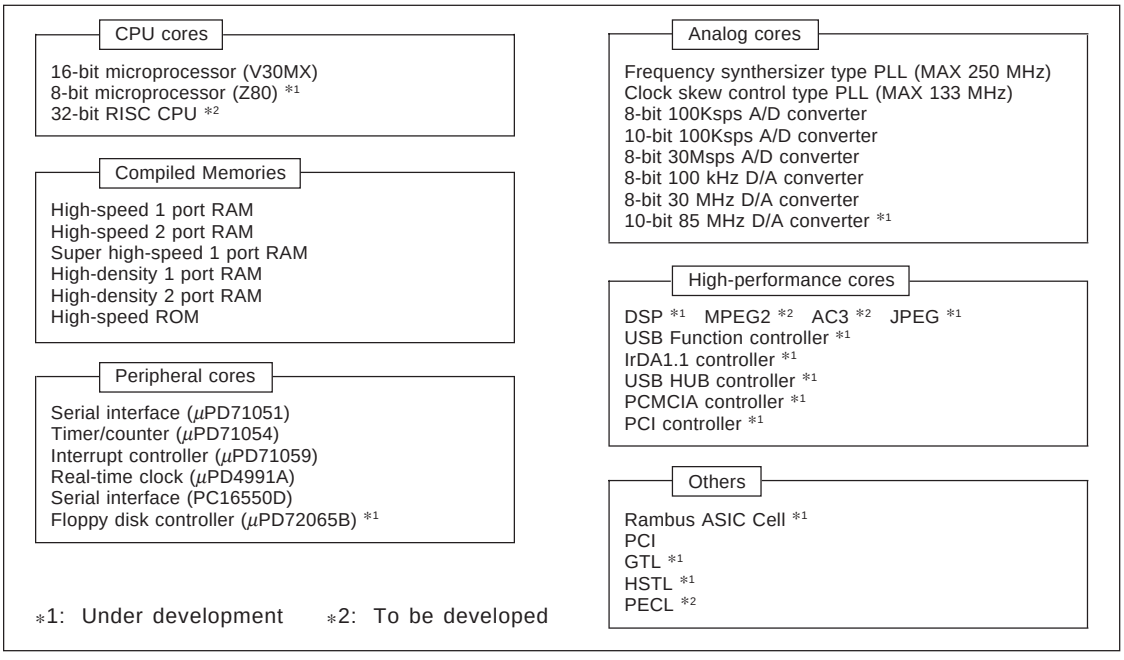
- 0.35 μm Cell-Based IC VX/VM type
  - · Maximum number of usable gates is 1.7M gates.
  - · Delay time is 35% smaller than the 0.5 μm model.
  - · Power consumption is 45% lower than the 0.5 μm model.
- Equipped with high-performance cores, for examples, 32-bit RISC CPU, DSP
- Lower supply voltage at 2.0 V is under development.

- Product overview

Family name		CB-C9 VX type		CB-C9 VM type	
Part number		**μPD82200**		**μPD82600**	
Derivative product *1		○			
Integration level *2		1.7M gate		1.6M gate	
Interface level		LVTTL level (5 V tolerant buffer included)		5 V CMOS/TTL level LVTTL level (5 V tolerant buffer included)	
I/O supply voltage		3.3 V ±0.3 V		5.0 V ±10%	3.3 V ±0.3 V
Internal supply voltage		3.3 V ±0.3 V	2.0 V ±0.2 V	3.3 V ±0.3 V	
Delay time	Internal gate	114 ps [F/O = 2, L = std] *3	207 ps [F/O = 2, L = std] *3	114 ps [F/O = 2, L = std] *3	114 ps [F/O = 2, L = std] *3
	Input buffer	169 ps [F/O = 2, L = std]	—	690 ps [F/O = 2, L = std]	169 ps [F/O = 2, L = std]
	Output buffer	864 ps [C_L = 15 pF, I_{OL} = 18 mA]	—	2045 ps [C_L = 15 pF, I_{OL} = 18 mA]	864 ps [C_L = 15 pF, I_{OL} = 18 mA]
Power consumption μW/MHz/gate		0.3	0.12	0.3	
Output drive capability		I_{OL} = 3, 6, 9, 12, 18, 24 mA [3 V interface] I_{OL} = 1, 2, 3, 6, 9 mA [5 V tolerant]			
		—		I_{OL} = 1, 2, 3, 6, 9, 12, 18 mA [5 V full swing]	

∗1: ROM cord are variable.
∗2: Number of usable gates when not mounting hard macros (2-input NAND Gate conversion)
∗3: power gate

- Core overview



PC16550D is a product name of National Semiconductor Corp.

Embedded Array

CMOS Embedded Array

■ EA-C9HD family

[3-Metal layer]

Part number		μPD65443	μPD65444	μPD65445	μPD65446	μPD65448	μPD65449	μPD65451	μPD65454	μPD65456	μPD65458
Inte-gration	Internal gates *1	76720	103032	128872	207000	314104	440832	592020	840768	1104432	1626628
	Usable gate counts *2	42196	56668	70880	113850	172757	242458	325611	462422	607438	894645
	Pad number *3	172	196	216	268	324	380	436	516	588	708
Delay time	Internal gate	131 ps (Fan-out = 1, Standard wiring length)									
	Input buffer	229 ps									
	Output buffer	1396 ps (C_L = 15 pF, I_{OL} = 9 mA)									
Power consumption		0.524 μA/MHz/gate (internal gate)									
Supply voltage		3.3 V ±0.3 V									

∗1: gate = 2-input NAND

∗2: Usable gate counts depend on mounting MACRO condition.

∗3: V_{DD}, GND terminals included. Number of pins which can be actually used depends on package.

- Compiled Macro
  High-speed 1-port RAM
  High-speed 2-port RAM
  High-density 1-port RAM
  High-density 2-port RAM
  High-speed ROM

Analog Master

The Analog Master is a semi-custom LSI for creating analog circuits on a master wafer by inter-connecting pre-defused elements (bipolar transistors, resistors, and capacitors, already formed on the wafter) with the user-defined wiring.

■ Feature

■ Very short development period

Four to eight weeks from layout design to ES (Engineering Sample) production

■ Suitable in case of small lot production

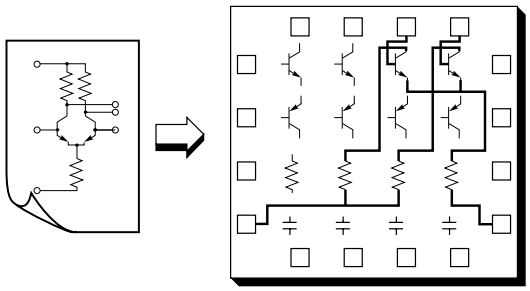
A few thousand LSIs per lot (six months) in case of simulation level interface (the number depends on product)

■ Powerful line-up

3 families (13 masters) are provided to cover wide range of operating frequency, withstand voltage, and number of elements used.

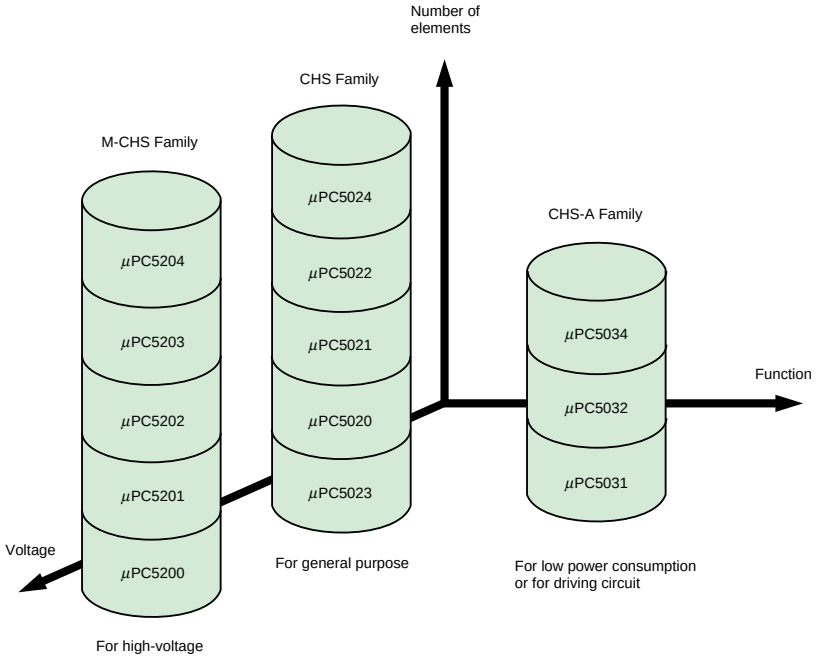
■ Various macro libraries

To reduce user's works, variety of macros are provided such as operational amplifier, comparator, regulator, etc.



bipolar transistors, resistors,
and capacitors, already formed
on master wafer.

Line-up and features

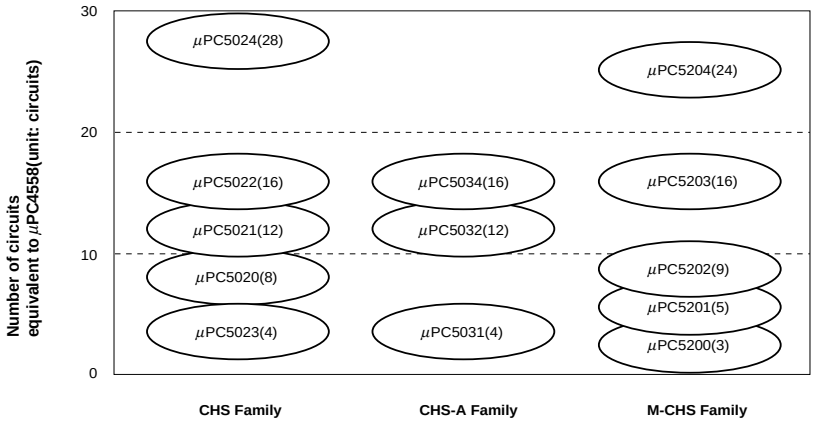


Analog Master

■ Master size

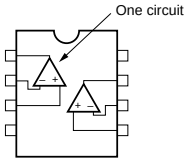
The scale of the circuit that can be integrated on the analog master differs depending on the model of each family.

The scale of the circuit that can be integrated shown below is roughly calculated in terms of the number of circuits where one circuit is equivalent to the general-purpose operational amplifier μPC4558[Note 1].



(Numbers in the parentheses indicate the number of circuits[Note 2] that can be integrated.)

Notes 1. One of the two operational amplifiers in the general-purpose operational amplifier μPC4558 package (8-pin DIP) is counted as one circuit.



2. The number of circuits shown above is a guideline in which only operational amplifiers are integrated, and does not include operational amplifier peripheral circuits (such as feedback circuits).
When selecting a model, estimate the circuit scale by taking these peripheral circuits into consideration.

■ Product list

Item \ Family name	CHS Family					CHS-A Family			M-CHS Family				
Technology	High-speed bipolar technology								High-voltage bipolar technology				
Absolute maximum ratings	14 V								44 V				
Features	• High-speed transistors • Adequate for circuit operating at up to 12 V and 50 MHz					• Internal transistor for output driver • Many-pin masters • Total internal resistance higher than the CHS Family			• Adequate for circuit operating at up to 40 V and 10 MHz				
Part number	μPC5020	μPC5021	μPC5022	μPC5023	μPC5024	μPC5031	μPC5032	μPC5034	μPC5200	μPC5201	μPC5202	μPC5203	μPC5204
Number of Pads	28	32	50	22	80	36	56	82	24	28	40	52	62
Total number of elements	1628	2328	3042	727	6152	576	1472	2252	654	1030	1787	3088	4562
NPN Transistors (For output driver)	260	368	476	129	836	118 (13)	298 (18)	446 (26)	105	163	283	484	710
PNP Transistors	224	320	416	116	672	105	280	450	84	140	252	448	672
Resistors	1116	1600	2098	468	4560	338	854	1266	456	712	1225	2108	3108
Capacitors	28	40	52	14	84	15	40	90	9	15	27	48	72
Macro library	Operational amplifier, Comparator, Regulator, Switch, Timer					Operational amplifier, Comparator, Super white TEG *			Operational amplifier, Comparator, Regulator, Switch				
Package	DIP, S-DIP, SOP, S-SOP, QFP, TQFP					S-DIP, SOP, S-SOP, QFP, TQFP			DIP, S-DIP, SOP, S-SOP, QFP, TQFP				

*: Provided with an analog circuit for use microcontroller peripherals.

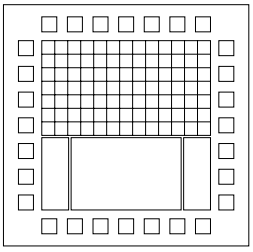
Mixed Signal ASIC

NEC offers mixed digital ASICs that employ the BiCMOS process with a process rule of 0.6 μm to 0.35 μm.

Furthermore, NEC supports analog IP cores through the use of a leading-edge 0.35-μm BiCMOS process, applying its rich analog circuit design know-how.

An extensive lineup of products is available to answer the wide-ranging needs of users.

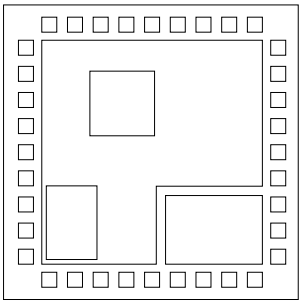
5-V power supply
for small-scale circuits



MA-8 Family
(μPD680××)

0.6-μm BiCMOS process
Gate array
+ Analog master slice form

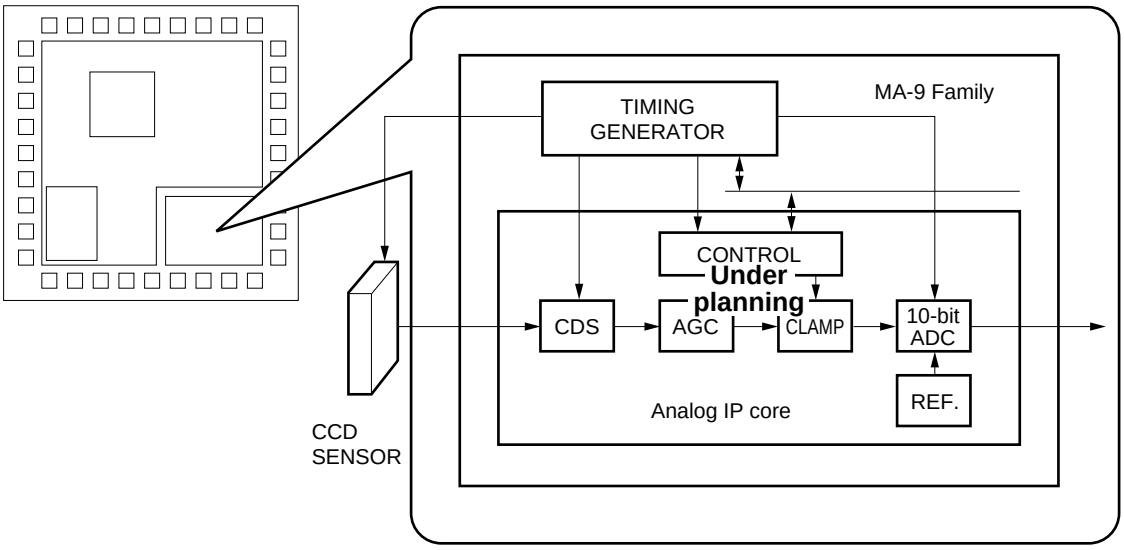
3.3-V power supply
for large-scale systems



MA-9 Family
(μPD681×××)

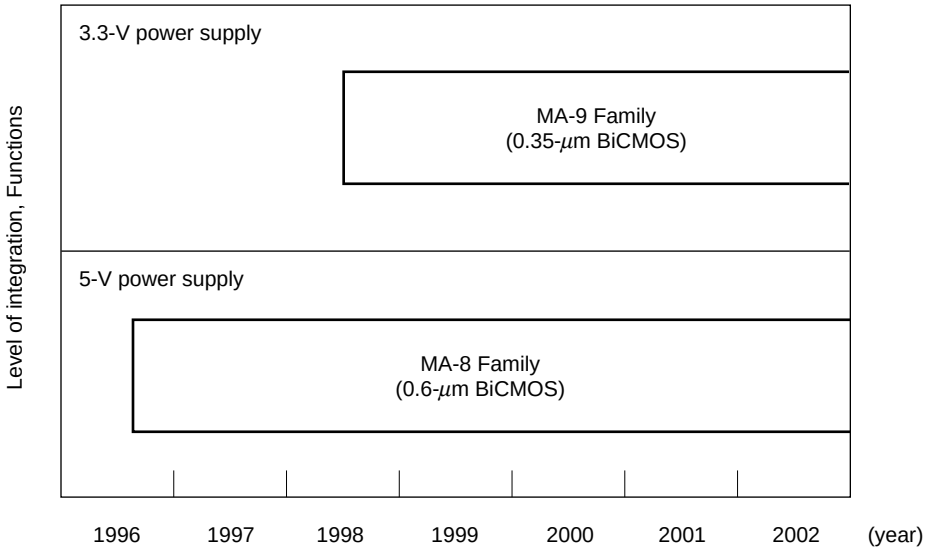
0.35-μm BiCMOS process
Cell-based form

MA-9 Family Analog IP Core

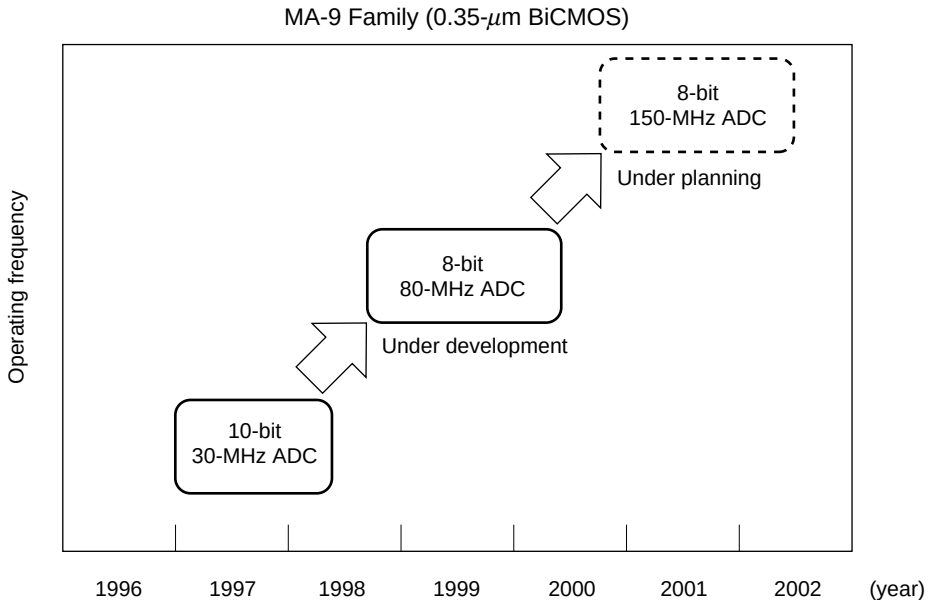


Mixed Signal ASIC

Mixed Signal ASIC Family Evolution



Mixed Signal ASIC Analog Core Evolution



Mixed Signal ASIC

■ MA-8 Family

The MA-8 Family (μPD6780××) consists of top-level mixed signal ASICs that employ the latest high-cost-performance BiCMOS process.

This family achieves the integration of mixed digital/analog circuits on a single chip, which will be pivotal for the development of the multimedia market.

■ Feature

■ Support of digital/analog mixed circuits

By employing the latest BiCMOS process, the MA-8 Family realizes the integration of 0.6-μm CMOS gate array and analog ASIC (analog master) on a single chip.

■ Analog block element configuration prioritizing circuit functions

Analog circuits that mix bipolar transistors and CMOS transistors can be created through the use of the BiCMOS process:

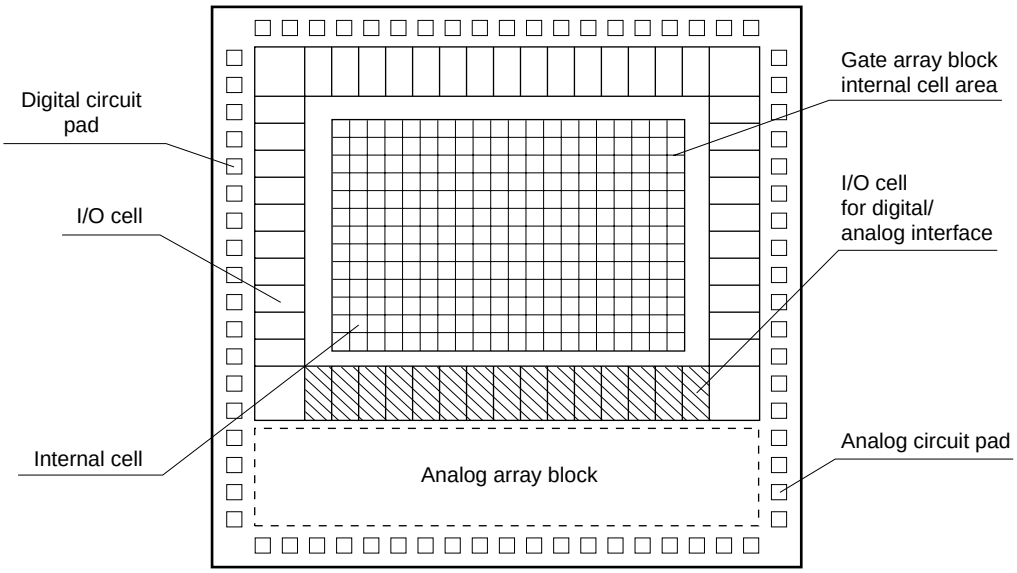
High input impedance operational amplifiers
Sample and hold circuits
Analog switches, etc.

■ Simple design and short development time

The logic block can be easily developed with OPENCAD™ (NEC's original CAE tool).

Furthermore, a short development time can be achieved, which is another advantage of ASICs.

■ Chip Configuration



Mixed Signal ASIC

■ Product list

Part number			μPD68000	μPD68001	μPD68010	μPD68011	μPD68031	μPD68041	μPD68061	μPD68070
Process			High-performance BiCMOS process							
Maximum Supply Voltage			6.0 V (absolute maximum rating)							
Number of Pads			68	68	108	84	96	108	128	168
Logic Block										
Number of Gates			1520	1520	1680	2880	6720	9520	16512	28320
Number of Usable Gates[Note]			912	912	1008	1728	4032	5712	9907	16992
Number of Logic I/Os			23	23	31	32	44	53	68	88
Analog block										
Total Number of elements			912	1264	4614	2016	2016	2304	2736	5017
	NPN Transistor	(Standard)	150	210	756	336	336	384	456	816
		(Large type)	25	35	126	56	56	64	76	136
	PNP Transistor (Lateral)		200	280	1008	448	448	512	608	1088
	Polysilicon Resistor	(1 KΩ)	150	210	756	336	336	384	456	816
		(5 KΩ)	300	420	1512	672	672	768	912	1632
Capacitor (5 pF)			25	35	126	56	56	64	76	136
P-ch MOS Transistor			50	70	252	112	112	128	152	272
Analog SW			4	0	24	0	0	0	0	15
P-ch MOS Transistor for analog			4	2	28	0	0	0	0	58
N-ch MOS Transistor for analog			4	2	26	0	0	0	0	48
Digital/Analog Junction Block										
Digital/Analog I/F (Maximum)			14	14	36	14	14	14	14	24

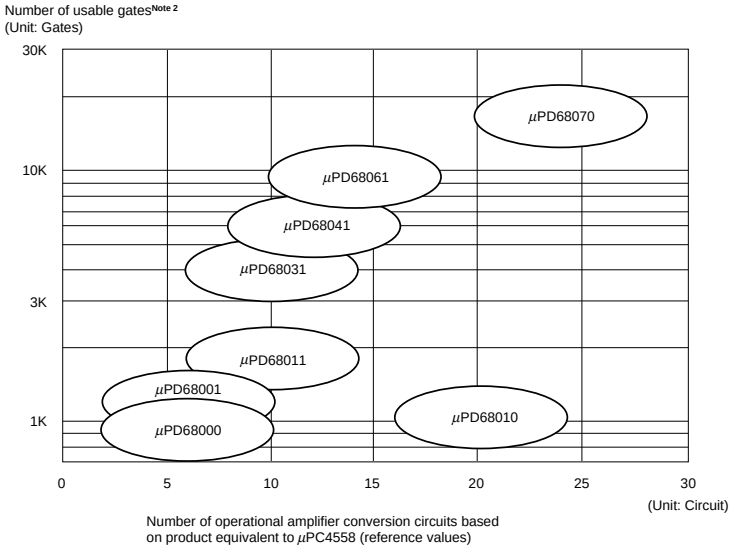
Note Gate utilization ratio: 60%

Remark The last digit in the part number of MA-8 Family products indicates the presence/absence of analog MOS transistors and analog switches in the analog array block, as follows:
Last digit = 0 → Mounted Last digit = 1 → Not mounted

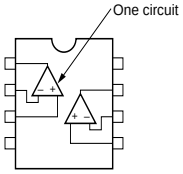
■ Master size

The figure below shows the master block size in terms of the number of usable gates and the number of circuits[Note 1] calculated as general-purpose operational amplifier circuits corresponding to the μPC4558.

MA-8 Family Master Sizes



Notes 1. The two operational amplifiers mounted on the package (8-pin DIP) of the μPC4558 general-purpose amplifier are each treated as one circuit.
Moreover, the numbers of conversion circuits indicated in the figure above are merely reference values for the case when only operational amplifiers are mounted on the analog array block, and operational amplifier peripheral circuits (such as feedback circuits) are not included.
In particular when selecting products, estimate the circuit scale considering their peripheral circuits.



2. Gate utilization ratio: 60%

Mixed Signal ASIC

■ MA-9 Family

The MA-9 Family (μPD681××) consists of leading-edge mixed signal ASICs that aim for system-on-a-chip through the use of a leading-edge 0.35-μm BiCMOS process pioneered by NEC.

While aiming for ultra-high integration for large-scale digital/analog mixed systems in the field of multimedia, the MA-9 Family realizes high-speed, high-accuracy stable signal processing.

■ Feature

■ Support of analog IP core

NEC supports analog IP core, using its rich analog circuit design resources.

■ Mixed signal ASICs aiming for system-on-a-chip

Large-scale mixed signal systems can now be created on a single chip using the cell-based form.

NEC supports mixed signal cores that include an analog IP core, high-speed ADC and DAC and other analog cores, digital/analog mixed circuits, etc.

■ Leading-edge BiCMOS process

High-speed digital circuits and high-accuracy, sophisticated analog circuits can now be realized on a single chip by employing NEC's leading-edge 0.35-μm BiCMOS process.

■ Low power consumption

A low power consumption is achieved for LSIs by employing a low-voltage operation process (3.3 V)

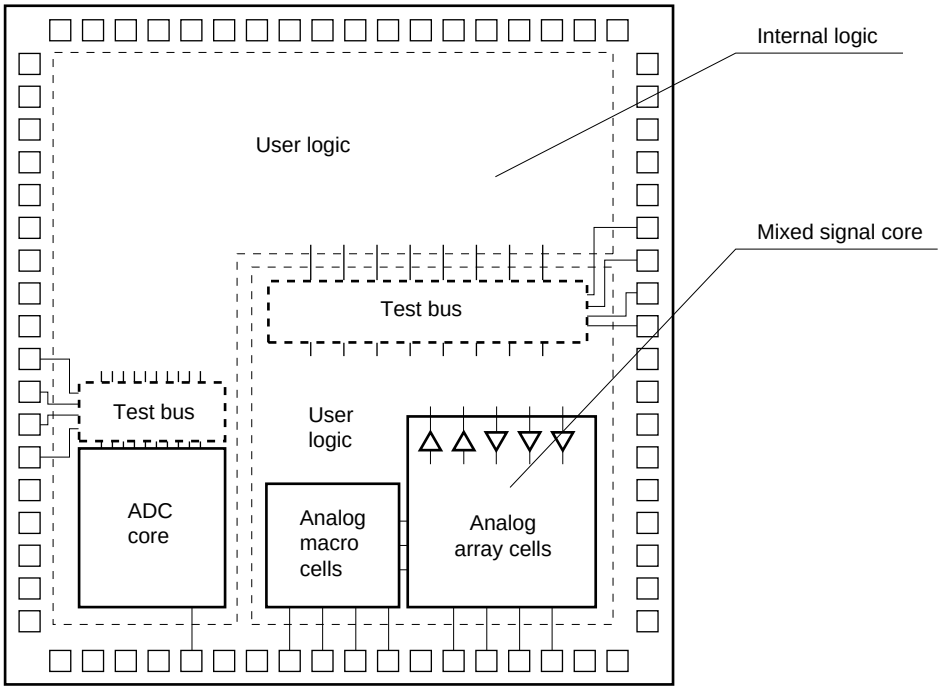
■ Flexible mixed signal development environment

NEC's development environment for cell-based ICs can be used for the internal logic.

For mixed signal cores, NEC supports a development environment using analog arrays that enables a short TAT comparable to that of embedded arrays.

Moreover, for circuit blocks, for which analog characteristics particularly matter, NEC supports a development environment that enables the use of macro cells.

■ Chip Configuration

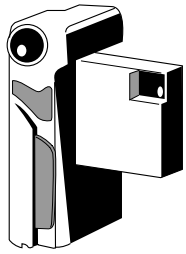


Mixed Signal ASIC

■ Analog IP cores

■ Analog Front-End Cores for CCD Signal Processing

Through the use of bipolar transistors, which have outstanding analog characteristics compared to CMOS transistors, NEC offers cores with superb low-noise characteristics made possible only by the use of the BiCMOS process.

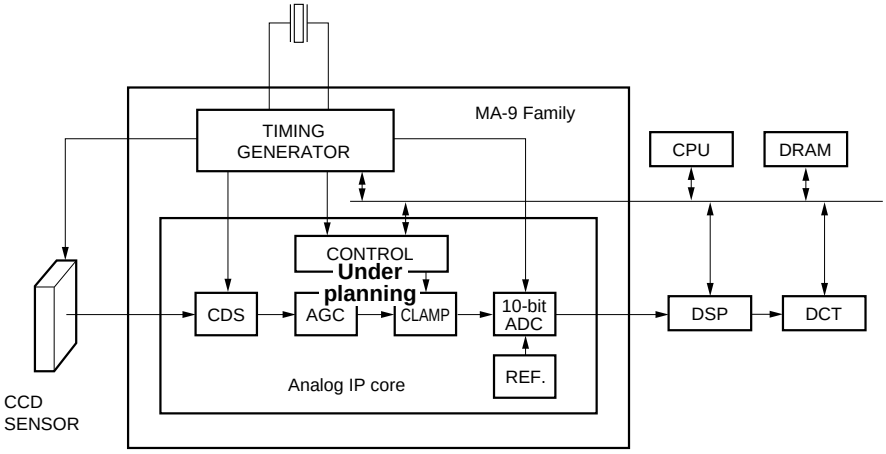


Under planning

Features

- ○ CDS optimum for digital still cameras (correlated double sampling)
- ○ High-gain AGC for higher sensitivity
- ○ Low-consumption-power-type 10-bit ADC and reference voltage
- ○ Controller circuit inside DAC for analog circuit characteristics correction

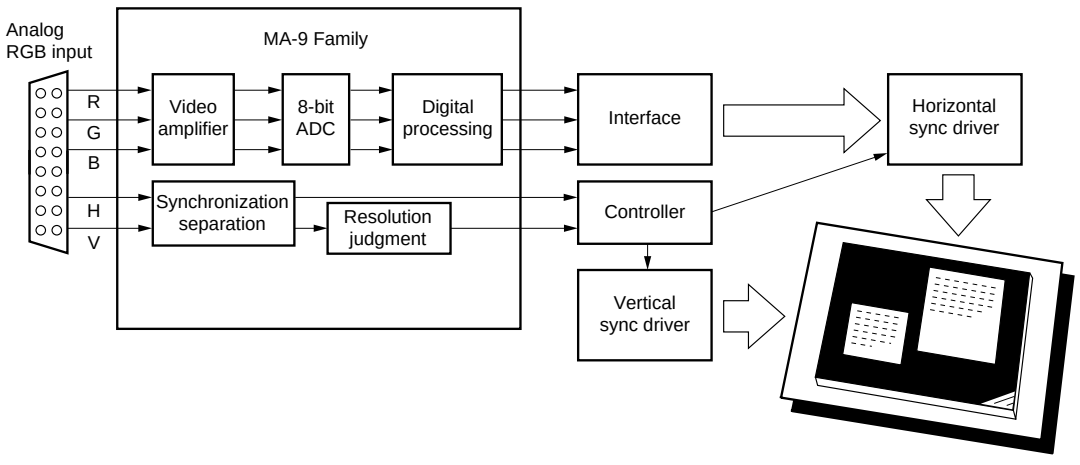
Application Example (Digital Still Camera)



■ High-Speed ADC Cores for Video

For video	8-bit 80-MHz	**Under development**

Application Example (LDC Display Video Signal Input)



Mixed Signal ASIC

■ Product overview

■ Logic Block

Product Name		μPD681××
Process		0.35-μm BiCMOS process
Supply Voltage		3.3 V ±0.3 V (I/O block, internal gates)
Maximum Integration (logic only)		1.7 million gates (usable)
Interface Level		LVTTL
Delay Time	Internal gates[Note 1]	114 ps (TYP)
	Input buffer[Note 2]	169 ps (TYP)
	Output buffer[Note 3]	864 ps (TYP)

Notes 1. Value assuming 2-input NAND power gate, fanout 2, and standard wiring length.

2. Value assuming fanout 2 and standard wiring length.

3. Value assuming load capacitance 15 pF, I_{OL} = 18 mA.

■ Analog Block

Product Name		μPD681××
Process		0.35-μm BiCMOS process
Supply Voltage		3.3 V ±0.3 V
Transistors	NPN type	f_T = 10 GHz, h_{FE} = 70 (all TYP)
	PNP type (Vertical type)	f_T = 2 GHz, h_{FE} = 30 (all TYP)
	MOS	N-ch type, P-ch type for analog circuit
Polysillicon Resistor[Note]		Absolute precision: ±20%, relative precision: ±2%
Capacitor (MIM type)[Note]		Absolute precision: ±20%, relative precision: ±2%

Note Values indicated are for reference only. The relative precision applies only to when the pin is positioned in an adjacent location.

Particular Purpose IC

DSP 156
CCD Linear Image Sensor 157
Communication IC 159
Mass Storage IC 163
Global Positioning System IC 164
Display IC 164
Motor Driver 167
Power MOS IC for Voltage Regulator 167
Power Switch for USB Application 168
Image IC 168
MPEG IC 168
Audio IC 169
TV IC 171
Multisync Monitor IC 173
Digital Image IC 174
On-Screen Character Display IC 175
Video Camera IC 176
Remote Control IC 176
Clock IC 176
Rotary Encoder IC 177
A/D Converter IC 177
D/A Converter IC 177
Line Driver Receiver IC 178

DSP

		μPD77016	μPD77015	μPD77017	μPD77018A
Instruction cycle (ns)		30	30	30	16.7
Data type *		fixed	fixed	fixed	fixed
Multiplier precision (bits)		40 + 16 × 16 → 40	40 + 16 × 16 → 40	40 + 16 × 16 → 40	40 + 16 × 16 → 40
ALU precision (bit)		40	40	40	40
Internal memory (words × bits)	Program ROM	–	4K × 32	12K × 32	24K × 32
	Program RAM	1.5K × 32	256 × 32	256 × 32	256 × 32
	Data ROM	–	4K × 16	8K × 16	24K × 16
	Data RAM	4K × 16	2K × 16	4K × 16	6K × 16
External memory (words x bits)		48K×32 (program) 96K×16 (data)	32K×16 (data)	32K×16 (data)	32K×16 (data)
Package		• 160-pin QFP	• 100-pin TQFP	• 100-pin TQFP	• 100-pin TQFP • 116-pin FBGA

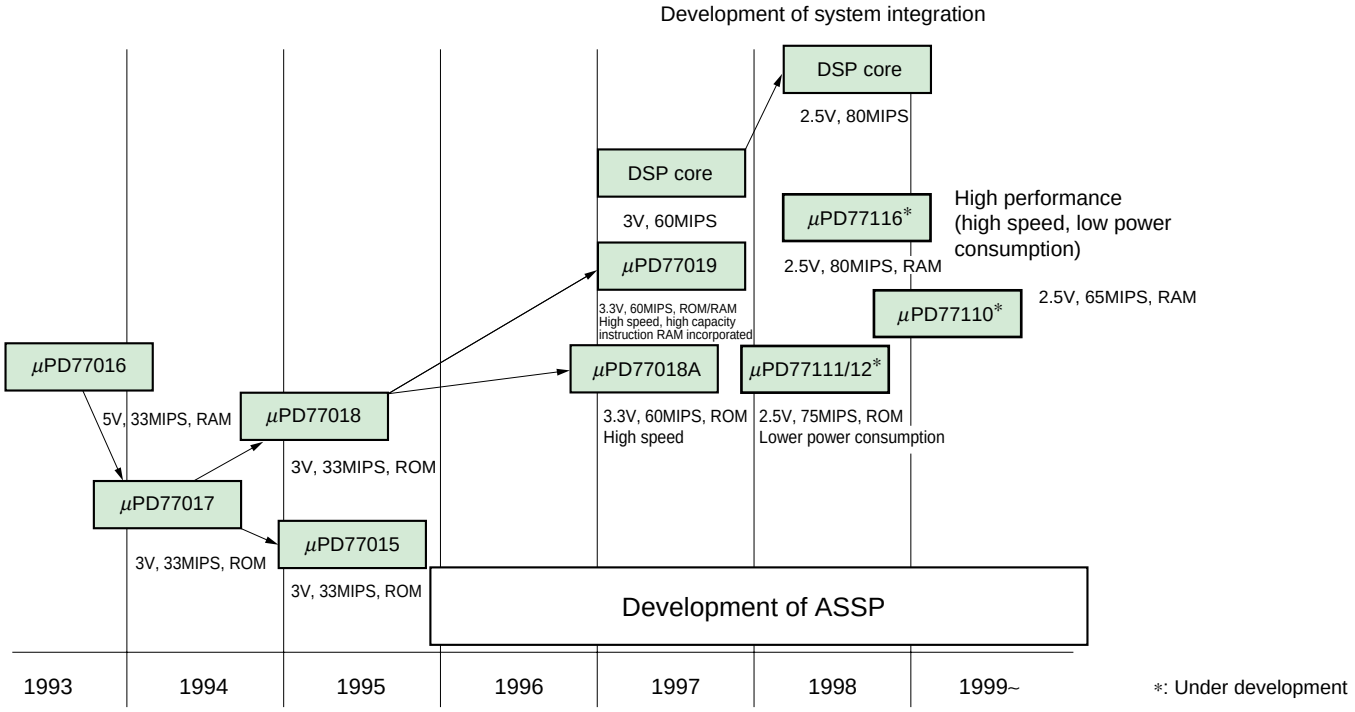
*: fixed: Fixed point decimal

		μPD77019	μPD77019-013	μPD77111
Instruction cycle (ns)		16.7	16.7	13.3
Data type *		fixed	fixed	fixed
Multiplier precision (bits)		40 + 16 × 16 → 40	40 + 16 × 16 → 40	40 + 16 × 16 → 40
ALU precision (bit)		40	40	40
Internal memory (words × bits)	Program ROM	24K × 32	–	32K × 32
	Program RAM	4K × 32	4K × 32	1K × 32
	Data ROM	24K × 16	–	32K × 16
	Data RAM	6K × 16	6K × 16	6K × 16
External memory (words x bits)		32K×16 (data)	32K×16 (data)	–
Package		• 100-pin TQFP	• 100-pin TQFP	• 80-pin TQFP • 80-pin FBGA

*: fixed: Fixed point decimal

DSP

■ DSP Development Plan



Remark ASSP: Application Specific Standard Product

CCD Linear Image Sensor

■ Monochrome

Part number	Pixel number	Pixel size (μm)	Sensitivity (TYP.) (V/lx·s)	Data rate (MAX.) (MHz)	Package	Features
μPD3753	2088	14 × 14	90	2	• 22-pin plastic DIP (400 mil)	Single 5 V power supply, Single-sided readout, Clamp circuit
μPD3734A	2660	11 × 11	70	5		Sample & Hold, Clamp circuit, High sensitive and low noise amplifier, Low image lag
μPD3739	5000	7 × 7	9.0	40	• 22-pin CERDIP (400 mil)	μPD35H71A pin compatible, High speed, Low image lag, 2-output (differential phase or same phase)
μPD3737	5150		7.5	20		Middle speed, Low noise, Low image lag

Light source of the Sensitivity: Daylight color fluorescent lamp

CCD Linear Image Sensor

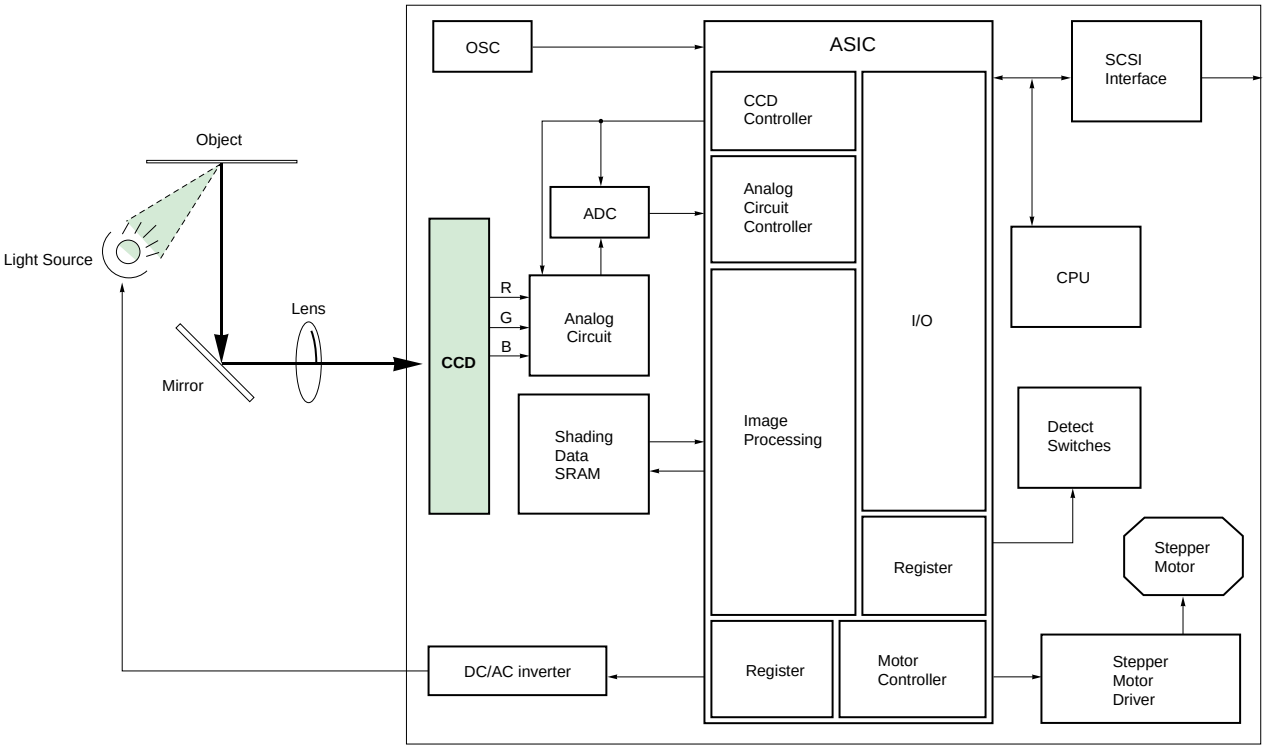
■ Color

Part number	Pixel number	Pixel size (μm)	Sensitivity (TYP.) (V/lx·s)	Data rate (MAX.) (MHz)	Package	Features
μPD3700	160 × 3	11 × 33	R 14.4 G 13.1 B 7.90	3	• 20-pin plastic SDIP (300 mil)	Sample & Hold, Clamp circuit, Compensation output, Dot sequential color filter type, Sample & Hold pulse generator, Clamp pulse generator, Reset pulse generator, Single-sided readout, Miniature package
μPD3720A	2700 × 3	10.5 × 10.5	R 20.2 G 18.5 B 11.1	3	• 22-pin plastic DIP (400 mil)	High sensitivity, Low noise, 4-line spacing (42 μm), 3-output
μPD3794	2700 × 3	8 × 8	R 14.6 G 13.3 B 8.00	4		High sensitivity, Low noise, single-sided readout, 1-output by the RGB selector, 4-line spacing (32 × μm)
μPD3729	5000 × 3	10 × 10	R 6.20 G 5.40 B 6.80	30	• 24-pin ceramic DIP (400 mil)	High speed, Low noise, Clamp circuit, 2-output/color (same phase) total 6-output, 4-line spacing (40 μm)
μPD3799★	5300 × 3	7 × 7	R 11.2 G 10.2 B 6.10	4	• 32-pin plastic DIP (400 mil)	High sensitivity, Low noise, Clamp pulse generator, Clamp circuit, 4-line spacing (28 μm), 3-output, Single-sided readout
μPD3797	5300 × 3		R 11.2 G 10.2 B 6.10	3	• 22-pin ceramic DIP (CERDIP) (400 mil) • 32-pin plastic DIP (400 mil)	High sensitivity, Low noise, Sample & Hold, Clamp circuit, 4-line spacing (28 μm), 3-output, Single-side readout
μPD3728	7300 × 3	10 × 10	R 6.20 G 5.40 B 6.80	40	• 36-pin ceramic DIP (600 mil)	High speed, Low noise, Clamp circuit, 2-output/color (same phase) total 6-output, 4-line spacing (40 μm)
μPD3719	10600 × 3	7 × 7	R 9.70 G 8.80 B 5.30	2		15 V power supply, Clamp circuit, 10-line spacing (70 μm), Single-sided readout, High S/N, High saturation voltage (4.0 V MIN.), 3-output

★: Under development

Light source of the Sensitivity: 3200K halogen lamp + C-500S (infrared cut filter, t = 1 mm)

■ CCD Linear Image Sensor System Block Diagram

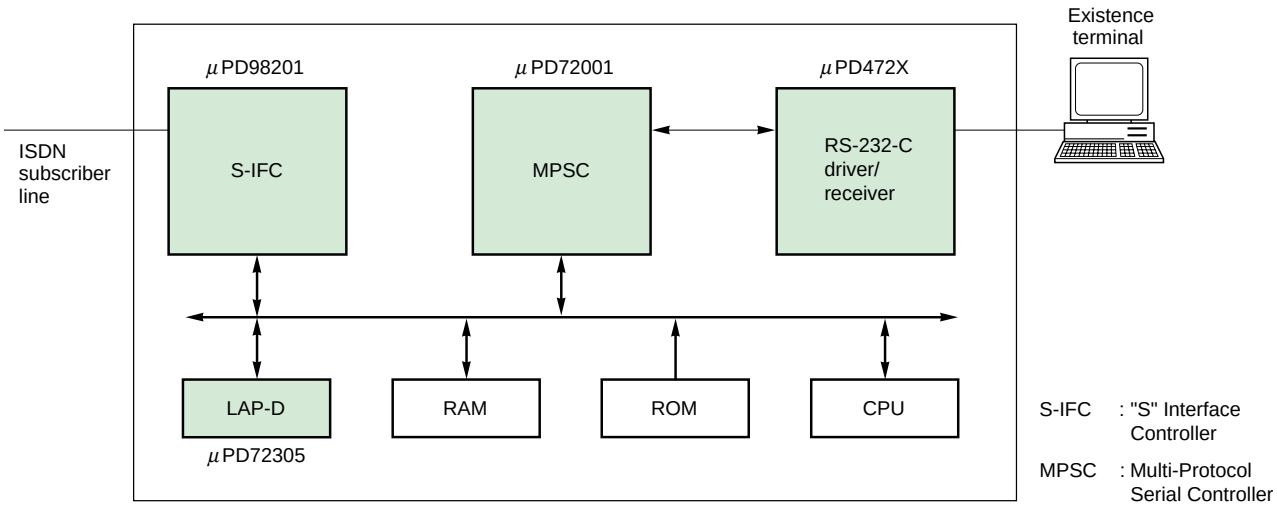


Communication IC

■ Communication LSI

Part number	Function	Features	Package
μPD72001-11	Multi-protocol serial controller	Allows communication in ASYNC, COP, and BOP mode (2 ch) μPD72001-11: 5 V Operation μPD72001-A8: 3.3 V Operation	• 40-pin DIP • 52-pin QFP • 52-pin QFJ • 44-pin TQFP
μPD72001-A8			• 40-pin DIP • 44-pin TQFP • 52-pin QFP
μPD72002-11	Multi-protocol serial controller	Allows communication in ASYNC, COP, and BOP mode (1 ch)	• 40-pin DIP • 44-pin QFP • 44-pin QFJ • 44-pin TQFP
μPD72103A	HDLC controller	HDLC frame control, link/separation are possible, internal DMA controller 8 Mbps operation	• 68-pin QFJ • 80-pin QFP
μPD72105	LAN controller	Accommodates CSMA/ACK	• 48-pin DIP • 52-pin QFJ
μPD72107	Packet communication control LSI (X.25 communication control)	Can accommodate high speed transmission (4 MHz) Conforms ITU-T recommendation X.25 (LAP-B '84 ver)	• 64-pin shrink DIP • 68-pin QFJ • 80-pin QFP
μPD72305	LAP-D protocol processing LSI	Supports LAP-D protocol of ITU-T recommendation	• 64-pin shrink DIP • 68-pin QFJ • 80-pin QFP
μPD98201	S interface transceiver for NT or TE	4-wire full duplexed communication (basic access) Conforms to ITU-T recommendation I430	• 64-pin QFP
μPD98203	S interface transceiver for NT	Conforms to ITU-T recommendation I430	• 20-pin DIP
μPD77523	G.728/722/711 VOICE CODEC	• Conforms to ITU-T recommendation G.728/722/711 • V_{DD} = 3 V	• 100-pin TQFP
μPD6708	IE Bus™ controller/driver	Transmit data buffer: 4 bytes Receive data buffer: 20 bytes	• 16-pin SOP (300 mil)
μPD72042A **μPD72042B**		Transmit data buffer: 33 bytes Receive data buffer: 40 bytes	• 16-pin SOP (375 mil)

[ISDN Terminal Adapter]



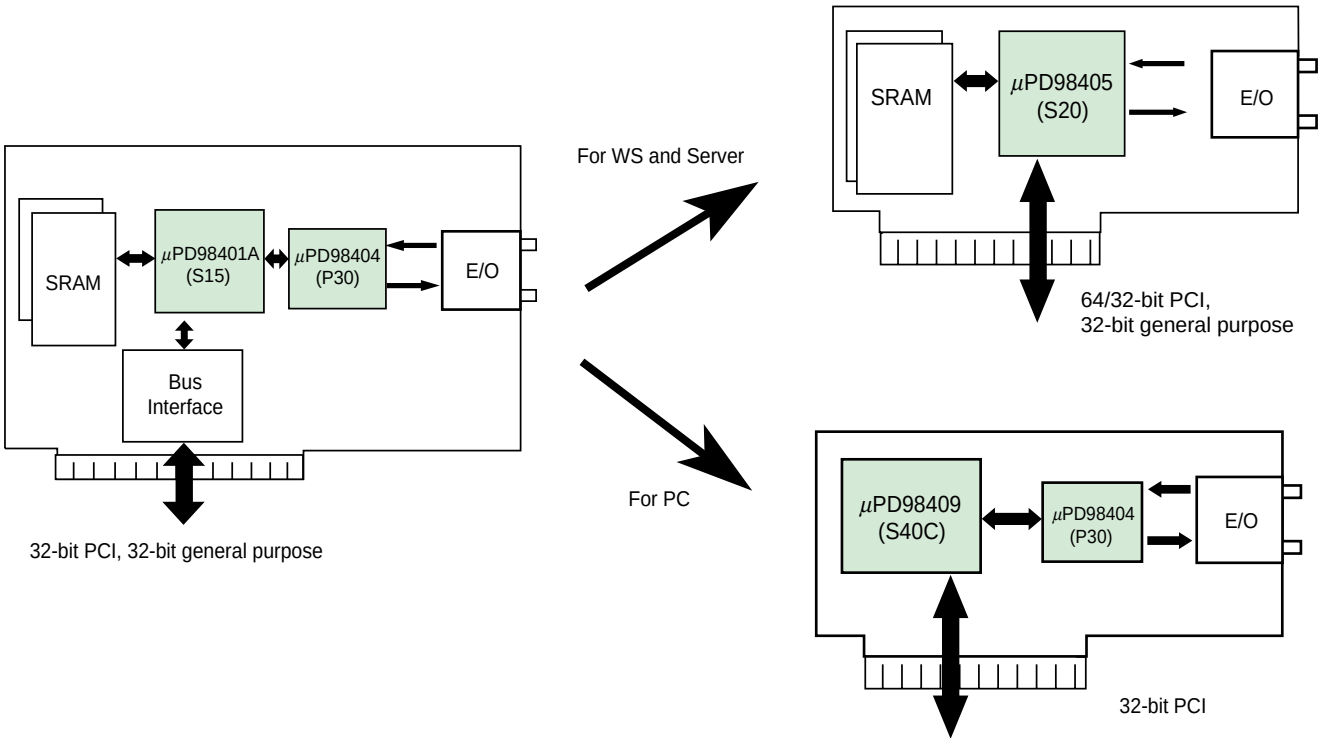
Communication IC

■ Chip Set for ATM

Part number	Function	Features	Package
μPD98401A	ATM SAR Chip	• Upward compatible with μPD98401 • Conforms to ATM Forum recommendations • Implements AAL-5 SAR sublayer and ATM layer functions	• 208-pin QFP (Fine Pitch)
μPD98402A	ATM SONET Framer	• ATM physical layer having the TC sublayer function • Conforms to ATM Forum UNI specifications • SONET STS-3c frame format	• 160-pin QFP (Fine Pitch)
μPD98404★	ATM Advanced SONET Framer	• Supports TC sublayer in PHY layer • Implements SONET STS-3c/SDH STM-1 (155 Mbps) framer function • Integrated Clock recovery/Clock synthesis function • UTOPIA Level 2 interface	• 144-pin QFP (Fine Pitch)
μPD98405★	ATM Integrated SAR controller	• SONET STS-3c/SDH STM-1 (155M bps), Framer and SAR controller with ABR function in a single-chip • Support both PCI bus and generic bus	• 304-pin QFP (Fine Pitch)
μPD98408	6 port 25M ATM PHY	• 25.6 Mbps ATM PHY (PMD & TC) function for 6 circuits • UTOPIA Level 2 interface	• 208-pin QFP (Fine Pitch)
μPD98409	ATM Light SAR Controller	• Implements AAL-5 SAR sublayer and ATM layer function • Supports PCI bus interface • MPEG packet transfer engine	• 240-pin QFP (Fine Pitch)
μPD98410	1.2 Gbps ATM Switch	• ATM switching function in a single-chip • 1.2 Gbps with non blocking • ABR traffic control	• 580-pin BGA
μPD98411★	ATM QUAD SONET FRAMER	• Implements 4 ports SONET STS-3c/SDH-1 (155M bps) framer function on chip	• 240-pin QFP (Fine Pitch)

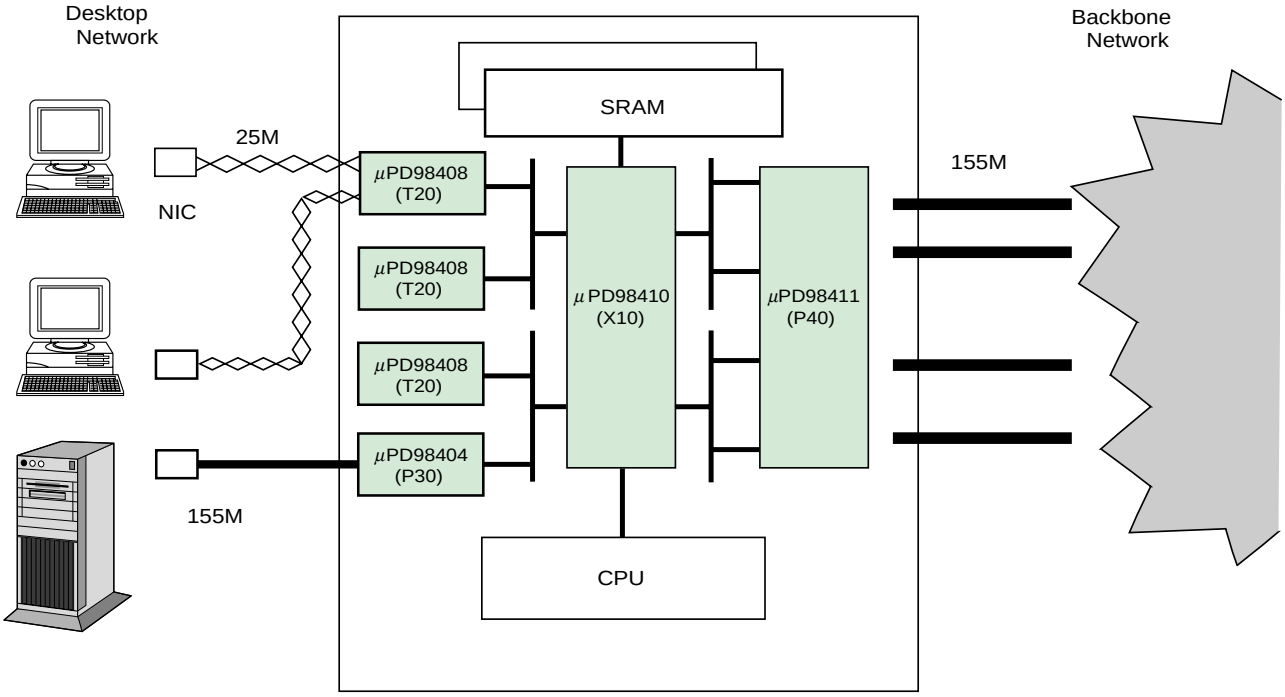
★: Under development

PCI Bus Supporting ATM-NIC



Communication IC

ATM Switch



■ Ethernet LSI

Part number	Function	Features	Package
μPD98421★	High Speed Address Search Engine	• CAM (64-bit × 8K Entry) • Implements three search modes	• 240-pin QFP (Fine Pitch)
μPD98431★	8-port 10/100 Mbps Ethernet Controller	• 8-port 10/100 Mbps Ethernet MAC • MII and 10 Mbps serial interface	• 352-pin BGA

★: Under development

■ Standard USB

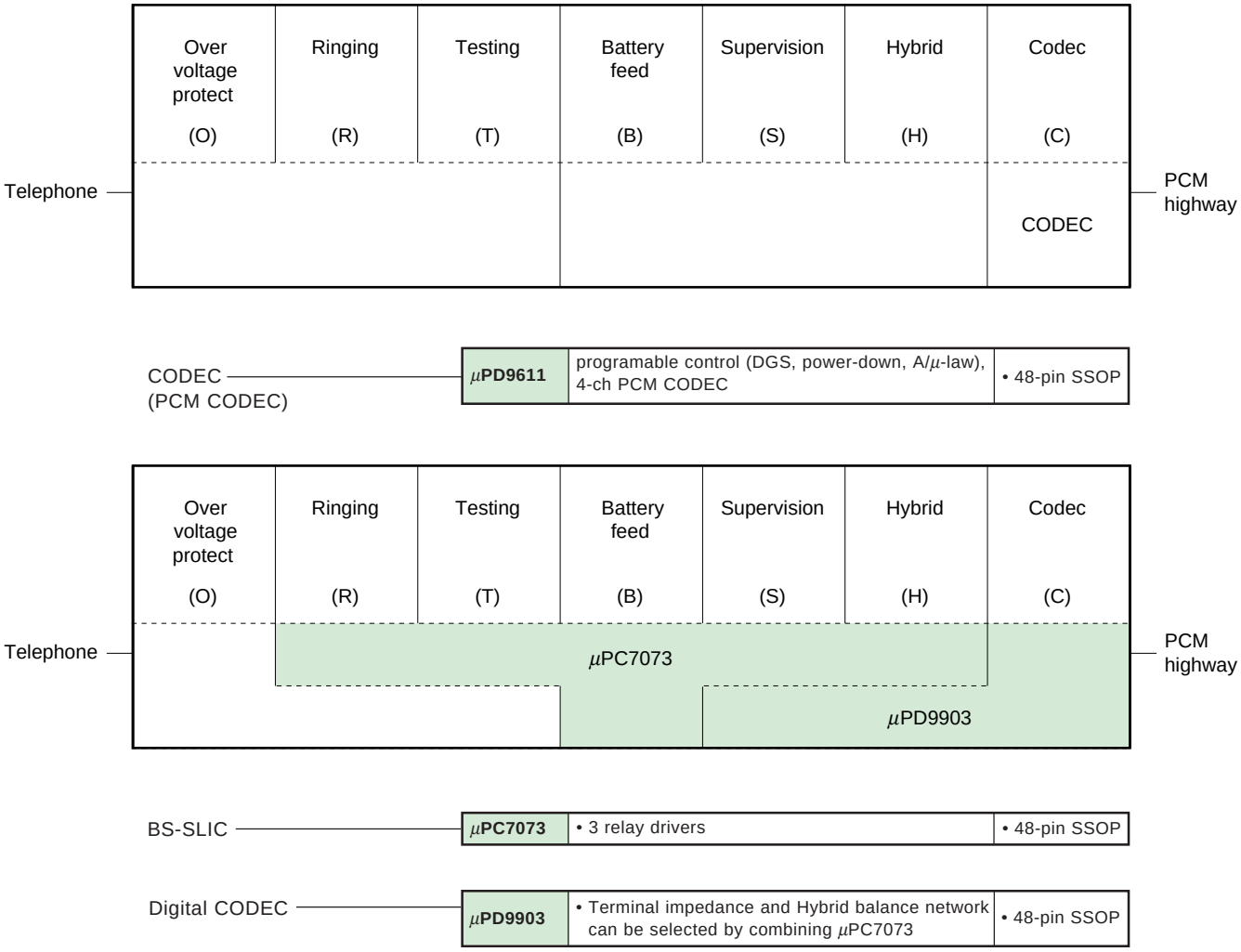
Part number	Function	Features	Package
μPD72011	USB Hub Controller	• All USB HUB function in a single-chip • Compliance with USB Specification 1.0 • Supports Self-powered and Bus-powered • 4 or 5 down-port selectable	• 42-pin SDIP • 44-pin QFP

■ Power Switch for USB Application

Part number	Function	Internal circuit	Input voltage (V)	Output current (A)	Over current detect (A)	Package	Main application
μPD16855	Pch High-side switch for USB	2	4.0 to 5.5	0.5	0.6 to 1.25	• 8-pin SOP (225 mil)	Note PC CRT monitor LCD monitor

Communication IC

■ Analog Line Card LSI



Mass Storage IC

Part number	Function	Features	Package
μPD72611	SCSI-2 controller	Conforms to ANSI X3T9.2/86-109 Rev. 10c High speed synchronous transfer 10 MBytes/s	• 100-pin QFP
μPD6378	CD ROM decoder LSI	Conforms to CD-ROM XA format	• 100-pin QFP
μPD63701	CD ROM decoder LSI	Supports 512K bytes of external DRAM	• 100-pin QFP
μPC2102	RF amplifier for floppy disk controller	Low power Internal PLA, D/A	• 20-pin SSOP
μPD63724	1 chip CD-ROM controller	Digital Servo, Data Processor, CD-ROM Decoder, Host I/F on 1 chip Supports 24X speed operation Automatic adjustment function Wide capture range High speed CAV audio playback Supports PIO mode 4, DMA mode 2 (Enhanced IDE/ATAPI) Power management	• 160-pin LQFP
μPC2575	RF amplifier for 24X speed CD-ROM	RF amplifier, error amplifier for 24X speed CD-ROM	• 36-pin SSOP
μPD63724A	1 chip CD-ROM controller	Digital Servo, Data Processor, CD-ROM Decoder, Host I/F on 1 chip Supports 24X speed operation Automatic adjustment function Wide capture range High speed CAV audio playback Supports PIO mode 4, DMA mode 2 (Enhanced IDE/ATAPI) Power management Digital audio interface output	• 160-pin LQFP

Global Positioning System IC

■ RF IC for GPS receiver

Type	Part number	Function (frequency unit: MHz)	V_{CC} (V)	I_{CC} (mA)	CG (dB)	Package
General purpose wideband separate IC	**μPC2756**	RF downconverter with osc. Tr	2.7 to 3.3	6	14	• 6-pin mini-mold • 6-pin super mini-mold
	μPC2753	IF downconverter with gain control amplifier	2.7 to 3.3	6.5	60 to 79	• 20-pin plastic SSOP (225 mil)
Clock frequency specific 1 chip IC	**μPB100x**★	RF/IF downconverter + PLL synthesizer TCXO = 18.414/1stIF = 28.644/ 2ndIF = 1.023	2.7 to 3.3	37.5	72 to 92	• 30-pin plastic SSOP (300 mil)
	μPB100x★	RF/IF downconverter + PLL synthesizer TCXO = 16.368/1stIF = 61.380/ 2ndIF = 4.092	2.7 to 3.3	37.5	72 to 92	• 30-pin plastic SSOP (300 mil)

Typical performance. (T_A = +25°C, V_{CC} = 3.0 V)

★: Under development

Display IC

■ Fluorescent indicator panel (FIP) driver IC

Part number	Drives	Number of outputs	Withstand voltage (V)	Output current (mA)	Output type	Package	Main applications
μPD6320 **μPD6321**	Static	39	18	1	CMOS	• 52-pin QFP	Car stereo
μPD6323	Static	21	25	5	Pch-O.D	• 28-pin DIP	Automotive
μPD6323B			40				
μPD6340	Dynamic	20	80	25	NPN+Nch	• 52-pin QFP	ECR, PPC
μPD6700	Static	47	18	1	CMOS	• 56-pin QFP	Car stereo
μPD16304	Dynamic	40	200	20	CMOS	• 80-pin QFP*	Graphic FIP module
μPD16306	Dynamic	64	80	50	CMOS	• 100-pin QFP	
μPD16306A	Dynamic	64	80	25	CMOS	• 100-pin QFP	
μPD16310	Dynamic	40	80	50	CMOS	• 80-pin QFP*	
μPD16326	Dynamic	32	130	20	CMOS	• 80-pin QFP*	
μPD16326A	Dynamic	32	150	20	CMOS	• 44-pin QFP	

*: Three direction lead

■ Fluorescent indicator panel (FIP) controller/driver IC

Part number	Drives	Duty (Max.)	Number of outputs		Number of characters*	Logic voltage (V)/ Drive voltage (V)	Package	Main application	Remarks
			Segments	Grids					
μPD16311	Segment	1/16	12	16	192 segments	5/35	• 52-pin QFP	VCR	Included key scan
μPD16312	Segment	1/11	11	11	121 segments	5/35	• 44-pin QFP	VCR	
μPD16313	Segment	1/11	11	11	121 segments	5/35	• 44-pin QFP	VCR	Included key scan (μPD16312 mirror)
μPD16314★	DOT character	1/24	80	24	24 × 2 characters	5/50	• 144-pin QFP	OA, FA	

*: At maximum duty

★: Under development

Display IC

■ Liquid crystal display (LCD) controller/driver IC

Part number	Drives	Duty (MAX.)	Number of outputs		Number of characters*1	Logic voltage (V)/ Drive voltage (V)	Package	Main application	Remarks
			Segments	Commons					
μPD7225	Segment	1/4	32	4	128 segments	2.7 to 5.5/ 2.7 to 5.5	• 52-pin QFP • 56-pin QFP		
μPD16430A			60	4	240 segments	3.5 to 6/ 3.5 to 14	• 80-pin QFP	Car audio	
μPD16431A			56	4	224 segments	2.7 to 5.5/ 2.7 to 6.5			Included key scan
μPD16432B	DOT character	1/15			12 × 2 characters + 60 pictographs	2.7 to 5.5/12	• 100-pin QFP		
μPD16434		1/16	42	16	8 × 2 characters	2.7 to 5.5/12	• 80-pin QFP		
μPD16435 μPD16435A	Graphic	1/73	119	73	119 × 71 dots + 119 × 2 pictographs	2.7 to 5.5/10*2	• TCP		Included DC/DC Converter
μPD16435B★							• CHIP		
μPD16670	DOT character	1/36	60	33	12 × 4 characters + 60 pictographs	2.7 to 3.6/12*2	• CHIP	Mobile phone	
μPD16675A	Graphic		128	34	128 × 32 dots + 128 × 2 pictographs	2.7 to 5.5/10*2		Pager	
μPD16676		1/16 1/32	61	16	61 × 16 dots	2.7 to 5.5/12			
μPD16677★		1/32	96	32	96 × 32 dots + 20 pictographs	2.7 to 3.6/12*2		Mobile phone	Included DC/DC Converter 4-gray scale display
μPD16680		1/53	100	53	100 × 51 dots + 100 × 2 pictographs				Included DC/DC Converter
μPD16682★		1/65	132	65	132 × 64 dots + 132 pictographs	2.4 to 5.5/12*2	• CHIP • TCP		

*1: At maximum duty

*2: Internal DC/DC Converter

★: Under development

Display IC

■ Liquid crystal display (LCD) driver IC

Part number	Drives	Number of outputs	Withstand voltage (V)	Logic voltage (V)	Package	Main applications
μPD6320 μPD6321	Static	39	18	5.0	• 52-pin QFP	Car stereo
μPD16448A	TFT (source) Analog	240	5.0	3.3	• TCP	LCD-TV
μPD16780★		288/300	5.0	3.3/5.0		LCD-TV
μPD16633	TFT (source) 64 gray scale	312	10.5	3.3		XGA
μPD16633A		312	8.5	3.3		XGA
μPD16633B		312	10.5	3.3		XGA
μPD16634A		300	8.5	3.3		SVGA
μPD16636		312	10.5	3.3		XGA
μPD16637		384	13.0	3.3		XGA/SXGA
μPD16638A		300/309	8.5	3.3		SVGA/XGA
μPD16640A		300/309	3.3/5.0	3.3		SVGA/XGA
μPD16640B★		300/309	3.3/5.0	3.3		SVGA/XGA
μPD16640C		300/309	3.3/5.0	3.3		SVGA/XGA
μPD16644		384	5.0	3.3		XGA
μPD16647★		402	3.3/5.0	3.3		SVGA
μPD16710		300/309	12.0	3.3		SVGA/XGA
μPD16712★		384	13.0	3.3		XGA/SXGA
μPD16714		300	10.5	3.3		SVGA
μPD16715		384	13.5	3.3		XGA/SXGA
μPD16732★		384	8.5	3.3		XGA/SXGA
μPD16740★	TFT (source) 64/256 gray scale	384	8.5	3.3		XGA/SXGA
μPD16650	TFT (gate)	120/128	40	3.3/5.0		–
μPD16652★		192/200	31	3.3		–
μPD16654★		150/154	40	3.3		–
μPD16655★		240	31	3.3		–
μPD16656/7★		256	40	3.3		–
μPD16661A	STN (column) 4 gray scale	160	5	3.3		PDA (1/8 VGA to VGA)
μPD16666A	STN (row)	240	36	5.0		PDA (1/8 VGA to VGA)
μPD16662	STN (column) 4 gray scale	240	5	3.3		PDA (1/8 VGA to 1/2 VGA)
μPD16667	STN (row)	160	36	5.0		PDA (1/8 VGA to 1/2 VGA)
μPD16663★	STN (column) 16 gray scale	240	5	3.3		PDA (1/8 VGA to 1/2 VGA)

★: Under development

■ Plasma display, EL display driver IC

Part number	Drives	Number of outputs	Withstand voltage (V)	Output current (mA)	Output type	Package
μPD16305	AC-PDP row	40	200	400	CMOS	• 80-pin QFP*
μPD16306B	AC-PDP/EL column	64	80	50		• 100-pin QFP
μPD16306A		64	80	25		• 100-pin QFP
μPD16310	AC-PDP/EL	40	80	50		• 80-pin QFP*
μPD16327	AC-PDP column	64	100	40		• 100-pin QFP
μPD16331		40	100	150		• 80-pin QFP
μPD16334		96	80	50		• For COB
μPD16335		96	80	50		• For COB
μPD16337		64	150	40		• 100-pin QFP

*: Three direction lead

Display IC

■ Other driver IC

Part number	Number of outputs	Withstand voltage (V)	Output current (mA)	Output type	Package	Main applications
μPD6320 μPD6321	4 × 4	18	20	NPN+Nch	• 52-pin QFP	Car stereo (LED)
μPD6345	8	40	100	NPN-O.C	• 16-pin DIP • 16-pin SOP	Thermal head, LED, solenoid

NPN-O.C: NPN open-collector output (sink driver)

Motor Driver

Part number	Function	Recommended operation voltage range (V)		Output current (A)	Package	Main application
		Control part	Output part			
μPD16804	DC motor driver	3 to 6	Up to 7.5	±0.5	• 16-pin SOP	Camera, DSC
μPD16805		3 to 6	Up to 7.5	±1.0	• 16-pin SOP	
μPD16823		2.5 to 6	Up to 7.5	±0.5	• 20-pin SOP	
μPD16803	stepping motor driver	4 to 6	4 to 13.2	±0.38	• 20-pin SOP	FDD (2 aspects excitation)
μPD16808		4 to 6	4 to 13.2	±0.45	• 20-pin SOP	FDD (1-2 aspects excitation)
μPD16813		4 to 6		±0.31	• 16-pin SOP	FDD (2 aspects excitation)
μPD16814		4 to 6		±0.415	• 16-pin SOP	FDD (1-2 aspects excitation)
μPD16818		2.7 to 6		±0.38	• 20-pin SOP • 20-pin SSOP	FDD (2 aspects excitation)
μPD16833		2.7 to 5.5		±0.3	• 30-pin SSOP	camcorder 4 ch-H bridge
μPD16833A		2.7 to 5.5		±0.3	• 30-pin SSOP	camcorder 4 ch-H bridge
μPD16835		2.7 to 5.5	4.8 to 11	±0.15	• 38-pin SSOP	camcorder microstep drive
μPD16836	3 aspects blush-less DC motor driver	4.5 to 5.5	10.8 to 13.2	±1.3 peak	• 24-pin SOP (375 mil)	CD-ROM PWM drive
μPD16837	stepping motor driver	4 to 6	10.8 to 13.2	±1.0 peak	• 30-pin SSOP (0.8 mm pitch)	CD-ROM/DVD actuator drive
μPD16856	3 aspects blush-less DC motor driver	4.5 to 5.5	10.8 to 13.2	±2.0 peak	• 30-pin SSOP (0.8 mm pitch)	CD-ROM PWM drive
μPD16857★	stepping motor driver	3.0 to 3.6	10.8 to 13.2	±1.0 peak	• 30-pin SSOP	DVD-RAM
μPD16858	3 aspects blush-less DC motor driver	4.5 to 5.5		±1.5 peak	• 30-pin SSOP	CD-ROM PWM drive

★: Under development

Power MOS IC for Voltage Regulator

Part number	Function	Input voltage (V)	Output voltage (V)	Output current (mA)	Package	Main application
μPD16901	Step-up type DC/DC converter for PCMCIA	4.5 to 5.5	11.72 to 12.28	140	• 14-pin SOP	Note PC

Power Switch for USB Application

Part number	Function	Internal circuit	Input voltage (V)	Output current (A)	Over current detect (A)	Package	Main application
μPD16855	Pch High-side switch for USB	2	4.0 to 5.5	0.5	0.6 to 1.25	• 8-pin SOP (225 mil)	Note PC CRT monitor LCD monitor

Image IC

Part number	Function	Package
μPD72254★	Graphics LSI	• 208-pin QFP

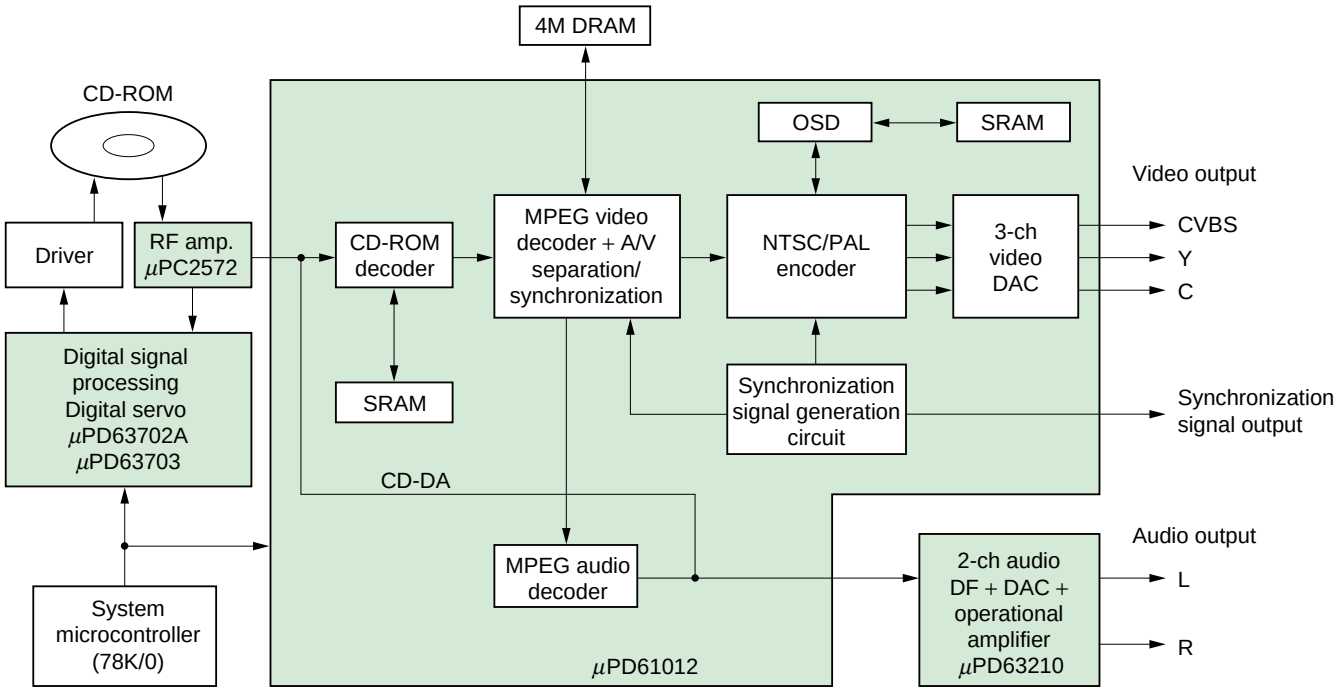
★: Under development

MPEG IC

Part number	Function	Package
μPD61012	MPEG1 Audio/Video decoder for Video-CD	• 100-pin LQFP
μPD61030★	Signal processing for STB, includes MPEG2 Audio/Video decoder, TS demultiplexer, Graphics and VR4110CPU	• 304-pin QFP

★: Under development

■ Video CD



Audio IC

■ Radio frequency amplification

Part number	Recommended operating voltage range (V)	Function					Recommended application	Package
		AM tuner	FM front end	FM IF amplification	FM detection	Other functions	Car radio	
μPC2533	7.5 to 8.5	●				Double super heterodyne DTS use	●	• 36-pin SOP

■ Pre-amplifier

Part number	Recommended operating voltage range (V)	Function				Recommended application		Package
		Play back amplifier	Recording amplifier	MIC amplifier	Other functions	Radio cassette recorder	Car radio	
μPC1228	6 to 16	●			Dual	●	●	• 8-pin SIP

■ Power amplifier

Part number	Recommended operating voltage range (V)	Electrical characteristics (typical value)			Function	Recommended application			Package
		P_O (W)	V_{CC} (V)	R_L (Ω)		Home stereo	Radio cassette recorder	Car radio	
μPC1316	3 to 16	1.2	9	8	Dual		●		• 14-pin DIPTAB
		0.5	6	8	Low pop noise				

■ D/A converter for audio system

Part number	Recommended operating voltage range (V)	Function	Recommended application			Package
			Home stereo	Portable	Car stereo	
μPD6376	4.5 to 5.5	16-bit D/A converter	●	●	●	• 16-pin SOP
μPD6379 **μPD6379A**	4.5 to 5.5	16-bit D/A converter	●	●	●	• 8-pin SOP
μPD6379L **μPD6379AL**	3.0 to 5.5					
μPD63200	4.5 to 5.5	18-bit D/A converter	●	●	●	• 16-pin SOP
μPD63210	4.5 to 5.5	16-bit D/A converter with digital filter	●	●	●	• 28-pin SOP
μPD63210L	3.0 to 5.5					

■ Sound CODEC

Part number	Function	Features	Package
μPD63310	Sound codec	• ΔΣ type ADCs and DACs are incorporated. • Mixers are incorporated into the analog input section and analog output section. • Low-noise microphone amplifiers are incorporated. • Sampling frequency: 2 to 48 kHz (master clock: 256 fs) • Single power source: +3 to 5.5 V • Operating temperature range: –20 to +80°C	• 80-pin TQFP

Audio IC

■ Device kit for CD

Part number	Recommended operating voltage range (V)	Function	Recommended application			Package
			Home stereo	Portable	Car stereo	
μPC2572	4.5 to 5.5	RF amp, error amp, APC for CD player	●		●	• 38-pin SOP
μPD63702A	4.5 to 5.5	Digital servo, data processor, DAC for CD player	●		●	• 80-pin QFP
μPD63703	4.5 to 5.5	Digital servo, data processor, ATT for CD player	●		●	• 80-pin QFP
μPD63710	4.5 to 5.5	RF amp, servo, data processor, ATT, DAC for CD player	●		●	• 100-pin LQFP

■ Other functions

Part number	Recommended operating voltage range (V)	Function	Recommended application			Package
			Home stereo	Tape deck	Radio cassette recorder	
μPC1237	25 to 60	Stereo power amplifier protection circuit	●			• 8-pin SIP

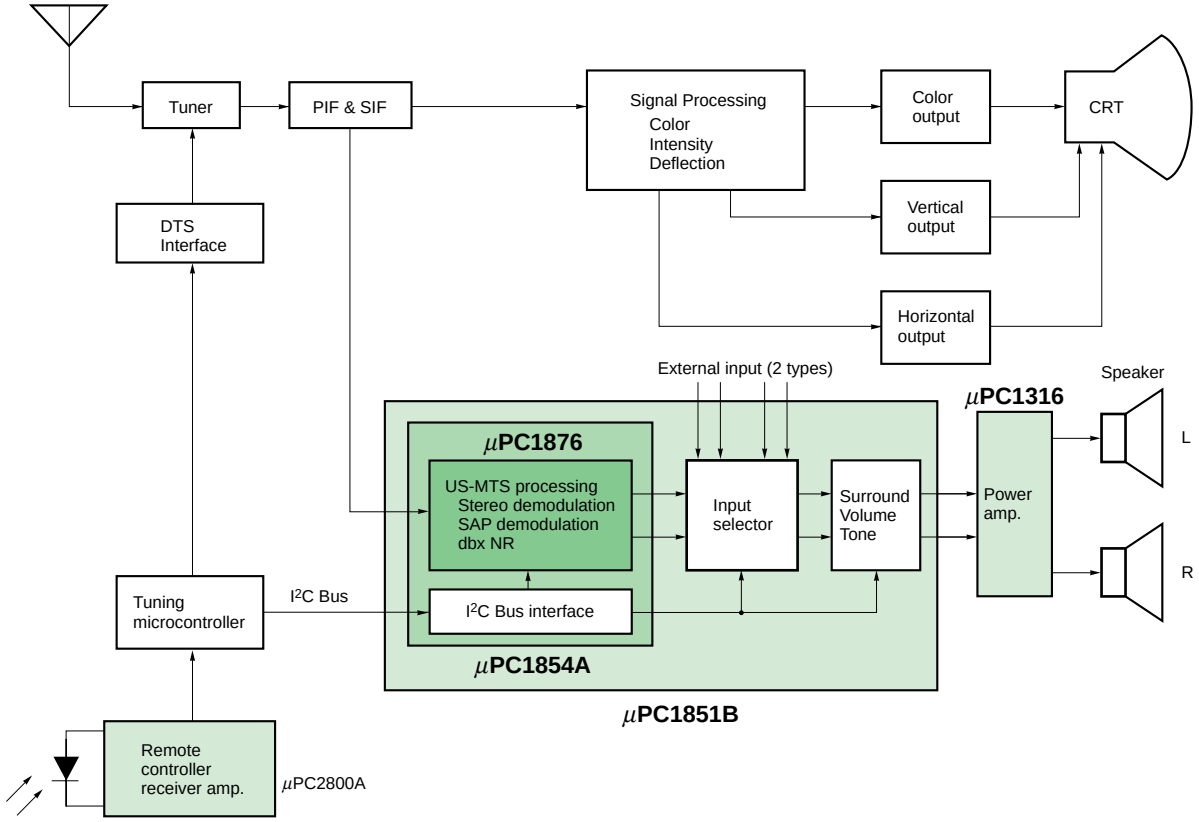
TV IC

■ US MTS Decoder

US MTS: Multi Television Sound for USA specifications

Part number	Function	Package
μPC1851B	I²C Bus, Input Selector (2 ch), dbx NR, SAP Decoder, Tone/Volume control	• 42-pin SDIP
μPC1876	dbx NR, SAP Decoder	• 42-pin SSOP
μPC1854A	I²C Bus, dbx NR, SAP Decoder, Normal Output	• 28-pin SDIP • 28-pin SOP

■ US MTS (TV) System Block Diagram



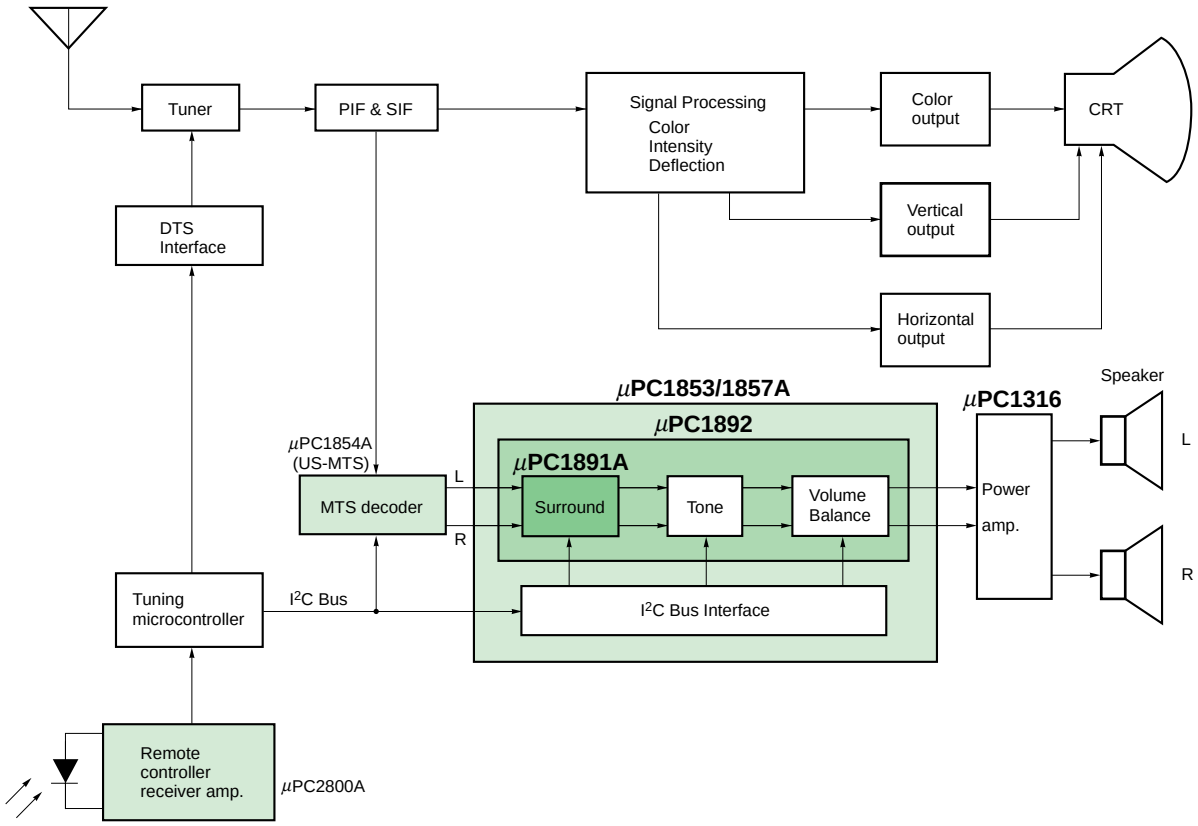
Remark DTS: Digital Tuning System
MTS: Multichannel Television Sound
PIF: Picture Intermediate Frequency
SIF: Sound Intermediate Frequency

■ Matrix Surround

Part number	Function	Package
μPC1891A	Matrix Surround (Phase Shift)	• 20-pin DIP
μPC1892	Matrix Surround (Phase Shift), Tone/Balance/Volume, L + R output	• 30-pin SDIP
μPC1853	I²C bus, BASS boost, Matrix Surround, Tone/Balance/Volume, Audio output, L + R output	• 30-pin SDIP
μPC1857A	I²C bus, Matrix Surround, Tone/Balance/Volume, Mixer input	• 30-pin SDIP

TV IC

■ Matrix Surround (TV) System Block Diagram



Remark DTS: Digital Tuning System
MTS: Multichannel Television Sound
PIF: Picture Intermediate Frequency
SIF: Sound Intermediate Frequency

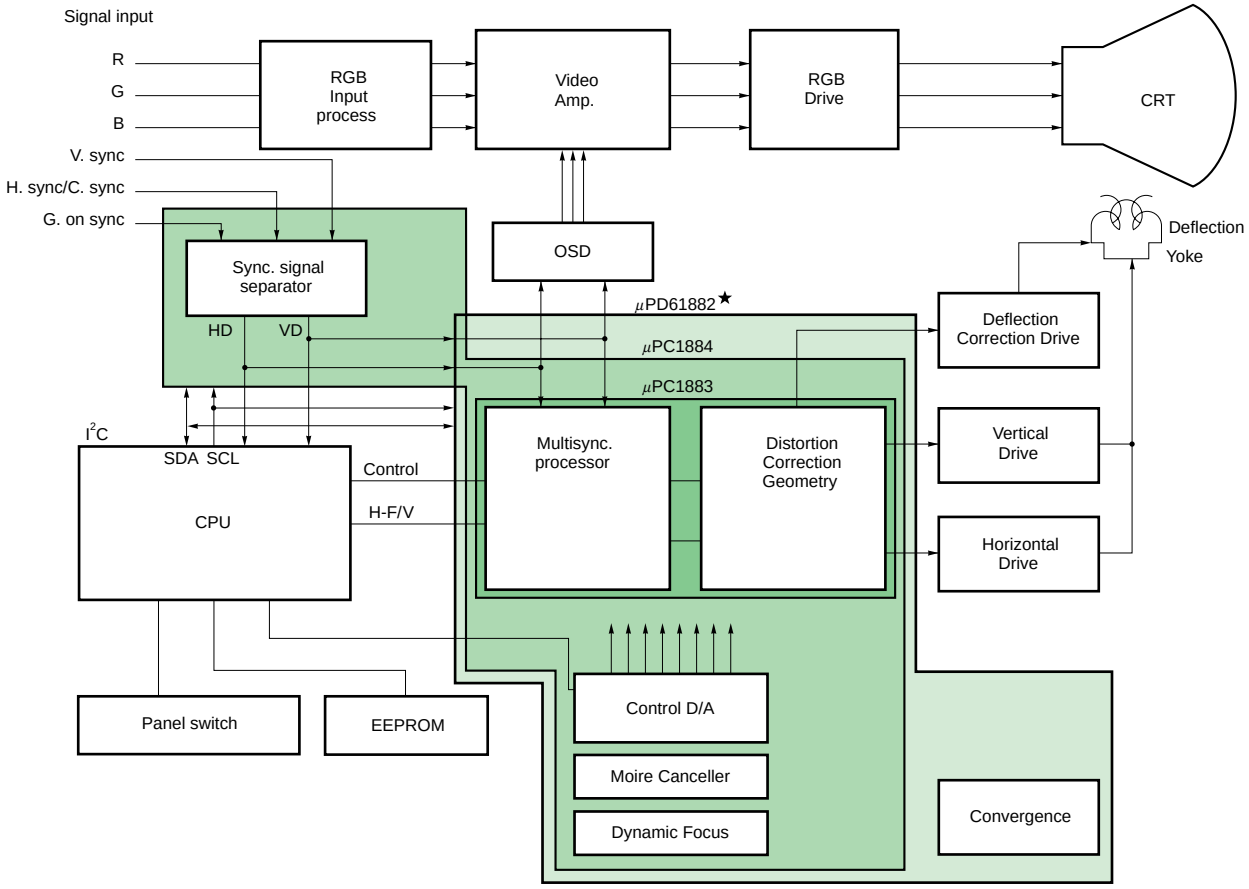
■ Digital Satellite Broadcasting

Part number	Function	Package
μPD61511	A/D Converter, QPSK demodulation, Viterbi decoder, Reed-Solomon error correction	• 100-pin QFP
μPC2781	AGC amplifier + Quadrature demodulator	• 20-pin SSOP
μPC2782	L band frequency down converter + IF-AGC amplifier	• 20-pin SSOP
μPC2795	L band frequency down converter	• 8-pin SSOP (175 mil)
μPC3205	AGC amplifier + Quadrature demodulator	• 20-pin SSOP

■ Digital CATV

Part number	Function	Package
μPC2798GR	QAM IF down converter (AGC amplifier + MIX/OSC + Video amplifier)	• 20-pin SSOP
μPC3206GR	50 dB AGC amplifier + Video amplifier	• 20-pin SSOP
μPC3211GR	AGC amplifier for Return-pass	• 20-pin SSOP

Multisync Monitor IC



★: Under development

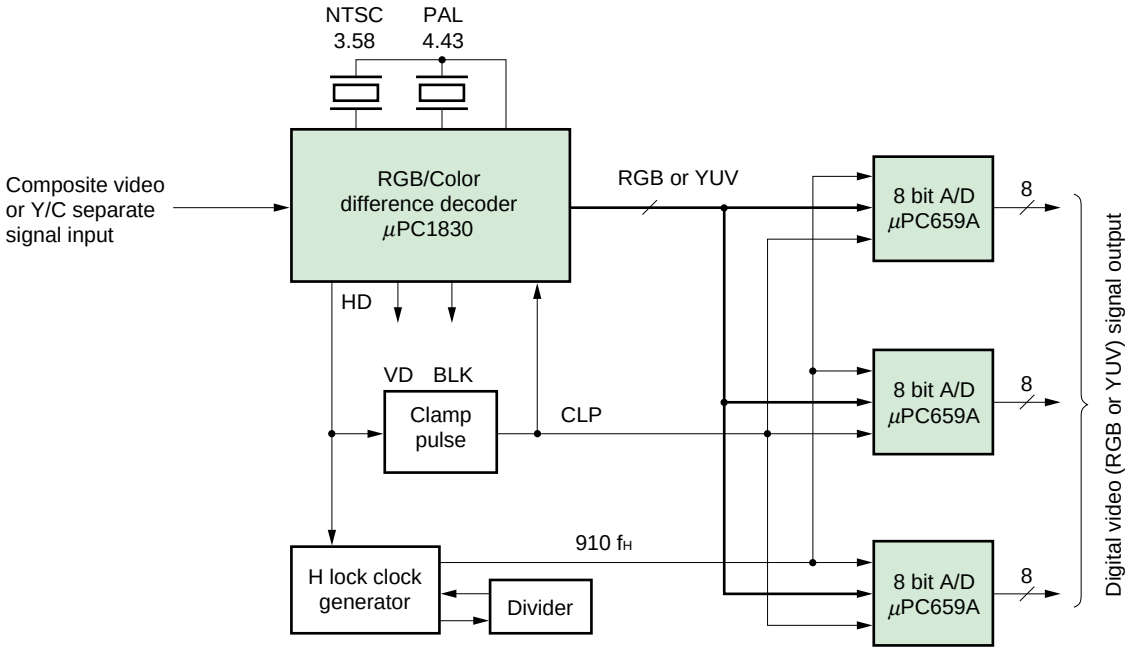
■ Multisync monitor IC

Part number	Function	Package
μPC1883	Horizontal Vertical sync. processor, Picture geometry processor	• 30-pin SDIP
μPC1884	Horizontal Vertical sync. processor, Picture geometry processor (I^2C bus)	• 30-pin SDIP
μPD61882★	Horizontal Vertical sync. processor, Picture geometry processor (I^2C bus)	• 64-pin QFP

★: Under development

Digital Image IC

■ Video capture system block diagram



■ Analog Frontend IC

Part number	Function	Package
μPC1830	5 V operation chroma decoder NTSC/PAL, 525/625 line, 50/60 Hz selectable, Matrix circuit	• 42-pin SOP

On-Screen Character Display IC

■ Video type

Function \ Part number	μPD6464A	μPD6465
Number of character types	128 (ROM)	256 (ROM)
Number of display characters	288 (12 lines × 24 digits)	
Character configuration	12 × 18	
Character color	Single (white, the luminance level of a character can be set in 2 stages.)	
Character size	1 dot/1H, 2H (per line)	
Internal video signal color	White/Black/Blue/Green	
Background	None/Fringed/Squared/Solid (per field)	
Applicable video signal mode	NTSC/PAL/PAL-M/SECAM	
Supply Voltage	4.5 to 5.5 V	
Package	• 24-pin SDIP (300 mil) • 24-pin SOP (375 mil)	

• μPD6464A adapts PAL-N.

■ RGB type

Function \ Part number	μPD6461	μPD6462	μPD6466
Number of character types	256 (ROM)	128 (ROM)	512 (ROM)
Number of display characters	288 (12 lines × 24 digits)		
Character configuration	12 × 18		
Character color	8 colors		
Character size (per line)	1 dot/1H, 2H		1 dot/1H, 2H, 3H, 4H
Character color reverse (per character)	Black character/No friged		Black character/No friged White character/Friged
Background (per frame)	None Square (8 colors) Solid (8 colors)		
Fringed (per frame)	Two colors (Black, White)		
Supply voltage	2.7 to 5.5 V		
Package	• 20-pin SSOP (300 mil) • 24-pin SOP (375 mil)	• 20-pin SSOP (300 mil)	• 20-pin SSOP (300 mil) • 24-pin SOP (375 mil)

• Making ideally suited for camcoders (μPD6461, μPD6462)
• μPD6461 and μPD6462 are commands conpertible and pins conpertible for 20-pin SSOP package.
• μPD6466 has a blue background and character mirror reverse function.

Video Camera IC

■ CCD driver circuit

Part number	Function	Package
μPD16501	Area sensor vertical drive interface	• 16-pin SOP (300 mil)
μPD16502	Area sensor vertical drive interface	• 20-pin SOP (300 mil)
μPD16510	Area sensor vertical, VOD shutter driver, for low voltage logic	• 22-pin SSOP (225 mil)
μPD16520★	Area sensor vertical driver 10ch (3 state: 6, 2 state: 4), VOD shutter driver, for low voltage logic	• 38-pin SSOP (300 mil)

★: Under development

Remote Control IC

■ Infrared remote control preamplifiers IC

Part number	Supply voltage	Features	Package	Remarks
μPC2800A	5 V ± 10%	Internal high trap circuit	• 8-pin SOP	Active low type

■ Infrared remote control IC

Part number	**μPD6121G-001**	**μPD6122G-001**
Operating voltage range	2.0 to 3.3 V	
Operation clock (f_{OSC})	400 to 500 kHz ceramic oscillator	
Transmission format	Leader | Custom code 16-bit | Data code 8-bit | Data code 8-bit	
Modulation mode	P.P.M. 0 ····⎍⎍⎍⎍⎍⎍ 1···⎍⎍⎍⎍⎍⎍ 38 kHz • carrier modulation(f_{OSC} = 455 kHz)	
Custom code	16-bit set*	
Data code	32 × 2	64 × 2
Number of keys	32	64
Package	• 20-pin SOP	• 24-pin SOP

*: All codes except code “00FFH” are managed by NEC.

Clock IC

Part number	I/O	Alarm	Supply voltage (V)	Circuit Current (μA)	Package
μPD4990A	Serial	×	2.0 to 5.5	20	• 14-pin DIP • 16-pin SOP
μPD4991A	4-bit · Parallel	○		14	• 18-pin DIP • 20-pin SOP
μPD4992	8-bit · Parallel	×	2.4 to 5.5	6	• 20-pin DIP • 20-pin SOP

Rotary Encoder IC

Part number	Axis	Counter-bit	Supply voltage (V)	Package
μPD4701A	2	12	5.0	• 24-pin DIP • 24-pin SOP
μPD4702	1	8		• 20-pin DIP • 20-pin SOP
μPD4704	1 (Expansion for μPD4702)			

A/D Converter IC

■ A/D Converter for Video processing

Part number	Resolution (bits)	Linearity (%FSR)	Sampling time (MSPS)	Supply voltage (V)	Data format	Output format	Package	Remarks
μPC659A	8	0.2	20	+5	Binary	8-bit parallel	• 24-pin SOP	Parallel approximation Video processing

D/A Converter IC

■ D/A Converter

Part number	Resolution (bits)	Linearity (%FSR)	Settling time (μs)	Supply voltage (V)	Input format	Output format	Package	Remarks
μPD6325 **μPD6335**	6 (4 circuits)	Monotonicity	200	+5 to +15 +5	Binary	Voltage output	• 16-pin DIP • 16-pin SOP	Serial input, CMOS
μPD6326 **μPD6336**	6 (8 circuits)	Monotonicity	200	+5 to +15 +5	Binary	Voltage output	• 16-pin DIP	Serial input, CMOS
μPD6221	8 (8 circuits)	0.31	100	+5	Binary	Voltage output	• 20-pin DIP • 20-pin SOP	I^2C-bus compatible Internal Output CMOS' Buffer
μPD6222	8 (12 circuits)	0.31	100	+5	Binary	Voltage output	• 24-pin SDIP • 24-pin SOP	I^2C-bus compatible Internal Output CMOS' Buffer

■ D/A Converter for Video processing

Part number	Resolution (bits)	Linearity (%FSR)	Sampling rate (MSPS)	Supply voltage (V)	Input format	Output format	Package	Remarks
μPC667	10	0.1	60	+5	Binary	Voltage output	• 30-pin SDIP	Parallel input Video processing
μPC662	8 (3 circuits)	0.2	35	+5	Binary	Voltage output	• 48-pin QFP	Parallel input, 3 circuits Video processing
μPC664	8 (2 circuits)	0.2	35	+5	Binary	Voltage output	• 36-pin SOP	Parallel input, 2 circuits Video processing
μPC665	8	0.2	35	+5	Binary	Voltage output	• 16-pin SOP	Parallel input Video processing
μPC666	6 (3 circuits)	0.8	35	+5	Binary	Voltage output	• 36-pin SOP	Parallel input, 3 circuits Video processing

Line Driver Receiver IC

Part number	Driver	Receiver	Capacitor (μF)	Supply voltage (V)	Package
μPD4711B	2	2	1.0 to 4.7 (4 pcs)	5.0	• 20-pin DIP/SOP
μPD4712C	4	4	4.7 to 47 (4 pcs)		• 28-pin DIP/SOP
μPD4712D	4	4			• 28-pin DIP/SOP
μPD4713A	3	3			• 24-pin DIP/SOP
μPD4714A	3	5			• 28-pin DIP/SOP
μPD4715A	5	3			• 28-pin DIP/SOP
μPD4721	2	2	0.33 to 4.7 (4 or 5 pcs)	3.3/5.0	• 20-pin SSOP
μPD4722	4	4	0.47 to 4.7 (4 or 5 pcs)		• 30-pin SSOP
μPD4723	3	3	0.33 to 4.7 (4 or 5 pcs)		• 30-pin SSOP
μPD4724	3	5			• 30-pin SSOP
μPD4726	4	7	1.0 to 4.7 (4 pcs)	5.0	• 36-pin SSOP

General Purpose Linear IC

Operational Amplifier 180

Comparator 181

Fixed Output Voltage, 3-Terminal Regulator 181

Variable Output Voltage Regulator 182

Regulator with System Reset 182

Regulator with ON/OFF Function 183

High Precision Reference Voltage 183

Switching Regulator Control Circuit 183

Functional Block 184

Operational Amplifier

Part number		Function*1			Recommended power supply voltage*2 (V)	Common mode input voltage range TYP. (V) T_A = +25 °C	Input stage transistor	GND Input/ Output on single power	Low V_{IO} Low T_A drift	Low input bias current	High speed	Wide band	Low noise	Low power	Packages*3			Number of pins
Communication/industry use	General use	Single	Dual	Quad											C	G2	HA	
μPC151	μPC741	○			±7.5 to ±16	V^-+2 to V^+–0.5	NPN		○						○	○		8
μPC251	μPC1458		○												○	○		8
μPC354		○			±3 to ±16	V^-+1 to V^+–1	NPN		◎	◎			○					8
μPC454			○															14
μPC815		○			±3 to ±20	V^-+1.5 to V^+–2.5	NPN		◎	○			◎		○			8
μPC816		○			±3 to ±20	V^-+1.5 to V^+–2.5	NPN		◎	○	○	◎	◎		○			8
μPC802	μPC4250	○			±1 to ±16	V^-+0.2 to V^+–0.6	PNP		○	○				◎	○	○		8
μPC801	μPC4081	○			±5 to ±16	V^-+2.3 to V^+	J–FET			◎	○	○			○	○		8
μPC803	μPC4082		○												○	○		8
μPC804	μPC4084			○											○			14
μPC811		○			±5 to ±16	V^-+3 to V^+–1	J–FET		○	◎	○	○	○		○	○		8
μPC812			○												○	○		8
μPC813		○			±5 to ±16	V^-+3 to V^+–1	J–FET		○	◎	◎	○	○		○	○		8
μPC814			○												○	○		8
μPC821	μPC4071	○			±5 to ±16	V^-+2 to V^+	J–FET			◎	○	○	○		○	○		8
μPC822	μPC4072		○												○	○	○	8*4
μPC824	μPC4074			○											○	○		14
μPC831	μPC4061	○			±2 to ±16	V^-+2 to V^+	J–FET			◎				○	○	○		8
μPC832	μPC4062		○												○	○		8
μPC834	μPC4064			○											○	○		14
μPC258	μPC4558		○		±4 to ±16	V^-+1 to V^+–1	PNP		○			○	○		○	○		8
μPC458	μPC4741			○											○	○		14
μPC259	μPC4560		○		±4 to ±16	V^-+1 to V^+–1	PNP		○			○	○		○	○		8
	μPC4556		○		±4 to ±16	V^-+1 to V^+–1	PNP		○		○	◎	○		○	○		8
	μPC4557		○		±4 to ±16	V^-+1 to V^+–1	PNP		○			○	○		○			8
	μPC4559		○		±4 to ±16	V^-+1 to V^+–1	PNP		○			○	○		○			8
	μPC4570		○		±4 to ±16	V^-+1 to V^+–1	PNP		○		○	◎	◎		○	○	○	8*4
	μPC4574			○											○	○		14
	μPC4572		○		±2 to ±7	V^-+1 to V^+–1	PNP		○		○	◎	◎		○	○	○	8*4
μPC1251	μPC358		○		+3 to ±30	GND to V^+–1.5	PNP	○		○				○	○	○	○	8*4
μPC451	μPC324			○											○	○		14
μPC452	μPC3403			○	+3 to ±32	GND to V^+–1.5	PNP	○		○					○	○		14
μPC842			○		±3 to ±32	GND to V^+–1.8	PNP	○			○	○			○	○		8
μPC844				○											○	○		14

○: Recommended for designing ◎: High performance ◎: Very high performance

*1 : Single type has a offset adjust pin.

*2 : For product indicated by +/– voltage, single supply voltage operation is possible if the input/output voltage range is observed.

*3 : C, G2 or HA shows Package Type, as follows.
C : Plastic DIP (300 mil)
G2 : Plastic SOP (225 mil)
HA : 9-pin plastic slim SIP

*4 : HA has 9 pins.

Comparator

Part number		Function			Recommended power supply voltage (V)	Common mode input voltage range TYP. (V) T_A = +25 °C	Input stage transistor	GND Input on single power	High speed	Low power	Output circuit type		Packages*1			Number of pins
Communication/ industry use	General use	Single	Dual	Quad							Open-collector	Emitter-follower	C	G2	HA	
μPC271	**μPC311**	○			+5 to +32	V^-+0.3 to V^+–1.2	PNP		○		○	○	○	○		8
μPC272	**μPC319**		○		+5 to +32	V^-+2 to V^+–2	NPN		○		○		○	○		14
μPC277	**μPC393**		○		+2 to +32	GND to V^+–1.5	PNP	○		○	○		○	○	○	8*2
μPC177	**μPC339**			○									○	○		14

*1: C, G2 or HA shows Package Type, as follows.
C : Plastic DIP (300 mil)
G2 : Plastic SOP (225 mil)
HA : 9-pin plastic slim SIP
*2: HA has 9 pins.

Fixed Output Voltage, 3-Terminal Regulator

Type	Part number	Output current (A)	Output voltage (V)													Absolute maximum ratings		Package	Remarks
			3	3.3	4	5	6	7	8	9	9.3	10	12	15	18	24	Input voltage (V)	Total power dissipation (W)*1		
Positive voltage output	**μPC78L00**	0.1				○	○	○	○								30	0.7/2*3	• TO-92 • SOT-89	
												○	○	○			35			
	μPC78N00	0.3				○			○			○	○	○	○		35	12.5	• TO-126	
																○	40			
	μPC78M00A	0.5				○	○	○	○	○		○	○	○	○		35	15	• MP-45*2	Improved version of μPC78M00
																○	40			
	μPC7800A	1.0				○			○		○		○	○	○		35	15	• MP-45*2	Improved version of μPC7800
																○	40			
	μPC29L00	0.1	○	○	○	○											16	0.7/2*3	• TO-92 • SOT-89	Low dropout voltage type
	μPC29M00	0.5	○	○		○	○	○	○	○		○	○				20	1.0/2.0*4 15	• MP-3 • MP-3Z*5 • MP-45*2	Low dropout voltage type
	μPC2900	1.0	○	○		○	○	○	○	○		○	○				20	1.0/2.0*4 15	• MP-3 • MP-3Z*5 • MP-45*2	Low dropout voltage type
	μPC24A00	2.0				○							○	○			36	20	• MP-45*2	Low dropout voltage type
	μPC24M00A	0.5				○	○	○	○	○		○	○	○	○		36	15	• MP-45*2	Low dropout voltage type
	μPC2400A	1.0				○	○	○	○	○		○	○	○	○		36	15	• MP-45*2	Low dropout voltage type
Negative voltage output	**μPC79L00**	0.1				○			○								–30	0.7	• TO-92	
													○	○			–35			
	μPC79N00	0.3				○			○				○	○	○		–35	12.5	• TO-126	
																○	–40			
	μPC79M00	0.5				○			○				○	○	○		–35	15	•MP-45*2	
																○	–40			
	μPC7900A	1.0				○			○				○	○	○		–35	15	• MP-45*2	Improved version of μPC7900
																○	–40			

*1: Limited by internal circuit
*2: TO-220 Plastic insulated package
*3: With 16 $cm^2 \times 0.7$ mm ceramic substrate
*4: With 7.5 $cm^2 \times 0.7$ mm ceramic substrate
*5: Surface mount package of MP-3

Variable Output Voltage Regulator

Type	Part number	Output current (A)	Output voltage range (V)	Absolute maximum ratings		Package	Remarks
				Input voltage (V)	Total power dissipation (W)		
Positive voltage output	**μPC141/305**∗3	0.05	4.5 to 30	40	0.35, 0.5/0.35, 0.44	• 8-pin DIP (plastic/ceramic) • 8-pin SOP	
	μPC317	1.5	1.3 to 30	40	15∗1	• MP-45∗2	3-pin regulator
	μPC1093	0.15	2.5 to 36	37	0.48, 0.7, 2∗4, 0.2	• 8-pin SOP • TO-92 • SOT-89 • 5-pin mini mold	Shunt regulator
	μPC1943	0.05	1.26 to 24	25	1.6∗4	• SOT-89	Shunt regulator for Low Voltage
	μPC1944	0.05	1.26 to 24	25	0.385, 0.56, 1.6	• 8-pin SOP • TO-92 • SOT-89	Shunt regulator for Low Voltage
Negative voltage output	**μPC337**	1.5	–1.3 to –30	–40	15∗1	• MP-45∗2	3-pin regulator

∗1 : Limited by internal circuit
∗2 : TO-220 Plastic insulated package
∗3 : μPC141 is for communication/industry use.
∗4 : When mounted on 16 cm^2 (0.7 mm thick) ceramic board

Regulator with System Reset

Type	Part number	Output current (A)	Output voltage (V)	Reset start voltage (V)	Reset output logic		Absolute maximum ratings		Package	Remarks
					Active low	Active high	Input voltage (V)	Total power dissipation (W)		
Positive voltage output	**μPC2251**	0.1	3	2.85	◯		12	1.2∗	• TO-126 (4-pin)	Low dropout type
	μPC2252	0.1	3	2.85		◯	12	1.2∗	• TO-126 (4-pin)	Low dropout type
	μPC2253	0.1	5	2.85	◯		12	1.2∗	• TO-126 (4-pin)	Low dropout type
	μPC2254	0.1	5	2.85		◯	12	1.2∗	• TO-126 (4-pin)	Low dropout type
	μPC2255	0.1	5	4.75	◯		12	1.2∗	• TO-126 (4-pin)	Low dropout type
	μPC2256	0.1	5	4.75		◯	12	1.2∗	• TO-126 (4-pin)	Low dropout type
	μPC2260	0.5	5	4.85	◯		35	20∗	• TO-220 (5-pin)	Low dropout type
Supervisory for Microprocessor	**μPC2270A**	–	–	4.3	◯	◯	8	0.35	• 8-pin DIP	Manual Reset Input
								0.44	• 8-pin SOP	
								0.35	• 9-pin Slim SIP	
	μPC1074A	0.01	2 ~ 5.18	Adjustable	◯		40	0.5	•16-pin SOP	Watch-dog Timer

∗: Limited by internal circuit

Regulator with ON/OFF Function

Part number	Output current (A)	Output voltage (V)	ON/OFF voltage (V)		Absolute maximum ratings		Package
			Output ON	Output OFF	Input voltage (V)	Total power dissipation (W)	
μPC29S78	0.1	7.8	2.0	0.8	20	1.2	• TO-126 (4-pin)
μPC29S10	0.1	10	2.0	0.8	20	1.2, 0.48	• TO-126 (4-pin) • 8-pin SOP

High Precision Reference Voltage

Part number	Input voltage range (V)	Output voltage (V)	Output current (mA)	Total power dissipation (mW)	Output voltage vs. temperature (ppm/°C)	Package
μPC1060	4.5 to 40	2.5 ±0.025	10	350, 500	40	• 8-pin DIP

Switching Regulator Control Circuit

Part number	Input voltage range (V)	Absolute maximum ratings		Package	Output circuit operation mode	Applications
		Output current (mA)	Total power dissipation (W)			
μPC494	7 to 40	250	1, 0.78*2, 0.65*2	• 16-pin DIP • 16-pin SOP*1	Push-pull/single selectable	General purpose
μPC1094	11 to 24	1200 (peak)	0.57, 0.55	• 14-pin DIP • 14-pin SOP	Totem pole circuit configuration Single mode	Can operate up to 500 kHz General purpose
μPC1099	11.5 to 24	1200 (peak)	1, 0.694	• 16-pin DIP • 16-pin SOP	Totem pole circuit configuration Single mode	Can operate up to 500 kHz General purpose
μPC1905	12 to 30	1200 (peak)	1, 0.694	• 16-pin DIP • 16-pin SOP	Totem pole circuit configuration Single mode	Can operate up to 500 kHz General purpose
μPC1906	12 to 30	1200 (peak)	1, 0.694	• 16-pin DIP • 16-pin SOP	Totem pole circuit configuration Single mode	Can operate up to 500 kHz General purpose
μPC1900	12 to 30	1200 (peak)	1.225, 0.775	• 24-pin DIP • 24-pin SOP	Totem pole circuit configuration 2 outputs	Can operate up to 500 kHz General purpose
μPC1100	3.6 to 40	25	1, 0.694	• 16-pin DIP • 16-pin SOP	2 outputs (synchronous control possible) If one output is shorted, both outputs will be turned OFF.	DC/DC converter
μPC1150	3.6 to 40	25	1, 0.694	• 16-pin DIP • 16-pin SOP	2 outputs (synchronous control possible) If one output is shorted, only the output will be turned OFF.	DC/DC converter
μPC1933★	2.5 to 20	20	0.4	• 8-pin SOP	Open drain 1 output	DC/DC converter
μPC1934★	2.5 to 20	20	0.4	• 16-pin SSOP • 16-pin T-SSOP	Open drain 2 outputs	DC/DC converter
μPC1935	2.5 to 20	20	0.4	• 16-pin T-SSOP	Open drain 3 outputs	DC/DC converter

*1 : μPC494G is 375 mil. μPC494GS is 300 mil.

*2 : When mounted on 5 × 5 cm² (1.6 mm thick) glass epoxy board.

★: Under development

Functional Block

Function	Part number	Features	Package
Analog Multiplexer	**μPD5205**	Single-pole 8 position mode/double-pole 4 position mode Supply Voltage: 44 V, ON Resistance: 270 Ω TYP.	• 24-pin shrink DIP • 24-pin SOP
Precision Timer	**μPC1555**	CR Timer, Operating Temperature: –20 ~ +80°C Supply Voltage: 4.5 ~ 16 V, Free Running Frequency: 0.1 ~ 100 kHz	• 8-pin DIP • 8-pin SOP
	μPD5555	CMOS CR Timer, CMOS Type of μPC1555 Supply Voltage: 3 ~ 16 V, Free Running Frequency: 0.1 ~ 500 kHz	• 8-pin DIP • 8-pin SOP
	μPD5556	CMOS CR Timer Dual Type of μPD5555	• 14-pin DIP • 14-pin SOP

Transistor/Diode/Thyristor

Transistor 186

- Quick Reference by Package 186
- Quick Reference Table by Function/Application 196

Field Effect Transistor 202

- Small Signal FET 202
- Power MOS FET 205

Transistor with Internal Resistor 215

Transistor for Array 220

Transistor Array 221

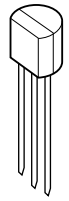
Zener Diode 224

Noise Clipping Diode 226

Thyristor 227

Quick Reference by Package

■ TO-92 Type Transistor


TO–92

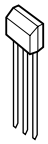
V_{CEO} (V) / I_C (A)	~15	~30	~50	~70	~100	~150	~200	~250	~400
~20 m		2SC1674							
~50 m	2SA1206**					2SA988 2SA992 2SC1841 2SC1845			
~100 m			2SA733 2SA987 2SA990 2SC945 2SC1840 2SC1842 2SC1843 AA1[ ] AN1[ ]		2SA675[1)]		2SA1376 2SA1376A 2SC3478 2SC3478A	2SA1544	
~200 m	2SC2901**		2SC3622* (150 mA) 2SC3622A* (150 mA)						
~500 m			2SC3615* (300 mA)	2SA953 2SC2002	2SA954 2SC2003				2SA1625
~1.0		2SA952 2SC2001 2SC3616* (25 V/ 700 mA) AB1[ ] (25 V/ 700 mA) AP1[ ] (25 V/ 700 mA)	2SB1116 2SD1616	AD1[ ] AD2[ ]*** AR1[ ] 2SB1116A (60 V) 2SD1616A (60 V)*** 2SD1701 (60 V)	2SD1698 (80 V)				
~2.0		2SD1513 (16 V/2.0 A) 2SB1068 (16 V/2.0 A) AQ1[ ] (20 V/2.0 A)							
~3.0		2SB1300 (16 V/3 A)							

1) V_{CES} ☐ : Darlington transistor, ∗: High h_{FE} transistor, ∗∗: High speed switching, ∗∗∗: Contains internal zener diode

Quick Reference by Package

■ SST Type Transistor

V_{CEO} (V) / I_C (A)	~15	~20	~30	~50	~120
~20 m	2SC2786		2SC2787		
~50 m	2SA1459**				2SC2784
~100 m				2SA1175 2SC2785 2SC3623* BA1[ ] BN1[ ]	
~200 m	2SC3732**			2SA1458** (40 V) 2SC3623A* (150 mA) 2SC3731** (40 V)	
~700 m		2SB810 2SD1020	BB1[ ] (25 V) BP1[ ] (25 V)		

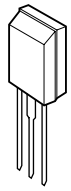


SST

*: High h_{FE} transistor, **: High speed switching

Quick Reference by Package

■ SP-8 Type Transistor



SP–8

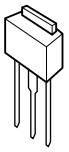
V_{CEO} (V) / I_C (A)	~16	~25	~50	~80	~140	~160	~300	~400	~600
~50 m					2SA915 2SC1940	2SA916 2SC1941			
~200 m							2SC3209 2SC4000 (250 V/ 100 mA)		
~500 m					2SA1221 2SC2958	2SA1222 2SC2959			
~700 m			2SB605 2SD571 2SC2721						
~1.0		2SB564 2SD471	2SA1460** 2SB734 2SC3733** 2SD774 2SD1582*	2SB984 2SB1093 2SD1312 2SD1579 2SD1697 (800 mA) 2SD1700*** (60 V/ 800 mA) 2SD1843*** (60 V)					2SA1627
~2.0	2SB733 2SD773	2SD1581*	2SD2463	2SD1779* (60 V) 2SD1780* (60 V) CE1[ ]*** (60 V) CE2[ ]* (60 V)				2SA1626	
~3.0		2SB1117			2SB1318 (100 V)				
~5.0		2SA1897							

□: Darlington transistor, *: High h_{FE} transistor, **: High speed switching, ***: Internal zener diode

Quick Reference by Package

■ MP-3 Type Transistor

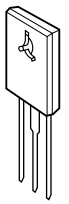
V_{CEO} (V) / $I_{C(DC)}$ (A)	~20	~40	~60	~100	~150	~300	~400	~600
~0.5							2SA1400 2SC3588	
~1.0			2SB963 2SD1286					2SA1413 2SC3632
~2.0	2SD1583	2SD992	2SD1164		2SB768 2SD1033	2SC2885 2SC2946	2SA1412 2SC3631	
~3.0		2SB962 ◎	2SD1584 2SB1261 ◎ 2SD1899 ◎					
~5.0			2SA1385 ◎ 2SC3518 ◎ 2SA1648 ◎	2SA1647 ◎ 2SC4331 ◎			2SC4346	
~10	2SA1615 ◎	2SA1649 ◎						


MP–3

▭ : Darlington transistor, (: Complementary pair, ○: Single High h_{FE}, ◎: Low V_{CE}(sat)

■ MP-5 (TO-126) Type Transistor

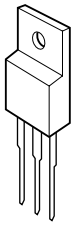
V_{CEO} (V) / $I_{C(DC)}$ (A)	~45	~60	~80	~120	~160	~300	~400	~600
~1.0						2SA1546 2SC4001		
~0.5						2SC2688	2SA1156 2SC2752	
~1.0		2SD1630 *		2SD415				2SA1486 2SC3840
~2.0	2SD1695 *	2SB794 2SD985	2SB795 2SD986	2SC2690	2SC2690A			
~3.0	2SD794 2SB772 ◎ 2SD882 ◎	2SD794A 2SB1150 * 2SD1693 * 2SD1694 ○ 2SB1217 ◎ 2SD1818 ◎		2SB1149 2SD1692 2SC4342				
~5.0		2SB1151 ◎ 2SD1691 ◎						


MP–5
(TO–126)

▭ : Darlington transistor, (: Complementary pair, ∗: Internal zener diode between C-B, ◎: Low $V_{CE(sat)}$,
○: Single Hight h_{FE}

Quick Reference by Package

■ MP-25 (TO-220) Type Transistor

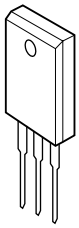


MP–25
(TO–220)

V_{CEO} (V) / $I_{C(DC)}$ (A)	~40	~60	~80	~100	~150	~300	~400
~0.5						2SC1505	
~2.0				2SA1008 2SC2331	2SB546A 2SD401A	2SA1009	2SA1009A 2SC2333
~5.0		2SA1069	2SA1069A	2SB601 2SD560 2SC2517			2SC2518
~7.0	2SA1129	2SB707	2SB708	2SA1010 2SC2334 2SA1645 ◎			2SC2335
~7.0				2SA1646 ◎			

▭ : Darlington transistor, (: Complementary pair, ∗: Internal zener diode between C-B, ◎ : Low $V_{CE(sat)}$

■ MP-40 Type Transistor

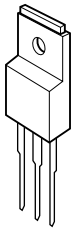


MP–40

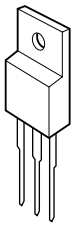
V_{CEO} (V) / $I_{C(DC)}$ (A)	~60	~100	~400
~5.0	2SD1392∗	2SD1308	
~7.0	2SC4305		2SC3158
~8.0		2SD1309	
~10		2SA1261 2SC3157 2SC4350	2SC3159

▭ : Darlington transistor, (: Complementary pair, ∗: Internal zener diode between C-B

Quick Reference by Package



MP–45
(MP-25 Insulated type)



MP–45F
(MP-25 Insulated type)

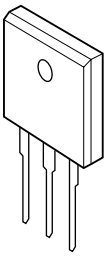
■ MP-45, MP-45F (TO-220 Insulated Type) Transistor

V_{CEO} (V) / Package / $I_{C(DC)}$ (A)	~60		~100		~300	~400
	MP-45	MP-45F	MP-45	MP-45F	MP-45	MP-45
~2.0			2SA1395 2SC3567			2SC3569
~3.0	2SB1094 2SD1585 2SD1593○	2SB1453 2SD2164○				
~5.0	2SA1394 2SA1441◎ 2SC3691◎ 2SC4351*	2SA1741◎ 2SC4549◎	2SB1098 2SD1589 2SA1650◎	2SB1430 2SD2161	2SD1592	2SC3570
~6.0			2SD1594○	2SD2165○		
~7.0	2SB1097 2SD1588 2SA1442◎ 2SC3692◎	2SA1742◎ 2SC4550◎	2SC4062○	2SC4553○		2SC3571
~8.0			2SB1099 2SD1590	2SB1431 2SD2162		
~10	2SA1443◎ 2SC3693◎	2SA1743◎ 2SC4551◎	2SA1396 2SC3568 2SA1652◎ 2SC4336◎ 2SA1720 2SB1100 2SD1591	2SB1432 2SD2163		2SC3572
~15	2SA1444◎ 2SC3694	2SA1744◎ 2SC4552◎	2SC4063○	2SC4554○		

▭ : Darlington transistor, (: Complementary pair, * : Internal zener diode between C-B, ◎ : Low $V_{CE(sat)}$,
○ : Single High h_{FE}

Quick Reference by Package

■ MP-80 Type Transistor

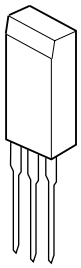


MP–80

$I_{C(DC)}$ (A) \ V_{CEO} (V)	~100	~400
~5.0	2SC2908	
~1.0	2SB897 2SD1210	2SC2749
~15	2SC2750 2SD1296	2SC2751

□: Darlington transistor, ∗: MP-88

■ MP-10 Type Transistor



MP–10

$I_{C(DC)}$ (A) \ V_{CEO} (V)	~60	~100
2.5		2SC4814 ○
~5	(2SA1843 ◎ 2SC4815 ◎	(2SA1840 2SC4810 2SA1845 ◎
~7		2SC4813 ○
~8		(2SA1841 2SC4811
~10		2SA1847 ◎

□: Darlington transistor, (: Complementary pair, ○: Low V_{CE}(sat) High h_{FE}, ◎: Low V_{CE}(sat)

Quick Reference by Package



Mini mold
SC-59

■ Mini-Mold Transistor

V_{CEO} (V) / I_C (mA)	~15	~20	~30	~40	~50	~60	~80	~120	~200	~300
~20		2SC2223								2SD2383
~30				2SA1226						
~50	2SA1462		2SC1009A					2SA811A 2SA1247 2SC1622A 2SC3115	2SC1653 (130 V) 2SC1654 (160 V)	
~100					2SA812 2SC1623 FA1[ ] FN1[ ]				2SA1330 2SC3360	
~150					2SC3624 2SC3624A					
~200	2SC3735	2SC1621		2SA1461 2SC3734						
~300						2SB736 2SD780	2SB736A 2SD780A			
~500				2SA1464 2SC3739						
~700			2SB624 (–25 V) 2SD596 (25 V) FB1[ ] (25 V) FP1[ ] (–25 V)							

Quick Reference by Package



Power mini mold
SC-62

■ Power Mini-Mold Transistor

V_{CEO} (V) / I_C (A)	~20	~25	~50	~60	~80	~100	~120	~140	~300
~50 m								2SA1173 2SC2780	
~200 m									2SC3554
~300 m			2SC3617		2SB800 2SD1001				
~0.7		2SC3618	2SB799 2SD1000			2SB805 2SD1006	2SB806 2SD1007		
~1.0		2SB798 2SD999	2SA1463 2SB1115 2SC3736 2SD1615 2SD1702	2SB1115A 2SD1615A HD1[ ] HD2[ ] HR1[ ]	2SB804 2SD1005 2SD1699				
~2.0	2SB1114 2SD1614 HQ1[ ]	2SD1950							
~3.0	2SB1628			2SB1572 2SD2403					
~5.0			2SB1571 2SD2402						

: Darlington connection

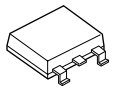


MP-2
SC-84

■ MP-2 Type Transistor

V_{CEO} (V) / I_C (A)	~60	~600
~1.0		2SA1871 2SC4942
~5.0	2SB1578 2SD2425	

Quick Reference by Package



MP-3

■ MP-3 Type (SC-63) Transistor

V_{CEO} (V) / I_C (A)	~20	~30	~45	~60	~200	~400	~600
~0.5						2SA1400-Z 2SC3588-Z	
~1.0				2SB963-Z 2SD1286-Z			2SA1413-Z 2SC3632-Z
~2.0	2SD1583-Z			2SD1164-Z (1.5 A)	2SB768 (150 V) 2SD1033 (150 V) 2SC2946(1)	2SA1412-Z 2SC3631-Z	
~3.0		2SB962-Z		2SB1261-Z 2SD1584-Z 2SD1899-Z			
~5.0				2SA1385-Z 2SC3518-Z			
~10	2SA1615-Z						

: Darlington connection

Small mini mold
SC-70

■ Small Mini-Mold Transistor

V_{CEO} (V) / I_C (mA)	~15	~20	~30	~40	~50	~120
~20		2SC4178				
~50	2SA1610		2SC4179			2SA1612 2SC4180
~100					2SA1611 2SC4177 GA1[ ] GN1[ ]	
~150					2SC4181	
~200	2SC4176	2SC4175				
~500			2SB1475 (–25 V) 2SD2228 (25 V)	2SA1608 2SC4173		

■ Ultra Small Mini-Mold Transistor

V_{CEO} (V) / I_C (mA)	~50
~100	2SA1836 2SC4783

Ultra small mini mold

Quick Reference Table by Function/Application

■ Small Signal Transistor (1/2)

	TO-92	SP-8	SST	SMALL MINI MOLD	MINI MOLD	POWER MINI
FM/FM RF RF/MIX/CON	2SC1674		2SC2786	2SC4178	2SC2223	
FM AM/FM/AM MIX/OSC/IF			2SC2787	2SC4179	2SC1009A	
Audio Frequency Amplification	2SA733		2SA1175	2SA1611	2SA812	
	2SA987				(2SA812)	
	2SA988			2SA1612	2SA811A	
	2SC945		2SC2785	2SC4177	2SC1623	
	2SC1840				(2SC1623)	
	2SC1841				2SC1622A	
	2SC3622/A		2SC3623/A	2SC4181	2SC3624/A	
Low Noise Amplification	2SA992		2SA1174		2SA1247	
	2SC1845		2SC2784		2SC3115	
Driver Output	2SA952		2SB810		2SB624	
	2SA953				2SB736	
	2SA954				2SB736A	2SB800
	2SA1376				2SA1330	
		2SB564	2SB811			2SB798
		2SB605				2SB799
	2SB1068					2SB1114
	2SB1116					2SB1115
	2SB1116A					2SB1115A
	2SC2001		2SD1020		2SD596	
	2SC2002				2SD780	
	2SC2003				2SD780A	2SD1001
	2SC3478				2SC3360	
		2SC1940			2SC1653	2SC2780
		2SC1941			2SC1654	
		2SC3209				2SC3554
	2SC3615					2SC3617
	2SC3616					2SC3618
		2SD471				2SD999
		2SD571				2SD1000
	2SD1513					2SD1614
	2SD1616					2SD1615
	2SD1616A					2SD1615A
	2SD1698	2SD1697				2SD1699
	2SD1701	2SD1700				2SD1702
		2SD1581				2SD1950

Quick Reference Table by Function/Application

■ Small Signal Transistor (2/2)

	TO-92	SP-8	SST	SMALL MINI MOLD	MINI MOLD	POWER MINI
Switching	2SA1206		2SA1459	2SA1610	2SA1462	
				2SA1608	2SA1464	
		2SA1460				2SA1463
			2SA1458		2SA1461	
				2SC4173	2SC3739	
	2SC2901		2SC3732	2SC4176	2SC3735	
		2SC3733				2SC3736
			2SC3731		2SC3734	

Quick Reference Table by Function/Application

Power Transistor

■ Low $V_{CE(sat)}$ Transistor

Characteristics V_{CEO}, $I_{C(DC)}$	SP-8	MP-2	MP-3	MP-5 (TO-126)	MP-25	MP-10	MP-45	MP-45F
16 V, 3 A		2SB1581						
20 V, 10 A	2SA1897 (5A)		2SA1615					
25 V, 3 A	2SB1117							
30 V, 3 A			2SB962	2SB772 2SD882				
30 V, 10 A			2SA1649					
60 V, 3 A			2SB1261 2SD1899	2SB1217 2SD1818				
60 V, 5 A		2SB1578 2SD2425	2SA1385 2SC3518	2SB1151 2SD1691		2SA1843 2SC4815		
			2SA1648				2SA1441 2SC3691	2SA1741 2SC4549
60 V, 7 A							2SA1442 2SC3692	2SA1742 2SC4550
60 V, 10 A							2SA1443 2SC3693	2SA1743 2SC4551
60 V, 15 A							2SA1444 2SC3694	2SA1744 2SC4552
100 V, 5 A			2SA1647 2SC4331			2SA1845	2SA1650	
100 V, 7 A					2SA1645			
100 V, 10 A						2SA1847	2SA1652 2SC4336	

■ High h_{FE} Transistor

Characteristics V_{CEO}, $I_{C(DC)}$	SP-8	MP-3	MP-5 (TO-126)	MP-10	MP-45	MP-45F
20 V, 2 A		2SD1583				
25 V, 2 A	2SD1581					
50 V, 1 A	2SD1582					
60 V, 3 A		2SD1584	2SD1694		2SD1593	2SD2164
100 V, 2.5 A				2SC4814*		
100 V, 6 A					2SD1594	2SD2165
100 V, 7.5 A				2SC4813*	2SC4062	2SC4553*
100 V, 15 A					2SC4063	2SC4554*

*: Low VCE(sat) & High hFE

Quick Reference Table by Function/Application

■ Low Voltage, High Speed Switching Transistor

Characteristics V_{CEO}, $I_{C(DC)}$	SP-8	MP-25	MP-45	MP-80
45 V, 1 A	2SA1460 2SC3733			
60 V, 5 A		2SA1069	2SA1394	
100 V, 2 A		2SA1008 2SC2331	2SA1395 2SC3567	
100 V, 5 A		2SC2517		
100 V, 7 A		2SA1010 2SC2334		
100 V, 10			2SA1396 2SC3568	
100 V, 15 A				2SC2750

■ High Voltage, High Speed Switching Transistor

Characteristics V_{CEO}, $I_{C(DC)}$	MP-3	MP-5 (TO-126)	MP-25 (TO-220)	MP-40	MP-45	MP-80
200 V, 2 A	2SC2885 2SC2946 2SC2946(1)					
350 V, 2 A			2SA1009			
400 V, 0.5 A	2SA1400 2SC3588	2SA1156 2SC2752				
400 V, 2 A	2SC3631		2SA1009A 2SC2333		2SC3569	
400 V, 5 A	2SC4346		2SC2518		2SC3570	
400 V, 7 A				2SC3158	2SC3571	
400 V, 10 A				2SC3159	2SC3572	2SC2749
400 V, 15 A						2SC2751
600 V, 2 A	2SA1413 2SC3632	2SA1486 2SC3840				

Quick Reference Table by Function/Application

■ Audio Frequency Amplification Transistor

Characteristics V_{CEO}, $I_{C(DC)}$	MP-3	MP-5 (TO-126)	MP-25 (TO-220)	MP-45
45 V, 3 A		2SD794		
60 V, 3 A		2SD794A		2SB1094 2SD1585
60 V, 7 A			2SB707	2SB1097 2SD1588
80 V, 0.8 A		2SD414		
80 V, 7 A			2SB708	
100 V, 0.8 A		2SD415		
150 V, 2 A	2SB768 2SD1033		2SB546A 2SD401A	

■ High Frequency Amplification Transistor

Characteristics V_{CEO}, $I_{C(DC)}$	SP-8	MP-5 (TO-126)	MP-25 (TO-220)
40 V, 7 A			2SA1129
120 V, 1.2 A		2SC2690	
160 V. 1.2 A		2SC2690A	
250 V, 0.1 A	2SC4000	2SA1546 2SC4001	
300 V, 0.2 A	2SC2802	2SC2688	2SC1505

Quick Reference Table by Function/Application

■ Darlington Transistor

Characteristics V_{CEO}, $I_{C(DC)}$	SP-8	MP-3	MP-5 (TO-126)	MP-25 (TO-220)	MP-10	MP-40	MP-45	MP-45F	MP-80
31 V, 2 A	2SD2463*		2SD1695*						
60 V, 0.8 A	2SD1700*								
60 V, 1 A	2SD1843	2SB963 2SD1286	2SD1630*						
60 V, 1.5 A		2SD1164	2SB794 2SD985						
60 V, 3 A			2SB1150* 2SD1693*						
60 V, 5 A						2SD1392*	2SC4351◎		
80 V, 0.8 A	2SD1697								
80 V, 1.5 A	2SB1093 2SD1579		2SB795 2SD986						
100 V, 3 A			2SB1149 2SD1692 2SC4342◎						
100 V, 5 A				2SB601 2SD560	2SA1840 2SC4810	2SD1308	2SB1098◎ 2SD1589◎	2SB1430 2SD2161	
100 V, 8 A					2SA1841 2SC4811	2SD1309	2SB1099 2SD1590	2SB1431 2SD2162	
100 V, 10 A							2SA1720◎ 2SB1100 2SD1591	2SB1432 2SD2163	2SB897 2SD1210
100 V, 15 A									2SD1296
100 V, 25 A									2SD1297
300 V, 5 A							2SD1592		

◎: High speed darlington transistor, ∗: Internal zener diode between C-B

Field Effect Transistor

Small Signal FET

■ **2SK type (MOS type) (1/2)**

Part number	Package	Absolute maximum rating (T_A = 25 °C)				Electrical characteristics (T_A = 25 °C)							Applications
		V_{DSS} (V)	I_D		P_T (W)	$\|y_{fs}\|$ (S)			$R_{DS(on)}$ (Ω)				
			DC (A)	Pulse (A)		V_{DS} (V)	I_D (A)	MIN.	V_{GS} (V)	I_D (A)	TYP.	MAX.	
2SK679A	TO-92	30	±0.5	±1.5	0.75	10	0.5	0.4	4	0.5	0.6	1.0	Switching
2SK680A	SC-62	30	±1.0	±2.0	1.0	10	0.5	0.4	4	0.5	0.6	1.0	
2SK681A	SP-8	30	±1.0	±2.0	1.0	10	0.5	0.4	4	0.5	0.6	1.0	
2SK1132	SST	50	0.1	0.2	0.25	5	0.02	0.03	4	0.02	30	50	
2SK1133	SC-59	50	0.1	0.2	0.2	5	0.02	0.03	4	0.02	30	50	
2SK1272	TO-92	60	±1.0	±2.0	0.75	10	0.5	0.4	4	0.5	—	1.0	
2SK1273	SC-62	60	±2.0	±4.0	2.0	10	0.5	0.4	4	0.5	—	1.0	
2SK1274	SP-8	60	±1.5	±3.0	1.0	10	0.5	0.4	4	0.5	—	1.0	
2SK1398	SST	50	0.1	0.2	0.25	3	0.01	0.02	2.5	0.005	20	40	
2SK1399	SC-59	50	0.1	0.2	0.2	3	0.01	0.02	2.5	0.005	20	40	
2SK1482	TO-92	30	±1.5	±3.0	0.75	10	0.5	0.4	4	0.5	0.19	0.8	
2SK1483	SC-62	30	±2.0	±4.0	2.0	10	0.5	0.4	4	0.5	0.19	0.8	
2SK1484	TO-92	100	±0.5	±1.0	0.75	10	0.5	0.4	4	0.5	0.62	1.2	
2SK1485	SC-62	100	±1.0	±2.0	2.0	10	0.5	0.4	4	0.5	0.62	1.2	
2SK1580	SC-70	16	0.1	0.2	0.15	3	0.01	0.02	2.5	0.001	9	15	
2SK1581	SC-59	16	0.2	0.4	0.2	3	0.01	0.02	2.5	0.001	3.2	5	
2SK1582	SC-59	30	0.2	0.4	0.2	5	0.01	0.02	4	0.01	2.2	5	
2SK1583	SC-62	16	±0.5	±1.0	2.0	5	0.3	0.4	2.5	0.3	1.2	2.0	
2SK1584	SC-62	30	±0.5	±1.0	2.0	5	0.3	0.4	2.5	0.3	1.2	2.0	
2SK1585	SC-62	16	±1.0	±2.0	2.0	5	0.5	0.4	2.5	0.5	0.8	1.2	
2SK1586	SC-62	30	±1.0	±2.0	2.0	5	0.5	0.4	4	0.5	—	1.0	
2SK1587	SC-62	16	±2.0	±4.0	2.0	5	1.0	0.4	2.5	1.0	0.55	0.8	
2SK1588	SC-62	16	±3.0	±6.0	2.0	3	1.0	0.4	2.5	1.0	0.34	0.5	
2SK1589	SC-59	100	0.1	0.2	0.2	5	0.01	0.02	4	0.01	19	30	
2SK1590	SC-59	60	0.2	0.4	0.2	5	0.01	0.02	4	0.01	3.2	6	
2SK1591	SC-59	100	0.2	0.4	0.2	5	0.01	0.02	4	0.01	5.8	8	
2SK1592	SC-62	60	±0.5	±1.0	2.0	10	0.5	0.4	4	0.5	1.6	2.5	
2SK1593	SC-62	100	±0.5	±1.0	2.0	10	0.5	0.4	4	0.5	4.0	6.0	
2SK1656	SST	30	0.1	0.2	0.25	3	0.01	0.02	2.5	0.01	25	45	
2SK1657	SC-59	30	0.1	0.2	0.2	3	0.01	0.02	2.5	0.01	25	45	
2SK1658	SC-70	30	0.1	0.2	0.15	3	0.01	0.02	2.5	0.01	25	45	
2SK1824	3-pin ultra super mini	30	0.1	0.2	0.2	3	0.01	0.02	2.5	0.001	7	13	
2SK1958	SC-70	16	0.1	0.2	0.15	3	0.01	0.02	1.5	0.001	30	50	
2SK1959	SC-62	16	±2.0	±4.0	2.0	3	1.0	1.0	1.5	0.05	0.08	0.8	
2SK1960	SC-62	16	±3.0	±6.0	2.0	3	1.5	2.0	1.5	0.1	0.35	0.8	
2SK2053	MP-2	16	±5	±10	2.0	3	2.5	0.4	1.5	0.5	0.2	0.4	
2SK2054	MP-2	60	±3	±6	2.0	10	1.5	2.0	4	1.5	0.18	0.25	

Field Effect Transistor

■ 2SK type (MOS type) (2/2)

Part number	Package	Absolute maximum rating (T_A = 25 °C)				Electrical characteristics (T_A = 25 °C)							Applications
		V_{DSS} (V)	I_D		P_T (W)	$\|y_{fs}\|$ (S)			$R_{DS(on)}$ (Ω)				
			DC (A)	Pulse (A)		V_{DS} (V)	I_D (A)	MIN.	V_{GS} (V)	I_D (A)	TYP.	MAX.	
2SK2055	MP-2	100	±2	±4	2.0	10	1.0	2.0	4	1.0	0.28	0.45	Switching
2SK2070	SP-8	100	±1	±2	2.0	10	1.0	2.0	4	1.0	0.28	0.45	
2SK2090	SC-70	50	±0.1	±0.2	0.15	3	0.01	0.02	2.5	0.01	20	40	
2SK2109	SC-62	60	±0.5	±1.0	2.0	10	0.3	0.4	4	0.3	0.55	1.5	
2SK2110	SC-62	100	±0.5	±1.0	2.0	10	0.3	0.4	4	0.3	0.90	2.0	
2SK2111	SC-62	60	±1.0	±2.0	2.0	10	0.5	0.4	4	0.5	0.35	1.0	
2SK2112	SC-62	100	±1.0	±2.0	2.0	10	0.5	0.4	4	0.5	0.67	1.5	
2SK2157	MP-2	30	±5	±10	2.0	10	2.5	2.0	4	2.5	0.08	0.15	
2SK2158	SC-59	50	0.1	0.2	0.2	3	0.01	0.02	1.5	0.001	21	50	
2SK2159	SC-62	60	±2	±4	2.0	10	1.0	0.4	1.5	0.1	0.26	0.7	
2SK2541	SST	50	±0.1	±0.2	0.2	3	0.01	0.02	1.5	0.001	21	50	
2SK2857★	SC-62	60	±4	±16	2.0	10	1	1	4	1.5	0.15	0.22	
2SK2858★	SC-70	30	±0.1	±0.4	0.15	3	0.01	0.02	2.5	0.001	10	15	

★: Under development

Field Effect Transistor

■ 2SJ type (MOS type)

Part number	Package	Absolute maximum rating (T_A = 25 °C)				Electrical characteristics (T_A = 25 °C)							Applications
		V_{DSS} (V)	I_D		P_T T_C = 25 °C (W)	$\|y_{fs}\|$ (S)			$R_{DS(on)}$ (Ω)				
			DC (A)	Pulse (A)		V_{DS} (V)	I_D (A)	MIN.	V_{GS} (V)	I_D (A)	TYP.	MAX.	
2SJ165	SST	−50	∓0.1	∓0.2	0.25	−5	−0.02	0.03	−4	−0.02	22	50	Switching
2SJ166	SC-59	−50	∓0.1	∓0.2	0.2	−5	−0.02	0.03	−4	−0.02	22	50	
2SJ178	TO-92	−30	∓1.0	∓2.0	0.75	−10	−0.5	0.4	−4	−0.5	—	1.5	
2SJ179	SC-62	−30	∓1.5	∓3.0	2.0	−10	−0.5	0.4	−4	−0.5	—	1.5	
2SJ180	SP-8	−30	∓1.0	∓2.0	1.0	−10	−0.5	0.4	−4	−0.5	—	1.5	
2SJ184	SST	−50	∓0.1	∓0.2	0.25	−5	−0.02	0.02	−2.5	−0.05	25	40	
2SJ185	SC-59	−50	∓0.1	∓0.2	0.2	−5	−0.02	0.02	−2.5	−0.05	25	40	
2SJ196	TO-92	−60	∓1.0	∓2.0	0.75	−10	−0.5	0.4	−4	−0.5	0.89	1.5	
2SJ197	SC-62	−60	∓1.5	∓3.0	2.0	−10	−0.5	0.4	−4	−0.5	0.89	1.5	
2SJ198	TO-92	−100	∓0.5	∓1.0	0.75	−10	−0.5	0.4	−4	−0.5	1.7	2.0	
2SJ199	SC-62	−100	∓1.0	∓2.0	2.0	−10	−0.5	0.4	−4	−0.5	1.7	2.0	
2SJ202	SC-70	−16	∓0.1	∓0.2	0.15	−3	−0.01	0.02	−2.5	−0.001	—	60	
2SJ203	SC-59	−16	∓0.2	∓0.4	0.2	−3	−0.01	0.02	−2.5	−0.001	16	20	
2SJ204	SC-59	−30	∓0.2	∓0.4	0.2	−3	−0.01	0.02	−2.5	−0.001	8.8	15	
2SJ205	SC-62	−16	∓0.5	∓1.0	2.0	−5	−0.3	0.4	−2.5	−0.3	2.2	5.0	
2SJ206	SC-62	−30	∓0.5	∓1.0	2.0	−5	−0.3	0.4	−2.5	−0.3	2.8	3.0	
2SJ207	SC-62	−16	∓1.0	∓2.0	2.0	−5	−0.5	0.4	−2.5	−0.5	1.5	2.0	
2SJ208	SC-62	−16	∓2.0	∓4.0	2.0	−5	−1.0	0.4	−2.5	−1.0	0.6	1.5	
2SJ209	SC-59	−100	∓0.1	∓0.2	0.2	−5	−0.01	0.02	−4	−0.01	60	100	
2SJ210	SC-59	−60	∓0.2	∓0.4	0.2	−5	−0.01	0.02	−4	−0.01	10	15	
2SJ211	SC-59	−100	∓0.2	∓0.4	0.2	−5	−0.01	0.02	−4	−0.01	18	30	
2SJ212	SC-62	−60	∓0.5	∓1.0	2.0	−10	−0.5	0.4	−4	−0.5	1.8	4.0	
2SJ218	SC-62	−100	∓0.5	∓1.0	2.0	−10	−0.5	0.4	−4	−0.5	2.7	5.0	
2SJ243	3-pin ultra super mini	−30	∓0.1	∓0.2	0.2	−3	−0.01	0.02	−2.5	−0.0005	55	100	
2SJ353	SP-8	−60	∓1.5	∓3.0	1.0	−10	−1	1.0	−4	−0.8	0.58	0.68	
2SJ355	SC-62	30	2	4	2.0	−10	−1	1.0	4.0	1.0	0.50	0.60	
2SJ356	SC-62	60	2	4	2.0	−10	−1	1.0	4.0	1.0	0.65	0.95	
2SJ357	MP-2	30	3	6	2.0	−10	−1	1.8	4.0	1.5	0.23	0.35	
2SJ358	MP-2	60	3	6	2.0	−10	−1	1.8	4.0	1.5	0.29	0.40	
2SJ411	SP-8	−30	∓ 5	∓20	1.0	−10	− 2.5	3.0	−4	−2.5	0.096	0.24	
2SJ460	SST	−50	∓ 0.1	∓0.2	0.25	−3	− 0.01	0.01	−2.5	−0.003	75	100	
2SJ461	SC-59	−50	∓ 0.1	∓0.2	0.2	−3	− 0.01	0.01	−2.5	−0.003	75	100	
2SJ462	MP-2	−12	∓ 2.5	∓5.0	2.0	−3	−1	1.5	−2.5	−0.5	0.23	0.3	
2SJ463A	SC-70	−30	∓ 0.1	∓0.4	0.15	−3	− 0.01	0.02	−2.5	−0.001	23	60	

Field Effect Transistor

Power MOS FET

■ **2SK type (1/4)**

Part number	Package	Absolute maximum rating (T_A = 25 °C)				Electrical characteristics (T_A = 25 °C)							Applications
		V_{DSS} (V)	I_D		P_T T_C = 25 °C (W)	$\|y_{fs}\|$ (S)			$R_{DS(on)}$ (Ω)				
			DC (A)	Pulse (A)		V_{DS} (V)	I_D (A)	MIN.	V_{GS} (V)	I_D (A)	TYP.	MAX.	
2SK1059	MP-3	60	±5	±20	20	10	3.0	4	4	3.0	0.15	0.22	Switching
2SK1060	MP-3	100	±5	±20	20	10	3.0	4	4	3.0	0.22	0.38	
2SK1122	MP-88	100	±40	±160	100	10	20	12	4	20	0.05	0.07	
2SK1123	MP-88	60	±40	±160	100	10	20	12	4	20	0.03	0.05	
2SK1198	MP-45	700	±2	±8	35	10	1.0	1	10	1.0	2.5	3.2	
2SK1271	MP-88	1400	±5	±10	150	20	3.0	1.5	10	3.0	3.5	4.0	
2SK1282	MP-3	60	±3	±12	20	10	2.0	2.4	4	2.0	0.2	0.3	
2SK1283	MP-5	60	±3	±12	20	10	2.0	2.4	4	2.0	0.2	0.3	
2SK1284	MP-3	100	±3	±12	20	10	2.0	2.4	4	2.0	0.32	0.45	
2SK1285	MP-5	100	±3	±12	20	10	2.0	2.4	4	2.0	0.32	0.45	
2SK1286	MP-45F	60	±15	±60	35	10	8.0	7	4	8.0	0.075	0.095	
2SK1287	MP-45F	100	±15	±60	35	10	8.0	7	4	8.0	0.12	0.14	
2SK1288	MP-25	100	±20	±80	60	10	8.0	7	4	8.0	0.12	0.14	
2SK1289	MP-45F	60	±25	±100	35	10	15	12	4	15	0.045	0.06	
2SK1290	MP-25	60	±30	±120	60	10	15	12	4	15	0.045	0.06	
2SK1292	MP-45F	100	±20	±100	35	10	15	12	4	15	0.07	0.085	
2SK1293	MP-25	100	±30	±120	60	10	15	12	4	15	0.07	0.085	
2SK1294	MP-45F	60	±40	±160	35	10	20	12	4	20	0.03	0.05	
2SK1295	MP-45F	100	±30	±160	35	10	20	12	4	20	0.06	0.075	
2SK1491	MP-88	250	±25	±100	120	10	13	7.0	10	13	0.12	0.15	
2SK1492	MP-88	250	±35	±140	140	10	18	10	10	18	0.08	0.1	
2SK1493	MP-25	450	±3.0	±12	50	10	2.0	1.0	10	2.0	2.2	2.8	
2SK1494	MP-25	500	±3.0	±12	50	10	2.0	1.0	10	2.0	2.4	3.0	
2SK1495	MP-25	450	±7.0	±28	50	10	4.0	3.0	10	4.0	0.7	0.9	
2SK1496	MP-25	450	±7.0	±28	50	10	4.0	3.0	10	4.0	0.8	1.0	
2SK1497	MP-88	450	±20	±80	130	10	10	7.5	10	10	0.28	0.35	
2SK1498	MP-88	500	±20	±80	130	10	10	7.5	10	10	0.32	0.40	
2SK1499	MP-88	450	±25	±100	160	10	13	8.0	10	13	0.20	0.25	
2SK1500	MP-88	500	±25	±100	160	10	13	8.0	10	13	0.22	0.27	
2SK1501	MP-25	900	±4.0	±8.0	70	10	2.0	1.0	10	2.0	2.8	4.0	
2SK1502	MP-88	900	±7.0	±14	120	10	4.0	2.0	10	4.0	1.7	2.0	
2SK1594	MP-45F	30	±20	±80	30	10	10	7.0	4	10	0.05	0.08	
2SK1596	MP-45F	30	±40	±160	35	10	20	20	4	20	0.02	0.03	
2SK1748	MP-3	60	±8.0	±32	20	10	4.0	5.0	4	4.0	0.11	0.16	
2SK1749	MP-88	60	±50	±200	150	10	25	20	4	25	0.022	0.025	
2SK1750	MP-25	450	±5.0	±20	50	10	2.5	1.0	10	2.5	1.1	1.4	
2SK1751	MP-25	500	±5.0	±20	50	10	2.5	1.0	10	2.5	1.2	1.5	
2SK1752	MP-88	450	±10	±40	100	10	5.0	3.5	10	5.0	0.7	0.9	

Field Effect Transistor

■ 2SK type (2/4)

Part number	Package	Absolute maximum rating (T_A = 25 °C)				Electrical characteristics (T_A = 25 °C)							Applications
		V_{DSS} (V)	I_D		P_T T_C = 25 °C (W)	$\|y_{fs}\|$ (S)			$R_{DS(on)}$ (Ω)				
			DC (A)	Pulse (A)		V_{DS} (V)	I_D (A)	MIN.	V_{GS} (V)	I_D (A)	TYP.	MAX.	
2SK1753	MP-88	500	±10	±40	100	10	5.0	3.5	10	5.0	0.8	1.0	Switching
2SK1756	MP-88	450	±15	±60	120	10	8.0	5.0	10	8.0	0.4	0.5	
2SK1757	MP-88	500	±15	±60	120	10	8.0	5.0	10	8.0	0.5	0.6	
2SK1758	MP-45F	600	±2.0	±8.0	30	10	1.0	0.5	10	1.0	2.8	4.2	
2SK1760	MP-88	900	±5.0	±10	100	20	3.0	1.0	10	3.0	3.1	4.0	
2SK1784	MP-88	450	±12	±48	100	10	6.0	1.5	10	6.0	0.5	0.6	
2SK1785	MP-88	500	±12	±48	100	10	6.0	1.5	10	6.0	0.6	0.7	
2SK1793	MP-25	900	±3.0	±6.0	75	20	2.0	0.8	10	2.0	6.2	7.5	
2SK1794	MP-88	900	±6.0	±12	100	20	3.0	2.0	10	3.0	1.8	2.8	
2SK1795	MP-88	900	±8.0	±16	140	20	4.0	1.0	10	4.0	1.3	1.6	
2SK1796	MP-88	900	±10	±20	150	20	5.0	1.5	10	5.0	1.0	1.2	
2SK1850	MP-10	60	±10	±40	1.8	10	5.0	7.0	4	5.0	0.08	0.095	
2SK1851	MP-10	60	±15	±60	1.8	10	7.5	12	4	7.5	0.045	0.06	
2SK1852	MP-10	100	±10	±40	1.8	10	5.0	7.0	4	5.0	0.15	0.2	
2SK1853	MP-10	100	±15	±60	1.8 *	10	7.5	12	4	7.5	0.08	0.1	
2SK1953	MP-45F	600	±2.0	±6.0	25 *	20	1.0	0.5	10	1.0	4.2	5.0	
2SK1954	MP-3	180	±4.0	±16	20 *	10	2.0	0.5	10	2.0	0.5	0.65	
2SK1988	MP-45F	450	±2.5	±10	30 *	10	1.5	0.9	10	1.5	2.2	2.8	
2SK1989	MP-45F	500	±2.5	±10	30	10	1.5	0.9	10	1.5	2.4	3.0	
2SK1990	MP-45F	450	±4.5	±18	30	10	2.5	1.5	10	2.5	1.1	1.4	
2SK1991	MP-45F	500	±4.5	±18	30	10	2.5	1.5	10	2.5	1.2	1.5	
2SK1992	MP-45F	450	±6.0	±24	35	10	3.0	2.8	10	3.0	0.7	0.9	
2SK1993	MP-45F	500	±6.0	±24	35	10	3.0	2.8	10	3.0	0.8	1.0	
2SK1994	MP-45F	900	±2.0	±4.0	30	20	1.0	0.6	10	1.0	6.2	7.5	
2SK1995	MP-45F	900	±3.0	±6.0	35	20	2.0	1.0	10	2.0	3.2	4.0	
2SK2040	MP-3	600	±2.0	±6.0	20	20	1.0	0.5	10	1.0	4.2	5.0	
2SK2131	MP-45F	150	±15	±60	35	10	8.0	10	10	8.0	0.1	0.12	
2SK2132	MP-10	180	±4	±16	1.8	10	2.0	0.5	10	2.0	0.52	0.65	
2SK2133	MP-25	250	±16	±64	75	10	8.0	4.0	10	8.0	0.21	0.26	
2SK2134	MP-25	200	±13	±52	70	10	7.0	2	10	7.0	0.32	0.4	
2SK2135	MP-45F	200	±14	±56	35 *	10	7.0	4	10	7.0	0.14	0.18	
2SK2136	MP-25	200	±20	±80	75	10	10	4	10	10	0.14	0.18	
2SK2137	MP-45F	600	±4	±16	30	10	20	1	10	2.0	2.0	2.4	
2SK2138	MP-25	600	±5	±20	70	10	2.5	1	10	2.5	2.0	2.4	
2SK2139	MP-45F	600	±5	±20	35	10	2.5	1.5	10	2.5	1.25	1.5	
2SK2140	MP-25	600	±7	±28	75	10	3.5	1.5	10	3.5	1.25	1.5	
2SK2141	MP-45F	600	±6	±24	40	10	3.0	2.0	10	3.0	0.9	1.1	
2SK2234	MP-45F	500	±8	±32	40	10	4.0	3.0	10	4.0	0.5	0.6	
2SK2275	MP-45F	900	±3.5	±7.0	35	20	2.0	4.0	10	2.0	2.4	2.8	
2SK2341	MP-45F	250	±11	±44	35	10	6.0	3.0	10	6.0	0.21	0.26	

*: T_A = 25°C

Field Effect Transistor

■ 2SK type (3/4)

Part number	Package	Absolute maximum rating (T_A = 25 °C)				Electrical characteristics (T_A = 25 °C)							Applications
		V_{DSS} (V)	I_D		P_T T_C = 25 °C (W)	$\|y_{fs}\|$ (S)			$R_{DS(on)}$ (Ω)				
			DC (A)	Pulse (A)		V_{DS} (V)	I_D (A)	MIN.	V_{GS} (V)	I_D (A)	TYP.	MAX.	
2SK2353	MP-45F	450	±4.5	±18	30	10	2.5	1.0	10	2.5	1.0	1.4	Switching
2SK2354	MP-45F	500	±4.5	±18	30	10	2.5	1.0	10	2.5	1.1	1.5	
2SK2355	MP-25	450	±5.0	±20	50	10	2.5	1.0	10	2.5	1.0	1.4	
2SK2356	MP-25	500	±5.0	±20	50	10	2.5	1.0	10	2.5	1.1	1.5	
2SK2357	MP-45F	450	±6.0	±24	35	10	3.0	3.0	10	3.0	0.7	0.9	
2SK2358	MP-45F	500	±6.0	±24	35	10	3.0	3.0	10	3.0	0.8	1.0	
2SK2359	MP-25	450	±7.0	±28	75	10	4.0	3.0	10	4.0	0.7	0.9	
2SK2360	MP-25	500	±7.0	±28	75	10	4.0	3.0	10	4.0	0.8	1.0	
2SK2361	MP-88	450	±10	±40	100	10	5.0	3.0	10	5.0	0.7	0.9	
2SK2362	MP-88	500	±10	±40	100	10	5.0	3.0	10	5.0	0.8	1.0	
2SK2363	MP-45F	450	±8.0	±32	35	10	5.0	4.0	10	5.0	0.4	0.5	
2SK2364	MP-45F	500	±8.0	±32	35	10	5.0	4.0	10	5.0	0.5	0.6	
2SK2365	MP-25	450	±10	±40	75	10	6.0	3.0	10	6.0	0.4	0.5	
2SK2366	MP-25	500	±10	±40	75	10	6.0	3.0	10	6.0	0.5	0.6	
2SK2367	MP-88	450	±15	±60	120	10	8.0	5.0	10	8.0	0.4	0.5	
2SK2368	MP-88	500	±15	±60	120	10	8.0	5.0	10	8.0	0.5	0.6	
2SK2369	MP-88	450	±20	±80	120	10	10	7.5	10	10	0.3	0.35	
2SK2370	MP-88	500	±20	±80	120	10	10	7.5	10	10	0.32	0.4	
2SK2371	MP-88	450	±25	±100	160	10	13	8.0	10	13	0.2	0.25	
2SK2372	MP-88	500	±25	±100	160	10	13	8.0	10	13	0.22	0.27	
2SK2409	MP-45F	60	±40	±160	35	10	20	20	4	20	0.03	0.05	
2SK2410	MP-45F	60	±30	±120	35	10	15	15	4	15	0.04	0.06	
2SK2411	MP-25	60	±30	±120	75	10	15	15	4	15	0.04	0.06	
2SK2412	MP-45F	60	±20	±80	30	10	10	7	4	10	0.067	0.095	
2SK2413	MP-10	60	±10	±40	1.8*	10	5	7	4	10	0.07	0.095	
2SK2414	MP-3	60	±10	±40	20	10	5	7	4	10	0.07	0.095	
2SK2415	MP-3	60	±8.0	±32	20	10	4	5	4	4	0.1	0.15	
2SK2461	MP-45F	100	±20	±80	35	10	10	12	4	10	0.07	0.1	
2SK2462	MP-45F	100	±15	±60	30	10	8	7	4	8	0.12	0.17	
2SK2476	MP-45F	800	±3.0	±9.0	40	20	2	1	10	2	3.4	5.0	
2SK2477	MP-88	800	±10	±30	150	20	5	3.5	10	5	0.65	1.0	
2SK2478	MP-45F	900	±2.0	±8.0	30	20	1	0.6	10	1	5.0	7.5	
2SK2479	MP-25	900	±3.0	±8.0	70	20	2	0.8	10	2	5.6	7.5	
2SK2480	MP-45F	900	±3.5	±12	35	20	2	1	10	2	3.2	4.0	
2SK2481	MP-25	900	±4.0	±12	70	20	2	1	10	2	3.2	4.0	
2SK2482	MP-88	900	±5.0	±12	100	20	3	1	10	3	3.2	4.0	
2SK2483	MP-45F	900	±3.5	±10.5	40	20	2	1	10	2	2.2	2.8	
2SK2484	MP-25	900	±5.0	±10	75	20	3	2	10	3	2.2	2.8	
2SK2485	MP-88	900	±6.0	±12	100	20	3	2	10	3	2.2	2.8	

*: T_A = 25°C

Field Effect Transistor

■ 2SK type (4/4)

Part number	Package	Absolute maximum rating (T_A = 25 °C)				Electrical characteristics (T_A = 25 °C)							Applications
		V_{DSS} (V)	I_D		P_T T_C = 25 °C (W)	$\|y_{fs}\|$ (S)			$R_{DS(on)}$ (Ω)				
			DC (A)	Pulse (A)		V_{DS} (V)	I_D (A)	MIN.	V_{GS} (V)	I_D (A)	TYP.	MAX.	
2SK2486	MP-88	900	±7.0	±18	120	20	4	2.5	10	4	1.4	2.0	Switching
2SK2487	MP-88	900	±8.0	±20	140	20	4	3.0	10	4	1.1	1.6	
2SK2488	MP-88	900	±10	±30	150	20	5	6	10	5	1.0	1.2	
2SK2498	MP-45F	60	±50	±200	35	10	25	20	4	25	0.011	0.014	
2SK2499	MP-25	60	±50	±200	75	10	25	20	4	25	0.011	0.014	
2SK2510	MP-45F	60	±40	±160	35	10	20	13	4	20	0.024	0.03	
2SK2511	MP-88	60	±40	±160	80	10	20	10	4	20	0.032	0.04	
2SK2512	MP-45F	60	±45	±180	35	10	23	15	4	23	0.016	0.023	
2SK2513	MP-25	60	±45	±180	75	10	23	15	4	23	0.016	0.023	
2SK2514	MP-88	60	±50	±200	150	10	25	15	4	25	0.016	0.023	
2SK2515	MP-88	60	±50	±200	150	10	25	20	4	25	0.011	0.014	
2SK2723	MP-45F	60	±25	±100	25	10	13	8	4	13	0.045	0.06	
2SK2724	MP-45F	60	±35	±140	30	10	18	10	4	18	0.033	0.04	
2SK2826	MP-25	60	±70	±280	100	10	35	20	4	35	0.007	0.0105	
2SK2941	MP-25	30	±35	±140	60	10	18	8	4	18	0.022	0.033	
2SK2981	MP-3	30	±20	±80	20	10	10	6	4.5	10	0.030	0.040	
2SK2982	MP-3	30	±30	±120	30	10	15	8	4.5	15	0.013	0.0165	
2SK2983	MP-25	30	±30	±120	50	10	15	9	4.5	15	0.018	0.027	
2SK2984	MP-25	30	±40	±160	60	10	20	18	4.5	20	0.0085	0.013	
2SK3055	MP-45F	60	±30	±120	25	10	15	8.0	4	15	0.042	0.06	
2SK3056	MP-25	60	±32	±128	34	10	16	8.0	4	16	0.042	0.06	
2SK3057	MP-45F	60	±45	±180	30	10	23	13	4	23	0.021	0.03	
2SK3058	MP-25	60	±55	±220	58	10	28	13	4	28	0.021	0.03	
2SK3059	MP-45F	60	±50	±200	30	10	25	15	4	25	0.016	0.023	
2SK3060★	MP-25	60	±70	±280	70	10	35	15	4	35	0.016	0.023	
2SK3061	MP-45F	60	±70	±280	35	10	35	20	4	35	0.0098	0.014	
2SK3062	MP-25	60	±70	±280	100	10	35	20	4	35	0.0098	0.014	

★: Under development

Field Effect Transistor

■ 2SJ type

Part number	Package	Absolute maximum rating (T_A = 25 °C)				Electrical characteristics (T_A = 25 °C)							Applications
		V_{DSS} (V)	I_D		P_T T_C = 25 °C (W)	$\|y_{fs}\|$ (S)			$R_{DS(on)}$ (Ω)				
			DC (A)	Pulse (A)		V_{DS} (V)	I_D (A)	MIN.	V_{GS} (V)	I_D (A)	TYP.	MAX.	
2SJ128	MP-3	−100	∓2	∓8	20	−10	−1	1	−4	−0.8	1.1	1.5	Switching
2SJ132	MP-3	−30	∓2	∓8	20	−10	−1	1	−4	−0.8	0.47	0.6	
2SJ133	MP-3	−60	∓2	∓8	20	−10	−1	1	−4	−0.8	0.7	1.3	
2SJ134	MP-25	−100	∓6	∓24	40	−10	−3.5	1	−4	−3.5	0.5	0.9	
2SJ135	MP-45	−100	∓5	∓20	30	−10	−3.5	1	−4	−3.5	0.5	0.9	
2SJ136	MP-25	−60	∓12	∓48	40	−10	−6.5	2	−4	−6.5	0.45	0.5	
2SJ137	MP-45	−60	∓10	∓40	30	−10	−6.5	2	−4	−6.5	0.45	0.5	
2SJ138	MP-25	−100	∓12	∓48	60	−10	−6.5	2	−4	−6.5	0.3	0.45	
2SJ139	MP-45	−100	∓10	∓40	35	−10	−6.5	2	−4	−6.5	0.3	0.45	
2SJ140	MP-25	−60	∓19	∓76	60	−10	−10	5	−4	−8	0.3	0.4	
2SJ141	MP-45	−60	∓13	∓52	35	−10	−10	5	−4	−8	0.3	0.4	
2SJ142	MP-45	−100	∓13	∓52	35	−10	−10	5	−4	−8	0.25	0.3	
2SJ143	MP-45	−60	∓16	∓64	35	−10	−10	5	−4	−10	0.22	0.25	
2SJ302	MP-25	− 60	∓16	∓64	75	−10	−8	6.0	−4	−6	0.13	0.24	
2SJ303	MP-45F	− 60	∓14	∓56	35	−10	−7	5.0	−4	−6	0.13	0.24	
2SJ324	MP-3	−30	∓2.0	∓8.0	20	−10	−1.0	1.0	−4	−0.8	0.40	0.52	
2SJ325	MP-3	−30	∓4.0	∓16	20	−10	−2.0	3.0	−4	−1.6	0.15	0.24	
2SJ326	MP-3	−60	∓2.0	∓8.0	20	−10	1.0	1.0	−4	−0.8	0.5	0.68	
2SJ327	MP-3	−60	∓4.0	∓16	20	−10	2.0	3.0	−4	−1.6	0.22	0.34	
2SJ328	MP-25	−60	∓20	∓80	75	−10	−10	8.0	−4	−8	0.085	0.11	
2SJ329	MP-45F	−60	∓15	∓60	35	−10	−8	8.0	−4	−6	0.085	0.11	
2SJ330	MP-45F	−60	∓20	∓80	35	−10	−10	10	−4	−8	0.065	0.09	
2SJ331	MP-88	−60	∓30	∓120	150	−10	−15	15	−4	−12	0.04	0.055	
2SJ448	MP-45F	−250	∓4	∓16	30	−10	−2	1	−10	−2	1.5	2.0	
2SJ449	MP-45F	−250	∓6	∓24	35	−10	−3	2	−10	−3	0.55	0.8	
2SJ492	MP-25	−60	∓20	∓80	70	−10	−10	5.0	− 4	−10	0.12	0.19	
2SJ493	MP-45F	−60	∓16	∓64	30	−10	−8	5.0	− 4	−8	0.12	0.19	
2SJ494	MP-45F	−60	∓20	∓80	35	−10	−10	8	−4	−10	0.061	0.088	
2SJ495	MP-45F	−60	∓30	∓120	35	−10	−15	12	−4	−15	0.038	0.056	

Field Effect Transistor

■ Low Voltage Power MOS FET (N3-L series)

DC drain current $I_{D(DC)}$ (A)	Drain to source voltage V_{DSS} (V)			Package
	30	60	100/[150]	
~ 4.0	△ 2SJ324 (0.25/0.52) △ 2SJ325 (0.11/0.24)	△ 2SK1282 (0.18/0.24) ◇ 2SK1283 (0.18/0.24) △ 2SJ326 (0.37/0.68) △ 2SJ327 (0.17/0.34)	△ 2SK1284 (0.32/0.40) ◇ 2SK1285 (0.32/0.40)	△ : MP-3 ◇ : MP-5 ○ : MP-25 ● : MP-45F ◎ : MP-10 ⧈ : MP-88
~ 8.0		◎ 2SK1748 (0.11/0.16)		
~ 10		◎ 2SK1850 (70 m/95 m)	◎ 2SK1852 (0.15/0.20)	
~ 15		2SK1851 (45 m/60 m) ● 2SK1286 (70 m/95 m) ● 2SJ303 (0.1/0.24) ● 2SJ329 (60 m/0.11)	◎ 2SK1853 (80 m/0.10) ● 2SK1288 (0.15/0.20) [● 2SK2131 (0.12/0.20)]	
~ 20	● 2SK1594 (50 m/80 m)	○ 2SK1287 (70 m/95 m) ○ 2SJ302 (0.1/0.24) ○ 2SJ328 (60 m/0.11) ● 2SJ330 (50 m/90 m)	○ 2SK1289 (0.15/0.20) ● 2SK1292 (80 m/0.10)	
~ 25		● 2SK1290 (45 m/60 m)		
~ 30		⧈ 2SJ331 (30 m/55 m)	○ 2SK1293 (80 m/0.10) ● 2SK1295 (50 m/70 m)	
~ 40	● 2SK1596 (20 m/30 m)	● 2SK1294 (27 m/50 m) ⧈ 2SK1123 (27 m/50 m)	⧈ 2SK1122 (50 m/70 m)	
~ 50		⧈ 2SK1749 (18 m/25 m)		

( ): $R_{DS(on)\ MAX}$ (Ω) (@V_{GS} = 10 V/V_{GS} = 4 V)

■ Low Voltage Power MOS FET (N4-L series/N5-L series)

DC drain current $I_{D(DC)}$ (A)	Drain to source voltage V_{DSS} (V)			Package
	30	60	100	
8.0		△ 2SK2415 (0.1/0.15)		△ : MP-3 ◇ : MP-5 ○ : MP-25 ● : MP-45F ◎ : MP-10 ⧈ : MP-88
10		◎ 2SK2413 (70 m/95 m) △ 2SK2414 (70 m/95 m)		
15			● 2SK2462 (0.14/0.17)	
20		● 2SK2412 (70 m/95 m) ○ 2SJ492 (0.1/0.19)* ● 2SJ494 (50 m/88 m)*	● 2SK2461 (80 m/0.1)	
25		● 2SK2723 (40 m/60 m)*		
30		● 2SK2410 (40 m/60 m) ○ 2SK2411 (40 m/60 m) ● 2SJ495 (30 m/56m)*		
35	○ 2SK2941 (20 m/33 m)*	● 2SK2724 (27 m/40 m)*		
40		● 2SK2409 (27m/40 m) ● 2SK2510 (20 m/30 m)* ⧈ 2SK2511 (27 m/40 m)*		
45		● 2SK2512 (15 m/23 m)* ○ 2SK2513 (15 m/23 m)*		
50		● 2SK2498 (9 m/14 m)* ○ 2SK2499 (9 m/14 m)* ⧈ 2SK2514 (15 m/23 m)* ⧈ 2SK2515 (9 m/14 m)*		

( ): $R_{DS(on)\ MAX}$ (Ω) (@V_{GS} = 10 V)

* : N5-L series

Field Effect Transistor

■ Low Voltage Power MOS FET (N6-L series)

DC drain current $I_{D(DC)}$ (A)	Drain to source voltage V_{DSS} (V)		Package
	30	60	
to 20	△ 2SK2981 (27 m/40 m)*		△: MP-3 ○: MP-25 ●: MP-45F
to 30	△ 2SK2982 (12.5 m/16.5 m)* ○ 2SK2983 (20 m/27 m)*	● 2SK3055 (40 m/60 m)	
to 40	○ 2SK2984 (10 m/13 m)*	○ 2SK3056 (40 m/60 m)	
to 50		● 2SK2057 (20 m/30 m) ● 2SK3059 (15 m/23 m)	
to 60		○ 2SK2058 (20 m/30 m)	
to 70		● 2SK3061 (9 m/14 m) ○ 2SK3062 (9 m/14 m) ○ 2SK2826 (6.5 m/10.5 m)	

( ): $R_{DS(on)\ MAX}$ (Ω) (@V_{GS} = 10 V/V_{GS} = 4 V)
* : (@V_{GS} = 10 V/V_{GS} = 4.5 V)

■ High Voltage Power MOS FET (N3-H series)

DC drain current $I_{D(DC)}$ (A)	Drain to source voltage V_{DSS} (V)					Package
	250	450	500	600	900	
2.0				● 2SK1758 (4.2)	● 2SK1994 (7.5)	△: MP-3 ○: MP-25 ●: MP-45F ⧈: MP-88
2.5		● 2SK1988 (2.8)	● 2SK1989 (3.0)			
3.0		○ 2SK1493 (2.8)	○ 2SK1494 (3.0)		○ 2SK1793 (7.5) ● 2SK1995 (4.0)	
3.5					● 2SK2275 (2.8)	
4.0					○ 2SK1501 (4.0)	
4.5		● 2SK1990 (1.4)	● 2SK1991 (1.5)			
5.0		○ 2SK1750 (1.4)	○ 2SK1751 (1.5)		⧈ 2SK1760 (4.0)	
6.0		● 2SK1992 (0.9)	● 2SK1993 (1.0)		⧈ 2SK1794 (2.8)	
7.0		○ 2SK1495 (0.9)	○ 2SK1496 (1.0)		⧈ 2SK1502 (2.0)	
8.0			● 2SK2234 (0.6)		⧈ 2SK1795 (1.6)	
10		⧈ 2SK1752 (0.9)	⧈ 2SK1753 (1.0)		⧈ 2SK1796 (1.2)	
12		⧈ 2SK1784 (0.6)	⧈ 2SK1785 (0.7)			
15		⧈ 2SK1756 (0.5)	⧈ 2SK1757 (0.6)			
20		⧈ 2SK1497 (0.35)	⧈ 2SK1498 (0.4)			
25	⧈ 2SK1491 (0.15)	⧈ 2SK1499 (0.25)	⧈ 2SK1500 (0.27)			
35	⧈ 2SK1492 (0.10)					

( ): $R_{DS(on)\ MAX}$ (Ω) (@V_{GS} = 10 V)

Field Effect Transistor

■ High Voltage Power MOS FET (N4-H series/P4-H series)

DC drain current $I_{D(DC)}$ (A)	Drain to source voltage V_{DSS} (V)						Package
	[180]/200	250	450	500	600	[800]/900	
2.0					● 2SK1953 (5.0) △ 2SK2040 (5.0)	● 2SK2478 (7.5)	△ : MP-3 ○ : MP-25 ◎ : MP-10 ● : MP-45F ⊡ : MP-88
3.0						[● 2SK2476 (5.0)] ○ 2SK2479 (7.5) ● 2SK2480 (4.0)	
4.0	[△ 2SK1954 (0.65) ◎ 2SK2132 (0.65)]	● 2SJ448 (2.0)			● 2SK2137 (2.4)	○ 2SK2481 (4.0) ● 2SK2483 (2.8) (3.5 A)	
4.5			● 2SK2353 (1.4)	● 2SK2354 (1.5)			
5.0			○ 2SK2355 (1.4)	○ 2SK2356 (1.5)	○ 2SK2138 (2.4) ● 2SK2139 (1.5)	⊡ 2SK2482 (4.0) ○ 2SK2484 (2.8)	
6.0		● 2SJ449 (0.8)	● 2SK2357 (0.9)	● 2SK2358 (1.0)	● 2SK2141 (1.1)	⊡ 2SK2485 (2.8)	
7.0			○ 2SK2359 (0.9)	○ 2SK2360 (1.0)	○ 2SK2140 (1.5)	⊡ 2SK2486 (2.0)	
8.0			● 2SK2363 (0.5)	● 2SK2364 (0.6)		⊡ 2SK2487 (1.6)	
10			⊡ 2SK2361 (0.9) ○ 2SK2365 (0.5)	⊡ 2SK2362 (1.0) ○ 2SK2366 (0.6)		[⊡ 2SK2477 (1.0)] ⊡ 2SK2488 (1.2)	
11		● 2SK2341 (0.26)					
13	○ 2SK2134 (0.4)						
14	● 2SK2135 (0.18)						
15			⊡ 2SK2367 (0.5)	⊡ 2SK2368 (0.6)			
16		○ 2SK2133 (0.26)					
20	○ 2SK2136 (0.18)		⊡ 2SK2369 (0.35)	⊡ 2SK2370 (0.4)			
25			⊡ 2SK2371 (0.25)	⊡ 2SK2372 (0.27)			

( ): $R_{DS(on)\ MAX}$ (Ω) (@V_{GS} = 10 V)

Field Effect Transistor

■ 8-pin SOP Type Power MOS FET

On-Resistance R_{DS} (on) (mΩ) (TYP.) @V_{GS} = 10 V	Drain to source voltage V_{DSS} (V)	
	20	30
~90		μPA1750 (P-ch Dual)
~70		μPA1710 (P-ch Single) μPA1710A (P-ch Single) μPA1754 (N-ch Dual)
~50		μPA1713 (P-ch Single)
~40	μPA1752 (N-ch Dual)*	μPA1711 (P-ch Single)
~30	μPA1753 (N-ch Dual)* μPA1756 (N-ch Dual)*	μPA1751 (N-ch Dual) μPA1714 (P-ch Single) μPA1755 (N-ch Dual) μPA1758 (N-ch Dual)*
~20	μPA1701 (N-ch Single)* μPA1701A (N-ch Single) μPA1757 (N-ch Dual)*	μPA1700 (N-ch Single) μPA1700A (N-ch Single) μPA1712 (P-ch Single) μPA1705 (N-ch Single)*
~10		μPA1702 (N-ch Single) μPA1703 (N-ch Single) μPA1704 (N-ch Single) μPA1706 (N-ch Single) μPA1707 (N-ch Single) μPA1715 (P-ch Single)

*: 2.5 V drive is possible. Otherwise 4 V drive is possible.

■ Power SOP8 series

Part number	Configuration	Absolute maximum rating (T_A = 25 °C)				Electrical characteristics (T_A = 25 °C)				Applications
		V_{DSS} (V)	I_D		P_T T_C = 25 °C (W)	V_{GS} (V)	I_D (A)	$R_{DS(on)}$ (Ω)		
			DC (A)	Pulse (A)				TYP.	MAX.	
μPA1700	N-ch Single	30	±7	±28	2.0*1	4	3.5	0.040	0.050	Switching
μPA1700A	N-ch Single	30	±7	±28	2.0*1	4	3.5	0.028	0.05	
μPA1701	N-ch Single	20	±7	±28	2.0*1	2.5	3.5	0.03	0.04	
μPA1701A	N-ch Single	20	±7	±28	2.0*1	2.5	3.5	0.026	0.04	
μPA1702	N-ch Single	30	±8	±32	2.0*1	4	4	0.018	0.025	
μPA1703	N-ch Single	30	±10	±40	2.0*1	4	5.0	0.012	0.017	
μPA1704	N-ch Single	30	±10	±40	2.0*1	2.5	5.0	0.0125	0.016	
μPA1705	N-ch Single	30	±7	±28	2.0*1	4.5	3.5	0.031	0.04	
μPA1706	N-ch Single	30	±13	±52	2.0*1	4	7.0	0.008	0.012	
μPA1707	N-ch Single	30	±10	±40	2.0*1	4	5.0	0.014	0.021	
μPA1710	P-ch Single	−30	∓5	∓20	2.0*1	−4	−2.5	0.11	0.16	
μPA1710A	P-ch Single	−30	∓5	∓20	2.0*1	−4	−2.0	0.091	0.16	
μPA1711	P-ch Single	−30	∓7	∓28	2.0*1	−4	−3.5	0.049	0.07	
μPA1712	P-ch Single	−30	∓8	∓32	2.0*1	−4	−4.0	0.027	0.048	
μPA1713★	P-ch Single	30	∓5	∓20	2.0*1	−4	2.5	0.10	0.16	
μPA1714★	P-ch Single	−30	∓7	∓28	2.0*1	−4	3.5	0.049	0.07	
μPA1715★	P-ch Single	−30	∓11	∓44	2.0*1	−4.5	−6.0	0.012	0.017	
μPA1750	P-ch Dual	−30	∓3.5	∓14	2.0*2	−4	−1.8	0.125	0.18	
μPA1751	N-ch Dual	30	±5	±20	2.0*2	4	2.5	0.044	0.064	
μPA1752	N-ch Dual	20	±5	±20	2.0*2	2.5	2.5	0.042	0.061	
μPA1753	N-ch Dual	20	±6	±24	2.0*2	2.5	3.0	0.028	0.04	
μPA1754	N-ch Dual	30	±7	±28	2.0*2	4	3.5	0.029	0.053	
μPA1755★	N-ch Dual	−30	±7	±28	2.0*2	4.5	3.5	0.028	0.045	
μPA1756	N-ch Dual	30	±6	±24	2.0*2	2.5	3.0	0.028	0.04	
μPA1757	N-ch Dual	20	±7	±28	2.0*2	2.5	3.5	0.022	0.032	
μPA1758	N-ch Dual	30	±6	±24	2.0*2	2.5	3.0	0.026	0.04	

*1: T_A = 25°C, Mounted on ceramic substrate of 1200 mm² × 0.7 mm

*2: T_A = 25°C, 2 circuits, Mounted on ceramic substrate of 2000 mm² × 1.1 mm

★: Under development

Field Effect Transistor

■ 8-pin TSSOP Type Power MOS FET

On-Resistance $R_{DS(on)}$ (mΩ) (MAX.) @V_{GS} = 4.5 V	Drain to source voltage V_{DSS} (V)		
	12	20	30
~210		μPA1851 (P-ch Dual)	
~190			μPA1853 (P-ch Dual)
~115	μPA1850 (P-ch Dual)		
~75		μPA1811 (P-ch Single)	μPA1812 (P-ch Single)
~60	μPA1854 (P-ch Dual) μPA1810 (P-ch Single)		
~40		μPA1852 (N-ch Dual)	μPA1800 (N-ch Single)
~30			μPA1814 (P-ch Single)★
~25	μPA1813 (P-ch Single)	μPA1801 (N-ch Single) μPA1855 (N-ch Dual)★ μPA1802 (N-ch Single) μPA1815 (P-ch Single)★	μPA1803 (N-ch Single)★

★: Under development

■ 8-pin TSSOP Type Power MOS FET

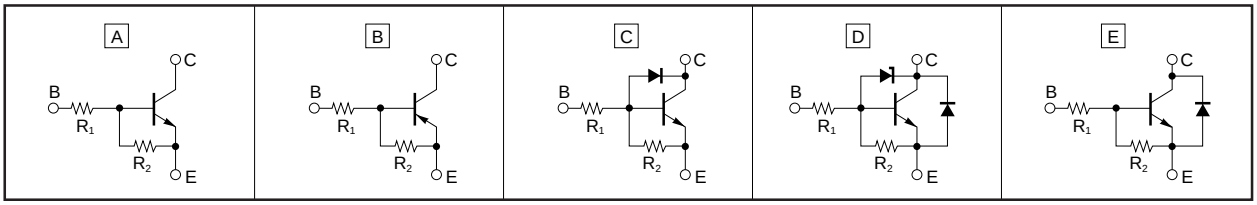
Part number	Configuration	Absolute maximum rating (T_A = 25 °C)				Electrical characteristics (T_A = 25 °C)				Applications
		V_{DSS} (V)	V_{GSS} (V)	$I_{D(DC)}$ (A)	P_T* (W)	$R_{DS(on)}$ (mΩ) MAX.				
						V_{GS} = 2.5 V	V_{GS} = 4 V	V_{GS} = 4.5 V	V_{GS} = 10 V	
μPA1800	N-ch Single	30	±20	±5	2	–	45	39	27	Switching
μPA1801	N-ch Single	20	±8	±6	2	34	25	24	–	
μPA1802	N-ch Single	20	±12	±7	2	32	25	23	–	
μPA1803★	N-ch Single	30	±20	±7	2	–	–	21	14	
μPA1810	P-ch Single	–12	–10/+5	±4	2	100	60	55	–	
μPA1811	P-ch Single	–20	–12/+5	±4	2	120	80	75	–	
μPA1812	P-ch Single	–30	–20/+5	±5	2	–	90	70	40	
μPA1813	P-ch Single	–12	–10/+5	±5	2	40	30	25	–	
μPA1814★	P-ch Single	–30	–20/+5	±7	2	–	–	30	19	
μPA1815★	P-ch Single	–20	–12/+3	±7	2	30	23	22	–	
μPA1850	P-ch Dual	–12	–10/+5	±2.5	2	200	130	115	–	
μPA1851	P-ch Dual	–20	–20/+5	±2.5	2	–	250	210	105	
μPA1852	N-ch Dual	20	±12	±5	2	60	45	40	–	
μPA1853	P-ch Dual	–30	–20/+5	±2.5	2	–	220	190	85	
μPA1854	P-ch Dual	–12	–10/+5	±2.5	2	105	70	60	–	
μPA1855★	N-ch Dual	20	±12	±6	2	29	24	23	–	

★: Under development

*: Mounted on ceramic substrate of 50 cm^2 × 1.1 mm

Transistor with Internal Resistor

■ Equivalent circuit



■ AA1[ ], AN1[ ], BA1[ ], BN1[ ] series (TO-92, SST)

Part number		Equivalent circuit and applications	Absolute maximum ratings (T_A = 25 °C)						Electrical characteristics (T_A = 25 °C)					
TO-92	SST		V_{CBO} (V)	V_{CEO} (V)	V_{EBO} (V)	I_C (mA)	P_T (mW)	T_j (°C)	R_1 (kΩ)	R_2 (kΩ)	h_{FE}	V_{CE} (V)	I_C (mA)	Remarks
AA1A4M	**BA1A4M**	Refer to equivalent circuit A Switching	60	50	10	100	250	150	10	10	35 to 100	5.0	5.0	
AA1A4P	**BA1A4P**		60	50	5	100	250	150	10	47	35 to 340	5.0	5.0	
AA1F4M	**BA1F4M**		60	50	10	100	250	150	22	22	60 to 195	5.0	5.0	
AA1F4N	**BA1F4N**		60	50	5	100	250	150	22	47	85 to 340	5.0	5.0	
AA1L4M	**BA1L4M**		60	50	10	100	250	150	47	47	85 to 340	5.0	5.0	
AA1L4L	**BA1L4L**		60	50	15	100	250	150	47	22	60 to 195	5.0	5.0	
AA1L3Z	**BA1L3Z**		60	50	5	100	250	150	4.7	–	135 to 600	5.0	5.0	
AA1A4Z	**BA1A4Z**		60	50	5	100	250	150	10	–	135 to 600	5.0	5.0	
AA1F4Z	**BA1F4Z**		60	50	5	100	250	150	22	–	135 to 600	5.0	5.0	
AA1L4Z	**BA1L4Z**		60	50	5	100	250	150	47	–	135 to 600	5.0	5.0	
AA1L3M	**BA1L3M**		60	50	10	100	250	150	4.7	4.7	20 to 80	5.0	5.0	
AA1L3N	**BA1L3N**		60	50	5	100	250	150	4.7	10	35 to 100	5.0	5.0	
AA1A3Q	**BA1A3Q**		60	50	5	100	250	150	1.0	10	35 to 100	5.0	5.0	
AN1A4M	**BN1A4M**	Refer to equivalent circuit B	–60	–50	–10	–100	250	150	10	10	35 to 100	–5.0	–5.0	
AN1A4P	**BN1A4P**		–60	–50	–5	–100	250	150	10	47	85 to 340	–5.0	–5.0	
AN1F4M	**BN1F4M**		–60	–50	–10	–100	250	150	22	22	60 to 195	–5.0	–5.0	
AN1F4N	**BN1F4N**		–60	–50	–5	–100	250	150	22	47	85 to 340	–5.0	–5.0	
AN1L4M	**BN1L4M**		–60	–50	–10	–100	250	150	47	47	85 to 340	–5.0	–5.0	
AN1L4L	**BN1L4L**		–60	–50	–15	–100	250	150	47	22	60 to 195	–5.0	–5.0	
AN1L3Z	**BN1L3Z**		–60	–50	–5	–100	250	150	4.7	–	135 to 600	–5.0	–5.0	
AN1A4Z	**BN1A4Z**		–60	–50	–5	–100	250	150	10	–	135 to 600	–5.0	–5.0	
AN1F4Z	**BN1F4Z**		–60	–50	–5	–100	250	150	22	–	135 to 600	–5.0	–5.0	
AN1L4Z	**BN1L4Z**		–60	–50	–5	–100	250	150	47	–	135 to 600	–5.0	–5.0	
AN1L3M	**BN1L3M**		–60	–50	–10	–100	250	150	4.7	4.7	20 to 80	–5.0	–5.0	
AN1L3N	**BN1L3N**		–60	–50	–5	–100	250	150	4.7	10	35 to 100	–5.0	–5.0	
AN1A3Q	**BN1A3Q**		–60	–50	–5	–100	250	150	1.0	10	35 to 100	–5.0	–5.0	

Transistor with Internal Resistor

■ AB1[ ], AP1[ ], BB1[ ], BP1[ ] series (TO-92, SST)

Part number		Equivalent circuit and applications	Absolute maximum ratings (T_A = 25 °C)							Electrical characteristics (T_A = 25 °C)					
			V_{CBO}	V_{CEO}	V_{EBO}	I_C	P_T (mW)		T_j	R_1	R_2	h_{FE}	V_{CE}	I_C	Re-
TO-92	SST		(V)	(V)	(V)	(A)	TO-92	SST	(°C)	(kΩ)	(kΩ)		(V)	(A)	marks
AB1A4A	**BB1A4A**	Refer to equivalent circuit A Switching	30	25	10	0.7	750	250	150	–	10	300 ~	2.0	0.5	
AB1L2Q	**BB1L2Q**		30	25	10	0.7	750	250	150	0.47	4.7	300 ~	2.0	0.5	
AB1A3M	**BB1A3M**		30	25	10	0.7	750	250	150	1.0	1.0	100 ~	2.0	0.5	
AB1F3P	**BB1F3P**		30	25	10	0.7	750	250	150	2.2	10	300 ~	2.0	0.5	
AB1J3P	**BB1J3P**		30	25	10	0.7	750	250	150	3.3	10	300 ~	2.0	0.5	
AB1L3N	**BB1L3N**		30	25	10	0.7	750	250	150	4.7	10	300 ~	2.0	0.5	
AB1A4M	**BB1A4M**		30	25	10	0.7	750	250	150	10	10	300 ~	2.0	0.5	
AP1A4A	**BP1A4A**	Refer to equivalent circuit B Switching	–25	–25	–10	–0.7	750	250	150	–	10	100 ~	–2.0	–0.5	
AP1L2Q	**BP1L2Q**		–25	–25	–10	–0.7	750	250	150	0.47	4.7	100 ~	–2.0	–0.5	
AP1A3M	**BP1A3M**		–25	–25	–10	–0.7	750	250	150	1.0	1.0	100 ~	–2.0	–0.5	
AP1F3P	**BP1F3P**		–25	–25	–10	–0.7	750	250	150	2.2	10	100 ~	–2.0	–0.5	
AP1J3P	**BP1J3P**		–25	–25	–10	–0.7	750	250	150	3.3	10	100 ~	–2.0	–0.5	
AP1L3N	**BP1L3N**		–25	–25	–10	–0.7	750	250	150	4.7	10	100 ~	–2.0	–0.5	
AP1A4M	**BP1A4M**		–25	–25	–10	–0.7	750	250	150	10	10	100 ~	–2.0	–0.5	

■ AQ1[ ] series (TO-92)

Part number	Equivalent circuit and applications	Absolute maximum ratings (T_A = 25 °C)						Electrical characteristics (T_A = 25 °C)					
		V_{CBO}	V_{CEO}	V_{EBO}	I_C	P_T	T_j	R_1	R_2	h_{FE}	V_{CE}	I_C	Remarks
TO-92		(V)	(V)	(V)	(A)	(mW)	(°C)	(kΩ)	(kΩ)		(V)	(A)	
AQ1A4A	Refer to equivalent circuit B Switching	–20	–20	–10	–2.0	750	150	–	10	150 ~	–2.0	–1.0	
AQ1F2Q		–20	–20	–10	–2.0	750	150	0.22	2.2	150 ~	–2.0	–1.0	
AQ1L2N		–20	–20	–10	–2.0	750	150	0.47	1.0	150 ~	–2.0	–1.0	
AQ1L2Q		–20	–20	–20	–10	750	150	0.47	4.7	150 ~	–2.0	–1.0	
AQ1A3M		–20	–20	–10	–2.0	750	150	1.0	1.0	150 ~	–2.0	–1.0	
AQ1F3M		–20	–20	–10	–2.0	750	150	2.2	2.2	150 ~	–2.0	–1.0	
AQ1F3P		–20	–20	–10	–2.0	750	150	4.7	10	150 ~	–2.0	–1.0	

Transistor with Internal Resistor

■ AD1[ ], AD2[ ], AR1[ ] series (TO-92)

Part number	Equivalent circuit and applications	Absolute maximum ratings (T_A = 25 °C)						Electrical characteristics (T_A = 25 °C)					
TO-92		V_{CBO} (V)	V_{CEO} (V)	V_{EBO} (V)	I_C (A)	P_T (mW)	T_j (°C)	R_1 (kΩ)	R_2 (kΩ)	h_{FE}	V_{CE} (V)	I_C (A)	Remarks
AD1A3M	Refer to equivalent circuit A Switching	80	60	10	1.0	750	150	1.0	1.0	200 ~	2.0	0.5	
AD1F3P		80	60	10	1.0	750	150	2.2	10	300 ~	2.0	0.5	
AD1L3N		80	60	10	1.0	750	150	4.7	10	300 ~	2.0	0.5	
AD1A4M		80	60	10	1.0	750	150	10	10	300 ~	2.0	0.5	
AD1L2Q		80	60	10	1.0	750	150	0.47	4.7	300 ~	2.0	0.5	
AD1F2Q		80	60	10	1.0	750	150	0.22	2.2	300 ~	2.0	0.5	
AD1A4A		80	60	10	1.0	750	150	–	10	300 ~	2.0	0.5	
AD2A3M	Refer to equivalent circuit C Switching	60 ±10	60 ±10	10	1.0	750	150	1.0	1.0	200 ~	2.0	0.5	
AD2F3P		60 ±10	60 ±10	10	1.0	750	150	2.2	10	300 ~	2.0	0.5	
AD2L3N		60 ±10	60 ±10	10	1.0	750	150	4.7	10	300 ~	2.0	0.5	
AD2A4M		60 ±10	60 ±10	10	1.0	750	150	10	10	300 ~	2.0	0.5	
AD2L2Q		60 ±10	60 ±10	10	1.0	750	150	0.47	4.7	300 ~	2.0	0.5	
AD2F2Q		60 ±10	60 ±10	10	1.0	750	150	0.22	2.2	300 ~	2.0	0.5	
AD2A4A		60 ±10	60 ±10	10	1.0	750	150	–	10	300 ~	2.0	0.5	
AR1A3M	Refer to equivalent circuit B Switching	–60	–60	–10	–1.0	750	150	1.0	1.0	100 ~	–2.0	–0.5	
AR1F3P		–60	–60	–10	–1.0	750	150	2.2	10	100 ~	–2.0	–0.5	
AR1L3N		–60	–60	–10	–1.0	750	150	4.7	10	100 ~	–2.0	–0.5	
AR1A4M		–60	–60	–10	–1.0	750	150	10	10	100 ~	–2.0	–0.5	
AR1L2Q		–60	–60	–10	–1.0	750	150	0.47	4.7	100 ~	–2.0	–0.5	
AR1F2Q		–60	–60	–10	–1.0	750	150	0.22	2.2	100 ~	–2.0	–0.5	
AR1A4A		–60	–60	–10	–1.0	750	150	–	10	100 ~	–2.0	–0.5	

■ CE1[ ], CE2[ ] series (SP-8)

Part number	Equivalent circuit and applications	Absolute maximum ratings (T_A = 25 °C)						Electrical characteristics (T_A = 25 °C)				
SP-8		V_{CBO} (V)	V_{CEO} (V)	V_{EBO} (V)	I_C (A)	P_T (W)	T_j (°C)	R_1 (kΩ)	R_2 (kΩ)	h_{FE}	V_{CE} (V)	I_C (A)
CE1A3Q	Refer to D	60 ±10	60 ±10	15	±2.0	1.0	150	1.0	10	1000 to 3000	5.0	1.0
CE1F3P		60 ±10	60 ±10	15	±2.0	1.0	150	2.2	10	1000 to 3000	5.0	1.0
CE1N2R		60 ±10	60 ±10	15	±2.0	1.0	150	0.68	10	1000 to 3000	5.0	1.0
CE2A3Q	Refer to E	60	60	15	±2.0	1.0	150	1.0	10	1000 to 3000	5.0	1.0
CE2F3P		60	60	15	±2.0	1.0	150	1.0	10	1000 to 3000	5.0	1.0

Transistor with Internal Resistor

■ FA1[ ], FN1[ ], GA1[ ], GN1[ ] series (SC-59, SC-70)

Part number		Equivalent circuit and applications	Absolute maximum ratings (T_A = 25 °C)						Electrical characteristics (T_A = 25 °C)					
SC-59	SC-70		V_{CBO} (V)	V_{CEO} (V)	V_{EBO} (V)	I_C (mA)	P_T (mW)	T_j (°C)	R_1 (kΩ)	R_2 (kΩ)	h_{FE}	V_{CE} (V)	I_C (mA)	Remarks
FA1A4M	**GA1A4M**	Refer to equivalent circuit [A] Switching	60	50	10	100	200 (SC-59) 150 (SC-70)	150	10	10	35 to 100	5.0	5.0	
FA1A4P	**GA1A4P**		60	50	5	100		150	10	47	35 to 340	5.0	5.0	
FA1F4M	**GA1F4M**		60	50	10	100		150	22	22	60 to 195	5.0	5.0	
FA1F4N	**GA1F4N**		60	50	5	100		150	22	47	85 to 340	5.0	5.0	
FA1L4M	**GA1L4M**		60	50	10	100		150	47	47	85 to 340	5.0	5.0	
FA1L4L	**GA1L4L**		60	50	15	100		150	47	22	60 to 195	5.0	5.0	
FA1L3Z	**GA1L3Z**		60	50	5	100		150	4.7	–	135 to 600	5.0	5.0	
FA1A4Z	**GA1A4Z**		60	50	5	100		150	10	–	135 to 600	5.0	5.0	
FA1F4Z	**GA1F4Z**		60	50	5	100		150	22	–	135 to 600	5.0	5.0	
FA1L4Z	**GA1L4Z**		60	50	5	100		150	47	–	135 to 600	5.0	5.0	
FA1L3M	**GA1L3M**		60	50	10	100		150	4.7	4.7	20 to 80	5.0	5.0	
FA1L3N	**GA1L3N**		60	50	5	100		150	4.7	10	35 to 100	5.0	5.0	
FA1A3Q	**GA1A3Q**		60	50	5	100		150	1.0	10	35 to 100	5.0	5.0	
FN1A4M	**GN1A4M**	Refer to equivalent circuit [B] Switching	–60	–50	–10	–100	200 (SC-59) 150 (SC-70)	150	10	10	35 to 100	–5.0	–5.0	
FN1A4P	**GN1A4P**		–60	–50	–5	–100		150	10	47	85 to 340	–5.0	–5.0	
FN1F4M	**GN1F4M**		–60	–50	–10	–100		150	22	22	60 to 195	–5.0	–5.0	
FN1F4N	**GN1F4N**		–60	–50	–5	–100		150	22	47	85 to 340	–5.0	–5.0	
FN1L4M	**GN1L4M**		–60	–50	–10	–100		150	47	47	85 to 340	–5.0	–5.0	
FN1L4L	**GN1L4L**		–60	–50	–15	–100		150	47	22	60 to 195	–5.0	–5.0	
FN1L3Z	**GN1L3Z**		–60	–50	–5	–100		150	4.7	–	135 to 600	–5.0	–5.0	
FN1A4Z	**GN1A4Z**		–60	–50	–5	–100		150	10	–	135 to 600	–5.0	–5.0	
FN1F4Z	**GN1F4Z**		–60	–50	–5	–100		150	22	–	135 to 600	–5.0	–5.0	
FN1L4Z	**GN1L4Z**		–60	–50	–5	–100		150	47	–	135 to 600	–5.0	–5.0	
FN1L3M	**GN1L3M**		–60	–50	–10	–100		150	4.7	4.7	20 to 80	–5.0	–5.0	
FN1L3N	**GN1L3N**		–60	–50	–5	–100		150	4.7	10	35 to 100	–5.0	–5.0	
FN1A3Q	**GN1A3Q**		–60	–50	–5	–100		150	1.0	10	35 to 100	–5.0	–5.0	

■ FB1[ ] series (SC-59)

Part number	Equivalent circuit and applications	Absolute maximum ratings (T_A = 25 °C)						Electrical characteristics (T_A = 25 °C)					
		V_{CBO} (V)	V_{CEO} (V)	V_{EBO} (V)	I_C (A)	P_T (mW)	T_j (°C)	R_1 (kΩ)	R_2 (kΩ)	h_{FE}	V_{CE} (V)	I_C (A)	Remarks
FB1A4A	Refer to equivalent circuit [A] Switching	30	25	10	0.7	200	150	–	10	300	2.0	0.5	
FB1L2Q		30	25	10	0.7		150	0.47	4.7	300	2.0	0.5	
FB1A3M		30	25	10	0.7		150	1.0	1.0	100	2.0	0.5	
FB1F3P		30	25	10	0.7		150	2.2	10	300	2.0	0.5	
FB1J3P		30	25	10	0.7		150	3.3	10	300	2.0	0.5	
FB1L3N		30	25	10	0.7		150	4.7	10	300	2.0	0.5	
FB1A4M		30	25	10	0.7		150	10	10	300	2.0	0.5	
FP1A4A	Refer to equivalent circuit [B] Switching	–25	–25	–10	–0.7	200	150		10	100	–2.0	–0.5	
FP1L2Q		–25	–25	–10	–0.7		150	0.47	4.7	100	–2.0	–0.5	
FP1A3M		–25	–25	–10	–0.7		150	1.0	1.0	100	–2.0	–0.5	
FP1F3P		–25	–25	–10	–0.7		150	2.2	10	100	–2.0	–0.5	
FP1J3P		–25	–25	–10	–0.7		150	3.3	10	100	–2.0	–0.5	
FP1L3N		–25	–25	–10	–0.7		150	4.7	10	100	–2.0	–0.5	
FP1A4M		–25	–25	–10	–0.7		150	10	10	100	–2.0	–0.5	

Transistor with Internal Resistor

■ HD1[ ], HD2[ ], HR1[ ] series (SOT-89)

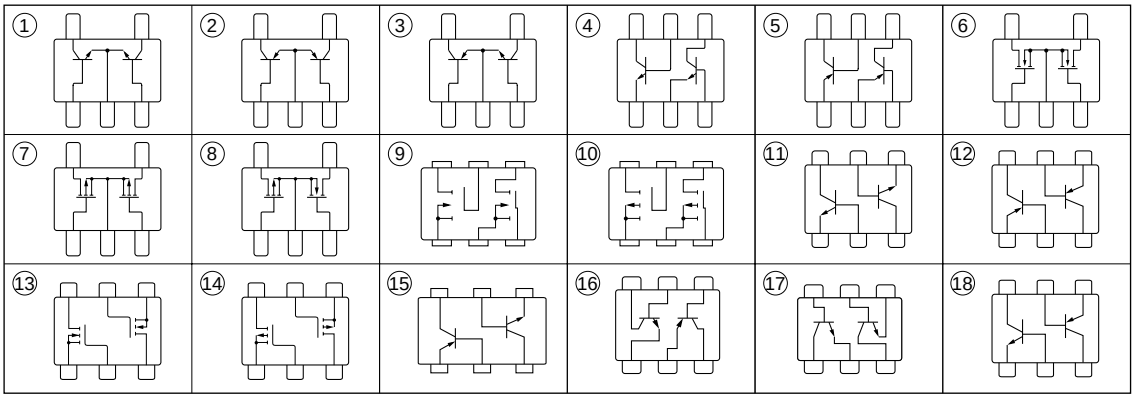
Part number	Equivalent circuit and applications	Absolute maximum ratings (T_A = 25 °C)						Electrical characteristics (T_A = 25 °C)					
		V_{CBO} (V)	V_{CEO} (V)	V_{EBO} (V)	I_C (A)	P_T (mW)	T_j (°C)	R_1 (kΩ)	R_2 (kΩ)	h_{FE}	V_{CE} (V)	I_C (A)	Remarks
HD1A3M	Refer to equivalent circuit A Switching	80	60	10	1.0	2.0	150	1.0	1.0	200	2.0	0.5	
HD1F3P		80	60	10	1.0		150	2.2	10	300	2.0	0.5	
HD1L3N		80	60	10	1.0		150	4.7	10	300	2.0	0.5	
HD1A4M		80	60	10	1.0		150	10	10	300	2.0	0.5	
HD1L2Q		80	60	10	1.0		150	0.47	4.7	300	2.0	0.5	
HD1F2Q		80	60	10	1.0		150	0.22	2.2	300	2.0	5.0	
HD1A4A		80	60	10	1.0		150	–	10	300	2.0	0.5	
HD2A3M	Refer to equivalent circuit C Switching	60±10	60±10	10	1.0	2.0	150	1.0	1.0	200	2.0	0.5	
HD2F3P		60±10	60±10	10	1.0		150	2.2	10	300	2.0	0.5	
HD2L3N		60±10	60±10	10	1.0		150	4.7	10	300	2.0	0.5	
HD2A4M		60±10	60±10	10	1.0		150	10	10	300	2.0	0.5	
HD2L2Q		60±10	60±10	10	1.0		150	0.47	4.7	300	2.0	0.5	
HD2F2Q		60±10	60±10	10	1.0		150	0.2	2.2	300	2.0	0.5	
HD2A4A		60±10	60±10	10	1.0		150	–	10	300	2.0	0.5	
HR1A3M	Refer to equivalent circuit B Switching	–60	–60	–10	–1.0	2.0	150	1.0	1.0	100	–2.0	–0.5	
HR1F3P		–60	–60	–10	–1.0		150	2.2	10	100	–2.0	–0.5	
HR1L3N		–60	–60	–10	–1.0		150	4.7	10	100	–2.0	–0.5	
HR1A4M		–60	–60	–10	–1.0		150	10	10	100	–2.0	–0.5	
HR1L2Q		–60	–60	–10	–1.0		150	0.47	4.7	100	–2.0	–0.5	
HR1F2Q		–60	–60	–10	–1.0		150	0.22	2.2	100	–2.0	–0.5	
HR1A4A		–60	–60	–10	–1.0		150	–	10	100	–2.0	–0.5	

■ HQ1[ ] series (SOT-89)

Part number	Equivalent circuit and applications	Absolute maximum ratings (T_A = 25 °C)						Electrical characteristics (T_A = 25 °C)					
		V_{CBO} (V)	V_{CEO} (V)	V_{EBO} (V)	I_C (A)	P_T (mW)	T_j (°C)	R_1 (kΩ)	R_2 (kΩ)	h_{FE}	V_{CE} (V)	I_C (A)	Remarks
HQ1A4A	Refer to equivalent circuit B Switching	–20	–20	–10	–2.0	2.0	150	–	10	150	–2.0	–1.0	
HQ1F2Q		–20	–20	–10	–2.0		150	0.22	2.2	150	–2.0	–1.0	
HQ1L2N		–20	–20	–10	–2.0		150	0.47	1.0	150	–2.0	–1.0	
HQ1L2Q		–20	–20	–20	–10		150	0.47	4.7	150	–2.0	–1.0	
HQ1A3M		–20	–20	–10	–2.0		150	1.0	1.0	150	–2.0	–1.0	
HQ1F3M		–20	–20	–10	–2.0		150	2.2	2.2	150	–2.0	–1.0	
HQ1F3P		–20	–20	–10	–2.0		150	4.7	10	150	–2.0	–1.0	

Transistor for Array

Equivalent Circuit



■ Bipolar Transistor

Part number	Equivalent circuit	V_{CEO} (V)	$I_{C(DC)}$ (mA)	P_T* (mW)	h_{FE}	V_{CE} (V)	I_C (mA)	Package
μPA570T	①	50	100	200	90 to 600	6	1	• 5-pin Super Mini-Mold SC-88A
μPA571T	②	−50	−100	200		−6	−1	
μPA574T	③	50/−50	100/−100	300		6/−6	1/−1	
μPA670T	⑪	50	100	200		6	1	• 6-pin Super Mini-Mold SC-88
μPA671T	⑫	−50	−100	200		−6	−1	
μPA500T	①	50	100	300		6	1	• 5-pin Mini-Mold SC-74A
μPA501T	②	−50	−100	300		−6	−1	
μPA504T	③	50/−50	100/−100	300		6/−6	1/−1	
μPA600T	④	50	100	300		6	1	• 6-pin Mini-Mold SC-74
μPA601T	⑤	−50	−100	300		−6	−1	
μPA604T	⑪	50	100	300		6	1	
μPA605T	⑫	−50	−100	300		−6	−1	
μPA608T	⑮	50 −16	100 −500	300	90 to 600 110 to 400	6 −1	1 −100	
μPA609T	⑯	40 −40	500 −500	300	75 to 300 75 to 300	1 −1	150 −150	
μPA673T	⑰	15	200	200	82 to 180	6	1	• 6-pin Super Mini-Mold SC-88
μPA674T	⑱	15 −15	200 −50	200	40 to 200 50 to 150	1 −1	10 −10	

*: Total

■ MOS FET

Part number	Equivalent circuit	V_{DSS} (V)	$I_{D(DC)}$ (mA)	P_T* (mW)	$R_{DS\ (on)}$ (Ω) MAX	V_{GS} (V)	I_D (mA)	Package
μPA572T	⑥	30	100	200	13	2.5	10	• 5-pin Super Mini-Mold SC-88A
μPA573T	⑦	−30	−100	200	60	−2.5	−10	
μPA502T	⑥	50	100	300	30	4	10	
μPA503T	⑦	−50	−100	300	100	−4	−10	• 5-pin Mini-Mold SC-74A
μPA505T	⑧	50/−50	100/−100	300	30/100	4/−4	10/−10	
μPA602T	⑨	50	100	300	30	4	10	• 6-pin Mini-Mold SC-74
μPA603T	⑩	−50	−100	300	100	−4	−10	
μPA606T	⑬	50	100	300	30	4	10	
μPA607T	⑭	−50	−100	300	100	−4	−10	
μPA610TA	⑩	−30	−100	300	60	−2.5	−1	
μPA611TA	⑨	30	100	300	15	2.5	1	
μPA672T	⑬	50	100	200	40	2.5	10	• 6-pin Super Mini-Mold SC-88

*: Total

Transistor Array

■ Signal transistor array

Part number	Number of circuits				Output voltage V_O (V)	Output current I_O (mA/Unit)	Surge absorber diode	Input impedance	Ative level		Input/output relationship		Output current relationship		Remarks	Number of pins
	5	6	7	8					"H"	"L"	Inverter type	Non-iverter type	Source type	Sink type		
μPA80C, GR			○		60	50	–	20 kΩ		○	○		○		Darlington	16
μPA81C			○		45	400∗2	–	20 kΩ	○		○			○	FIP driver	18
μPA2001C, GR			○		60	500	○	–	○		○			○	Darlington	16
μPA2002C, GR			○		60	500	○	7VZDi +10.5 kΩ	○		○			○	Darlington	16
μPA2003C, GR			○		60	500	○	2.7 kΩ	○		○			○	Darlington	16
μPA2004C, GR			○		60	500	○	10.5 kΩ	○		○			○	Darlington	16
μPA2981C				○	50	500	○	3 kΩ	○			○	○		Darlington output stage	18
μPA2982C				○	50	500	○	8.5 kΩ	○			○	○		Darlington output stage	18
μPA2987C			○		50	500	○	3 kΩ	○			○	○		Darlington output stage	16

∗1: PW ≦ 30 ms, duty cycle ≦ 10% (7 circuits operation)
∗2: PW ≦ 10 ms, duty cycle ≦ 10% (7 circuits operation)

■ Signal transistor array (Monolithic)

I_O (mA) / V_O (V)	500
50	μPA2981 μPA2982 μPA2987
60	μPA2001 μPA2002 μPA2003 μPA2004

Transistor Array

■ Power transistor array

Part number	Number of circuits	Collector-emitter voltage V_{CE} (V)	Collector current I_C (A)	Surge absorber diode	Input impedance	Output current relationship		Remarks	Number of pins
						Source type	Sink type		
μPA1428AH	4	60 ±10	±2.0	○	–		○	Darlington	10
μPA1434H	4	60	3.0	–	–		○	High beta	10
μPA1436AH	4	100	±3.0	–	–		○	Darlington	10
μPA1437H	4	–100	∓3.0	–	–	○		Darlington	10
μPA1453H	4	–60	5.0	–	–	○		Low $V_{CE(sat)}$	10
μPA1454H	4	100	5.0	–	–		○	High beta	10
μPA1456H	4	100	±5.0	–	–		○	Darlington	10
μPA1458H	4	60 ±10	±5.0	○	–		○	Darlington	10
μPA1476H	4	100 ±15	±2.0	○	–		○	Darlington	10
μPA1478H	4	31 ±4	±2.0	○	–		○	Darlington	10

■ Power transistor array (Multi-chip, 4 circuits)

V_{CEO} (V) \ $I_{C\ (DC)}$ (A)	Polarity	2.0	3.0	5.0
30	NPN	μPA1478H ○		
	PNP			
60	NPN	μPA1428AH ○	μPA1434H ●	μPA1458H ○
	PNP			μPA1453H ☆
100	NPN	μPA1476H ○	μPA1436AH	μPA1454H ● μPA1456H
	PNP		μPA1437H	

○: with Zenner Diode ☆: Low $V_{CE\ (sat)}$ ●: Single High h_{FE}

Transistor Array

■ Power MOS FET array

Part number	Number of circuits			V_{DSS} (V)	I_D (A/Unit)	Surge absorber diode	4 V driving	Active level		Input/output relationship		Output current relationship		Number of pins
	4	7	8					"H"	"L"	Inverter type	Non-inverter type	Source type	Sink type	
μPA1500BH	○			60	±3	○	○	○		○			○	12
μPA1501H	○			120	±3	○	○	○		○			○	12
μPA1520BH	○			30	±2	–	○	○		○			○	10
μPA1522H	○			60	±2	–	○	○		○			○	10
μPA1523BH	○			–60	∓2	–	○		○		○	○		10
μPA1524H	○			80	±2	–	○	○		○			○	10
μPA1526H	○			100	±2	–	○	○		○			○	10
μPA1527H	○			–100	∓2	–	○		○		○	○		10
μPA1550H	○			30	±5	–	○	○		○			○	10
μPA1552BH	○			60	±5	–	○	○		○			○	10
μPA1556AH	○			100	±5	–	○	○		○			○	10
μPA1572BH	○			60	±2	–	○	○		○			○	10
μPA1576H	○			100	±2	–	○	○		○			○	10

■ Power MOS FET array (Multi-chip, 4 circuits)

$I_{D\,(DC)}$ (A) / V_{DSS} (V)	2.0	3.0	5.0
30	• μPA1520BH (0.25)		• μPA1550H (0.15)
60	• μPA1522H (0.35) • μPA1572BH (0.8) • μPA1523BH (1.3)	• μPA1500BH (0.24)	• μPA1552BH (0.24)
80	• μPA1524H (1.0)		
100	• μPA1526H (0.6) • μPA1576H (1.5) • μPA1527H (1.5)		• μPA1556AH (0.33)
120		• μPA1501H (0.53)	

( ): $R_{DS(on)}$ (@V_{GS} = 4 V) [Ω MAX.] Underlined: P-channel

■ Monolithic MOS FET array

Part number	Number of circuits			V_{DSS} (V)	I_D (A/Unit)	Surge absorber diode	4 V driving	Active level		Input/output relationship		Output current relationship		Number of pins
	4	7	8					"H"	"L"	Inverter type	Non-inverter type	Source type	Sink type	
μPA1600CX/GS			○	30	0.5	–	○	○		○			○	20
μPA1601CX/GS		○		30	0.43	–	○	○		○			○	16
μPA1602CX/GS		○		30	0.43	–	○		○		○		○	16
μPA1603CX	○			30	0.87	○	○	○		○			○	16
μPA1604CX	○			30	0.87	○	○		○		○		○	16

■ Power MOS FET array (Monolithic)

$I_{D\,(DC)}$ (A) / V_{DSS} (V)	0.43	0.5	0.87
30	• μPA1601 (5.3 Ω, 7CH) (DIP, SOP) • μPA1602 (5.3 Ω, 7CH) (DIP, SOP)	• μPA1600 (4.0 Ω, 8CH) (DIP, SOP)	• μPA1603 (1.3 Ω, 4CH) (DIP, SOP)

Zener Diode

■ Zener Diode Quick Reference (1/2)

Vz (V)	P (W)							
TYP.	0.15		0.2				1.0	
2.0		RD2.0UM	RD2.0M	RD2.0MW	RD2.0S		RD2.0P	
2.2		RD2.2UM	RD2.2M	RD2.2MW	RD2.2S		RD2.2P	
2.4		RD2.4UM	RD2.4M	RD2.4MW	RD2.4S		RD2.4P	
2.7		RD2.7UM	RD2.7M	RD2.7MW	RD2.7S		RD2.7P	
3.0		RD3.0UM	RD3.0M	RD3.0MW	RD3.0S		RD3.0P	
3.3		RD3.3UM	RD3.3M	RD3.3MW	RD3.3S		RD3.3P	
3.6		RD3.6UM	RD3.6M	RD3.6MW	RD3.6S		RD3.6P	
3.9		RD3.9UM	RD3.9M	RD3.9MW	RD3.9S		RD3.9P	
4.3		RD4.3UM	RD4.3M	RD4.3MW	RD4.3S		RD4.3P	
4.7	RD4.7UJ	RD4.7UM	RD4.7M	RD4.7MW	RD4.7S	RD4.7SL	RD4.7P	RD4.7FM
5.1	RD5.1UJ	RD5.1UM	RD5.1M	RD5.1MW	RD5.1S	RD5.1SL	RD5.1P	RD5.1FM
5.6	RD5.6UJ	RD5.6UM	RD5.6M	RD5.6MW	RD5.6S	RD5.6SL	RD5.6P	RD5.6FM
6.2	RD6.2UJ	RD6.2UM	RD6.2M	RD6.2MW	RD6.2S	RD6.2SL	RD6.2P	RD6.2FM
6.8	RD6.8UJ	RD6.8UM	RD6.8M	RD6.8MW	RD6.8S	RD6.8SL	RD6.8P	RD6.8FM
7.5	RD7.5UJ	RD7.5UM	RD7.5M	RD7.5MW	RD7.5S	RD7.5SL	RD7.5P	RD7.5FM
8.2	RD8.2UJ	RD8.2UM	RD8.2M	RD8.2MW	RD8.2S	RD8.2SL	RD8.2P	RD8.2FM
9.1	RD9.1UJ	RD9.1UM	RD9.1M	RD9.1MW	RD9.1S	RD9.1SL	RD9.1P	RD9.1FM
10	RD10UJ	RD10UM	RD10M	RD10MW	RD10S	RD10SL	RD10P	RD10FM
11	RD11UJ	RD11UM	RD11M	RD11MW	RD11S	RD11SL	RD11P	RD11FM
12	RD12UJ	RD12UM	RD12M	RD12MW	RD12S	RD12SL	RD12P	RD12FM
13	RD13UJ	RD13UM	RD13M	RD13MW	RD13S	RD13SL	RD13P	RD13FM
15	RD15UJ	RD15UM	RD15M	RD15MW	RD15S	RD15SL	RD15P	RD15FM
16	RD16UJ	RD16UM	RD16M	RD16MW	RD16S	RD16SL	RD16P	RD16FM
18	RD18UJ	RD18UM	RD18M	RD18MW	RD18S	RD18SL	RD18P	RD18FM
20	RD20UJ	RD20UM	RD20M	RD20MW	RD20S	RD20SL	RD20P	RD20FM
22	RD22UJ	RD22UM	RD22M	RD22MW	RD22S	RD22SL	RD22P	RD22FM
24	RD24UJ	RD24UM	RD24M	RD24MW	RD24S	RD24SL	RD24P	RD24FM
27	RD27UJ	RD27UM	RD27M	RD27MW	RD27S	RD27SL	RD27P	RD27FM
30	RD30UJ	RD30UM	RD30M	RD30MW	RD30S	RD30SL	RD30P	RD30FM
33	RD33UJ	RD33UM	RD33M	RD33MW	RD33S	RD33SL	RD33P	RD33FM
36	RD36UJ	RD36UM	RD36M	RD36MW	RD36S	RD36SL	RD36P	RD36FM
39	RD39UJ	RD39UM	RD39M	RD39MW	RD39S	RD39SL	RD39P	RD39FM
43			RD43M		RD43S		RD43P	RD43FM
47			RD47M		RD47S		RD47P	RD47FM
51					RD51S		RD51P	RD51FM
56					RD56S		RD56P	
62					RD62S		RD62P	
68					RD68S		RD68P	
75					RD75S		RD75P	
82					RD82S		RD82P	
91					RD91S		RD91P	
100					RD100S		RD100P	
110					RD110S		RD110P	
120					RD120S		RD120P	
Package	2-pin ultra mini-mold		Mini-mold (SC·59)		2-pin super mini-mold		Power mini-mold (SC·62)	2-pin power mini-mold
	Low noise	General Purpose			General Purpose	Low noise		

Zener Diode

■ Zener Diode Quick Reference (2/2)

V_Z (V)	P (W)				
TYP.	0.25	0.4	0.4	0.5	1.0
2.0	RD2.0HS		RD2.0ES	RD2.0E	RD2.0F
2.2	RD2.2HS		RD2.2ES	RD2.2E	RD2.2F
2.4	RD2.4HS		RD2.4ES	RD2.4E	RD2.4F
2.7	RD2.7HS		RD2.7ES	RD2.7E	RD2.7F
3.0	RD3.0HS		RD3.0ES	RD3.0E	RD3.0F
3.3	RD3.3HS		RD3.3ES	RD3.3E	RD3.3F
3.6	RD3.6HS		RD3.6ES	RD3.6E	RD3.6F
3.9	RD3.9HS		RD3.9ES	RD3.9E	RD3.9F
4.3	RD4.3HS		RD4.3ES	RD4.3E	RD4.3F
4.7	RD4.7HS	RD4.7JS	RD4.7ES	RD4.7E	RD4.7F
5.1		RD5.1JS	RD5.1ES	RD5.1E	RD5.1F
5.6		RD5.6JS	RD5.6ES	RD5.6E	RD5.6F
6.2		RD6.2JS	RD6.2ES	RD6.2E	RD6.2F
6.8		RD6.8JS	RD6.8ES	RD6.8E	RD6.8F
7.5		RD7.5JS	RD7.5ES	RD7.5E	RD7.5F
8.2		RD8.2JS	RD8.2ES	RD8.2E	RD8.2F
9.1		RD9.1JS	RD9.1ES	RD9.1E	RD9.1F
10		RD10JS	RD10ES	RD10E	RD10F
11		RD11JS	RD11ES	RD11E	RD11F
12		RD12JS	RD12ES	RD12E	RD12F
13		RD13JS	RD13ES	RD13E	RD13F
15		RD15JS	RD15ES	RD15E	RD15F
16		RD16JS	RD16ES	RD16E	RD16F
18		RD18JS	RD18ES	RD18E	RD18F
20		RD20JS	RD20ES	RD20E	RD20F
22		RD22JS	RD22ES	RD22E	RD22F
24		RD24JS	RD24ES	RD24E	RD24F
27		RD27JS	RD27ES	RD27E	RD27F
30		RD30JS	RD30ES	RD30E	RD30F
33		RD33JS	RD33ES	RD33E	RD33F
36		RD36JS	RD36ES	RD36E	RD36F
39		RD39JS	RD39ES	RD39E	RD39F
43				RD43E	RD43F
47				RD47E	RD47F
51				RD51E	RD51F
56				RD56E	RD56F
62				RD62E	RD62F
68				RD68E	RD68F
75				RD75E	RD75F
82				RD82E	RD82F
91				RD91E	
100				RD100E	
110				RD110E	
120				RD120E	
Package	DO-34	DO-34	DO-34	DO-35	DO-41

Noise Clipping Diode

■ Surface-Mount Type Quick Reference

V_{BR} (V)	P (W)					Application
TYP.	0.15	0.2				
3.3	NNCD3.3C	NNCD3.3D	NNCD3.3E	NNCD3.3F	NNCD3.3G	• E.S.D. protection • Surge absorber
3.6	NNCD3.6C	NNCD3.6D	NNCD3.6E	NNCD3.6F	NNCD3.6G	
3.9	NNCD3.9C	NNCD3.9D	NNCD3.9E	NNCD3.9F	NNCD3.9G	
4.3	NNCD4.3C	NNCD4.3D	NNCD4.3E	NNCD4.3F	NNCD4.3G	
4.7	NNCD4.7C	NNCD4.7D	NNCD4.7E	NNCD4.7F	NNCD4.7G	
5.1	NNCD5.1C	NNCD5.1D	NNCD5.1E	NNCD5.1F	NNCD5.1G	
5.6	NNCD5.6C	NNCD5.6D	NNCD5.6E	NNCD5.6F	NNCD5.6G	
6.2	NNCD6.2C	NNCD6.2D	NNCD6.2E	NNCD6.2F	NNCD6.2G	
6.8	NNCD6.8C	NNCD6.8D	NNCD6.8E	NNCD6.8F	NNCD6.8G	
7.5	NNCD7.5C	NNCD7.5D	NNCD7.5E	NNCD7.5F	NNCD7.5G	
8.2	NNCD8.2C	NNCD8.2D	NNCD8.2E	NNCD8.2F		
9.1	NNCD9.1C	NNCD9.1D	NNCD9.1E	NNCD9.1F		
10	NNCD10C	NNCD10D	NNCD10E	NNCD10F		
11	NNCD11C	NNCD11D	NNCD11E	NNCD11F		
12	NNCD12C	NNCD12D	NNCD12E	NNCD12F		
27					NNCD27G	
Package	2-pin ultra mini-mold	2-pin super mini-mold	3-pin mini-mold (SC-59)		5-pin mini-mold (SC-74A quarto)	
			Single	Twin		

■ Surface-Mount Type (Low Capacitance) Quick Reference

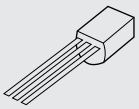
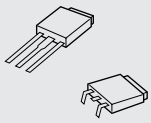
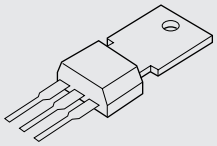
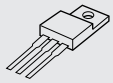
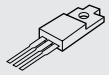
V_{BR} (V)	P (W)		Application
TYP.	0.2		
5.6	NNCD5.6LG	NNCD5.6LH	• E.S.D. protection
6.2	NNCD6.2LG	NNCD6.2LH	
6.8	NNCD6.8LG	NNCD6.8LH	
Package	5-pin mini-mold (SC-74A quad)	5-pin super mini-mold (SC-88A quad)	

■ Through Hole Type Quick Reference

V_{BR} (V)	P (W)		Application
TYP.	0.4	0.5	
3.3	NNCD3.3A	NNCD3.3B	• E.S.D. protection • Surge absorber
3.6	NNCD3.6A	NNCD3.6B	
3.9	NNCD3.9A	NNCD3.9B	
4.3	NNCD4.3A	NNCD4.3B	
4.7	NNCD4.7A	NNCD4.7B	
5.1	NNCD5.1A	NNCD5.1B	
5.6	NNCD5.6A	NNCD5.6B	
6.2	NNCD6.2A	NNCD6.2B	
6.8	NNCD6.8A	NNCD6.8B	
7.5	NNCD7.5A	NNCD7.5B	
8.2	NNCD8.2A	NNCD8.2B	
9.1	NNCD9.1A	NNCD9.1B	
10	NNCD10A	NNCD10B	
11	NNCD11A	NNCD11B	
12	NNCD12A	NNCD12B	
Package	DO-34	DO-35	

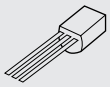
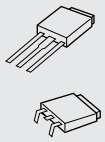
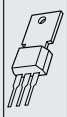
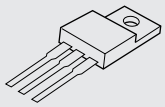
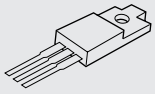
Thyristor

■ SCR

Electrical characteristics \ Package													 MP–25		 MP–45C	
		SOT-89	TO-92		MP-3			TO-202AA					TO-220AB		TO-220-like insulated type	
$I_{T(RMS)}$ (A)		0.47	0.47		4.7	8	12.6	4	4	4	4.7	4.7	8	12.6	8	12.6
$I_{T(AV)}$ (A)		0.3	0.3		3	5	8	2	2	2.5	3	3	5	8	5	8
I_{TSM} (A)		6	8		40	65	80	20	20	45	65	65	80	100	80	100
T_j (°C)		125	125		125			125					125		125	
V_{DRM} V_{RRM} (V)	200	03P2J	03P2M	–	–	–	–	–	–	2S2M	–	–	–	–	8P2M	–	8P2SM
	400	03P4J	03P4M	03P4MF*1	03P4MG	3P4J 3P4J-Z	5P4J 5P4J-Z	8P4J 8P4J-Z	2P4M	2S4M	2V5P4M	3P4MH	3S4M	5P4M	8P4M	5P4SM	8P4SM
	500	03P5J	–	–	03P5MG	–	–	–	2P5M	–	–	3P5MH	–	5P5M	–	–	–
	600	–	–	–	03P6MG	–	5P6J 5P6J-Z	–	2P6M	–	–	3P6MH	–	5P6M	–	5P6SM	–
I_{GT} (mA)		0.2	0.2	100 μA	3-50 μA	0.1	0.2	10	0.2	0.3	0.1	0.2	30	10	10	10	10
tq (TYP.) (μs)		25	25	40	60	30	80	50	30	15*2	80	80	5*2	50	100	50	100

*1: V_{RRM} = 5 V
*2: indicates MAX.

■ TRIAC

Electrical characteristics \ Package								 MP–25					 MP–45C					
		SOT-89	TO-92		MP-3		TO-202AA	TO-220AB					TO-220AB-like insulated type					
$I_{T(RMS)}$ (A)		1.0	0.8	1.0	3	5	3	5	8	10	12	16	3	5	8	10	12	16
$I_{TSM(50\ Hz)}$ (A)		7.0	7	9	30	50	30	50	80	80	100	150	30	50	80	80	100	150
T_j (°C)		125	125		125		125	125					125					
V_{DRM} V_{RRM} (V)	400	AC01DJM	AC0V 8DGM	AC01 DGM	AC03 DJM AC03 DJM-Z	AC05 DJM AC05 DJM-Z	AC03 DGM	AC05 DGM	AC08 DGM	AC10 DGM	AC12 DGM	AC16 DGM	AC03 DSM	AC05 DSM	AC08 DSM	AC10 DSM	AC12 DSM	AC16 DSM
	500	–	–	–	–	–	AC03 EGM	AC05 EGM	AC08 EGM	AC10 EGM	AC12 EGM	AC16 EGM	–	AC05 ESM	AC08 ESM	AC10 ESM	AC12 ESM	–
	600	–	–	–	AC03 FJM AC03 FJM-Z	AC05 FJM AC05 FJM-Z	AC03 FGM	AC05 FGM	AC08 FGM	AC10 FGM	AC12 FGM	AC16 FGM	AC03 FSM	AC05 FSM	AC08 FSM	AC10 FSM	AC12 FSM	AC16 FSM
I_{GT} (mA)	I	5	5	3	12	10	12	10	20	20	20	30	12	10	20	20	20	30
	II	10	10	5	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	III	5	5	3	12	10	12	10	20	20	20	30	12	10	20	20	20	30
	IV	10	10	3	12	10	12	10	20	20	20	30	12	10	20	20	20	30
$(dv/dt)_C$ (V/μs)	MIN.	0.5	1	0.5	5	5	5	5	10	10	10	10	5	5	10	10	10	10

[MEMO]

GaAs Device/Silicon Microwave Semiconductor

GaAs Device 230

- GaAs MMIC 230
- GaAs MCM 231
- Field Effect Transistor 232

Silicon Microwave Semiconductor 234

- Si MMIC 234
- Transistor, FET 241

GaAs MMIC

■ Analog IC

Part number	Function	Features	Package
μPG100	Wide band, low noise amplifier	f = 50 MHz to 3 GHz Gp ≧ 14 dB, N_F ≦ 3.5 dB	• 8-pin ceramic • Chip
μPG101	Wide band, medium output amplifier	f = 50 MHz to 3 GHz P_O (1dB) ≧ 16 dBm	
μPG103	Wide band, low noise amplifier	f = 50 MHz to 3 GHz Gp ≧ 10 dB, N_F ≦ 5 dB	
μPG105-1	S band, low noise amplifier	f = 2.7 to 4.2 GHz Gp ≧ 21 dB, N_F ≦ 2.2 dB	• 16-pin ceramic
μPG106	AGC amplifier	f = 100 kHz to 2.5 GHz Gp ≧ 16 dB, G_{AGC} ≧ 25 dB	• 8-pin ceramic
μPG107	SPDT switch	f = DC to 3.4 GHz L_{ins} ≦ 1.7 dB, t_{sw} = 5 ns (typ.)	• 8-pin ceramic • Chip
μPG110	Wide band, medium output amplifier	f = 2 to 8 GHz P_O (1dB) ≧ 10 dBm, Gp ≧ 12 dB	• 4-pin ceramic • Chip
μPG132	SPDT switch	f = 0.1 to 2.5 GHz, P_O (1 dB) = 28 dBm L_{ins} = 0.6 dB, I_{SO} = 23 dB at 2 GHz	• 8-pin SSOP
μPG133	SPDT switch	f = 0.1 to 2.5 GHz, P_O = 25 dBm	
μPG137 **μPG138**	SPDT switch	f = 0.1 to 2.5 GHz, P_O (1 dB) = 34 dBm L_{ins} = 0.55 dB, at 1 GHz	
μPG139	DPDT switch	f = 0.1 to 2.5 GHz, @1 GHz L_{ins} = 0.6 dB	
μPG152	SPDT switch	f = 0.1 to 2.5 GHz, P_O (1 dB) = 30 dBm L_{ins} = 0.6 dB, I_{SO} = 22 dB at 2 GHz	• 6-pin mini-mold
μPG153	SPDT switch	f = 0.1 to 2.5 GHz, P_O (1 dB) = 33 dBm L_{ins} = 0.7 dB, I_{SO} = 13 dB@f = 2 GHz	• 6-pin super mini-mold
μPG154	SPDT switch	f = 0.1 to 2.5 GHz, P_O (1 dB) = 30 dBm L_{ins} = 0.65 dB, I_{SO} = 21 dB@f = 2 GHz V_{DD} = 3 V	
μPG155	SPDT switch	f = 0.1 to 2.5 GHz, P_O (1 dB) = 34 dBm L_{ins} = 0.75 dB, I_{SO} = 16 dB@f = 2 GHz	
μPG170	PA driver, medium output	V_{DD} = 3.6 V, P_O (1 dB) = 14.5 dBm Gp = 11 dB, I_{DD} = 30 mA, P_{adj1} = –60 dBc @f = 1.5 GHz, Pout = +12 dBm	
μPG171	PA amplifier for PHS	V_{DD} = 3.0 V, I_{DD} = 160 mA Gp ≧ 20 dB, @P_O = 21.5 dBm, f = 1.9 GHz P_{adj1} = –55 dBc	• 8-pin SSOP
μPG172	PA driver with AGC amplifier	V_{DD} = 3.0 V, I_{DD} = 55 mA Gp ≧ 26 dB, P_{adj1} = –60 dBc, ΔG = 34 dB @Pout = +12 dBm, f = 1.5 GHz	
μPG173	PA driver	V_{DD} = 2.8 V, I_{DD} = 25 mA P_O = +9 dB @925 to 960 MHz	• 6-pin mini-mold
μPG174	PA driver	V_{DD} = 2.8 V, I_{DD} = 35 mA P_O = +10 dB @1429 to 1453 MHz	
μPG17×★	PA driver with AGC amplifier	V_{DD} = 3.0 V, I_{DD} = 30 mA ΔG = 30 dB	

★: Under development

GaAs MMIC

■ Prescaler IC

Part number	Function	Features	Package
μPG501	5 GHz, 1/4 static type prescaler	f = 1.5 to 5 GHz @Pin = +10 dBm	• 8-pin ceramic • Chip
μPG502	5 GHz, 1/2 static type prescaler	f = 1 to 5 GHz @Pin = +10 dBm	
μPG503	9 GHz, 1/4 dynamic type prescaler	f = 3.5 to 90 GHz @Pin = +10 dBm	
μPG504	9 GHz, 1/2 dynamic type prescaler	f = 2.0 to 9.0 GHz @Pin = +10 dBm	
μPG506	14 GHz, 1/8 dynamic type prescaler	f = 8 to 14 GHz @Pin = 6 dBm	
μPG508	14 GHz, 1/4 dynamic type prescaler	f = 8 to 14 GHz @Pin = 6 dBm	

GaAs MCM

Part number	Function	Features	Package
MC-7660	for PDC 900 MHz f = 925 to 960 MHz	V_{DD} = 3.5 V, P_O = 30.2 dBm Gp ≧ 22.2 dB, η_T = 53% typ.	$10 \times 10 \times 2.2$ mm
MC-7661	for PDC 1.5 GHz f = 1429 to 1453 MHz	V_{DD} = 3.5 V, P_O = 30.5 dBm Gp ≧ 20.5 dB, η_T = 53% typ.	
MC-7680★	for GSM class IV f = 880 to 915 MHz	V_{DD} = 3.0 V, P_O = 34.5 dBm Gp ≧ 23.5 dB, η_T = 50% typ.	
MC-7712	for 50 to 770 MHz *1 CATV Amp. (push-pull)	GL_2 = 18.0 dB min., NF_2 = 7.0 dB min. CTB = –55 dB max., @f = 770 MHz, V_{DD} = 24 V	7-pin special package
MC-7716		GL_2 = 21.5 dB min., NF_2 = 6.5 dB max. CTB = –55 dB max., @f = 770 MHz, V_{DD} = 24 V	
MC-7722	for 50 to 770 MHz *2 CATV Amp. (Power doubler)	GL_2 = 18.5 dB min., NF_2 = 7.0 dB min. CTB = –54 dB max., @f = 770 MHz, V_{DD} = 24 V	
MC-7726		GL_2 = 21.5 dB min., NF_2 = 6.5 dB max. CTB = –54 dB max., @f = 770 MHz, V_{DD} = 24 V	
MC-7852	for 50 to 860 MHz *1 CATV Amp. (push-pull)	GL_2 = 18.0 dB min., NF_2 = 7.0 dB max. CTB = –55 dB max., @f = 860 MHz, V_{DD} = 24 V	
MC-7862		GL_2 = 18.5 dB min., NF_2 = 7.0 dB max. CTB = –54 dB max., @f = 860 MHz, V_{DD} = 24 V	
MC-7856	for 50 to 860 MHz *2 CATV Amp. (Power doubler)	GL_2 = 21.5 dB min., NF_2 = 6.5 dB max. CTB = –55 dB max., @f = 860 MHz, V_{DD} = 24 V	
MC-7866		GL_2 = 21.5 dB min., NF_2 = 6.5 dB max. CTB = –54 dB max., @f = 860 MHz, V_{DD} = 24 V	
MC-7816	for 50 to 860 MHz *3 CATV Amp. (push-pull)	GL = 21.5 dB min., NF = 7.0 dB max. CTB = –52 dB max., @f = 860 MHz, V_{DD} = 24 V	
MC-7826	for 50 to 860 MHz *3 CATV Amp. (Power doubler)	GL = 21.5 dB min., NF = 7.0 dB max. CTB = –58 dB max., @f = 860 MHz, V_{DD} = 24 V	
MC-7900★	for PDC 900 MHz f = 889 to 960 MHz	V_{DD} = 3.5 V, P_O = 29.7 dBm Gp ≧ 21.7 dB, η_T = 55% typ.	$7.0 \times 7.0 \times 2.0$ mm
MC-7901★	for PDC 1.5 GHz f = 1429 to 1453 MHz	V_{DD} = 3.5 V, P_O = 30.5 dBm Gp ≧ 20.5 dB, η_T = 55% typ.	

*1: V_O = 44 dBmV, 110 ch Flat

*2: V_O = 50 dBmV, 110 ch 10 dB Tilt

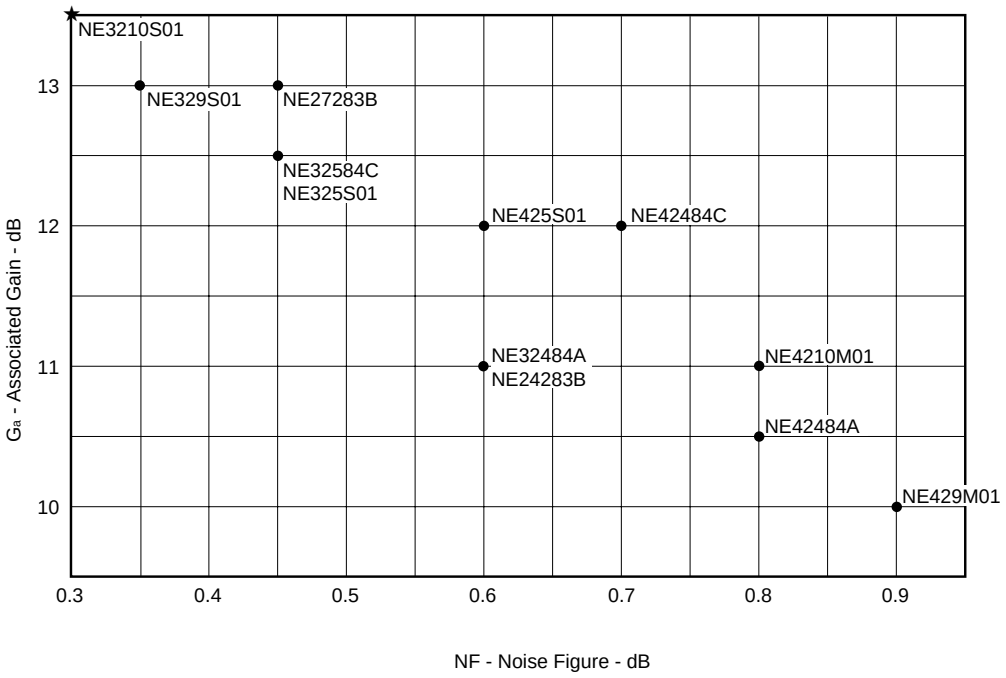
*3: V_O = 44 dBmV, 129 ch Flat

★: Under development

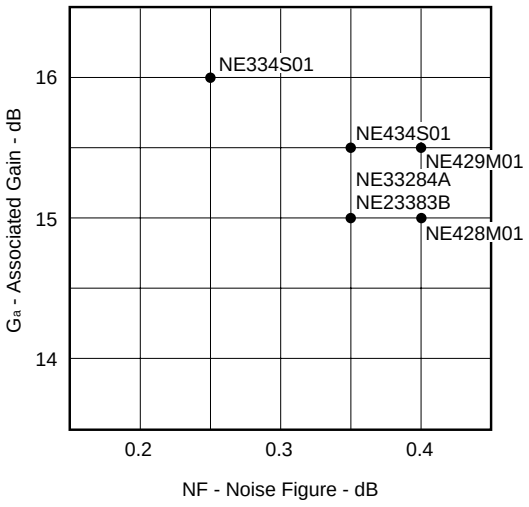
Field Effect Transistor

■ Low Noise GaAs HJ-FET Quick Reference Graph

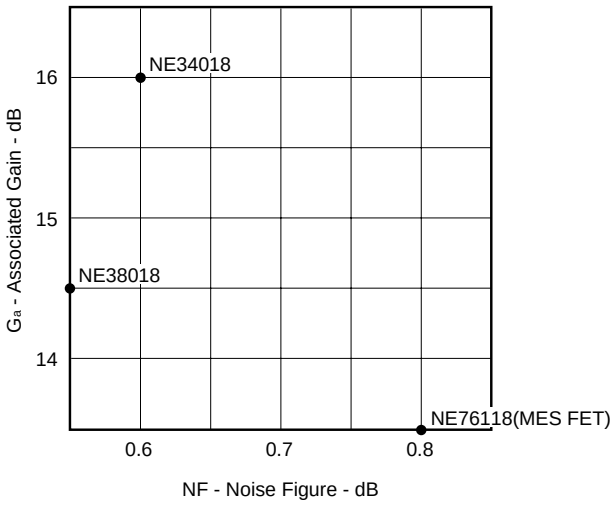
@ f = 12 GHz



@ f = 4 GHz



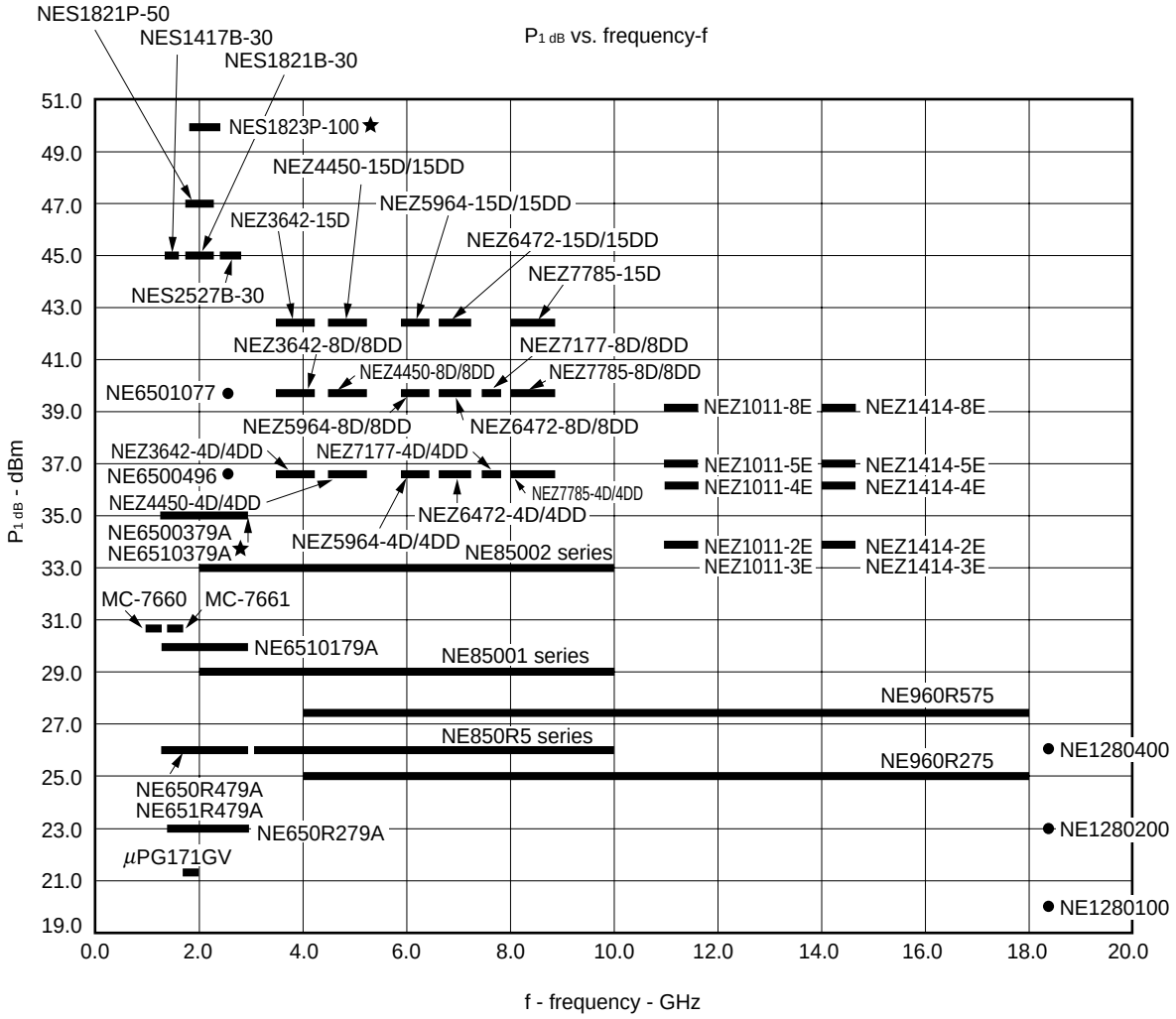
@ f = 2 GHz



● : In mass production
★ : Under development

Field Effect Transistor

■ Power GaAs FET Quick Reference Graph



★: Under development

Si MMIC

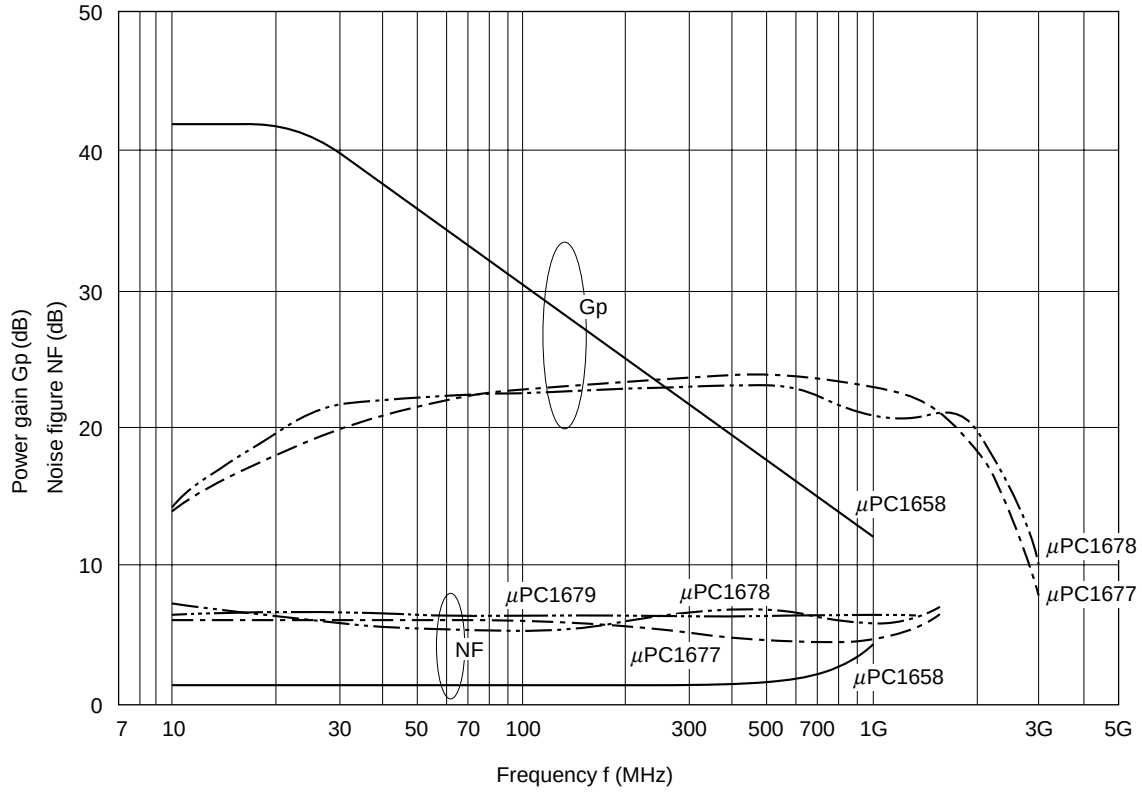
Part number	Function	Features	Package
μPC1652	General purpose UHF wide band amplifier	BW = 1.2 GHz, Gp = 18 dB, NF = 5.5 dB	• 8-pin SOP
μPC1655		BW = 1.0 GHz, Gp = 18 dB, NF = 5.5 dB	• 8-pin DIP
μPC1656		BW = 0.85 GHz, Gp = 19 dB, NF = 5.5 dB	
μPC1675		BW = 1.9 GHz, Gp = 12 dB, NF = 5.5 dB	• 4-pin mini-mold
μPC1676		BW = 1.2 GHz, Gp = 22 dB, NF = 4.5 dB	
μPC1688		BW = 1.1 GHz, Gp = 21 dB, NF = 4.0 dB	
μPC1658	Low noise HF wide band amplifier	Gp = 31 dB/f = 100 MHz, Gp = 17 dB/f = 500 MHz, GAIN DETERMINED BY EXTERNAL RESISTOR NF = 1.5 dB/f = 100 MHz, NF = 2.0 dB/f = 500 MHz	• 8-pin SOP • 8-pin DIP
μPC1663	Ultra high speed video amplifier	BW = 700 MHz/Ga = 20 dB, GAIN DETERMINED BY EXTERNAL RESISTOR INPUT NOISE 3 μVr.m.s.	• 8-pin DIP • 8-pin SOP/SSOP
μPC1677	Wide band, medium output amplifier	BW = 1.8 GHz, Gp = 24 dB, NF = 6 dB, P_O = 19.5 dBm	• 8-pin DIP
μPC1678		BW = 2.0 GHz, Gp = 23 dB, NF = 6 dB, P_O = 17.5 dBm	• 8-pin SOP/SSOP
μPC2708	Middle power wide band amplifier	BW = 2.9 GHz, $P_{o(sat)}$ = +10 dBm, Gp = 15 dB	• 6-pin mini/ super mini-mold
μPC2709		BW = 2.3 GHz, $P_{o(sat)}$ = +11.5 dBm, Gp = 23 dB	
μPC2710		BW = 1.0 GHz, $P_{o(sat)}$ = +13.5 dBm, Gp = 33 dB	
μPC2711	Wide band amplifier	BW = 2.9 GHz, Gp = 13 dB, NF = 5.0 dB	
μPC2712		BW = 2.6 GHz, Gp = 20 dB, NF = 4.5 dB	
μPC2713		BW = 1.2 GHz, Gp = 29 dB, NF = 3.2 dB	• 6-pin mini-mold
μPC2714	Low power consumption wide band amplifier	BW = 1.8 GHz, Icc = 4.5 mA, Vcc = 3.4 V	
μPC2715		BW = 1.2 GHz, Icc = 4.5 mA, Vcc = 3.4 V	
μPC2745	Wide band amplifier	BW = 2.7 GHz, Icc = 7.5 mA, Vcc = 3.0 V, Gp = 12 dB	• 6-pin mini/ super mini-mold
μPC2746		BW = 1.5 GHz, Icc = 7.5 mA, Vcc = 3.0 V, Gp = 19 dB	
μPC2747	900 MHz Band Low noise amplifier	BW = 0.1 to 1.8 GHz, Icc = 5.0 mA, Vcc = 3.0 V, Gp = 12 dB	
μPC2748		BW = 0.2 to 1.5 GHz, Icc = 6.0 mA, Vcc = 3.0 V, Gp = 19 dB	
μPC2749	1.9 GHz Low noise amplifier	BW = 2.9 GHz, Icc = 6 mA, Gp = 16 dB, NF = 4 dB	
μPC2762	Medium power amplifier	BW = 2.9 GHz, Icc = 26.5 mA, Gp = 13 dB, P_{1dB} = +8 dBm	
μPC2763		BW = 2.4 GHz, Icc = 27 mA, Gp = 20 dB, P_{1dB} = +9.5 dBm	
μPC2771		BW = 2.1 GHz, Icc = 36 mA, Gp = 21 dB, P_{1dB} = +11.5 dBm	
μPC2776	Middle power wide band amplifier	BW = 2.7 GHz, P_{1dB} = +6.5 dBm, Gp = 23 dB	
μPC2723	1.1 GHz AGC amplifier	BW = 1.1 GHz, Gpmax. = 13 dB, GCR = 38 dB, Vcc = 5.0 V	• 6-pin mini-mold
μPC2726	1.6 GHz differential amplifier	BW = 1.6 GHz, Icc = 11.5 mA, Gp = 15 dB, Vcc = 5.0 V	
μPC2791	UHF Wide band amplifier	BW = 1.9 GHz, Gp = 12 dB, NF = 5.5 dB	• 6-pin super mini-mold
μPC2792		BW = 1.2 GHz, Gp = 20 dB, NF = 3.5 dB	
μPC3210	Wide band amplifier	BW = 2.3 GHz, Gp = 20 dB, NF = 3.5 dB	
μPC8119	1.9 GHz Variable gain amplifier	Icc = 11 mA, GCR = 22 dB @1.9 MHz, Forward AGC	• 6-pin mini-mold
μPC8120		Icc = 11 mA, GCR = 22 dB @1.9 MHz, Reverse AGC	
μPC8104	1.9 GHz Up Converter + QPSK MOD	Vcc = 2.7 to 5.5 V, Icc = 28 mA	• 20-pin SSOP
μPC8105	400 MHz QPSK MOD	Vcc = 2.7 to 5.5 V, Icc = 16 mA	• 16-pin SSOP
μPC8110	900 MHz QPSK MOD	Vcc = 2.7 to 3.6 V, Icc = 24 mA	• 20-pin SSOP
μPC8125	Up converter with AGC + QPSK MOD	Vcc = 2.7 to 5.5 V, Icc = 36 mA, GCR = 40 dB, f_{LO1} = 220 to 270 MHz, f_{RFout} = 800 to 2000 MHz	
μPC8126	Direct QPSK MOD + Lo Pre – MIX.	Vcc = 2.7 to 3.6 V, Icc = 35 mA f_{RFout} = 800 to 1000 MHz	

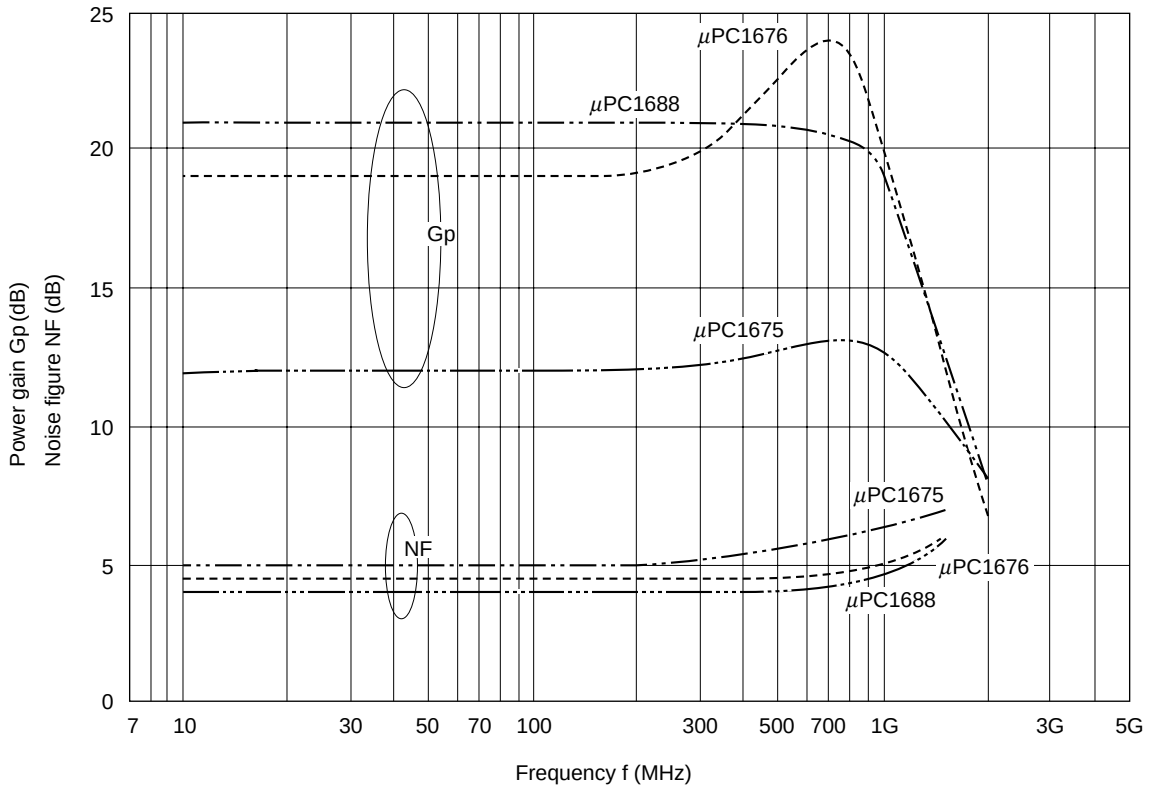
Si MMIC

Part number	Function	Features	Package
μPC8128	Low power consumption amplifier	BW = 1.9 GHz, Icc = 2.8 mA, Vcc = 3.0 V	• 6-pin super mini-mold
μPC8151		BW = 1.9 GHz, Icc = 4.2 mA, Vcc = 3.0 V	
μPC8152		BW = 1.9 GHz, Icc = 5.6 mA, Vcc = 3.0 V	
μPC8130	800 MHz - 1.5 GHz Variable gain amplifier	Icc = 11 mA, GCR = 40 dB_{MIN} @950 MHz, Reverse AGC	• 6-pin mini-mold
μPC8131		Icc = 11 mA, GCR = 40 dB_{MIN} @950 MHz, Forward AGC	
μPC8129	×2f MOD-LO QUAD. MOD+RF Up converter	Vcc = 2.7 to 5.5 V, Icc = 28 mA @3.0 V f_{MODout} = 100 to 400 MHz, f_{RFout} = 800 to 1900 MHz	• 20-pin SSOP

Si MMIC

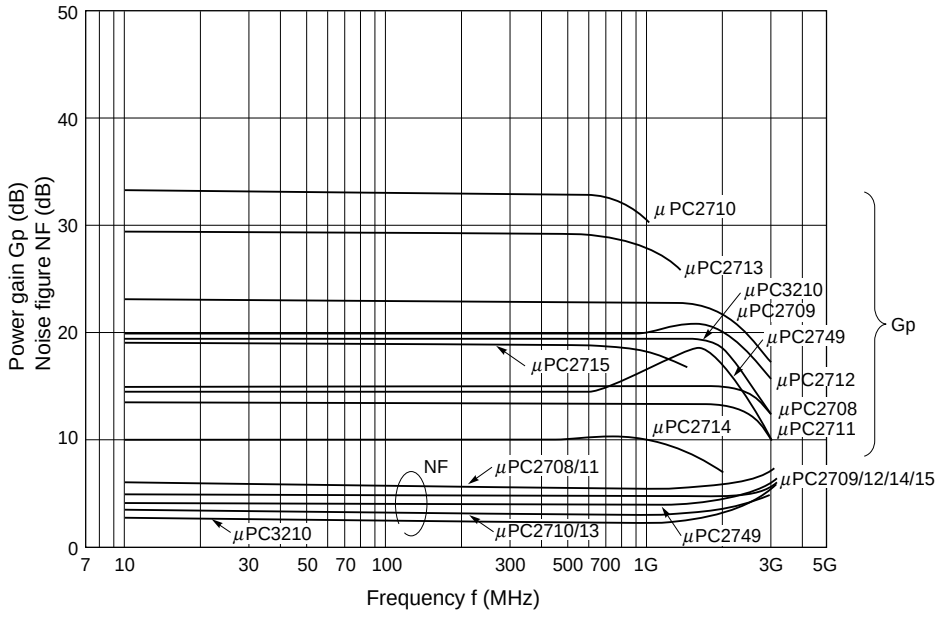
Power gain, noise figure vs. frequency



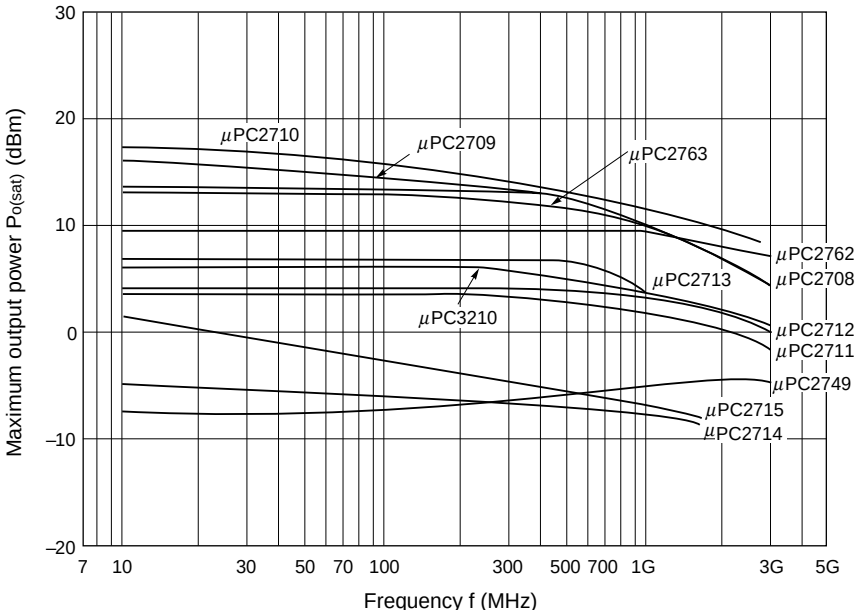


Si MMIC

Power gain, noise figure vs. frequency

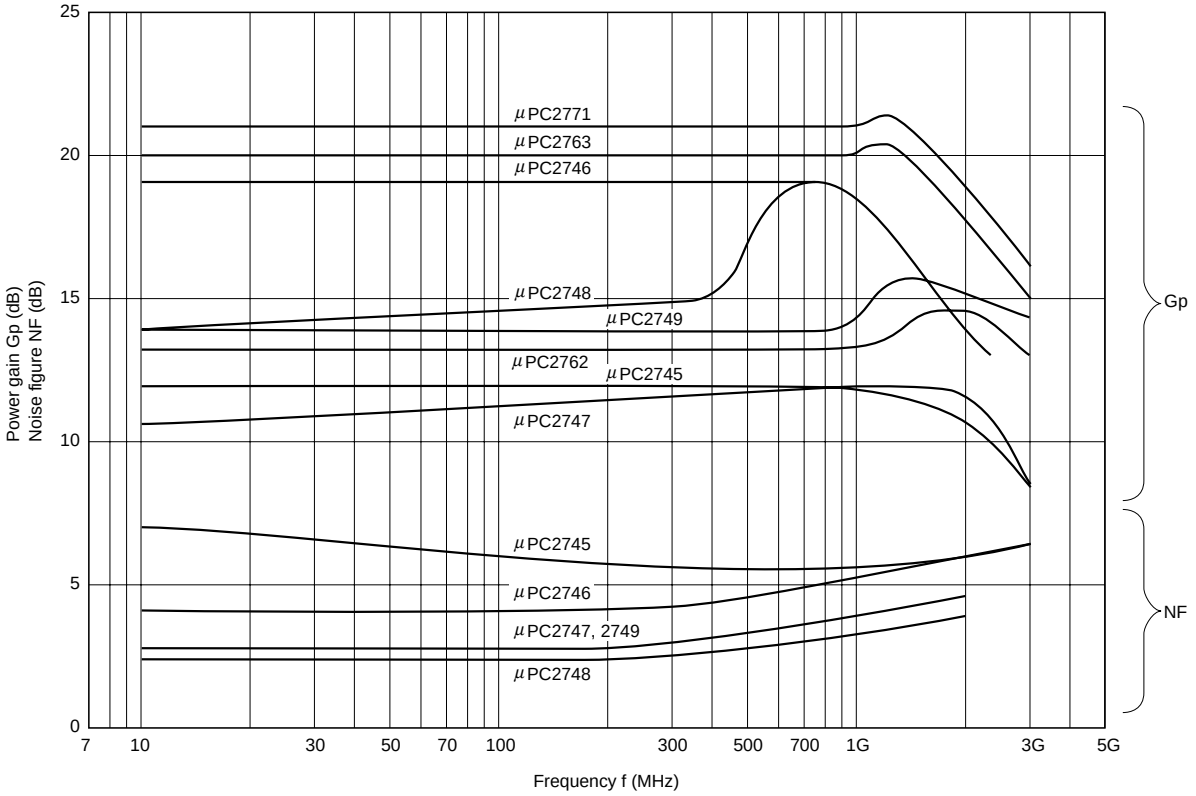


$P_{o(sat)}$ vs. frequency

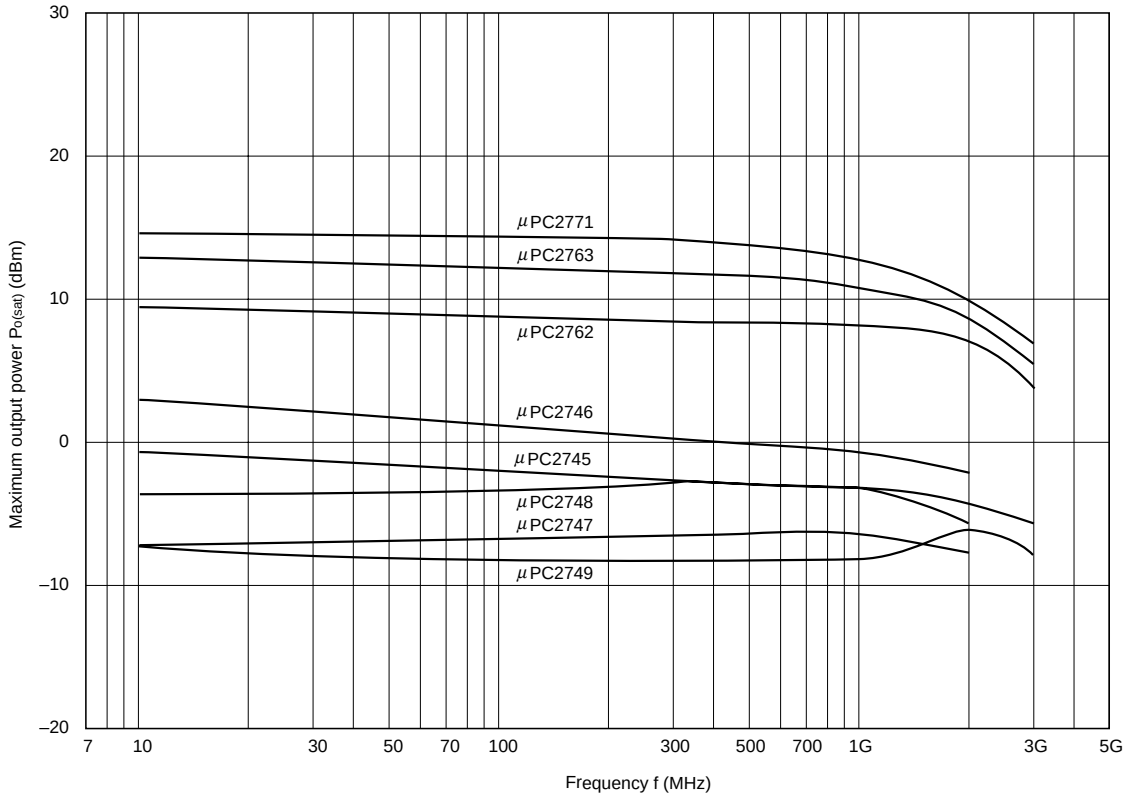


Si MMIC

Power gain, noise figure vs. frequency



$P_{o(sat)}$ vs. frequency



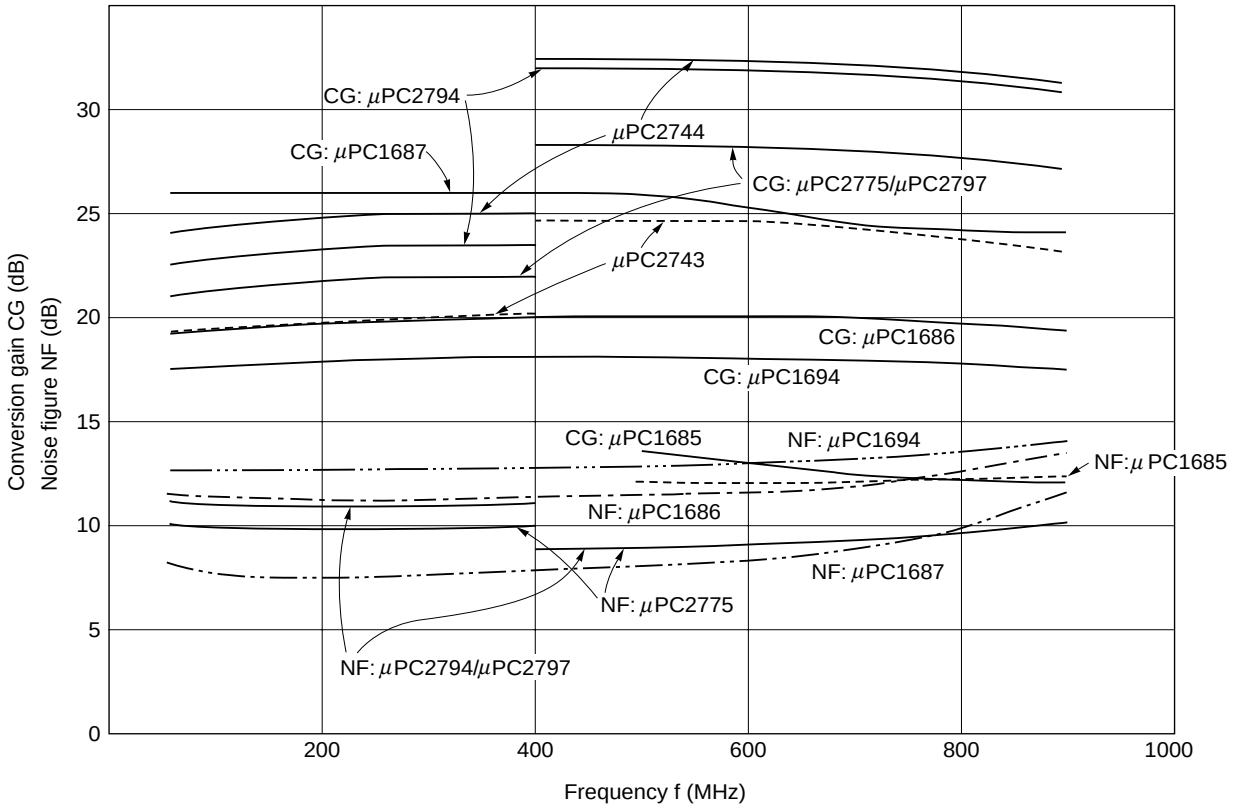
Si MMIC

Part number	Function	Features	Package
μPC1685	Down converter (MIX/OSC/IF amplifier)	CG = 12 dB, NF = 12.5 dB, CM = 86 dB μ, @900 MHz	• 8-pin SOP
μPC1686		CG = 20 dB, NF = 12 dB, CM = 94 dB μ, @470 MHz	• 8-pin SOP/SSOP
μPC1687		CG = 24 dB, NF = 10 dB, CM = 89 dB μ, @900 MHz	
μPC1694		CG = 18 dB, NF = 12.5 dB, CM = 103 dB μ, @470 MHz	• 14-pin SOP
μPC2743		CG = 20 dB, @470 MHz, CG = 23 dB @890 MHz	• 20-pin SOP
μPC2744		CG = 25 dB, @470 MHz, CG = 31 dB @890 MHz	
μPC2775		CG = 22 dB, @470 MHz, CG = 28 dB, @890 MHz	• 20-pin SOP/SSOP
μPC2794		CM = 96 dB μ, @470 MHz, CM = 92 dB μ, @890 MHz	• 20-pin SOP
μPC2797			• 20-pin SSOP
μPC3200			• 20-pin SOP
μPC3202		V_{CC} = 5 V, I_{CC} = 41 mA CM = 96 dB μ, @470 MHz, CM = 88 dB μ, @800 MHz	• 20-pin SSOP
μPC2721		CG = 20 dB, NF = 11 dB, f_{RF} = 0.9 to 2.0 GHz	• 8-pin SOP/SSOP
μPC2722		CG = 15 dB, NF = 11 dB, f_{RF} = 0.9 to 2.0 GHz	
μPC2795	Down converter	CG = 11 to 9.5 dB, NF = 13.5 to 14 dB, IM_3 = 55 dBc	• 8-pin SSOP
μPC2798	QAM IF Down Converter for Digital CATV	fin = 30 to 250 MHz, IIP_3 = –9 dBm, V_{CC} = 5 V V_{OUT} = 3 V_{P-P} (Differential, R_L = 1 KΩ)	• 20-pin SSOP
μPC2799	Up converter	V_{CC} = 5 V, I_{CC} = 60 mA, fin = 250 to 850 MHz	
μPC3206	AGC Amplifier + Video Amplifier	GCR = 50 dB (MIN.), f = to 100 MHz (TYP.), V_{CC} = 5 V	
μPC3211	AGC Amplifier	GCR = 55 dB (TYP.), V_{CC} = 9 V, IM_3 = 57 dBc (TYP.) @P_{out} = –10 dBm	
μPC2781	IQ Demo. with AGC	V_{CC} = 5 V, V_{OUT} = 1 V_{P-P} Δø = ±2 deg. ΔV = ±0.5 dB, fin = 440 to 520 MHz	
μPC3205			
μPC2731	AGC Amplifier + Down converter	CG = 14 dB, NF = 12 to 17 dB, f_{RF} = 0.9 to 2.0 GHz, GCR = 65 dB	• 20-pin SOP
μPC2782		CG = 10 to 11 dB, NF = 11 to 13 dB, IIP_3 = 0 to 4.5 dBm	• 20-pin SSOP
μPC2734	Down converter (MIX/OSC/IF Amp.)	CG = 10 to 13 dB, NF = 9 to 14 dB, f_{RF} = 0.9 to 2.1 GHz @f_{IF} = 402.8 MHz	
μPC2757	Down converter (MIX/Lo Amp./IF Amp.)	f_{RF}BW = 2.0 GHz, CG = 15 dB, Icc = 5.6 mA, Vcc = 3.0 V	• 6-pin mini/super mini-mold
μPC2758		f_{RF}BW = 2.0 GHz, CG = 15 dB, Icc = 11 mA, Vcc = 3.0 V	
μPC8112	Down converter	CG = 13 dB @f_{RFin} = 1.9 GHz, I_{CC} = 8.5 mA, V_{CC} = 3.0 V	
μPC2753		fin = DC to 400 MHz, CG_{MAX} = 79 dB, V_{CC} = 3 V, I_{CC} = 6.5 mA	• 20-pin SSOP
μPC8106	Up converter	CG = 10 dB, I_{CC} = 9 mA, OIP_3 = +5.5 dBm, V_{CC} = 3 to 5 V	• 6-pin mini/super mini-mold
μPC8109		CG = 7 dB, I_{CC} = 5 mA, OIP_3 = +1.5 dBm, V_{CC} = 3 to 5 V	
μPC8163	Higher IP_3 up converter	CG = 5.5 dB, I_{CC} = 16.5 mA, OIP_3 = 6.0 dBm, V_{CC} = 2.7 to 3.3 V	• 6-pin super mini-mold
μPC8100	Up/Down converter	Up converter + Down converter 1 chip IC	• 20-pin SSOP
μPB1502	Prescaler with power save mode	fmax. = 1.7 GHz, 64/65, 128/129, Vcc = 3.0 V	• 8-pin SOP
μPB1502(1)		fmax. = 2.0 GHz, 64/65, 128/129, Vcc = 3.0 V	
μPB1504	Prescaler	fmax. = 1.1 GHz, 64/65, 128/129, V_{CC} = 3 V, I_{CC} = 1.9 mA	
μPB1505		fmax. = 3.0 GHz, ÷256/128/64, V_{CC} = 5 V, I_{CC} = 14 mA	
μPB1506		fmax. = 3.0 GHz, ÷256/128/64, V_{CC} = 5.0 V	• 8-pin SSOP 175 mil
μPB1507			
μPB1508		fmax. = 3.0 GHz, 1/2, V_{CC} = 5.0 V	
μPB1509		fmax. = 700 MHz/800 MHz/1 GHz, 1/2, 1/4, 1/8	
μPB1510		fmax. = 3.0 GHz, 1/4, V_{CC} = 5.0 V	
μPA101	Transistor array	MULTI PLAYER	• 14-pin ceramic • 8-pin SOP
μPA102		2 DIFF AMP	• 14-pin ceramic • 14-pin SOP
μPA103		3 Trs+1 PIAR Tr	
μPA104		ADDER	

Si MMIC

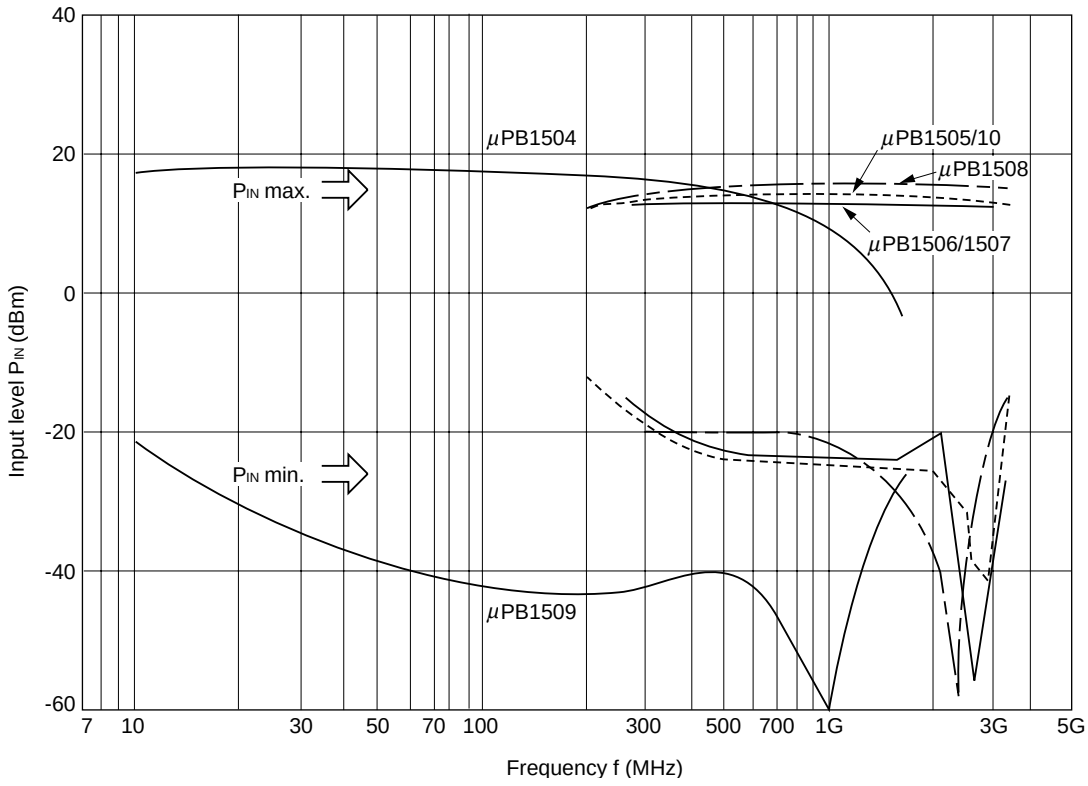
[Down converter]

Conversion gain, noise figure vs. frequency



[Prescaler]

Input level vs. frequency

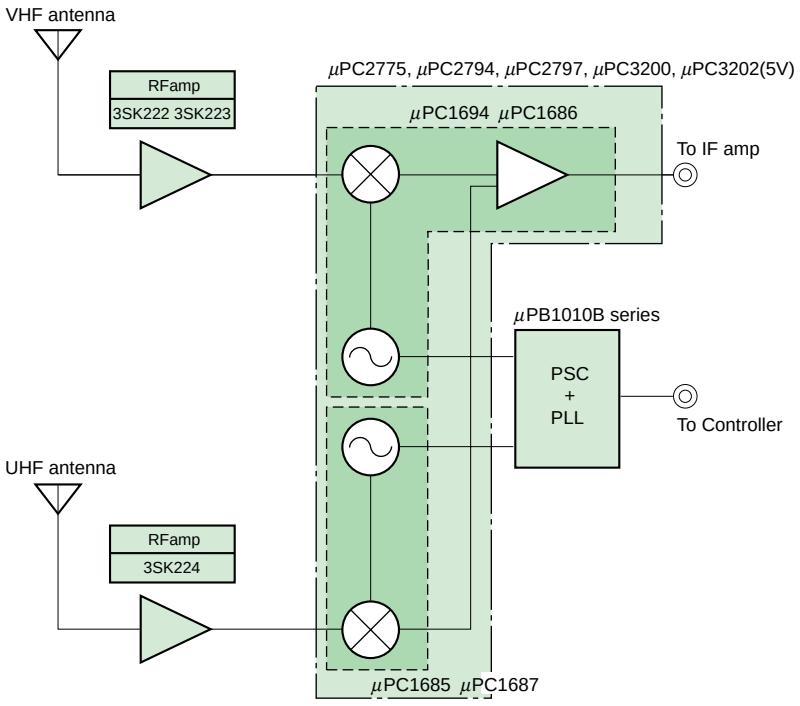


Transistor, FET

■ TV tuner

Application		Bi-polar transistor		Dual gate FET	
		Mini-mold (3-pin)	Small mini-mold (3-pin)	Mini-mold (4-pin)	Super mini-mold (4-pin)
VHF band	RF			**3SK131** **3SK222** **3SK223** **3SK230** **3SK252**	**3SK254**
	MIX	**2SC3545**	**2SC4182** **2SC4184**	**3SK131**	**3SK242**
	OSC	**2SC3545**	**2SC4182** **2SC4184**		
UHF band	RF (λ/4)		**2SC4183**	**3SK135A** **3SK224** **3SK231** **3SK253**	**3SK255**
	RF (λ/2)			**3SK134B**	
	RF (GaAs FET)			**3SK177** **3SK206**	
	MIX	**2SC3545** **2SC3841**	**2SC4183** **2SC4185** **2SC4184** **2SC4186**		
	OSC	**2SC3545** **2SC3841**	**2SC4182** **2SC4184** **2SC4185**		

Application Block Diagram

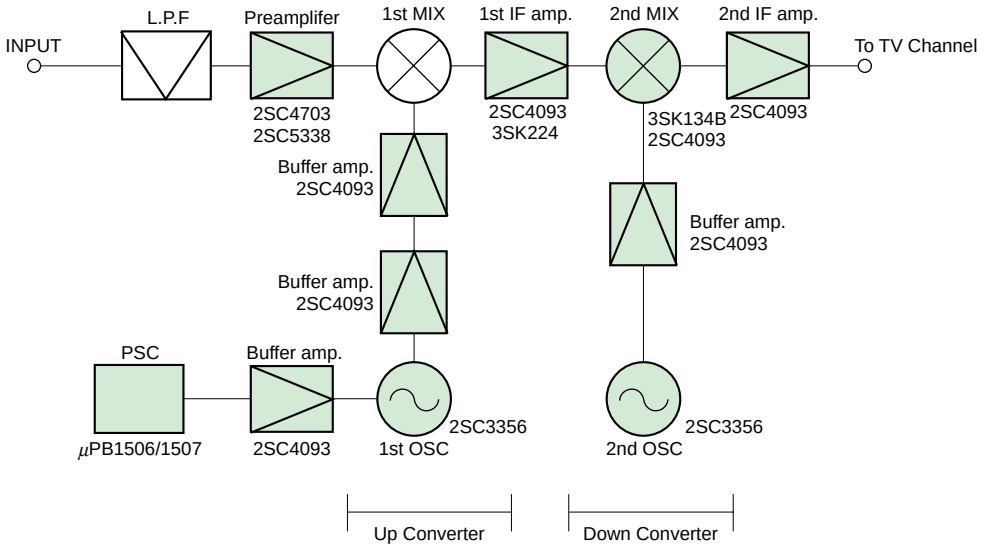


Transistor, FET

■ CATV converter

Application	Part number						Electrical characteristics	
	TO-92	Mini-mold		Super mini-mold		Ultrasuper mini-mold	fT (GHz)	NF (dB)
		3-pin	4-pin	3-pin	4-pin	3-pin		
Pre Amp.	**2SC3355**	**2SC3356**	**2SC4093**	**2SC4228**	**2SC5013**	**2SC5008**	7 (20 mA)	1.1 (1 GHz)
	2SC2570A	**2SC2351**	**2SC4092**				5 (20 mA)	1.5 (1 GHz)
Buffer Amp.				**2SC4185**			2 (5 mA)	3 (0.5 GHz)
	2SC2570A	**2SC2351**	**2SC4092**				5 (20 mA)	1.5 (1 GHz)
1st IF Amp.	**2SC3355**	**2SC3356**	**2SC4093**	**2SC4226**	**2SC5011**	**2SC5006**	7 (20 mA)	1.1 (1 GHz)
	2SC2570A	**2SC2351**	**2SC4092**				5 (20 mA)	1.5 (1 GHz)
1st and 2nd OSC		**2SC3545**		**2SC4184**			1.8 (5 mA)	
				2SC4185			2 (5 mA)	
		2SC4568		**2SC4570**		**2SC5005**	5.5 (5 mA)	
		2SC4569		**2SC4571**		**2SC5004**	5 (5 mA)	
2nd IF Amp.				**2SC4185**			2 (5 mA)	

Application Block Diagram



Transistor, FET

■ Low Noise Transistor Quick Reference Table

NF (dB) / Frequency (MHz)	2.5	3.5	4.0
500	———	2SC2148	———
2000	2SC3604	2SC3603	2SC2149
4000	2SC3587	———	———

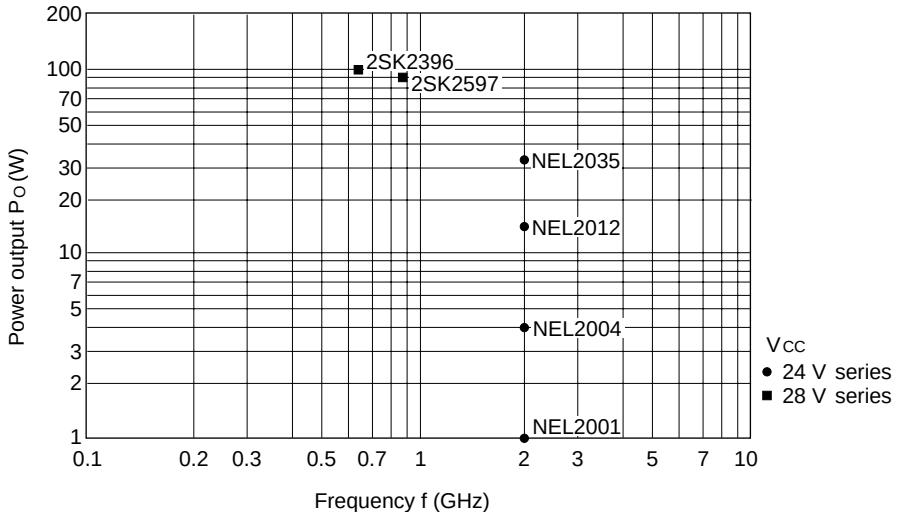
■ Power Transistor Quick Reference Table (for fixed radio station)

Output power / Frequency	10 W or higher
800 MHz or higher	2SK2396
900 MHz or higher	2SK2597
1900 MHz or higher	NEL2000 SERIES

■ Power Transistor Quick Reference Table (for mobile and portable radio station)

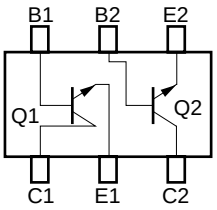
Output power / Frequency	0.1 W or higher	1 W or higher
1000 MHz or higher	2SC5288 2SC5289	

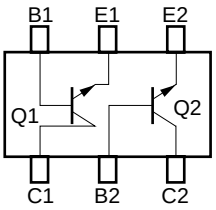
Output Power vs. Frequency

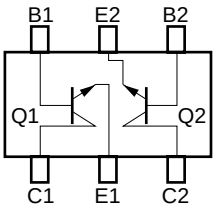


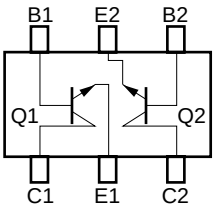
Transistor, FET

■ Twin Transistor









Same chip						Different chip		
for Pager		for VCO		for Mobile communication		for VCO		
Part number	2SC No. (X2)	Part number	2SC No. (X2)	Part number	2SC No. (X2)	Part number	2SC No.	
							Q1	Q2
μPA800T.TF	2SC5008	**μPA811T**	2SC5008					
μPA801T.TF	2SC5006	**μPA810T.TF**	2SC5006	**μPA821TF**	2SC5006	**μPA831TF**	2SC5006	2SC5007
μPA802T	2SC5007	**μPA812T**	2SC5007	**μPA822TF**	2SC5007	**μPA832TF**	2SC5006	2SC5010
μPA803T	2SC5005	**μPA813T**	2SC5005			**μPA833TF**	2SC5195	2SC5010
μPA804T.TF	2SC5004					**μPA834TF**	2SC5007	2SC5006
μPA805T	2SC5009					**μPA835TF**	2SC5010	2SC5006
μPA806T	2SC5010			**μPA826TF**	2SC5010	**μPA836TF**	2SC5010	2SC5195
μPA807T	2SC5181			**μPA827TF**	2SC5181	**μPA837TF**	2SC5181	2SC5195
μPA808T	2SC5186			**μPA828TF**	2SC5186	**μPA838TF**	2SC5195	2SC5186
μPA809T.TF	2SC5195	**μPA814T.TF**	2SC5195	**μPA829TF**	2SC5195	**μPA839TF**	2SC5008	2SC5006

Optical Device

Fiber Optic Communication/Light Beam Application Devices 246

- Laser Diodes 247
- Detectors 248

Photo Detectors 250

- PIN Photo-diode 250
- Light Receiving IC 250

Photocoupler 251

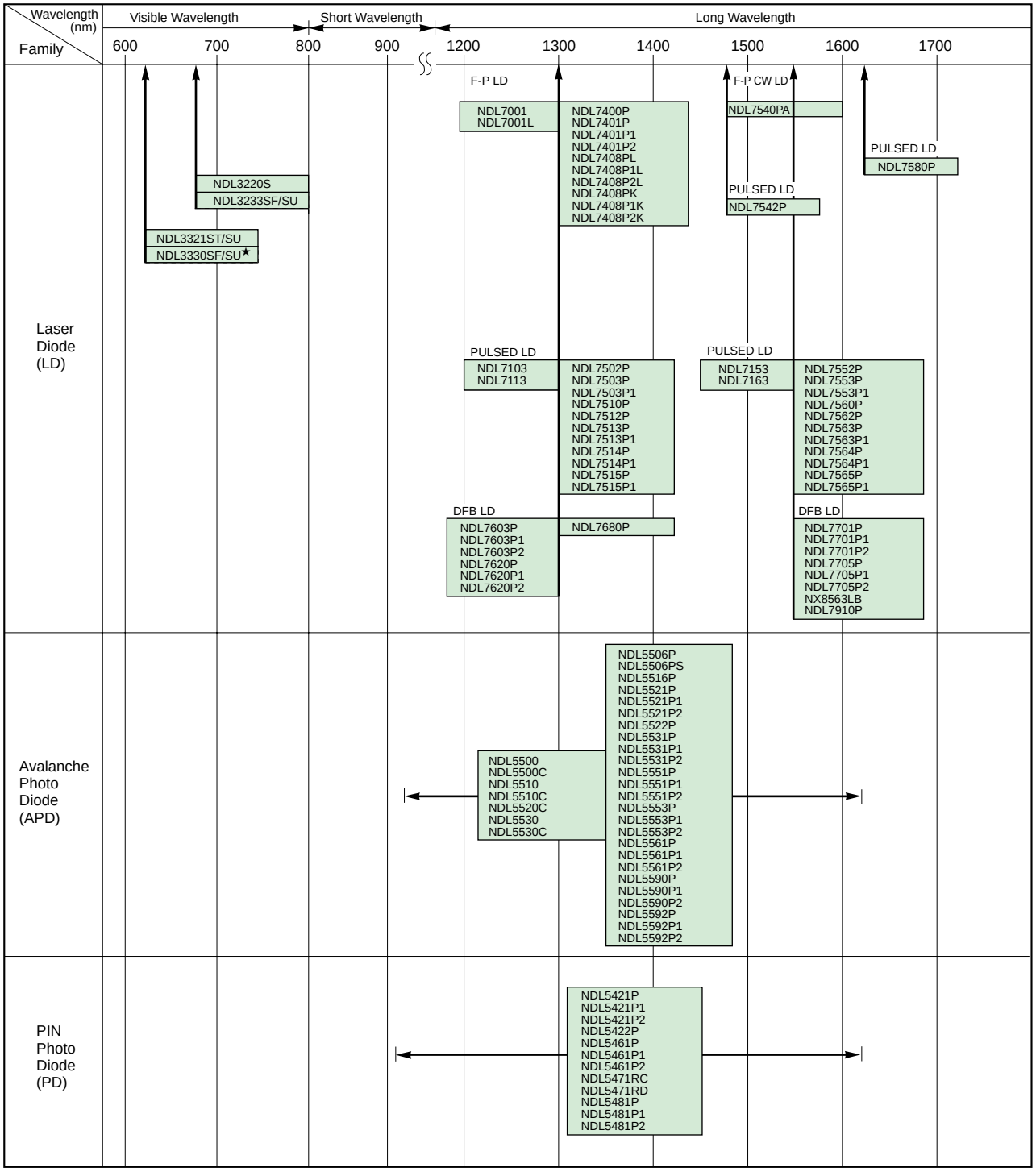
OCMOS FET™ 253

POF Link 254

Infra-Red Link Unit 254

Fiber Optic Communication/Light Beam Application Devices

■ Product map of optical communication/light beam application device



⟶ : Mark wavelength or operating range

★: Under development

Modules are available with FC-PC or SC-PC connector.

Fiber Optic Communication/Light Beam Application Devices

Laser Diodes

■ Visible LD for Beam Applications

Part number	Absolute maximum ratings			Typical characteristics						Remarks
	P_O/P_f (mW)	T_C (°C)	T_{stg} (°C)	I_{th} (mA)	I_{op} (mA)	P_o/P_f (mW)	λ_p (nm)	$\theta\perp$ (deg)	$\theta//$ (deg)	
				TYP.	TYP.	TYP.	TYP.	TYP.	TYP.	
NDL3220S	6	−10 to +60	−40 to +85	20	30	5	670	30	8	Low threshold current
NDL3233SF/SU	35	−10 to +70	−40 to +100	50	100	35	685	21	9	Magneto-optical disk application
NDL3321ST/SU	6	−10 to +70	−40 to +85	40	60	5	650	30	8	DVD application
NDL3330SF/SU★	35	−10 to +70	−40 to +100	50	100	30	660	21	9	DVD-RAM, High density Magneto-optical disk application

★: Under development

■ For Fiber Optic Communications

• Fabry Perrot LD

Part number	Absolute maximum ratings			Typical characteristics (25°C)					Remarks
	I_F (mA)	T_C (°C)	T_{stg} (°C)	P_o/P_f (mW)	I_{th} (mA)	λ_c (nm)	σ *2 (nm)	t_r/t_f (ns)	
				MIN.	TYP.	TYP.	TYP.	MAX.	
NDL7001	–	−40 to +85	−55 to +125	5.0	10	1310	1.0	0.5/0.5	with monitor PD, small package
NDL7001L	–	−40 to +85	−55 to +125	5.0	10	1310	1.0	0.5/0.5	with monitor PD, small package
NDL7400P	I_{th}+50	−20 to +65	−40 to +70	2.0*1	10	1310	1.0	1.0/1.0	with TEC
NDL7401P/P1/P2	I_{th}+50	−40 to +85	−40 to +85	2.0	10	1310	1.3	0.5/0.5	with monitor PD
NDL7408PL/P1L/P2L	I_{th}+50	−40 to +85	−40 to +85	0.2	10	1310	1.3	0.5/0.5	with monitor PD
NDL7408PK/P1K/P2K	I_{th}+50	−40 to +85	−40 to +85	1.0	10	1310	1.3	0.5/0.5	with monitor PD
NDL7800P★	I_{th}+50	−40 to +85	−40 to +85	0.2	5	1310	2.0	0.5/0.5	Ceramic mini DIL

∗1: TYP.

∗2: RMS (–20 dB)

★: Under development

• DFBLD

Part number	Absolute maximum ratings				Typical characteristics				Remarks
	I_F (mA)	P_o/P_f (mW)	T_C (°C)	T_{stg} (°C)	P_o/P_f (mW)	I_{th} (mA)	λ_p (nm)	t_r/t_f (ns)	
					MIN.	TYP.	TYP.	TYP.	
NDL7603P/P1/P2	150	5	−40 to +85	−40 to +85	2.0	50*1	1310	0.5/0.5	with monitor PD
NDL7620P/P1/P2	150	5	0 to +70	−40 to +85	2.0	45*1	1310	0.04/0.1	with monitor PD
NDL7701P/P1/P2	150	5	−20 to +85	−40 to +85	2.0	15	1550	0.5/0.5	with monitor PD
NDL7705P/P1/P2	I_{th} + 50	5	−40 to +85	−40 to +85	2.0	50*1	1550	0.5/0.5	with monitor PD
NX8563LB	300	–	−20 to +65	−40 to +85	10*2	20	ITU-T*3	–	with PMF fiber
NDL7910P	150	20	−20 to +70	−40 to +85	0.5	7	1550*4	0.125/ 0.125	Integrated Modulator for 2.5G

∗1: MAX.

∗2: TYP.

∗3: Wavelength selectable for ITU-T standards.

∗4: Wavelength selectable for ITU-T standards upon request.

■ For Analog Applications

Part number	Absolute maximum ratings				Typical characteristics (25°C)						Remarks
	I_F (mA)	P_f (mW)	T_C (°C)	T_{stg} (°C)	I_{th} (mA)	P_f (mW)	λ_p (nm)	RIN (dB/Hz)	CSO (dBc)	CTB (dBc)	
					TYP.	MIN.	TYP.	TYP.	TYP.	TYP.	
NDL7680P	150	15	−20 to +65	−40 to +70	20	3	1310	−155	−55	−60	with monitor PD, isolator and TEC

Fiber Optic Communication/Light Beam Application Devices

Laser Diodes

■ For EDFA Pumping and High Power OTDR Applications

Part number	Absolute maximum ratings			Typical characteristics						Remarks
	I_F (mA)	T_C (°C)	T_{stg} (°C)	I_{th} (mA)	I_{op} (mA)	P_o / P_f (mW)	λ_C (nm)	σ *1 (nm)	t_r / t_f (ns)	
				TYP.	TYP.	TYP.	TYP.	MAX.	MAX.	
NDL7540PA	600	−20 to +70	−40 to +85	40	500	110	1480	8	–	with TEC, thermistor and m-PD
NDL7542P	600*2	−20 to +65	−40 to +70	30	500*2	80*2	1480*2	10	2/2	with TEC
NDL7103	1200*2	−40 to +70	−55 to +125	35	1000*2	320*2	1310*2	7	2/2	ø5.6 can
NDL7113	600*2	−40 to +70	−55 to +125	20	400*2	175*2	1310*2	10	1/1	ø5.6 can
NDL7502P	1200*2	−20 to +65	−40 to +70	35	1000*2	190*2	1310*2	10	2/2	with TEC
NDL7503P/P1	1200*2	−20 to +60	−40 to +85	35	1000*2	180*2	1310*2	10	2/2	
NDL7510P	600*2	−20 to +65	−40 to +70	20	400*2	55*2	1310*2	10	1/1	with TEC
NDL7512P	600*2	−20 to +65	−40 to +70	20	400*2	110*2	1310*2	10	1/1	with TEC
NDL7513P/P1	600*2	−20 to +60	−40 to +85	20	400*2	110*2	1310*2	10	1/1	
NDL7514P/P1	600*2	−20 to +60	−40 to +85	20	400*2	50*2	1310*2	10	1/1	
NDL7515P/P1	600	−20 to +60	−40 to +85	20	400*2	30*2	1310*2	10	1/1	
NDL7153	1200*2	−40 to +70	−55 to +125	45	1000*2	240*2	1550*2	8	2/2	ø5.6 can
NDL7163	600*2	−40 to +70	−55 to +125	30	400*2	120*2	1550*2	10	1/1	ø5.6 can
NDL7552P	1200*2	−20 to +65	−40 to +70	40	1000*2	125*2	1550*2	10	2/2	with TEC
NDL7553P/P1	1200*2	−20 to +60	−40 to +85	45	1000*2	145*2	1550*2	10	2/2	
NDL7560P	600*2	−20 to +65	−40 to +70	20	400*2	30*2	1550*2	10	1/1	with TEC
NDL7562P	600*2	−20 to +65	−40 to +70	40	400*2	80*2	1550*2	10	1/1	with TEC
NDL7563P/P1	600*2	−20 to +60	−40 to +85	40	400*2	80*2	1550*2	10	1/1	
NDL7564P/P1	600*2	−20 to +60	−40 to +85	40	400*2	40*2	1550*2	10	1/1	
NDL7565P/P1	600	−20 to +60	−40 to +85	20	400	11	1310*2	10	1/1	
NDL7580P	1200*2	−20 to +65	−40 to +85	30	1000*2	100*2*3	1625	15	2/2	with TEC

*1: RMS (–20 dB)

*2: Pulse conditions; pulse width = 10 μs, duty = 1% (modules)
pulse width = 1 μs, duty = 1% (ø5.6 can)

*3: MIN.

Detectors

■ PIN Photo Diodes

Part number	Absolute maximum ratings				Detecting area size (μm)	Typical characteristics							Package	Remarks
	I_F (mA)	I_R (mA)	T_C (°C)	T_{stg} (°C)		I_D (nA)		C_t (pF)		S (A/W)		f_C (GHz)		
						V_R (V)	TYP.	V_R (V)	TYP.	λ (nm)	TYP.	MIN.		
NDL5471RC/RD	10	0.5	−40 to +85	−40 to +85	ø120	5	0.1	5	1.1	1300	0.89	1.5	Receptacle package	RC: FC type receptacle module RD: SC type receptacle module
										1550	1.00			
NDL5461P/P1/P2	10	–	−40 to +85	−40 to +85	ø80	5	0.1	5	1.0	1300	0.89	2.5	Coaxial module (With MMF)	P: Without flange P1: With flat mount flange P2: With vertical flange
										1550	0.94			
NDL5481P/P1/P2	10	–	−40 to +85	−40 to +85	ø80	10	0.1	10	0.7	1300	0.85	2.5	Coaxial module (With SMF)	for analog applications P: Without flange P1: With flat mount flange P2: With vertical flange
NDL5421P/P1/P2	10	0.5	−40 to +85	−40 to +85	ø50	5	0.1	5	0.7	1300	0.89	2.5	Coaxial module (With MMF)	P: Without flange P1: With flat mount flange P2: With vertical flange
										1550	0.94			
NDL5422P	–	0.5	−40 to +70	−40 to +85	ø50	5	0.1	–	–	–	–	2.5	BFY Package (6-pin) (With MMF)	With Pre-AMP $\bar{P}$ = –23 dBm typ. 2.5 Gb/s

Fiber Optic Communication/Light Beam Application Devices

Detectors

■ Avalanche Photo Diodes

Type	Part number	Absolute maximum ratings				Detecting area size (μm)	Typical characteristics									Package	Remarks
		I_F (mA)	I_R (mA)	T_C (°C)	T_{stg} (°C)		$V_{(BR)R}$ (V)	I_D (nA)		S (A/W)		M	f_C (GHz)	t_r / t_f (ns)		
							TYP.	V_R (V)	TYP.	λ (nm)	TYP.	TYP.	MIN.	TYP.		
InGaAs-APD	NDL5510	10	0.5	−40 to +70	−55 to +100	ø80	75	$V_{(BR)R}$×0.9	8	1300	0.89	40	0.7	–	CAN Package	
										1550	1.00					
	NDL5510C	10	0.5	−40 to +70	−55 to +100	ø80	75	$V_{(BR)R}$×0.9	8	1300	0.89	40	0.7	–	Chip on carrier	
										1550	1.00					
	NDL5500	10	0.5	−40 to +70	−55 to +100	ø50	70	$V_{(BR)R}$×0.9	5	1300	0.89	40	1.0	–	CAN Package	
										1550	1.00					
	NDL5500C	10	0.5	−40 to +70	−55 to +100	ø50	70	$V_{(BR)R}$×0.9	5	1300	0.89	40	1.0	–	Chip on carrier	
										1550	1.00					
	NDL5520C	10	0.5	−40 to +70	−55 to +100	ø50	55	$V_{(BR)R}$×0.9	5	1300	0.89	40	2.5	–	Chip on carrier	for 2.5 Gb/s (ø50)
										1550	1.00					
	NDL5530	10	0.5	−40 to +85	−55 to +100	ø30	70	$V_{(BR)R}$×0.9	5	1300	0.94	40	2.5	–	CAN Package	
										1550	0.96					
	NDL5530C	10	0.5	−40 to +85	−55 to +100	ø30	70	$V_{(BR)R}$×0.9	5	1300	0.94	40	1.0	–	Chip on carrier	
										1550	0.96					
	NDL5531P /P1/P2	10	0.5	−40 to +85	−40 to +85	ø30	70	$V_{(BR)R}$×0.9	5	1300	0.94	40	2.5	–	Coaxial Module	P: Without flange P1: With flat mount flange P2: With vertical flange
										1550	0.96					
	NDL5561P /P1/P2	10	0.5	−40 to +85	−40 to +85	ø80	75	$V_{(BR)R}$×0.9	8	1300	0.94	40	1.0	–	Coaxial Module (With MMF GI-62.5)	P: Without flange P1: With flat mount flange P2: With vertical flange
										1550	0.96					
	NDL5551P /P1/P2	10	0.5	−40 to +85	−40 to +85	ø50	70	$V_{(BR)R}$×0.9	5	1300	0.94	40	1.0	–	Coaxial module (With MMF)	P: Without flange P1: With flat mount flange P2: With vertical flange
										1550	0.96					
	NDL5521P /P1/P2	10	0.5	−40 to +85	−40 to +85	ø50	55	$V_{(BR)R}$×0.9	5	1300	0.94	40	2.5	–	Coaxial module (With MMF)	for 2.5 G Applications P: Without flange P1: With flat mount flange P2: With vertical flange
										1550	0.96					
	NDL5553P /P1/P2	10	0.5	−40 to +85	−40 to +85	ø50	70	M = 20	50	1300	0.89	20*1	–	0.5*2	Coaxial module (With MMF)	for OTDR Applications P: Without flange P1: With flat mount flange P2: With vertical flange
										1550	0.94					
	NDL5516P	10	0.5	−20 to +55	−40 to +85	ø80	75	$V_{(BR)R}$×0.9 I_C=1.1A@55°C	2	1300	0.89	40	0.7	–	DIP module (With MMF GI-62.5)	for OTDR Applications With TEC $\Delta T \geq 45K$
										1550	1.00					
	NDL5506P /PS	10	0.5	−20 to +55	−40 to +85	ø50	70	$V_{(BR)R}$×0.9 I_C=1.1A@55°C	2	1300	0.89	40	1.0	–	DIP module	for OTDR Applications With TEC $\Delta T \geq 45K$ P: With MMF, PS: With SMF
										1550	1.00					
	NDL5590P /P1/P2	–	0.5	−40 to +85	−40 to +85	ø50	70	$V_{(BR)R}$×0.9	5	1300	0.94	–	1.0	–	Coaxial module (With MMF)	With Pre-AMP $\overline{P}$ = – 36 dBm typ. 622 Mb/s P: Without flange P1: With flat mount flange P2: With vertical flange
										1550	0.96					
	NDL5592P /P1/P2	–	0.5	−40 to +85	−40 to +85	ø30	70	$V_{(BR)R}$×0.9	5	1300	0.94	–	2.5	–	Coaxial module (With SMF)	With Pre-AMP $\overline{P}$ = −33 dBm typ. 2.488 Gb/s P: Without flange P1: With flat mount flange P2: With vertical flange
										1550	0.96					
	NDL5522P	–	0.5	−40 to +70	−40 to +85	ø50	55	$V_{(BR)R}$×0.9	5	–	–	–	2.5	–	BFY Package (6-pin) (With MMF)	With Pre-AMP $\overline{P}$ = −33 dBm typ. 2.5 Gb/s

*1: MIN.
*2: MAX.

Photo Detectors

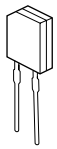
PIN Photo-diode

■ PIN photo-diode for remote control application

Part number	Features	Absolute maximum ratings (T_A = 25 °C)		Electrical characteristics (T_A = 25 °C)			Package (package No.)
		V_R (V)	P_C (mW)	Sensitivity (nA/lx) (V_R = 5 V)	I_D (nA) (V_R = 10 V)	t_r (ns) (R_L = 1 kΩ)	
PH302	Mold case	32	150	50*1 (6 (μA)*2)	~ 30	50	P10
PH302C	Mold case Visible ray cut filter	32	150	32*1 (5 (μA)*2)	~ 30	50	P10
PH310	Small mold case with lens Visible ray cut resin	32	150	32*1 (5 (μA)*2)	~ 10	30	P13
PH320	Small mold case with lens Visible ray cut resin	32	150	4.3 (μA)*2	~ 10	30	P13

*1: Light source color temperature 2854K
*2: Sensitivity (@H = 0.1 mW/cm^2) for infrared ray (λ_P = 940 nm)

Package (package appearance)

Package		
Package No.	P10	P13

Light Receiving IC

■ Photo-diode with built-in amplifier for Optical Storage (Custom device)

Part number	Absolute maximum ratings (T_A = 25 °C)				Light-electrical typical characteristics (T_A = 25 °C)						Remarks
	V_{CC} (V)	I_{CC} (mA)	P_D (mW)	T_{stg} (°C)	V_{CC} (V)	V_O Focus (mV)	V_O Tracking (mV)	S λ (nm)	S (A/W)	f_T (MHz)	
					TYP.	TYP.	TYP.		TYP.	TYP.	
PH5××★	9	–	100	–25 to +80	5	350	550	650	–	50	With internal amplifier

★: Under development

Photocoupler

■ General-purpose Tr. Photocoupler

Part number	Features	BV_{1-2}	I_F (mA)	V_{CEO} (V)	CTR (%)	Package	Remarks
PS2501-1 to -4	High isolation voltage	5 kVr.m.s.	80	80	80 to 600	• 4- to 16-pin DIP (multi)	Single transistor output
PS2502-1 to -4				40	200 ~		Darlington transistor output
PS2503-1 to -4			50	40	100 to 400		Single transistor output
PS2505-1 to -4			±80	80	80 to 600		AC input Single transistor output
PS2506-1 to -4				40	200 ~		AC input Darlington transistor output
PS2521-1 to -4	High isolation voltage Large current input	5 kVr.m.s.	150	80	20 to 80	• 4-pin DIP	Single transistor
PS2525-1 to -4			±150				Darlington transistor
PS2532-1 to -4	High isolation voltage (VDE0884 Option)	5 kVr.m.s. (3.75 kVr.m.s.)	80	300	1500 to 6500	• 4- to 16-pin DIP (multi)	Darlington transistor
PS2533-1 to -4				350	1500 to 6500		
PS2561-1 to -2				80	80 to 400		Single transistor
PS2562-1 to -2					200 ~		Darlington transistor
PS2565-1 to -2			±80		80 to 400		Single transistor
PS2566-1 to -2					200 ~		Darlington transistor
PS2581L1/L2	High isolation voltage Long creepage type (8 mm) (VDE0884 Standard)	5 kVr.m.s.	80	80	80 to 400	• 4-pin DIP	Single transistor
PS2601	High isolation voltage	5 kVr.m.s.	80	80	80 to 600	• 6-pin DIP	Single transistor With base
PS2602							Single transistor Without base
PS2603				40	200 ~		Darlington transistor With base
PS2604							Darlington transistor Without base
PS2605			±80	80	80 to 600		AC input Single transistor With base
PS2606							AC input Single transistor Without base
PS2607				40	200 ~		AC input Darlington transistor With base
PS2608							AC input Darlington transistor Without base
PS2621	High isolation voltage Large current input	5 kVr.m.s.	150	80	20 to 50	• 6-pin DIP	Single transistor With base
PS2622							Single transistor Without base
PS2625			±150				AC input Single transistor With base
PS2626							AC input Single transistor Without base
PS2633	High isolation voltage High V_{CEO}		80	300	1000 to 15000		Darlington transistor With base
PS2634							Darlington transistor Without base
PS2651	High isolation voltage (VDE0884 Option)			80	50 to 400	• 6-pin DIP (lead forming)	Single transistor With base
PS2652							Single transistor Without base
PS2653				40	200 ~		Darlington transistor With base
PS2654							Darlington transistor Without base

Photocoupler

■ SOP Photocoupler

Part number	Features	BV_{1-2}	I_F (mA)	V_{CEO} (V)	CTR (%)	Package	Remarks
PS2701-1,-2,-4	High isolation voltage	3.75 kVr.m.s.	50	40	50 to 300	• SOP (2.54 mm pitch)	Single transistor
PS2702-1,-2,-4					200 ~		Darlington transistor
PS2703-1,-2,-4				120	50 to 400		Single transistor
PS2705-1,-2,-4			±50	40	50 to 300		AC input, Single transistor
PS2706-1,-2,-4					200 ~		AC input, Darlington transistor
PS2707-1,-2,-4				120	50 to 400		AC input, Single transistor
PS2732-1,-2,-4	High isolation voltage High V_{CEO} (VDE0884 Option)		50	300	1500 to 6500		Darlington transistor
PS2733-1,-2,-4				350			
PS2801-1	High isolation voltage	2.5 kVr.m.s.	50	80	80 to 600	• SOP (1.27 mm pitch)	Single transistor
PS2802-1				40	200 ~		Darlington transistor
PS2805-1			±50	80	80 to 600		AC input, Single transistor
PS2806-1			50	40	200 ~		AC input, Darlington transistor
PS2801-4				80	80 to 600		Single transistor
PS2802-4				40	200 ~		Darlington transistor
PS2805-4			±50	80	80 to 600		AC input, Single transistor
PS2806-4			50	40	200 ~		AC input, Darlington transistor
PS2832-1★	High V_{CEO}		50	300	400 ~		Darlington transistor

★: Under development

■ High Speed Photocoupler

Part number	Features	BV_{1-2}	I_F (mA)	V_{CC} (V)	CTR (%)	Package	Remarks
PS8701	High speed 1 Mbps High CMR 10 kV/μs	2.5 kVr.m.s.	25	35	15 ~	• 5-pin SOP	Photodiode + transistor
PS8601	High speed 1 Mbps (VDE0884 Option)	5 kVr.m.s.			15 ~	• 8-pin DIP	Photodiode + transistor with base
PS8602							Photodiode + transistor

■ Analog Linear Photocoupler

Part number	Features	BV_{1-2}	I_F (mA)	V_R (V)	ΔK3 (%)	Package	Remarks
PS8741★	High linearity	1.5 kVr.m.s.	50	20	~ 1	• 16-pin SOP	for optical DAA

★: Under development

■ High Speed Photocoupler

Part number	Features	BV_{1-2}	I_F (mA)	V_{CC} (V)	t_{PHL}, t_{PLH} (ns)	Package	Remarks
PS9601	High speed 10 Mbps (IC output)	5 kVr.m.s.	30	7	50 (TYP.)	• 8-pin DIP	
PS9701	High speed 10 Mbps (IC output) (VDE0884 Option)	2.5 kVr.m.s.			50 (TYP.)	• 5-pin SOP	
PS9611★	High CMR High speed 10 Mbps					• 8-pin DIP	CMR 10 kV/μs
PS9711★						• 5-pin SOP	

★: Under development

■ Inverter Control Photocoupler

Part number	Features	BV_{1-2}	I_F (mA)	V_{CC} (V)	I_{O2} (A)	Package	Remarks
PS9634	High output, high noise immunity (VDE0884 Option)	5 kVr.m.s.	30	18	0.8	• 8-pin DIP	Built-in amplifier

OCMOS FET™

Part number	Features	V_{I-O} (V)	V_L (V)	I_L (mA)	R_{on} (Ω)	Package	Remarks
PS7112-1A	General use	1500	100	200	6	6-pin DIP (1-ch)	
PS7113-1A			100	350	2.5		
PS7122-1A			200	200	5		
PS7122A-1A			250	200	8		
PS7122A-1B			250	200	8		1b output
PS7141-1A			400	150	30		
PS7141-1B			400	150	30		1b output
PS7142-1A			400	150	10		
PS7160-1A			600	90	50		
PS7111-2A			100	100	6	8-pin DIP (2-ch)	
PS7113-2A			100	350	2.5		
PS7122-2A			200	200	5		
PS7122A-2A			250	200	8		
PS7122A-2B			250	200	8		1b + 1b output
PS7122A-1C			250	200	8		1a + 1b output
PS7141-2A			400	150	30		
PS7141-2B			400	150	30		1b + 1b output
PS7141-1C			400	150	30		1a + 1b output
PS7142-2A			400	200	12		1a + 1a output
PS7160-2A			600	90	50		
PS7200A-1A			40	100	12	4-pin SOP (1-ch)	Low capacitance 2.9 pF
PS7200B-1A			40	250	1.5		Low on-state resistance
PS7241-1A			400	120	35		
PS7241-1B			400	120	35		1b output
PS7211-2A			100	100	6	8-pin SOP (2-ch)	
PS7221-2A			200	120	8		
PS7241-2A			400	120	30		
PS7241-2B			400	120	30		1b + 1b output
PS7241-1C			400	120	30		1a + 1b output
PS7341-1A	High isolation voltage	3750	400	150	30	6-pin DIP (1-ch)	
PS7342-1A			400	200	10		
PS7360-1A			600	90	50		
PS7341C-1A			400	150	35		Current limit I_{LMT} 170 to 250 mA
PS7342C-1A			400	200	20		
PS7522-1A	Slow speed	1500	200	200	5	6-pin DIP	
PS7522-2A			200	200	5	8-pin DIP	
PS7241-AT1	Multi function	1500	200	120	30	8-pin SOP (2-ch)	OCMOS FET + photocoupler
PS7241-AT5			200	120	30		
PS7841-A15★			400	120	30	16-pin SOP	for DAA

★: Under development

POF Link

Part number	Absolute maximum rating (T_C = 25 °C)		Electrical characteristics (T_C = 25 °C)		
	V_{CC} (V)	T_C (°C)	Transmission speed (bps)	Transmission distance (m)	
				@SI-POF	@SI-HPCF
NL2100	0 to 6	0 to 70	156	50	100

Infra-Red Link Unit

Conform to IrDA Ver. 1.1 / 1.0

Part number	Absolute maximum rating (T_A = 25 °C)		Electrical characteristics (T_A = 25 °C)			Remarks
	V_{CC} (V)	Topt (°C)	Data rate (bps)	Transmission distance (m)	Viewing angle (deg.)	
NL1100	5	0 to +70	2.4 k to 4 M	~ 1.0	30	IrDA 1.1
NL1000★	3	0 to +70	2.4 k to 115.2 k	~ 1.0	30	IrDA 1.0

★: Under development

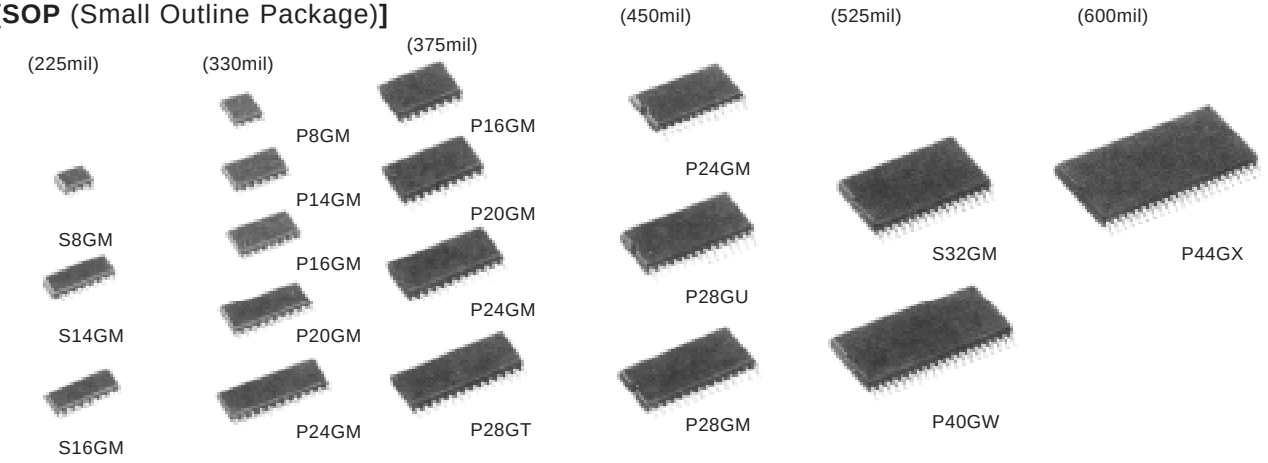
Packages

Surface Mount IC Packages 256
Through Hole IC Packages 272
Surface Mount Discrete Device Packages 276
Through Hole Discrete Device Packages 278

Remark Only typical packages are shown here.

[Surface Mount IC Packages]

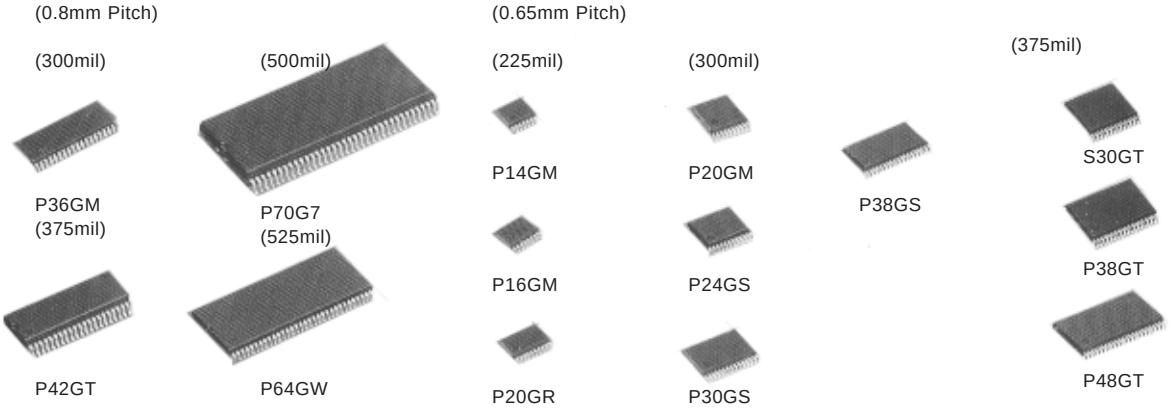
[SOP (Small Outline Package)]



Pin Pitch ; 1.27mm

Nominal Size	225mil	300mil	375mil	450mil	525mil	600mil
No. of Pins	8, 14, 16	8, 14, 16, 20, 24	16, 20, 24, 28	24, 28	32, 40	44
Code No. *	S8GM S14GM S16GM	P8GM P20GM P14GM P24GM P16GM	P16GM P28GT P20GM P24GM	P24GM P28GU P28GM	S32GM P40GW	P44GX

[SSOP (Shrink Small Outline Package)]



Pin Pitch ; 0.8mm

Nominal Size	300mil	375mil	500mil	525mil
No. of Pins	36	42	70	64
Code No. *	P36GM	P42GT	P70G7	P64GW

Pin Pitch ; 0.65mm

Nominal Size	225mil	300mil	375mil
No. of Pins	8, 14, 16, 20	20, 24, 30, 38	30, 38, 48
Code No. *	P14GM P16GM P20GR	P20GM P24GS P30GS P38GS	S30GT P38GT P48GT

(* Refer to <C13388E>)

[TSSOP (Thin Small Outline Package)**]**

(0.65mm Pitch)

(225mil)



S8GR

(300mil)



S20GS

(0.5mm Pitch)

(300mil)



S26GS

	Pin Pitch ; 0.65mm		Pin Pitch ; 0.5mm
Nominal Size	225mil	300mil	300mil
No. of Pins	8	20	26
Code No. *	S8GR	S20GS	S26GS

(* Refer to <C13388E>)

[TSOP(I) (Thin Small Outline Package) (Type I)]

(0.5mm Pitch) (0.55mm Pitch)

(6mm×16mm) (8mm×20mm) (10mm×20mm) (12mm×18mm) (8mm×13.4mm)



S24GX S32GZ S40GZ S48GY P28GW

Pin Pitch ; 0.5mm Pin Pitch ; 0.55mm

Nominal Size	6mm×16mm	8mm×13.4mm	8mm×20mm	10mm×20mm	12mm×18mm	8mm×13.4mm
No. of Pins	24	32	32	40	48	28
Code No. *	S24GX	Under Development	S32GZ	S40GZ	S48GY	P28GW

[TSOP(II) (Thin Small Outline Package) (Type II)]

(1.27mm Pitch) (0.8mm Pitch) (0.65mm Pitch)

(300mil) (400mil) (500mil) (400mil) (500mil) (400mil)



S26GS S28G5 S34G7 S44G5 S54G7 S70G5

(550mil)



S26G3 S28G5 (28/24pin) S44G5 (44/40pin) S48G8



S32G5 S50G5



S54G5

Pin Pitch ; 1.27mm Pin Pitch : 0.8mm Pin Pitch : 0.65mm

Nominal Size	300mil	400mil	500mil	400mil	500mil	550mil	400mil
No. of Pins	26	28, 32	34	44, 50, 54	54	48	70
Code No. *	S26GS S26G3	S28G5 S28G5 (28/24pin) S32G5	S34G7	S44G5 S44G5 (44/40pin) S50G5 S54G5	S54G7	S48G8	S70G5

(* Refer to <C13388E>)

[QFP (Quad Flat Package)**]** (Package height 2.0mm~)

(1.0mm Pitch)



P52GC



P64GF



S74GJ

(0.8mm Pitch)



P44GB



P64GC



S80GF



S94GJ



P120GD

(0.65mm Pitch)



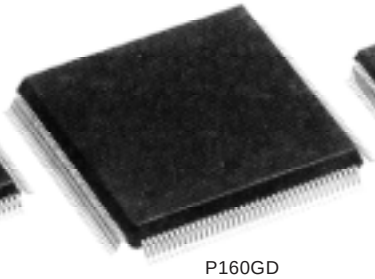
P48GB



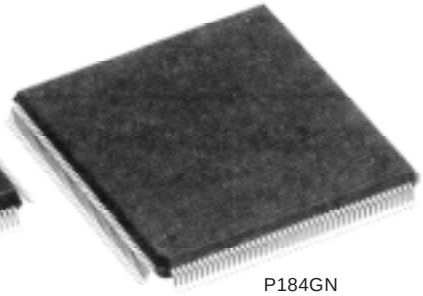
S80GC



P100GF



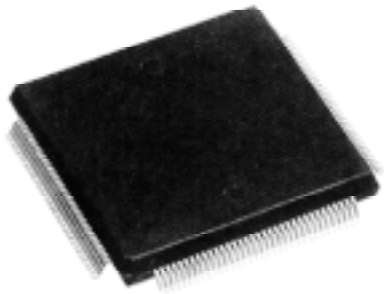
P160GD



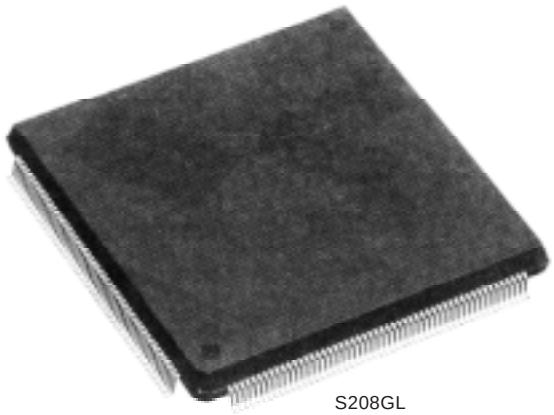
P184GN



S56GB



P136GD



S208GL

Pin Pitch ; 1.00mm / Pin Pitch ; 0.8mm

Nominal Size	□14mm	14mm×20mm	□20mm	□7mm	□10mm	□12mm	□14mm	14mm×20mm	□20mm	□24mm	□28mm
No. of Pins	52	64	74		44		64	80	94		120
Code No. *	P52GC	P64GF	S74GJ		P44GB		P64GC	S80GF	S94GJ		P120GD

Pin Pitch ; 0.65mm

Nominal Size	□7mm	□10mm	□12mm	□14mm	14mm×20mm	□20mm	□24mm	□28mm	□32mm	□36mm	□40mm
No. of Pins		48, 56		80	100			136, 160	184		208
Code No. *		P48GB S56GB		S80GC	P100GF			P136GD P160GD	P184GN		S208GL

(* Refer to <C13388E>)

(0.5mm Pitch)



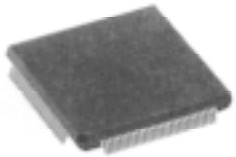
S48GA



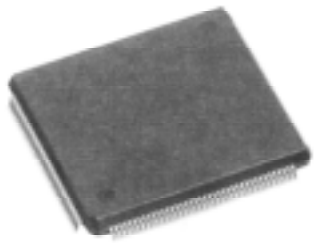
S72GB



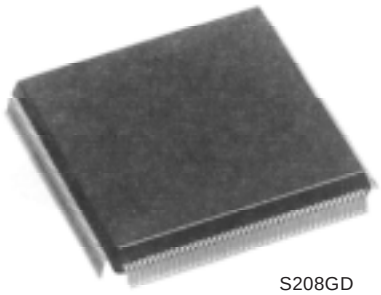
S88GK



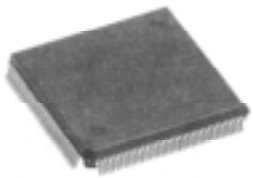
S120GJ



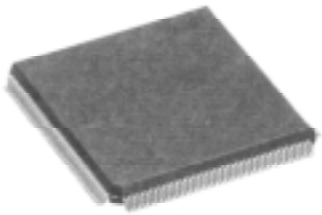
S160GM



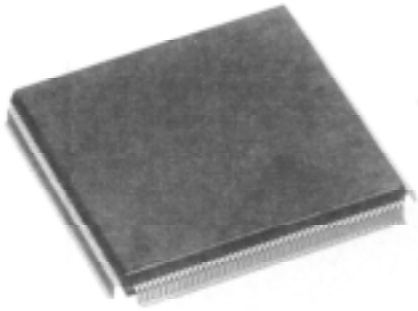
S208GD



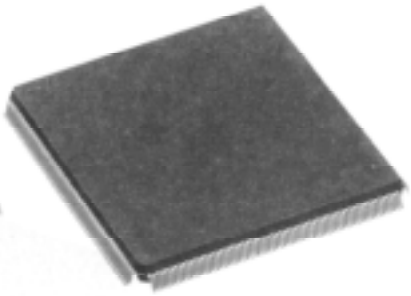
S144GJ



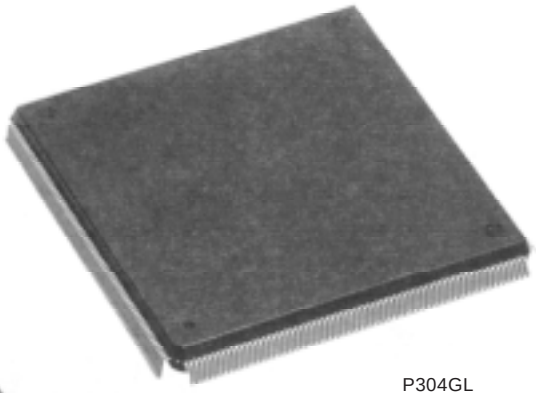
S176GM



P240GN

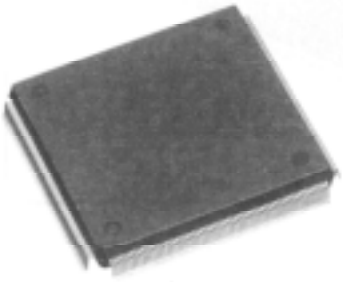


S272GP

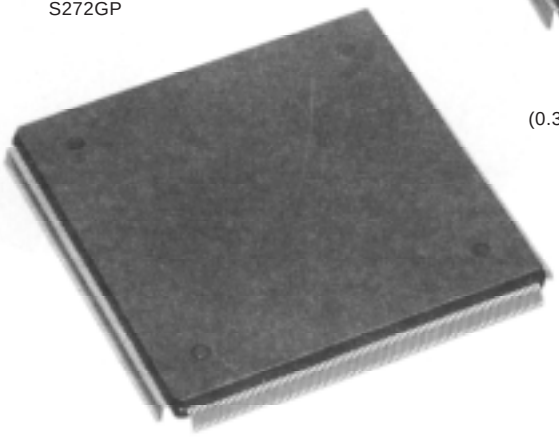


P304GL

(0.4mm Pitch)



S256GD



S376GL

(0.3mm Pitch)

Pin Pitch ; 0.5mm

Nominal Size	□7mm	□10mm	□12mm	□14mm	14mm×20mm	□20mm	□24mm	□28mm	□32mm	□36mm	□40mm
No. of Pins	48	72	88			120, 144	160, 176	208	240	272	304
Code No. *	S48GA	S72GB	S88GK			S120GJ S144GJ	S160GM S176GM	S208GD	P240GN	S272GP	P304GL

Pin Pitch ; 0.4mm

Nominal Size	□7mm	□10mm	□12mm	□14mm	□20mm	□24mm	□28mm	□32mm	□36mm	□40mm
No. of Pins							256			376
Code No. *							S256GD			S376GL

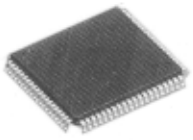
(* Refer to <C13388E>)

[LQFP (Low Profile Quad Flat Package)**]** (Package height 1.4mm Seating height 1.7mm MAX.)

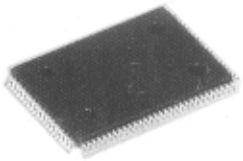
(0.65mm Pitch)



P64GK

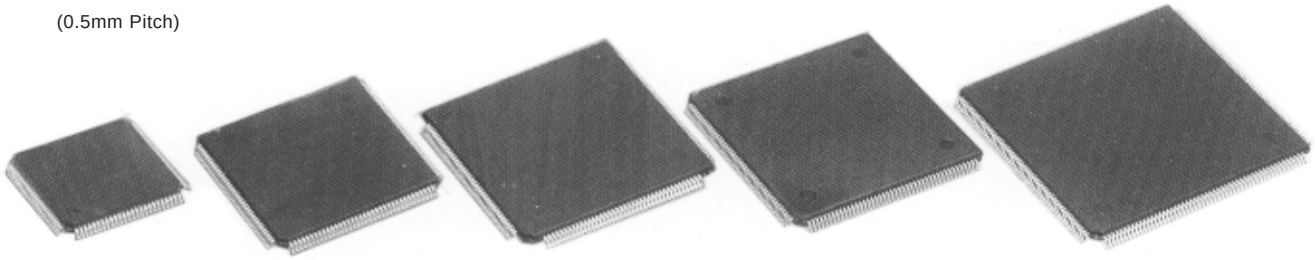


S80GC



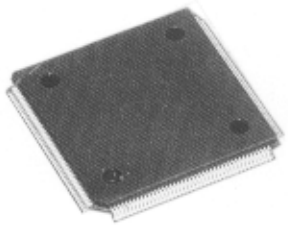
S100GF

(0.5mm Pitch)



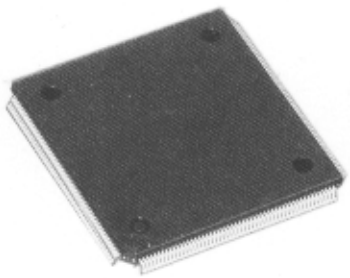
P100GC　S144GJ　S160GM　S176GM　S208GD

(0.4mm Pitch)



S176GJ

(0.3mm Pitch)



S216GM

Pin Pitch ; 0.65mm　　Pin Pitch ; 0.5mm

Nominal Size	□12mm	□14mm	14mm×20mm	□7mm	□10mm	□12mm	□14mm	□20mm	□24mm	□28mm	□32mm	□36mm
No. of Pins	64	80	100				100	144	160, 176	208	240	
Code No. *	P64GK	S80GC	S100GF				P100GC	S144GJ	S160GM S176GM	S208GD	Under Development	

Pin Pitch ; 0.4mm

Nominal Size	□7mm	□10mm	□12mm	□14mm	□20mm	□24mm	□28mm	□32mm	□36mm	□40mm
No. of Pins					176	216	256			
Code No. *					S176GJ	S216GM	S256GD			

(* Refer to <C13388E>)

[TQFP (Thin Quad Flat Package)**]** (Package height 1.0mm Seating height 1.27mm MAX.)

(0.5mm Pitch)



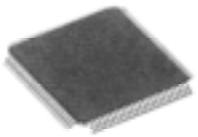
S48GA



S64GB



P80GK

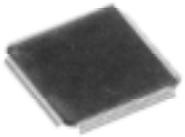


S100GC

(0.4mm Pitch)



S100GK



S120GC

Pin Pitch ; 0.5mm

Nominal Size	□7mm	□10mm	□12mm	□14mm	□20mm	□24mm	□28mm	□32mm	□36mm	□40mm
No. of Pins	48	64	80	100						
Code No. *	S48GA	S64GB	P80GK	S100GC						

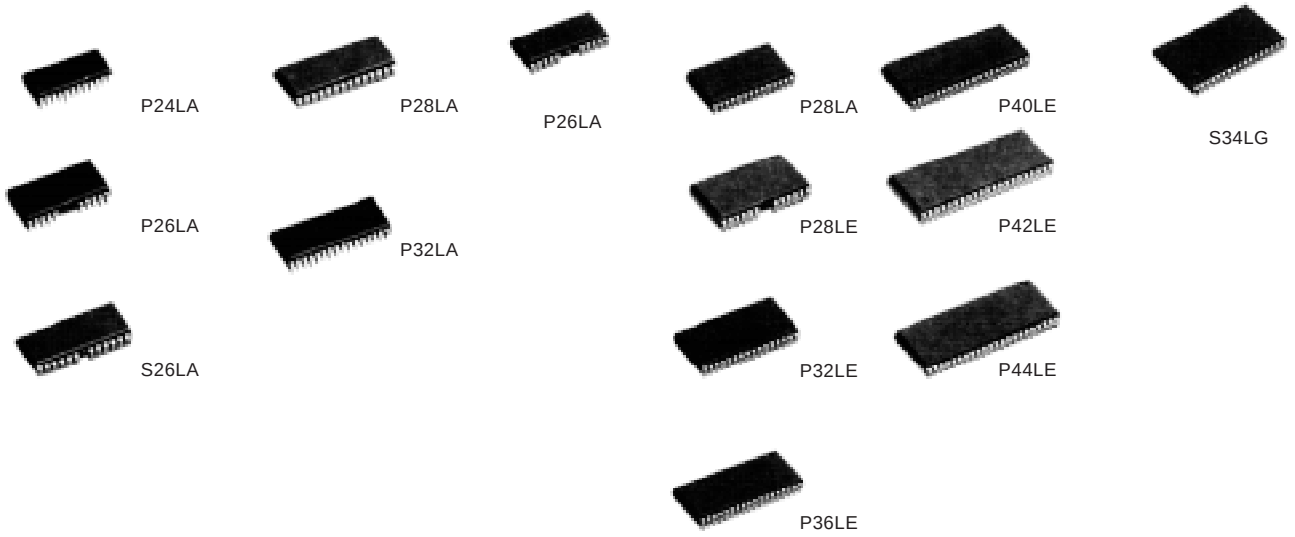
Pin Pitch ; 0.4mm

Nominal Size	□7mm	□10mm	□12mm	□14mm	□20mm	□24mm	□28mm	□32mm	□36mm	□40mm
No. of Pins			100	120						
Code No. *			S100GK	S120GC						

(* Refer to <C13388E>)

[SOJ (Small Outline J-leaded Package)]

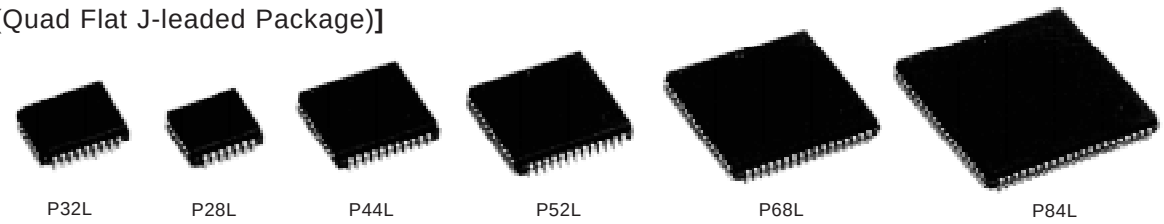
(300mil) (350mil) (400mil) (500mil)



Pin Pitch ; 1.27mm

Nominal Size	300mil					350mil	400mil							500mil
No. of Pins	24, 26, 28, 32					26	28, 32, 36, 40, 42, 44							34
Code No. *	P24LA	P26LA	S26LA	P28LA	P32LA	P26LA	P28LA	P28LE	P32LE	P36LE	P40LE	P42LE	P44LE	S34LG

[QFJ (Quad Flat J-leaded Package)]



Pin Pitch ; 1.27mm

Nominal Size	Rectangular	Square				
	550mil×450mil	□450mil	□650mil	□750mil	□950mil	□1150mil
No. of Pins	32	28	44	52	68	84
Code No. *	P32L	P28L	P44L	P52L	P68L	P84L

(* Refer to <C13388E>)

[QFN (Quad Flat Non-leaded Package)]

(Example: Window)



X32KW



X84KW-50A

(Example: Glass Sealed)



X22KH-50A



X24KH-50A



X28KH-50A

No. of Pins / Code No. *	32	44	64	68	74	80	84	94	100
Window	X32KW	X44KW-50A	X64KW-100A	X68KW-50A	X74KW-100A	X80KW-80A	X84KW-50A	X94KW-80A	X100KW-65A

No. of Pins / Code No. *	16	18	20	22	24	28	32	48	64
Glass Sealed	X16KH-A	X18KH-50A	X20KH-50A	X22KH-50A	X24KH-50A	X28KH-50A	X32KH-50A	X48KH-40A	X64KH-40A
Seam Weld					X24K-50A	X28K-50A			

[SVP (Surface Vertical Package)]



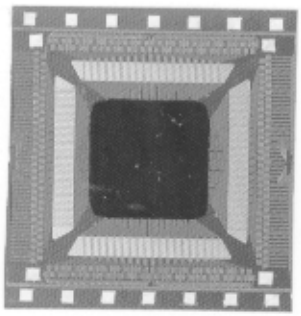
S32VN

No. of Pins	32
Code No. *	S32VN

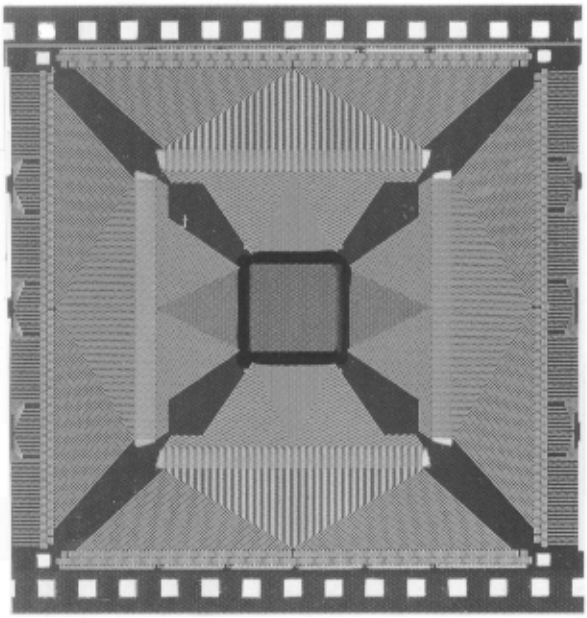
(* Refer to <C13388E>)

[TCP (Tape Carrier Package)**]**

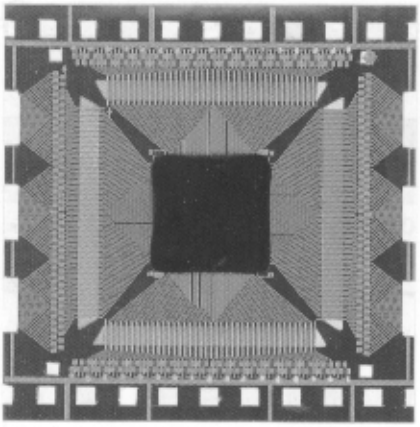
(QTP (Quad Tape Carrier Package)**)**



(35mm)

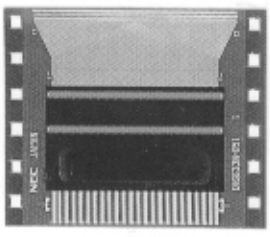


(70mm)

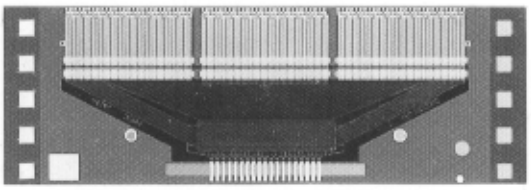


(48mm)

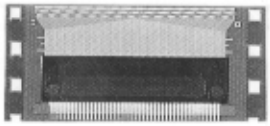
(LCD Driver, LCD Controller/Driver)



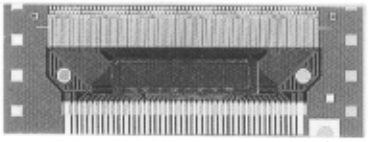
(35mm)



(70mm)

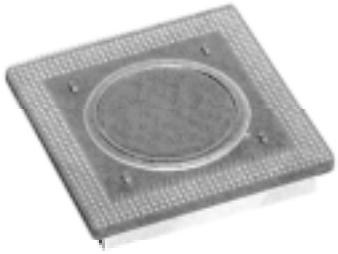


(35mm)



(48mm)

[PGA (Pin Grid Array)**]**



X528RF-50A

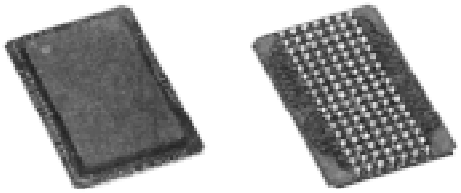
No. of Pins / Code No. *	528
Seam Weld	X528R-50A
Heat Sink	X528RF-50A

(* Refer to <C13388E>)

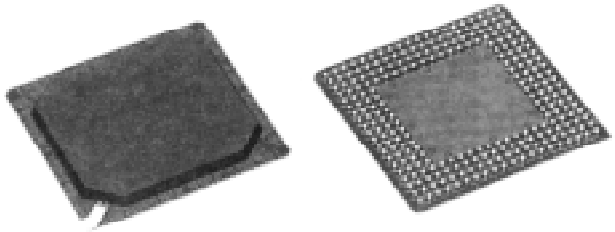
[BGA (Ball Grid Array)]

(Cavity Up Type)

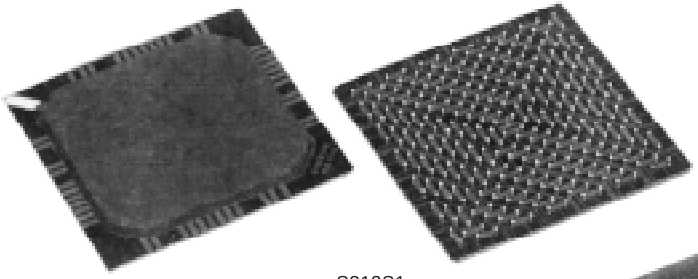
(1.27mm Pitch)



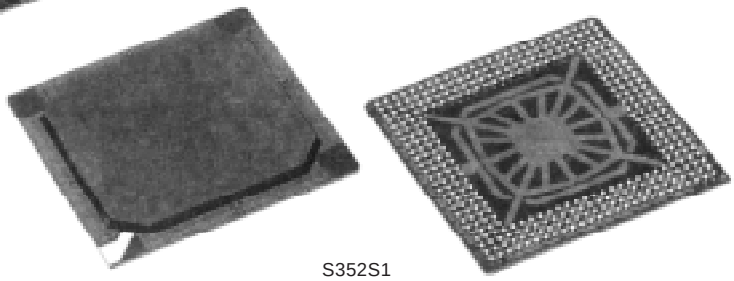
P119S1



S256S1

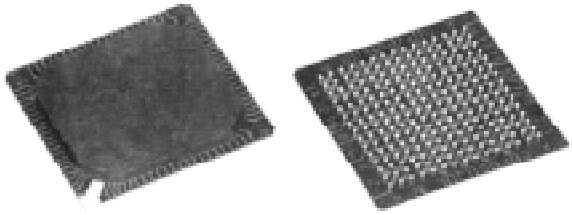


S313S1

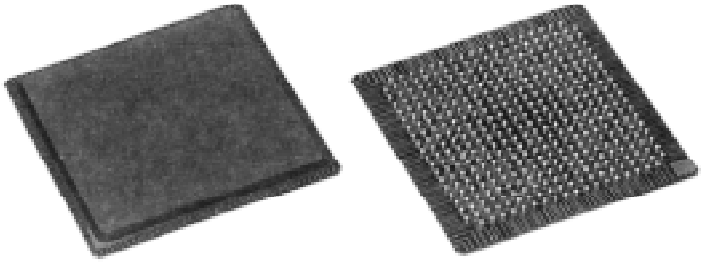


S352S1

(1.5mm Pitch)



S225S1



P396S1

Pin Pitch ; 1.27mm

Nominal Size	14mm×22mm	□27mm	□35mm	□40mm	□45mm
No. of Pins	119	256	313, 352		
Code No. *	P119S1	S256S1	S313S1 S352S1		

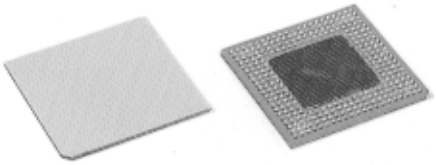
Pin Pitch ; 1.5mm

Nominal Size	14mm×22mm	□27mm	□35mm	□40mm	□45mm
No. of Pins		225	396		
Code No. *		S225S1	P396S1		

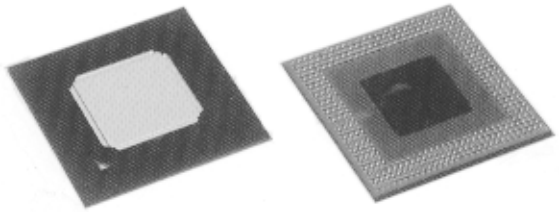
(* Refer to <C13388E>)

(Cavity Down Type)

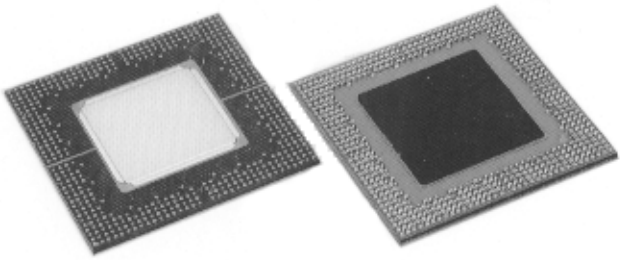
(1.27mm Pitch)



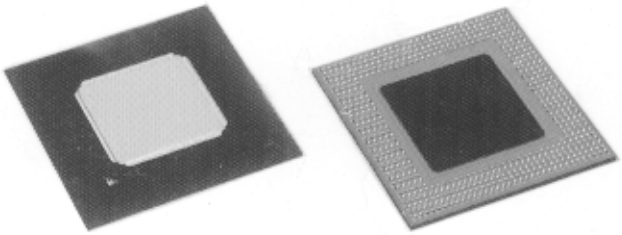
S272S2



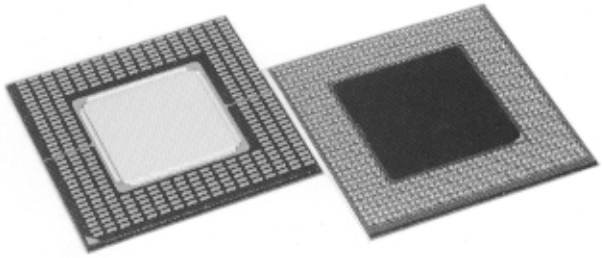
S416S2



S480S2



S580S2



S6722S2

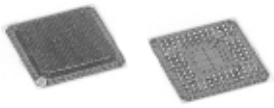
Pin Pitch ; 1.27mm

Nominal Size	□27mm	□35mm	□40mm	□45mm
No. of Pins	272		416	480, 580, 672
Code No. *	S272S2		S416S2	S480S2 S580S2 S672S2

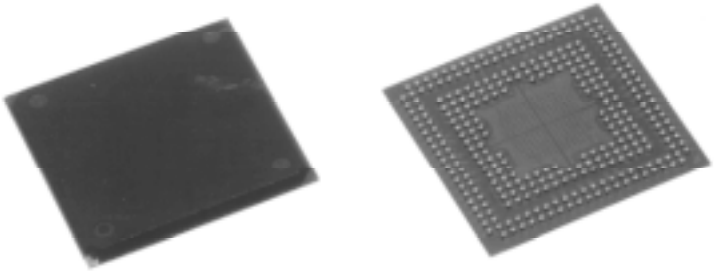
(* Refer to <C13388E>)

[FBGA]

(0.8mm Pitch)



P116S1

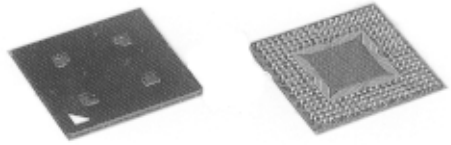


S304S1

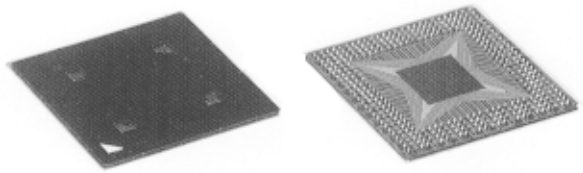
Pin Pitch ; 0.8mm

Nominal Size	□12mm	□19mm
No. of Pins	116	304
Code No.	P116S1	S304S1

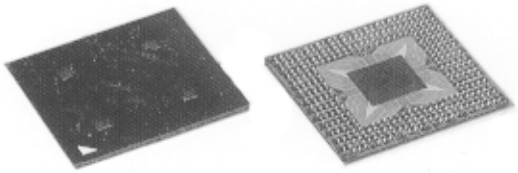
[T-BGA (Tape Ball Grid Array)]



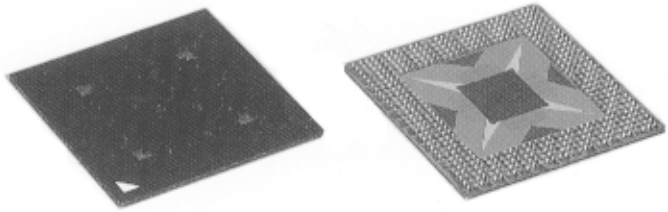
S256N7



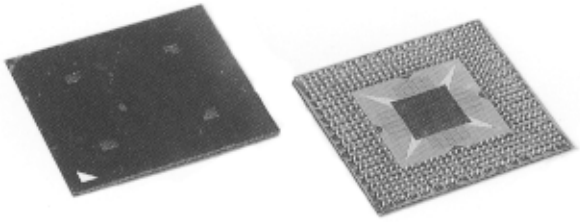
S352N7



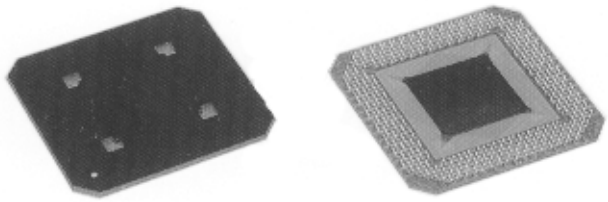
S420N7



S500N7



S576N7

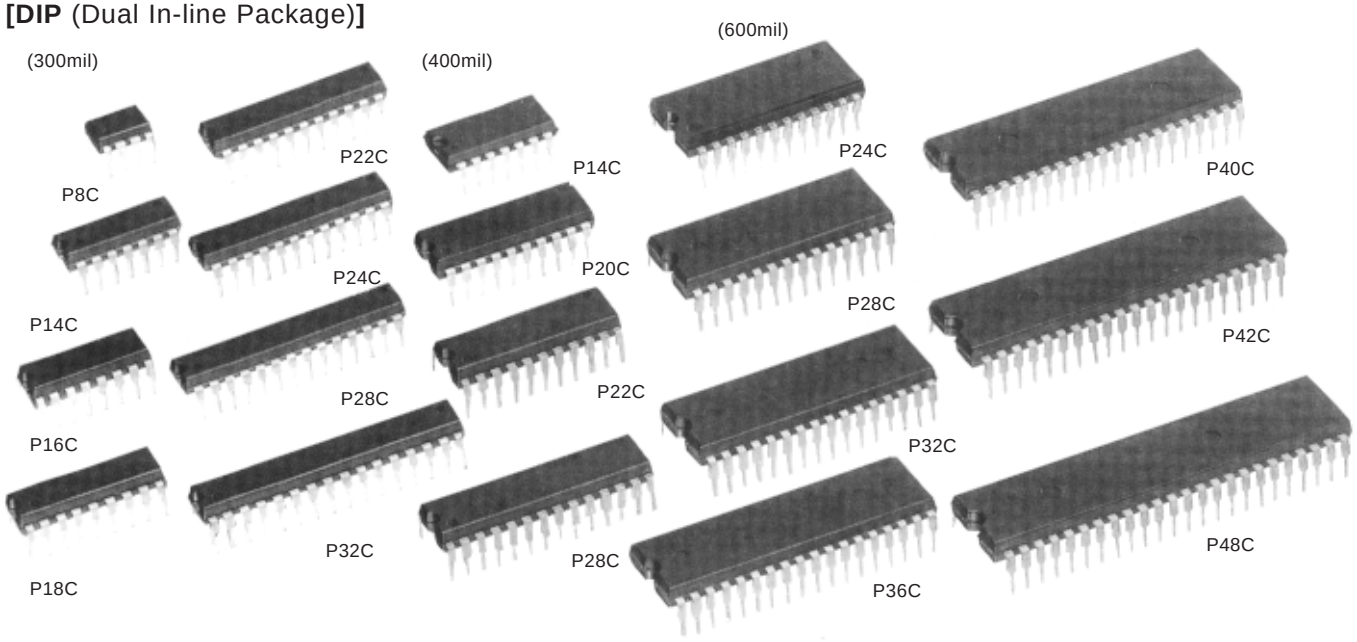


S696N7

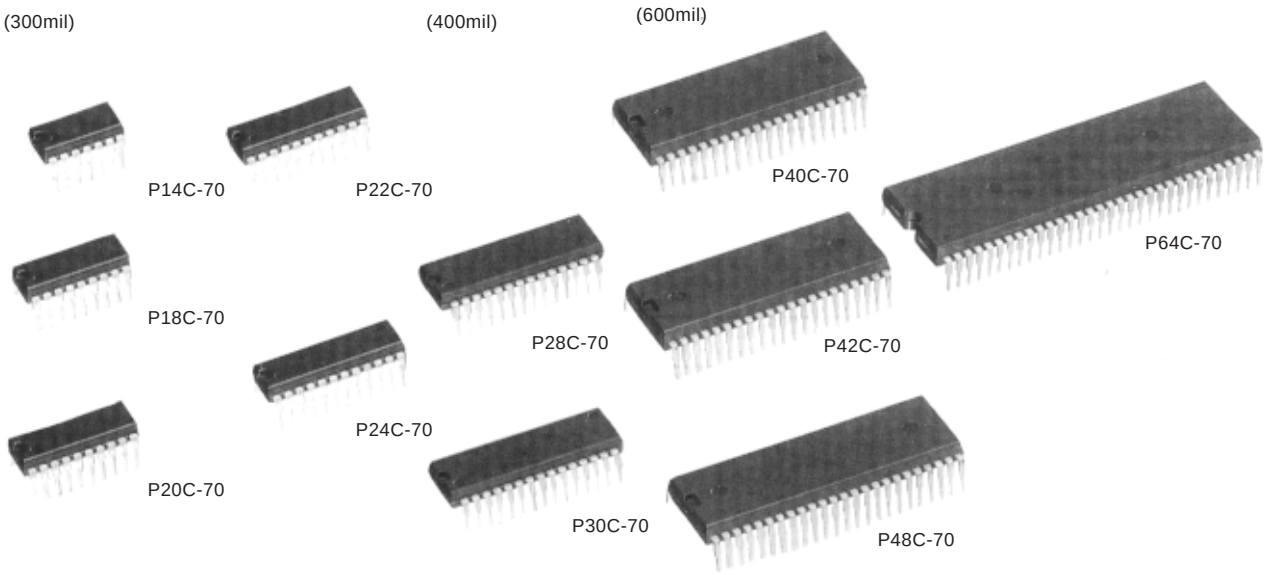
	Pin Pitch ; 1.27mm					Pin Pitch ; 1.0mm
Nominal Size	□27mm	□35mm		□40mm		□40mm
No. of Pins	256	352	420	500	576	696
Code No.	S256N7	S352N7	S420N7	S500N7	S576N7	S696N7

[Through Hole IC Packages]

[DIP (Dual In-line Package)]



[SDIP (Shrink Dual In-line Package)]



DIP Pin Pitch ; 2.54mm

Nominal Size	300mil	400mil	600mil
No. of Pins	8, 14, 16, 18, 20, 22, 24, 28, 32	14, 20, 22, 24, 28, 30	24, 28, 32, 36, 40, 42, 48
Code No. *	P8C P20C P14C P22C P16C P24C P18C P28C P32C	P14C P30C P20C P22C P28C	P24C P40C P28C P42C P32C P48C P36C

SDIP Pin Pitch ; 1.78mm

Nominal Size	300mil	400mil	600mil	750mil
No. of Pins	14, 16, 18, 20, 22, 24	28, 30	40, 42, 48	64
Code No. *	P14C-70 P24C-70 P18C-70 P20C-70 P22C-70	P28C-70 P30C-70	P40C-70 P42C-70 P48C-70	P64C-70

(* Refer to <C13388E>)

[Ceramic DIP (Ceramic Dual In-line Package)]

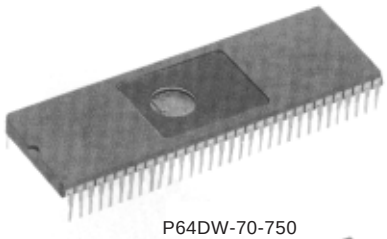
(Example: Window)



P28DW-100-300

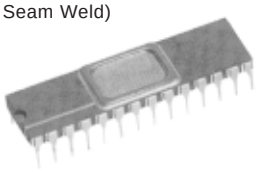


P40DW-100-300

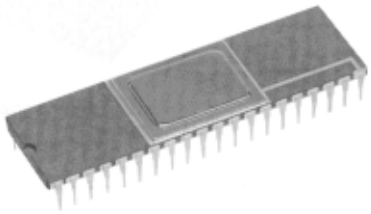


P64DW-70-750

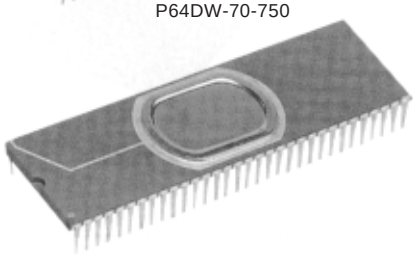
(Example: Seam Weld)



P28D-100-600



P40D-100-600



P64D-70-750

Pin Pitch ; 2.54mm

	Nominal Size	300mil	400mil	600mil
CerDIP	No. of Pins	8, 14, 16, 18, 20, 24	24, 28, 32, 40	22, 24, 28
	Code No. *	P8DH-100-300 P20DH-100-300 P14DH-100-300 P24DH-100-300 P16DH-100-300 P18DH-100-300	P22DH-100-400 P24DH-100-400 P28DH-100-400	P24DH-100-600 P28DH-100-600 P32DH-100-600 P40DH-100-600

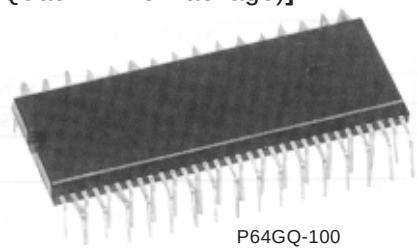
Pin Pitch ; 2.54mm / Pin Pitch ; 1.78mm

	Nominal Size	300mil	400mil	600mil	750mil
Seam Weld	No. of Pins	14, 16, 18, 20, 22	22, 24	24, 28, 40, 42, 48	64
	Code No. *	P14D-100-300 P16D-100-300 P18D-100-300 P20D-100-300 P22D-100-300	P22D-100-400 P24D-100-400	P24D-100-600 P28D-100-600 P40D-100-600 P42D-70-600 P48D-70-600	P64D-70-750

Pin Pitch ; 2.54mm / Pin Pitch ; 1.78mm

	Nominal Size	300mil	600mil	750mil
Window	No. of Pins	24	24, 28, 32, 40, 42	64
	Code No. *	P24DW-100-300	P24DW-100-600 P40DW-100-600 P28DW-100-600 P42DW-100-600 P32DW-100-600	P64DW-70-750

[QUIP (Quad In-line Package)]



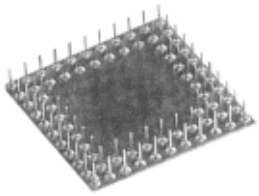
P64GQ-100

No. of Pins	Code No. *
64	P64GQ-100
64 (Window)	P64RQ-100

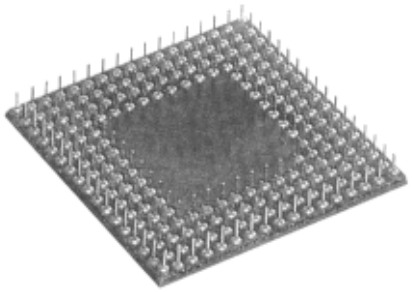
(* Refer to <C13388E>)

[PPGA (Plastic Pin Grid Array)]

(Example)



X72S-100A



X208S-100A

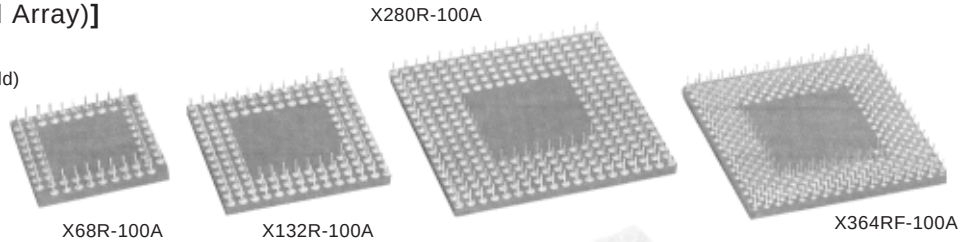
Pin Pitch ; 2.54mm

No. of Pins	72	84	120	132	175	176	208	280	447
Code No. *	X72S-100A X72SA-100A	X84SA-100A	X120S-100A X120SA-100A	X132S-100A X132SA-100A X132SF-100A	X175S-100A X175S-100A1	X176S-100A-1	X208S-100A	X280SF-100A	X447SF-100A

[PGA (Pin Grid Array)]

X280R-100A

(Example: Seam Weld)

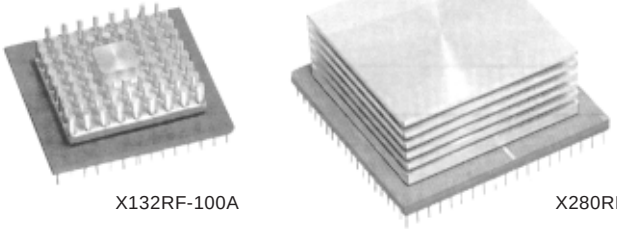


X68R-100A

X132R-100A

X364RF-100A

(Example: Heat Sink)



X132RF-100A

X280RF-100A

No. of Pins / Code No. *	68	72	84	132	144	176	179
Glass Sealed	X68RH-100A X68RW	X72RH-100A	X84RH-100A	X132RH-100A		X176RH-100A	
Seam Weld	X68R-100A	X72R-100A		X132R-100A	X144R-100A	X176R-100A	X179R-100A
Heat Sink		X72RF-100A		X132RF-100A		X176RF-100A	X179RF-100A

No. of Pins / Code No. *	208	280	364	477	528
Glass Sealed	X208RH-100A				
Seam Weld	X208R-100A	X280R-100A	X364R-100A	X447RJ-50A	X528R-50A1
Heat Sink	X208RF-100A	X280RF-100A	X364RF-100A	X447RF-50A	

(* Refer to <C13388E>)

[SIP (Single In-line Package)]

(Example)


P3J-127B


P3H-170B


P3H-230


P3HF-254B


P5HP-150B

[ZIP (Zigzag In-line Package)]

(Example) (400mil) (475mil)


P20V-254A


P24V-254A


P28V-254A

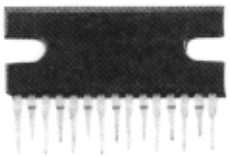

P40V-100

[V-DIP (Vertical Dual In-line Package)]

(Example)


P5VP-340B4


P15VP-254B


P15VP-400B

V-DIP

No. of Pins / Pin Pitch	5	11	14	15
2.0				P15VP-400B
1.7		P11VP-340B3	P15VP-254B	
1.27	P5VP-340B4			

SIP

No. of Pins / Pin Pitch	2	3	4	5
2.54		P3HP-254B P3HF-254B		
2.3		P3H-230		
2.0				
1.5				P5HP-150B
1.778		P3H-170B		
1.27		P3J-127B		

ZIP

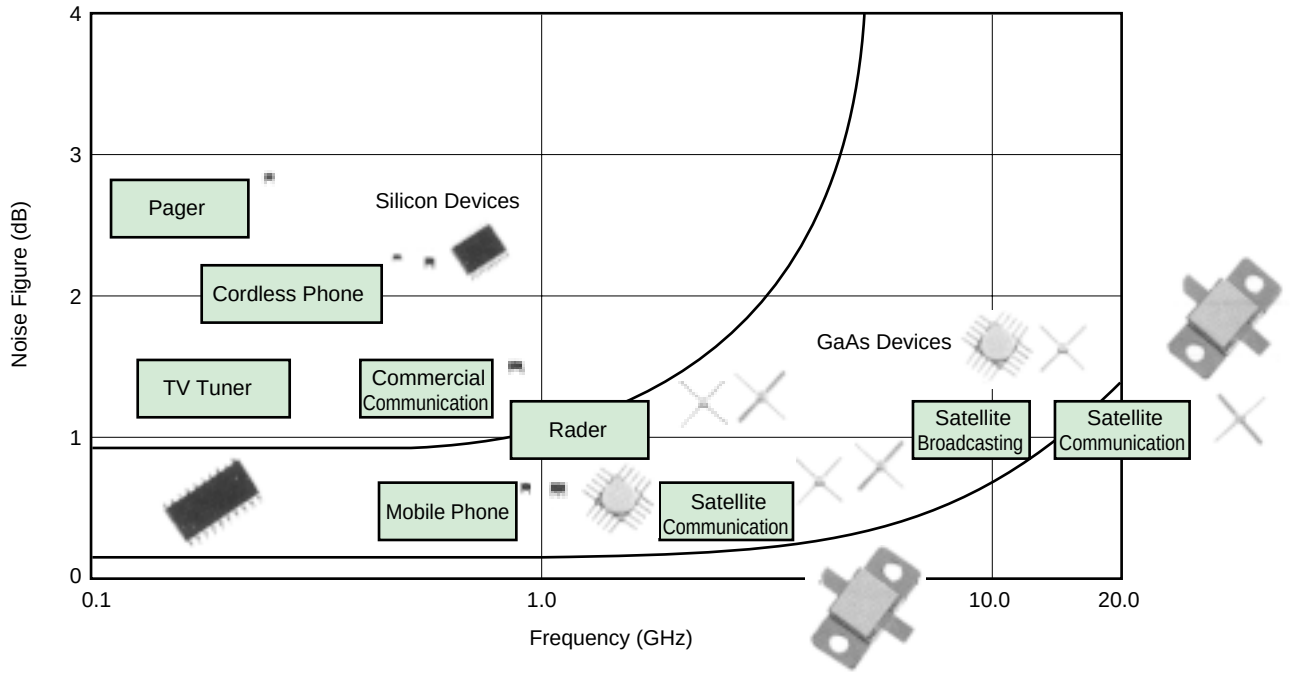
Nominal Size	325mil	400mil	475mil
No. of Pins	16, 20	20, 24, 28	24, 40
Code No. *	P16V-254A P20V-254A	P20V-254A P24V-254A P28V-254A	P24V-100 P40V-100 P40VF-70

(* Refer to <C13388E>)

[Surface Mount Discrete Device Packages]

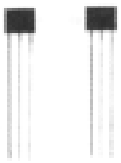
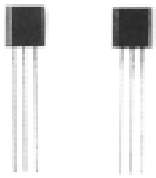
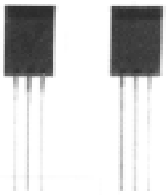
Mounted Area (mm^2)	Package Names \ Uses		Small Signal Tr/FET	Power MOS FET Power Tr	Zener Diodes	Noise Clipping Diodes	SCR	Triac	Metal Oxide Varistors
2.5	RC-3402								○
1.7	2pin Ultra Super Mini Mold				○	○			
2.6	3pin Ultra Super Mini Mold		○						
3.1	2pin Super Mini Mold				○	○			
4.2	6pin Super Mini Mold (Similar to SC-70)		○						
4.2	5pin Super Mini Mold (SC-70)		○			○ (SC-88A)			
8.1	3pin Super Mini Mold (Similar to SC-59)		○						
8.1	6pin Mini Mold (Similar to SC-59)		○						
8.1	5pin Mini Mold (Similar to SC-59)		○			○ (SC-74A)			
8.1	3pin Mini Mold (SC-59)		○		○	○			
11.8	2pin Power Mini Mold		○		○		○	○	
18	3pin Power Mini Mold		○	○	○				
31	MP-2			○					
65	MP-3Z (SC-63)			○			○	○	
125	MP-25Z (SC-46)			○			○	○	

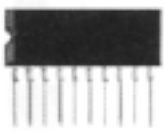
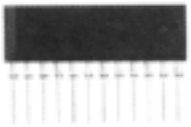
Radio Frequency Devices/Microwave Devices



	Microwave Devices
Ceramic Packages	 T-41 83A 84A T-61B K31

[Through Hole Discrete Device Packages]

	Small Signal Transistor Packages
Through Hole Packages	 SST  TO-92  SP-8

		Small Signal Transistor Packages
Through Hole Packages		 MP-5 (TO-126)  MP-3  MP-25  MP-25-S  MP-88
	Full Mold Packages	 MP-45  MP-45F
	Transistor Array	 10P SIP  12P SIP

INDEX

Part number	Page
[Numerals]	
03P2J	227
03P2M	227
03P4J	227
03P4M	227
03P4MF	227
03P4MG	227
03P5J	227
03P5MG	227
03P6MG	227
2P4M	227
2P5M	227
2P6M	227
2S2M	227
2S4M	227
2SA1008	190,199
2SA1009	190,199
2SA1009A	190,199
2SA1010	190,199
2SA1069	190,199
2SA1069A	190
2SA1129	190,200
2SA1156	189,199
2SA1173	194
2SA1174	196
2SA1175	187,196
2SA1206	186,197
2SA1221	188
2SA1222	188
2SA1226	193
2SA1247	193,196
2SA1261	190
2SA1330	193,196
2SA1376	186,196
2SA1376A	186
2SA1385	189,198
2SA1385-Z	195
2SA1394	191,199
2SA1395	191,199
2SA1396	191,199
2SA1400	189,199
2SA1400-Z	195
2SA1412	189
2SA1412-Z	195
2SA1413	189,199
2SA1413-Z	195
2SA1441	191,198
2SA1442	191,198
2SA1443	191,198
2SA1444	191,198
2SA1458	187,197
2SA1459	187,197
2SA1460	188,197,199
2SA1461	193,197
2SA1462	193,197
2SA1463	194,197
2SA1464	193,197
2SA1486	189,199
2SA1544	186
2SA1546	189,200
2SA1608	195,197
2SA1610	195,197
2SA1611	195,196
2SA1612	195,196
2SA1615	189,198
2SA1615-Z	195
2SA1625	186
2SA1626	188
2SA1627	188
2SA1645	190,198
2SA1646	190
2SA1647	189,198
2SA1648	189,198
2SA1649	189,198
2SA1650	191,198
2SA1652	191,198
2SA1720	191,201
2SA1741	191,198
2SA1742	191,198
2SA1743	191,198
2SA1744	191,198
2SA1836	195
2SA1840	192,201
2SA1841	192,201
2SA1843	192,198
2SA1845	192,198
2SA1847	192,198
2SA1871	194
2SA1897	188,198
2SA675	186
2SA733	186,196
2SA811A	193,196
2SA812	193,196
2SA915	188
2SA916	188
2SA952	186,196
2SA953	186,196
2SA954	186,196
2SA987	186,196
2SA988	186,196
2SA990	186
2SA992	186,196
2SB1068	186,196
2SB1093	188,201
2SB1094	191,200
2SB1097	191,200
2SB1098	191,201
2SB1099	191,201
2SB1100	191,201
2SB1114	194,196
2SB1115	194,196
2SB1115A	194,196
2SB1116	186,196
2SB1116A	186,196
2SB1117	188,198
2SB1149	189,201
2SB1150	189,201
2SB1151	189,198
2SB1217	189,198
2SB1261	189,198
2SB1261-Z	195
2SB1300	186
2SB1318	188
2SB1430	191,201
2SB1431	191,201
2SB1432	191,201
2SB1453	191
2SB1475	195
2SB1571	194
2SB1572	194
2SB1578	194,198
2SB1581	198
2SB1628	194
2SB546A	190,200
2SB564	188,196
2SB601	190,201
2SB605	188,196
2SB624	193,196
2SB707	190,200
2SB708	190,200
2SB733	188
2SB734	188
2SB736	193,196
2SB736A	193,196
2SB768	189,195,200
2SB772	189,198
2SB794	189,201
2SB795	189,201
2SB798	194,196
2SB799	194,196
2SB800	194,196
2SB804	194
2SB805	194
2SB806	194
2SB810	187,196
2SB811	196
2SB897	192,201
2SB962	189,198
2SB962-Z	195
2SB963	189,201
2SB963-Z	195
2SB984	188
2SC1009A	193,196
2SC1505	190,200
2SC1621	193
2SC1622A	193,196
2SC1623	193,196
2SC1653	193,196
2SC1654	193,196
2SC1674	186,196
2SC1840	186,196
2SC1841	186,196
2SC1842	186
2SC1843	186
2SC1845	186,196
2SC1940	188,196
2SC1941	188,196
2SC2001	186,196
2SC2002	186,196
2SC2003	186,196
2SC2148	243
2SC2149	243
2SC2223	193,196
2SC2331	190,199
2SC2333	190,199
2SC2334	190,199
2SC2335	190
2SC2351	242
2SC2517	190,199
2SC2518	190,199
2SC2570A	242
2SC2688	189,200
2SC2690	189,200
2SC2690A	189,200
2SC2721	188
2SC2749	192,199
2SC2750	192,199
2SC2751	192,199
2SC2752	189,199
2SC2780	194,196
2SC2784	187,196
2SC2785	187,196
2SC2786	187,196
2SC2787	187,196
2SC2802	200
2SC2885	189,199
2SC2901	186,197
2SC2908	192
2SC2946	189,199
2SC2946(1)	195,199
2SC2958	188
2SC2959	188
2SC3115	193,196
2SC3157	190
2SC3158	190,199
2SC3159	190,199
2SC3209	188,196
2SC3355	242
2SC3356	242
2SC3360	193,196
2SC3478	186,196
2SC3478A	186
2SC3518	189,198
2SC3518-Z	195
2SC3545	241,242
2SC3554	194,196
2SC3567	191,199
2SC3568	191,199
2SC3569	191,199
2SC3570	191,199
2SC3571	191,199
2SC3572	191,199
2SC3587	243
2SC3588	189,199
2SC3588-Z	195
2SC3603	243
2SC3604	243
2SC3615	186,196
2SC3616	186,196
2SC3617	194,196
2SC3618	194,196
2SC3622	186,196
2SC3622A	186,196
2SC3623	187,196
2SC3623A	187,196
2SC3624	193,196
2SC3624A	193,196
2SC3631	189,199
2SC3631-Z	195
2SC3632	189,199
2SC3632-Z	195
2SC3691	191,198
2SC3692	191,198
2SC3693	191,198
2SC3694	191,198
2SC3731	187,197
2SC3732	187,197
2SC3733	188,197,199
2SC3734	193,197
2SC3735	193,197
2SC3736	194,197
2SC3739	193,197
2SC3840	189,199
2SC3841	241
2SC4000	188,200
2SC4001	189,200
2SC4062	191,198
2SC4063	191,198
2SC4092	242
2SC4093	242
2SC4173	195,197
2SC4175	195
2SC4176	195,197
2SC4177	195,196
2SC4178	195,196
2SC4179	195,196
2SC4180	195
2SC4181	195,196
2SC4182	241
2SC4183	241
2SC4184	241,242
2SC4185	241,242
2SC4186	241
2SC4226	242
2SC4228	242
2SC4305	190
2SC4331	189,198
2SC4336	191,198
2SC4342	189,201
2SC4346	189,199
2SC4350	190
2SC4351	191,201
2SC4549	191,198
2SC4550	191,198
2SC4551	191,198
2SC4552	191,198
2SC4553	191,198
2SC4554	191,198
2SC4568	242
2SC4569	242
2SC4570	242
2SC4571	242
2SC4783	195
2SC4810	192,201
2SC4811	192,201
2SC4813	192,198
2SC4814	192,198
2SC4815	192,198
2SC4942	194
2SC5004	242
2SC5005	242
2SC5006	242
2SC5008	242
2SC5011	242
2SC5013	242
2SC5288	243
2SC5289	243
2SC945	186,196
2SD1000	194,196
2SD1001	194,196
2SD1005	194
2SD1006	194
2SD1007	194
2SD1020	187,196
2SD1033	189,195,200
2SD1164	189,201
2SD1164-Z	195
2SD1210	192,201
2SD1286	189,201
2SD1286-Z	195

Part number	Page
2SD1296	192,201
2SD1297	201
2SD1308	190,201
2SD1309	190,201
2SD1312	188
2SD1392	190,201
2SD1513	186,196
2SD1579	188,201
2SD1581	188,196,198
2SD1582	188,198
2SD1583	189,198
2SD1583-Z	195
2SD1584	189,198
2SD1584-Z	195
2SD1585	191,200
2SD1588	191,200
2SD1589	191,201
2SD1590	191,201
2SD1591	191,201
2SD1592	191,201
2SD1593	191,198
2SD1594	191,198
2SD1614	194,196
2SD1615	194,196
2SD1615A	194,196
2SD1616	186,196
2SD1616A	186,196
2SD1630	189,201
2SD1691	189,198
2SD1692	189,201
2SD1693	189,201
2SD1694	189,198
2SD1695	189,201
2SD1697	188,196,201
2SD1698	186,196
2SD1699	194,196
2SD1700	188,196,201
2SD1701	186,196
2SD1702	194,196
2SD1779	188
2SD1780	188
2SD1818	189,198
2SD1843	188,201
2SD1899	189,198
2SD1899-Z	195
2SD1950	194,196
2SD2161	191,201
2SD2162	191,201
2SD2163	191,201
2SD2164	191,198
2SD2165	191,198
2SD2228	195
2SD2383	193
2SD2402	194
2SD2403	194
2SD2425	194,198
2SD2463	188,201
2SD401A	190,200
2SD414	200
2SD415	189,200
2SD471	188,196
2SD560	190,201
2SD571	188,196
2SD596	193,196
2SD773	188
2SD774	188
2SD780	193,196
2SD780A	193,196
2SD794	189,200
2SD794A	189,200
2SD882	189,198
2SD985	189,201
2SD986	189,201
2SD992	189
2SD999	194,196
2SJ128	209
2SJ132	209
2SJ133	209
2SJ134	209
2SJ135	209
2SJ136	209
2SJ137	209
2SJ138	209
2SJ139	209
2SJ140	209
2SJ141	209
2SJ142	209
2SJ143	209
2SJ165	204
2SJ166	204
2SJ178	204
2SJ179	204
2SJ180	204
2SJ184	204
2SJ185	204
2SJ196	204
2SJ197	204
2SJ198	204
2SJ199	204
2SJ202	204
2SJ203	204
2SJ204	204
2SJ205	204
2SJ206	204
2SJ207	204
2SJ208	204
2SJ209	204
2SJ210	204
2SJ211	204
2SJ212	204
2SJ218	204
2SJ243	204
2SJ302	209,210
2SJ303	209,210
2SJ324	209,210
2SJ325	209,210
2SJ326	209,210
2SJ327	209,210
2SJ328	209,210
2SJ329	209,210
2SJ330	209,210
2SJ331	209,210
2SJ353	204
2SJ355	204
2SJ356	204
2SJ357	204
2SJ358	204
2SJ411	204
2SJ448	209,212
2SJ449	209,212
2SJ460	204
2SJ461	204
2SJ462	204
2SJ463A	204
2SJ492	209,210
2SJ493	209
2SJ494	209,210
2SJ495	209,210
2SK1059	205
2SK1060	205
2SK1122	205,210
2SK1123	205,210
2SK1132	202
2SK1133	202
2SK1198	205
2SK1271	205
2SK1272	202
2SK1273	202
2SK1274	202
2SK1282	205,210
2SK1283	205,210
2SK1284	205,210
2SK1285	205,210
2SK1286	205,210
2SK1287	205,210
2SK1288	205,210
2SK1289	205,210
2SK1290	205,210
2SK1292	205,210
2SK1293	205,210
2SK1294	205,210
2SK1295	205,210
2SK1398	202
2SK1399	202
2SK1482	202
2SK1483	202
2SK1484	202
2SK1485	202
2SK1491	205,211
2SK1492	205,211
2SK1493	205,211
2SK1494	205,211
2SK1495	205,211
2SK1496	205,211
2SK1497	205,211
2SK1498	205,211
2SK1499	205,211
2SK1500	205,211
2SK1501	205,211
2SK1502	205,211
2SK1580	202
2SK1581	202
2SK1582	202
2SK1583	202
2SK1584	202
2SK1585	202
2SK1586	202
2SK1587	202
2SK1588	202
2SK1589	202
2SK1590	202
2SK1591	202
2SK1592	202
2SK1593	202
2SK1594	205,210
2SK1596	205,210
2SK1656	202
2SK1657	202
2SK1658	202
2SK1748	205,210
2SK1749	205,210
2SK1750	205,211
2SK1751	205,211
2SK1752	205,211
2SK1753	206,211
2SK1756	206,211
2SK1757	206,211
2SK1758	206,211
2SK1760	206,211
2SK1784	206,211
2SK1785	206,211
2SK1793	206,211
2SK1794	206,211
2SK1795	206,211
2SK1796	206,211
2SK1824	202
2SK1850	206,210
2SK1851	206,210
2SK1852	206,210
2SK1853	206,210
2SK1953	206,212
2SK1954	206,212
2SK1958	202
2SK1959	202
2SK1960	202
2SK1988	206,211
2SK1989	206,211
2SK1990	206,211
2SK1991	206,211
2SK1992	206,211
2SK1993	206,211
2SK1994	206,211
2SK1995	206,211
2SK2040	206,212
2SK2053	202
2SK2054	202
2SK2055	203
2SK2057	211
2SK2058	211
2SK2070	203
2SK2090	203
2SK2109	203
2SK2110	203
2SK2111	203
2SK2112	203
2SK2131	206,210
2SK2132	206,212
2SK2133	206,212
2SK2134	206,212
2SK2135	206,212
2SK2136	206,212
2SK2137	206,212
2SK2138	206,212
2SK2139	206,212
2SK2140	206,212
2SK2141	206,212
2SK2157	203
2SK2158	203
2SK2159	203
2SK2234	206,211
2SK2275	206,211
2SK2341	206,212
2SK2353	207,212
2SK2354	207,212
2SK2355	207,212
2SK2356	207,212
2SK2357	207,212
2SK2358	207,212
2SK2359	207,212
2SK2360	207,212
2SK2361	207,212
2SK2362	207,212
2SK2363	207,212
2SK2364	207,212
2SK2365	207,212
2SK2366	207,212
2SK2367	207,212
2SK2368	207,212
2SK2369	207,212
2SK2370	207,212
2SK2371	207,212
2SK2372	207,212
2SK2396	243
2SK2409	207,210
2SK2410	207,210
2SK2411	207,210
2SK2412	207,210
2SK2413	207,210
2SK2414	207,210
2SK2415	207,210
2SK2461	207,210
2SK2462	207,210
2SK2476	207,212
2SK2477	207,212
2SK2478	207,212
2SK2479	207,212
2SK2480	207,212
2SK2481	207,212
2SK2482	207,212
2SK2483	207,212
2SK2484	207,212
2SK2485	207,212
2SK2486	208,212
2SK2487	208,212
2SK2488	208,212
2SK2498	208,210
2SK2499	208,210
2SK2510	208,210
2SK2511	208,210
2SK2512	208,210
2SK2513	208,210
2SK2514	208,210
2SK2515	208,210
2SK2541	203
2SK2597	243
2SK2723	208,210
2SK2724	208,210
2SK2826	208,211
2SK2857	203
2SK2858	203
2SK2941	208,210
2SK2981	208,211
2SK2982	208,211
2SK2983	208,211
2SK2984	208,211
2SK3055	208,211
2SK3056	208,211
2SK3057	208
2SK3058	208
2SK3059	208,211

Part number	Page
2SK3060	208
2SK3061	208,211
2SK3062	208,211
2SK679A	202
2SK680A	202
2SK681A	202
2V5P4M	227
3P4J	227
3P4J-Z	227
3P4MH	227
3P5MH	227
3P6MH	227
3S4M	227
3SK131	241
3SK134B	241
3SK135A	241
3SK177	241
3SK206	241
3SK222	241
3SK223	241
3SK224	241
3SK230	241
3SK231	241
3SK242	241
3SK252	241
3SK253	241
3SK254	241
3SK255	241
5P4J	227
5P4J-Z	227
5P4M	227
5P4SM	227
5P5M	227
5P6J	227
5P6J-Z	227
5P6M	227
5P6SM	227
8P2M	227
8P2SM	227
8P4J	227
8P4J-Z	227
8P4M	227
8P4SM	227
[A]	
AA1[ ]	186,215
AB1[ ]	186,216
AC01DGM	227
AC01DJM	227
AC03DGM	227
AC03DJM	227
AC03DJM-Z	227
AC03DSM	227
AC03EGM	227
AC03FGM	227
AC03FJM	227
AC03FJM-Z	227
AC03FSM	227
AC05DGM	227
AC05DJM	227
AC05DJM-Z	227
AC05DSM	227
AC05EGM	227
AC05ESM	227
AC05FGM	227
AC05FJM	227
AC05FJM-Z	227
AC05FSM	227
AC08DGM	227
AC08DSM	227
AC08EGM	227
AC08ESM	227
AC08FGM	227
AC08FSM	227
AC0V8DGM	227
AC10DGM	227
AC10DSM	227
AC10EGM	227
AC10ESM	227
AC10FGM	227
AC10FSM	227
AC12DGM	227
AC12DSM	227
AC12EGM	227
AC12ESM	227
AC12FGM	227
AC12FSM	227
AC16DGM	227
AC16DSM	227
AC16EGM	227
AC16FGM	227
AC16FSM	227
AD1[ ]	186,217
AD2[ ]	186,217
AN1[ ]	186,215
AP1[ ]	186,216
AQ1[ ]	186,216
AR1[ ]	186,217
[B]	
BA1[ ]	187,215
BB1[ ]	187,216
BN1[ ]	187,215
BP1[ ]	187,216
[C]	
CB-C7	141
CB-C8	143
CB-C9	144
CE1[ ]	188,217
CE2[ ]	188,217
CHS	147
CHS-A	147
[F]	
FA1[ ]	193,218
FB1[ ]	193,218
FN1[ ]	193,218
FP1[ ]	193
[G]	
GA1[ ]	195,218
GN1[ ]	195,218
[H]	
HD1[ ]	194,219
HD2[ ]	194,219
HQ1[ ]	194,219
HR1[ ]	194,219
[M]	
M-CHS	147
MC-22000	131
MC-22001	131
MC-22002	131
MC-22003	131
MC-22004	131
MC-22005	131
MC-22006	131
MC-22007	131
MC-22100	131
MC-22101	131
MC-22102	131
MC-22103	131
MC-22104	131
MC-22105	131
MC-22106	131
MC-22107	131
MC-4516CB64S	122
MC-4516CC725FA	120
MC-4516CC726F	120
MC-4516CD645FA	120
MC-4516CD646F	120
MC-4516CD64S	122
MC-4516DA726F	121
MC-4516DA72F	123
MC-4532CC726F	120
MC-4532CD646F	120
MC-4532DA726F	121
MC-454CA726F	120
MC-454CB645FA	120
MC-454CB646F	120
MC-454CB64S	122
MC-454DA726F	121
MC-458CA725FA	120
MC-458CA726F	120
MC-458CB645FA	120
MC-458CB646F	120
MC-458CB64SA	122
MC-458CD64S	122
MC-458DA726F	121
MC-458DA72F	123
MC-48R128AAG181F	123
MC-48R64AAG181F	123
MC-7660	231,233
MC-7661	231,233
MC-7680	231
MC-7712	231
MC-7716	231
MC-7722	231
MC-7726	231
MC-7816	231
MC-7826	231
MC-7852	231
MC-7856	231
MC-7862	231
MC-7866	231
MC-7900	231
MC-7901	231
μPA101	239
μPA102	239
μPA103	239
μPA104	239
μPA1428A	222
μPA1434	222
μPA1436A	222
μPA1437	222
μPA1453	222
μPA1454	222
μPA1456	222
μPA1458	222
μPA1476	222
μPA1478	222
μPA1500B	223
μPA1501	223
μPA1520B	223
μPA1522	223
μPA1523B	223
μPA1524	223
μPA1526	223
μPA1527	223
μPA1550	223
μPA1552B	223
μPA1556A	223
μPA1572B	223
μPA1576	223
μPA1600	223
μPA1601	223
μPA1602	223
μPA1603	223
μPA1604	223
μPA1700	213
μPA1700A	213
μPA1701	213
μPA1701A	213
μPA1702	213
μPA1703	213
μPA1704	213
μPA1705	213
μPA1706	213
μPA1707	213
μPA1710	213
μPA1710A	213
μPA1711	213
μPA1712	213
μPA1713	213
μPA1714	213
μPA1715	213
μPA1750	213
μPA1751	213
μPA1752	213
μPA1753	213
μPA1754	213
μPA1755	213
μPA1756	213
μPA1757	213
μPA1758	213
μPA1800	214
μPA1801	214
μPA1802	214
μPA1803	214
μPA1810	214
μPA1811	214
μPA1812	214
μPA1813	214
μPA1814	214
μPA1815	214
μPA1850	214
μPA1851	214
μPA1852	214
μPA1853	214
μPA1854	214
μPA1855	214
μPA2001	221
μPA2002	221
μPA2003	221
μPA2004	221
μPA2981	221
μPA2982	221
μPA2987	221
μPA500T	220
μPA501T	220
μPA502T	220
μPA503T	220
μPA504T	220
μPA505T	220
μPA570T	220
μPA571T	220
μPA572T	220
μPA573T	220
μPA574T	220
μPA600T	220
μPA601T	220
μPA602T	220
μPA603T	220
μPA604T	220
μPA605T	220
μPA606T	220
μPA607T	220
μPA608T	220
μPA609T	220
μPA610TA	220
μPA611TA	220
μPA670T	220
μPA671T	220
μPA672T	220
μPA673T	220
μPA674T	220
μPA80	221
μPA800	244
μPA801	244
μPA802	244
μPA803	244
μPA804	244
μPA805	244
μPA806	244
μPA807	244
μPA808	244
μPA809	244
μPA81	221
μPA810	244
μPA811	244
μPA812	244
μPA813	244
μPA814	244
μPA821	244
μPA822	244
μPA826	244
μPA827	244
μPA828	244
μPA829	244
μPA831	244
μPA832	244
μPA833	244
μPA834	244
μPA835	244
μPA836	244
μPA837	244
μPA838	244
μPA839	244

Part number	Page	Part number	Page	Part number	Page	Part number	Page	Part number	Page
μPB100x	164	μPC2270A	182	μPC3206	239	μPC802	180	μPD16633B	166
μPB1502	239	μPC2400A	181	μPC3206GR	172	μPC803	180	μPD16634A	166
μPB1502(1)	239	μPC24A00	181	μPC3210	234	μPC804	180	μPD16636	166
μPB1504	239	μPC24M00A	181	μPC3211	239	μPC8100	239	μPD16637	166
μPB1505	239	μPC251	180	μPC3211GR	172	μPC8104	234	μPD16638A	166
μPB1506	239	μPC2533	169	μPC324	180	μPC8105	234	μPD16640A	166
μPB1507	239	μPC2572	170	μPC337	182	μPC8106	239	μPD16640B	166
μPB1508	239	μPC2575	163	μPC339	181	μPC8109	239	μPD16640C	166
μPB1509	239	μPC258	180	μPC3403	180	μPC811	180	μPD16644	166
μPB1510	239	μPC259	180	μPC354	180	μPC8110	234	μPD16647	166
μPC1060	183	μPC2708	234	μPC358	180	μPC8112	239	μPD16650	166
μPC1074A	182	μPC2709	234	μPC393	181	μPC8119	234	μPD16652	166
μPC1093	182	μPC271	181	μPC4061	180	μPC812	180	μPD16654	166
μPC1094	183	μPC2710	234	μPC4062	180	μPC8120	234	μPD16655	166
μPC1099	183	μPC2711	234	μPC4064	180	μPC8125	234	μPD16656	166
μPC1100	183	μPC2712	234	μPC4071	180	μPC8126	234	μPD16657	166
μPC1150	183	μPC2713	234	μPC4072	180	μPC8128	235	μPD16661A	166
μPC1228	169	μPC2714	234	μPC4074	180	μPC8129	235	μPD16662	166
μPC1237	170	μPC2715	234	μPC4081	180	μPC813	180	μPD16663	166
μPC1251	180	μPC272	181	μPC4082	180	μPC8130	235	μPD16666A	166
μPC1316	169	μPC2721	239	μPC4084	180	μPC8131	235	μPD16667	166
μPC141	182	μPC2722	239	μPC4250	180	μPC814	180	μPD16670	165
μPC1458	180	μPC2723	234	μPC451	180	μPC815	180	μPD16675A	165
μPC151	180	μPC2726	234	μPC452	180	μPC8151	235	μPD16676	165
μPC1555	184	μPC2731	239	μPC454	180	μPC8152	235	μPD16677	165
μPC1652	234	μPC2734	239	μPC4556	180	μPC816	180	μPD16680	165
μPC1655	234	μPC2743	239	μPC4557	180	μPC8163	239	μPD16682	165
μPC1656	234	μPC2744	239	μPC4558	180	μPC821	180	μPD16710	166
μPC1658	234	μPC2745	234	μPC4559	180	μPC822	180	μPD16712	166
μPC1663	234	μPC2746	234	μPC4560	180	μPC824	180	μPD16714	166
μPC1675	234	μPC2747	234	μPC4570	180	μPC831	180	μPD16715	166
μPC1676	234	μPC2748	234	μPC4572	180	μPC832	180	μPD16732	166
μPC1677	234	μPC2749	234	μPC4574	180	μPC834	180	μPD16740	166
μPC1678	234	μPC2753	164,239	μPC458	180	μPC842	180	μPD16780	166
μPC1685	239	μPC2756	164	μPC4741	180	μPC844	180	μPD16803	167
μPC1686	239	μPC2757	239	μPC494	183	μPD16304	164	μPD16804	167
μPC1687	239	μPC2758	239	μPC5020	147	μPD16305	166	μPD16805	167
μPC1688	234	μPC2762	234	μPC5021	147	μPD16306	164	μPD16808	167
μPC1694	239	μPC2763	234	μPC5022	147	μPD16306A	164,166	μPD16813	167
μPC177	181	μPC277	181	μPC5023	147	μPD16306B	166	μPD16814	167
μPC1830	174	μPC2771	234	μPC5024	147	μPD16310	164,166	μPD16818	167
μPC1851B	171	μPC2775	239	μPC5031	147	μPD16311	164	μPD16823	167
μPC1853	171	μPC2776	234	μPC5032	147	μPD16312	164	μPD16833	167
μPC1854A	171	μPC2781	172,239	μPC5034	147	μPD16313	164	μPD16833A	167
μPC1857A	171	μPC2782	172,239	μPC5200	147	μPD16314	164	μPD16835	167
μPC1876	171	μPC2791	234	μPC5201	147	μPD16326	164	μPD16836	167
μPC1883	173	μPC2792	234	μPC5202	147	μPD16326A	164	μPD16837	167
μPC1884	173	μPC2794	239	μPC5203	147	μPD16327	166	μPD16855	161,168
μPC1891A	171	μPC2795	172,239	μPC5204	147	μPD16331	166	μPD16856	167
μPC1892	171	μPC2797	239	μPC659A	177	μPD16334	166	μPD16857	167
μPC1900	183	μPC2798	239	μPC662	177	μPD16335	166	μPD16858	167
μPC1905	183	μPC2798GR	172	μPC664	177	μPD16337	166	μPD16901	167
μPC1906	183	μPC2799	239	μPC665	177	μPD16430A	165	μPD17001	16
μPC1933	183	μPC2800A	176	μPC666	177	μPD16431A	165	μPD17003A	16
μPC1934	183	μPC2900	181	μPC667	177	μPD16432B	165	μPD17005	16
μPC1935	183	μPC29L00	181	μPC7073	162	μPD16434	165	μPD17010	16
μPC1943	182	μPC29M00	181	μPC741	180	μPD16435	165	μPD17012	16
μPC1944	182	μPC29S10	183	μPC7800A	181	μPD16435A	165	μPD17015	17
μPC2102	163	μPC29S78	183	μPC78L00	181	μPD16435B	165	μPD17016	16
μPC2251	182	μPC305	182	μPC78M00A	181	μPD16448A	166	μPD17017	16
μPC2252	182	μPC311	181	μPC78N00	181	μPD16501	176	μPD17071	17
μPC2253	182	μPC317	182	μPC7900A	181	μPD16502	176	μPD17072	17
μPC2254	182	μPC319	181	μPC79L00	181	μPD16510	176	μPD17073	17
μPC2255	182	μPC3200	239	μPC79M00	181	μPD16520	176	μPD17201A	26
μPC2256	182	μPC3202	239	μPC79N00	181	μPD16633	166	μPD17203A	26
μPC2260	182	μPC3205	172,239	μPC801	180	μPD16633A	166	μPD17204	26

Part number	Page
μPD17207	26
μPD17225	27
μPD17226	27
μPD17227	27
μPD17228	27
μPD17704	18
μPD17705	18
μPD17707	18
μPD17708	18
μPD17709	18
μPD17717	19
μPD17718	19
μPD17719	19
μPD178002	20
μPD178003	20
μPD178004A	20
μPD178006A	20
μPD178016A	20
μPD178018A	20
μPD178023	21
μPD178024	21
μPD178033	22
μPD178034	22
μPD178046	25
μPD178048	25
μPD178076	24
μPD178078	24
μPD178096	23
μPD178098	23
μPD178123	21
μPD178124	21
μPD178133	22
μPD178134	22
μPD178F048	25
μPD178F098	23,24
μPD178F124	21
μPD178F134	22
μPD178P018A	20
μPD17933A	17
μPD17934A	17
μPD17P001	16
μPD17P005	16
μPD17P010	16
μPD17P012	16
μPD17P073	17
μPD17P203A	26
μPD17P204	26
μPD17P207	26
μPD17P228	27
μPD17P709	18
μPD17P719	19
μPD23C128000L	128
μPD23C128040L	128
μPD23C16000AL	128
μPD23C16040AL	128
μPD23C24000	128
μPD23C24020	128
μPD23C32000A	128
μPD23C32000L	128
μPD23C32020A	128
μPD23C32040L	128
μPD23C32082L	128
μPD23C32202L	128
μPD23C64000	128
μPD23C64000AL	128
μPD23C64020	128
μPD23C64040AL	128
μPD23C8000LX	128
μPD23C8000X	128
μPD23C8001EJ	128
μPD26401	129
μPD26411	129
μPD29F008AL-X	130
μPD29F800AL-X	130
μPD30102	111
μPD30111	111
μPD30200	111
μPD30210	111
μPD30500	111
μPD30700	111
μPD3700	158
μPD3719	158
μPD3720A	158
μPD3728	158
μPD3729	158
μPD3734A	157
μPD3737	157
μPD3739	157
μPD3753	157
μPD3794	158
μPD3797	158
μPD3799	158
μPD42280	132
μPD431000A	124
μPD431000A-X	125
μPD431231L	127
μPD431532L	127
μPD431536L	127
μPD431632L	127
μPD431636L	127
μPD432232L	127
μPD43256B	124
μPD43256B-X	125
μPD43257B	124
μPD432836L	127
μPD434001A	126
μPD434001AL	126
μPD434004A	126
μPD434004AL	126
μPD434008A	126
μPD434008AL	126
μPD434016A	126
μPD434016AL	126
μPD441000L-X	125
μPD442000L-X	125
μPD4504161	117
μPD45128163	115
μPD45128441	115
μPD45128841	115
μPD4516161A	117
μPD4516421A	117
μPD4516821A	117
μPD45256163	114
μPD45256441	114
μPD45256841	114
μPD4564163	116
μPD4564323	116
μPD4564441	116
μPD4564841	116
μPD4565161	118
μPD4565421	118
μPD4565821	118
μPD45D128164	118
μPD45D128442	118
μPD45D128842	118
μPD464318L	127
μPD464336L	127
μPD464518L	127
μPD464536L	127
μPD4701A	177
μPD4702	177
μPD4704	177
μPD4711B	178
μPD4712C	178
μPD4712D	178
μPD4713A	178
μPD4714A	178
μPD4715A	178
μPD4721	178
μPD4722	178
μPD4723	178
μPD4724	178
μPD4726	178
μPD4811650	132
μPD481850	132
μPD485505	132
μPD485506	132
μPD488385	119
μPD4990A	176
μPD4991A	176
μPD4992	176
μPD5205	184
μPD5555	184
μPD5556	184
μPD61012	168
μPD61030	168
μPD6121G-001	176
μPD6122G-001	176
μPD6133	28
μPD6134	28
μPD61511	172
μPD61882	173
μPD61P34B	28
μPD6221	177
μPD6222	177
μPD63	29
μPD6320	164,166,167
μPD63200	169
μPD6321	164,166,167
μPD63210	169
μPD63210L	169
μPD6323	164
μPD6323B	164
μPD6325	177
μPD6326	177
μPD63310	169
μPD6335	177
μPD6336	177
μPD6340	164
μPD6345	167
μPD63701	163
μPD63702A	170
μPD63703	170
μPD63710	170
μPD63724	163
μPD63724A	163
μPD6376	169
μPD6378	163
μPD6379	169
μPD6379A	169
μPD6379AL	169
μPD6379L	169
μPD63A	29
μPD64	29
μPD6461	175
μPD6462	175
μPD6464A	175
μPD6465	175
μPD6466	175
μPD65443	145
μPD65444	145
μPD65445	145
μPD65446	145
μPD65448	145
μPD65449	145
μPD65451	145
μPD65454	145
μPD65456	145
μPD65458	145
μPD65611	134
μPD65621	134
μPD65625	134
μPD65630	134
μPD65636	134
μPD65640	134
μPD65643	134
μPD65646	134
μPD65650	134
μPD65653	134
μPD65654	134
μPD65656	134
μPD65658	134
μPD65662	134
μPD65664	134
μPD65672	134
μPD65676	134
μPD65800	135
μPD65801	135
μPD65802	135
μPD65803	135
μPD65804	135
μPD65806	135
μPD65808	135
μPD65810	135
μPD65811	135
μPD65812	135
μPD65813	135
μPD65837	136
μPD65839	136
μPD65840	136
μPD65841	136
μPD65842	136
μPD65843	136
μPD65845	136
μPD65846	136
μPD65848	136
μPD65849	136
μPD65850	136
μPD65851	136
μPD65852	136
μPD65853	136
μPD65855	136
μPD65858	136
μPD65860	136
μPD65861	136
μPD65862	136
μPD65863	136
μPD65865	136
μPD65866	137
μPD65868	137
μPD65869	137
μPD65870	137
μPD65871	137
μPD65872	137
μPD65873	137
μPD65875	137
μPD65878	137
μPD65880	135
μPD65881	135
μPD65882	135
μPD65883	135
μPD65884	135
μPD65885	135
μPD65887	135
μPD65889	135
μPD65890	135
μPD65893	135
μPD65943	137
μPD65944	137
μPD65945	137
μPD65946	137
μPD65948	137
μPD65949	137
μPD65951	137
μPD65954	137
μPD65956	137
μPD65958	137
μPD6604	28
μPD66P04B	28
μPD6700	164
μPD6708	159
μPD67821	138
μPD67822	138
μPD67823	138
μPD67824	138
μPD67825	138
μPD67826	138
μPD67827	138
μPD67828	138
μPD67829	138
μPD67830	138
μPD67831	138
μPD68000	151
μPD68001	151
μPD68010	151
μPD68011	151
μPD68031	151
μPD68041	151
μPD68061	151
μPD68070	151
μPD681××	154
μPD6P4B	29
μPD70108H	103
μPD70116H	103
μPD70208H	103
μPD70216H	103
μPD70236A	103
μPD703000	110
μPD703001	110
μPD703002	110

Part number	Page	Part number	Page	Part number	Page	Part number	Page	Part number	Page
μPD703003	109	μPD753012A	8	μPD78011F	49	μPD78076Y	71	μPD784054	90
μPD703003A	109	μPD753016A	8	μPD78011FY	50	μPD78078	70	μPD784214	91
μPD703004A	109	μPD753017A	8	μPD78011H	51	μPD78078Y	71	μPD784214Y	92
μPD703008	108	μPD753036	11,13	μPD78012F	49	μPD78081	73	μPD784215	91
μPD703008Y	108	μPD7531××	13	μPD78012FY	50	μPD78082	73	μPD784215Y	92
μPD703015	107	μPD753104	9	μPD78012H	51	μPD780831Y	80	μPD784216	91
μPD703015Y	107	μPD753106	9	μPD78013F	49	μPD780833Y	80	μPD784216Y	92
μPD703017	107	μPD753108	9	μPD78013FY	50	μPD780948	78	μPD784217	93
μPD703017Y	107	μPD7532××	13	μPD78013H	51	μPD780949	78	μPD784217Y	94
μPD703025A	109	μPD753204	10	μPD78014F	49	μPD780955	76	μPD784218	93
μPD703033	108	μPD753206	10	μPD78014FY	50	μPD78095B	74	μPD784218Y	94
μPD703033Y	108	μPD753208	10	μPD78014H	51	μPD78096B	74	μPD784224	95
μPD703100	106	μPD753304	12,13	μPD78015F	49	μPD780973	77	μPD784224Y	96
μPD703101	106	μPD754144	7,13	μPD78015FY	50	μPD780983	75	μPD784225	95
μPD703102	106	μPD754202	7	μPD78016F	49	μPD780984	75	μPD784225Y	96
μPD70320	103	μPD754244	7,13	μPD78016FY	50	μPD780986	75	μPD784907	100
μPD70325	103	μPD754264	7	μPD78018F	49	μPD780988	75	μPD784908	100
μPD70433	103	μPD75430×	13	μPD78018FY	50	μPD78098B	74	μPD784915	97
μPD705100	105	μPD754302	6	μPD780204	58	μPD78212	81	μPD784915A	97
μPD705101	105	μPD754304	6	μPD780205	58	μPD78213	81	μPD784915B	97
μPD705102	105	μPD75F4264	7	μPD780206	58	μPD78214	81	μPD784916A	97
μPD70732	105	μPD75P0016	3	μPD780208	58	μPD78217A	81	μPD784916B	97
μPD70741	105	μPD75P0076	5	μPD780226	59	μPD78218A	81	μPD784927	98
μPD70F3003	109	μPD75P0116	4	μPD780228	59	μPD78233	82	μPD784927Y	99
μPD70F3003A	109	μPD75P3018A	8	μPD780232	60	μPD78234	82	μPD784928	98
μPD70F3008	108	μPD75P3036	11	μPD780306	69	μPD78237	82	μPD784928Y	99
μPD70F3008Y	108	μPD75P3116	9	μPD780306Y	69	μPD78238	82	μPD784935	101
μPD70F3017	107	μPD75P3216	10	μPD780308	69	μPD78243	81	μPD784936	101
μPD70F3017Y	107	μPD75P4308	6	μPD780308Y	69	μPD78244	81	μPD784937	101
μPD70F3025A	109	μPD77015	156	μPD78042F	56	μPD78310A	83	μPD784953	102
μPD70F3033	108	μPD77016	156	μPD78043F	56	μPD78312A	83	μPD784955	102
μPD70F3033Y	108	μPD77017	156	μPD78044F	56	μPD78320	83	μPD789011	37
μPD70F3035	108	μPD77018A	156	μPD78044H	57	μPD78322	83	μPD789012	37
μPD70F3035Y	108	μPD77019	156	μPD78045F	56	μPD78323	83	μPD789022	38
μPD70F3037	108	μPD77019-013	156	μPD78045H	57	μPD78324	83	μPD789024	38
μPD70F3037Y	108	μPD77111	156	μPD78046H	57	μPD78327	84	μPD789025	38
μPD70F3102	106	μPD77523	159	μPD78052	61	μPD78328	84	μPD789026	38
μPD70P3000	110	μPD780016AY	55	μPD78052Y	62	μPD78330	84	μPD789046	39
μPD70P3002	110	μPD780018AY	55	μPD78053	61	μPD78334	84	μPD789101A	40
μPD72001-11	159	μPD780021	52	μPD78053Y	62	μPD78350	84	μPD789102A	40
μPD72001-A8	159	μPD780021Y	53	μPD78054	61	μPD78350A	84	μPD789104A	40
μPD72002-11	159	μPD780022	52	μPD78054Y	62	μPD78352A	84	μPD789111A	40
μPD72011	161	μPD780022Y	53	μPD78055	61	μPD78355	85	μPD789112A	40
μPD72042A	159	μPD780023	52	μPD78055Y	62	μPD78356	85	μPD789114A	40
μPD72042B	159	μPD780023Y	53	μPD78056	61	μPD78361A	86	μPD789121A	41
μPD72103A	159	μPD780024	52	μPD78056F	63	μPD78362A	86	μPD789122A	41
μPD72105	159	μPD780024Y	53	μPD78056FY	64	μPD78363A	86	μPD789124A	41
μPD72107	159	μPD780031	52	μPD78056Y	62	μPD78365A	86	μPD789131A	41
μPD7225	165	μPD780031Y	53	μPD78058	61	μPD78366A	86	μPD789132A	41
μPD72254	168	μPD780032	52	μPD78058F	63	μPD78368A	86	μPD789134A	41
μPD72305	159	μPD780032Y	53	μPD78058FY	64	μPD784020	87	μPD789144	42
μPD72611	163	μPD780033	52	μPD78058Y	62	μPD784021	87	μPD789146	42
μPD75000×	13	μPD780033Y	53	μPD78062	67	μPD784025	87	μPD789154	42
μPD750004	3	μPD780034	52	μPD78062Y	68	μPD784026	87	μPD789156	42
μPD750006	3	μPD780034Y	53	μPD78063	67	μPD784031	88	μPD789166	43
μPD750008	3	μPD780053	65	μPD78063Y	68	μPD784031Y	89	μPD789167	43
μPD75006×	13	μPD780053Y	66	μPD78064	67	μPD784035	88	μPD789176	43
μPD750064	5	μPD780054	65	μPD78064B	67	μPD784035Y	89	μPD789177	43
μPD750066	5	μPD780054Y	66	μPD78064Y	68	μPD784036	88	μPD789196AY	44
μPD750068	5	μPD780055	65	μPD780701Y	79	μPD784036Y	89	μPD789197AY	44
μPD75010×	13	μPD780055Y	66	μPD780702Y	79	μPD784037	88	μPD789216AY	45
μPD750104	4	μPD780056	65	μPD78070A	72	μPD784037Y	89	μPD789217AY	45
μPD750106	4	μPD780056Y	66	μPD78070AY	72	μPD784038	88	μPD789405A	46
μPD750108	4	μPD780058	65	μPD78074B	70	μPD784038Y	89	μPD789406A	46
μPD7530××	13	μPD780058Y	66	μPD78075B	70	μPD784044	90	μPD789407A	46
μPD7530××A	13	μPD780065	54	μPD78076	70	μPD784046	90	μPD789415A	46

Part number	Page
μPD789416A	46
μPD789417A	46
μPD789800	47
μPD789830	48
μPD78F0034	52
μPD78F0034Y	53
μPD78F0058	65
μPD78F0058Y	66
μPD78F0066	54
μPD78F0233	60
μPD78F0701Y	79
μPD78F0833Y	80
μPD78F0948	78
μPD78F0974	77
μPD78F0988	75
μPD78F4046	90
μPD78F4216	91
μPD78F4216Y	92
μPD78F4218	93
μPD78F4218Y	94
μPD78F4225	95
μPD78F4225Y	96
μPD78F4928	98
μPD78F4928Y	99
μPD78F4937	101
μPD78F4956	102
μPD78F9026A	38
μPD78F9046	39
μPD78F9116A	40
μPD78F9136A	41
μPD78F9156	42
μPD78F9177	43
μPD78F9197AY	44
μPD78F9217AY	45
μPD78F9418A	46
μPD78F9801	47
μPD78F9831	48
μPD78P0018AY	55
μPD78P018F	49
μPD78P018FY	50
μPD78P0208	58
μPD78P0308	69
μPD78P0308Y	69
μPD78P048A	56
μPD78P054	61
μPD78P058	61
μPD78P058F	63
μPD78P058FY	64
μPD78P058Y	62
μPD78P064	67
μPD78P064B	67
μPD78P078	70
μPD78P078Y	71
μPD78P083	73
μPD78P098B	74
μPD78P214	81
μPD78P218A	81
μPD78P238	82
μPD78P312A	83
μPD78P322	83
μPD78P324	83
μPD78P328	84
μPD78P334	84
μPD78P352	84
μPD78P356	85
μPD78P364A	86
μPD78P368A	86
μPD78P4026	87
μPD78P4038	88
μPD78P4038Y	89
μPD78P4908	100
μPD78P4916	97
μPD78P9014	37
μPD82200	144
μPD82600	144
μPD93000	141
μPD94000	141
μPD95100	141
μPD96100	141
μPD9611	162
μPD97000	143
μPD98201	159
μPD98203	159
μPD98401A	160
μPD98402A	160
μPD98404	160
μPD98405	160
μPD98408	160
μPD98409	160
μPD98410	160
μPD98411	160
μPD98421	161
μPD98431	161
μPD99000	143
μPD9903	162
μPG100	230
μPG101	230
μPG103	230
μPG105-1	230
μPG106	230
μPG107	230
μPG110	230
μPG132	230
μPG133	230
μPG137	230
μPG138	230
μPG139	230
μPG152	230
μPG153	230
μPG154	230
μPG155	230
μPG17×	230
μPG170	230
μPG171	230
μPG171GV	233
μPG172	230
μPG173	230
μPG174	230
μPG501	231
μPG502	231
μPG503	231
μPG504	231
μPG506	231
μPG508	231
[N]	
NDL3220	247
NDL3233	247
NDL3321	247
NDL3330	247
NDL5421	248
NDL5422	248
NDL5461	248
NDL5471	248
NDL5481	248
NDL5500	249
NDL5506	249
NDL5510	249
NDL5516	249
NDL5520	249
NDL5521	249
NDL5522	249
NDL5530	249
NDL5531	249
NDL5551	249
NDL5553	249
NDL5561	249
NDL5590	249
NDL5592	249
NDL7001	247
NDL7103	248
NDL7113	248
NDL7153	248
NDL7163	248
NDL7400	247
NDL7401	247
NDL7408	247
NDL7502	248
NDL7503	248
NDL7510	248
NDL7512	248
NDL7513	248
NDL7514	248
NDL7515	248
NDL7540	248
NDL7542	248
NDL7552	248
NDL7553	248
NDL7560	248
NDL7562	248
NDL7563	248
NDL7564	248
NDL7565	248
NDL7580	248
NDL7603	247
NDL7620	247
NDL7680	247
NDL7701	247
NDL7705	247
NDL7800	247
NDL7910	247
NE1280100	233
NE1280200	233
NE1280400	233
NE23383B	232
NE24283B	232
NE27283B	232
NE3210S01	232
NE32484A	232
NE32584C	232
NE325S01	232
NE329S01	232
NE33284A	232
NE334S01	232
NE34018	232
NE38018	232
NE4210M01	232
NE42484A	232
NE42484C	232
NE425S01	232
NE428M01	232
NE429M01	232
NE434S01	232
NE6500379A	233
NE6500496	233
NE6501077	233
NE650R279A	233
NE650R479A	233
NE6510179A	233
NE6510379A	233
NE651R479A	233
NE76118	232
NE85001	233
NE85002	233
NE850R5	233
NE960R275	233
NE960R575	233
NEL2000	243
NES1417B-30	233
NES1821B-30	233
NES1821P-50	233
NES1823P-100	233
NES2527B-30	233
NEZ1011	233
NEZ1414	233
NEZ3642	233
NEZ4450	233
NEZ5964	233
NEZ6472	233
NEZ7177	233
NEZ7785	233
NL1000	254
NL1100	254
NL2100	254
NNCD[ ]A	226
NNCD[ ]B	226
NNCD[ ]C	226
NNCD[ ]D	226
NNCD[ ]E	226
NNCD[ ]F	226
NNCD[ ]G	226
NNCD[ ]LG	226
NNCD[ ]LH	226
NX8563	247
[P]	
PH302	250
PH302C	250
PH310	250
PH320	250
PH5××	250
PS2501	251
PS2502	251
PS2503	251
PS2505	251
PS2506	251
PS2521	251
PS2525	251
PS2532	251
PS2533	251
PS2561	251
PS2562	251
PS2565	251
PS2566	251
PS2581	251
PS2601	251
PS2602	251
PS2603	251
PS2604	251
PS2605	251
PS2606	251
PS2607	251
PS2608	251
PS2621	251
PS2622	251
PS2625	251
PS2626	251
PS2633	251
PS2634	251
PS2651	251
PS2652	251
PS2653	251
PS2654	251
PS2701	252
PS2702	252
PS2703	252
PS2705	252
PS2706	252
PS2707	252
PS2732	252
PS2733	252
PS2801	252
PS2802	252
PS2805	252
PS2806	252
PS2832	252
PS7111-2A	253
PS7112-1A	253
PS7113-1A	253
PS7113-2A	253
PS7122-1A	253
PS7122-2A	253
PS7122A-1A	253
PS7122A-1B	253
PS7122A-1C	253
PS7122A-2A	253
PS7122A-2B	253
PS7141-1A	253
PS7141-1B	253
PS7141-1C	253
PS7141-2A	253
PS7141-2B	253
PS7142-1A	253
PS7142-2A	253
PS7160-1A	253
PS7160-2A	253
PS7200A-1A	253
PS7200B-1A	253
PS7211-2A	253
PS7221-2A	253
PS7241-1A	253
PS7241-1B	253
PS7241-1C	253
PS7241-2A	253
PS7241-2B	253
PS7241-AT1	253
PS7241-AT5	253
PS7341-1A	253
PS7341C-1A	253
PS7342-1A	253

Part number	Page	Part number	Page	Part number	Page	Part number	Page	Part number	Page
PS7342C-1A	253								
PS7360-1A	253								
PS7522-1A	253								
PS7522-2A	253								
PS7841-A15	253								
PS8601	252								
PS8602	252								
PS8701	252								
PS8741	252								
PS9601	252								
PS9611	252								
PS9634	252								
PS9701	252								
PS9711	252								
[R]									
RD[ ]E	225								
RD[ ]ES	225								
RD[ ]F	225								
RD[ ]FM	224								
RD[ ]HS	225								
RD[ ]JS	225								
RD[ ]M	224								
RD[ ]MW	224								
RD[ ]P	224								
RD[ ]S	224								
RD[ ]SL	224								
RD[ ]UJ	224								
RD[ ]UM	224								
[V]									
V20HL	103								
V25	103								
V25+	103								
V30HL	103								
V30MX	141								
V40HL	103								
V50HL	103								
V53A	103								
V55PI	103								
V810	105								
V821	105								
V830	105								
V831	105								
V832	105								
V850/SA1	107								
V850/SB1	108								
V850/SB2	108								
V850/SB3	108								
V850E/MS1	106								
V851	110								
V852	110								
V853	109								
V854	108								
VR10000	111								
VR4102	111								
VR4111	111								
VR4300	111								
VR4305	111								
VR4310	111								
VR5000	111								

For further information, please contact:

NEC Corporation
NEC Building
7-1, Shiba 5-chome, Minato-ku
Tokyo 108-8001, Japan
Tel: 03-3454-1111
Fax: 03-3798-6059

[North & South America]

NEC Electronics Inc.
2880 Scott Blvd.
Santa Clara, CA 95050-2554, U.S.A.
Tel: 408-588-6000
800-366-9782
Fax: 408-588-6130
800-729-9288

[Regional Sales Offices]

Central Region
Greenpoint Tower
2800 West Higgins
Road Suite 765
Hoffman Estates,
IL 60195, U.S.A.
Tel: 847-839-6300
Fax: 847-519-9329

Norcal Region
3033 Scott Blvd.
Santa Clara, CA 95054, U.S.A.
Tel: 408-588-5100
Fax: 408-588-5134

Eastern Region
901 N. Lake Destiny Drive
Suite 320
Maitland, FL 32751, U.S.A.
Tel: 407-875-1145
Fax: 407-875-0962

Western Region
One Embassy Centre
9020 S.W. Washington
Square Road
Suite 400
Tigard, OR 97223, U.S.A.
Tel: 503-672-4500
Fax: 503-643-5911

NEC do Brasil S.A.
Electron Devices Division
Rodovia Presidente Dutra, Km 214
07210-902-Guarulhos-SP Brasil
Tel: 55-11-6465-6810
Fax: 55-11-6465-6829

[Europe]

NEC Electronics (Germany) GmbH
Kanzlerstr. 2,
40472 Düsseldorf
Germany
Tel: 0211-650302
Fax: 0211-6503490

Munich Office
Arabellastr. 17
81925 München, Germany
Tel: 089-921003-0
Fax: 089-92100315

Stuttgart Office
Industriestr. 3
70507 Stuttgart, Germany
Tel: 0711-99010-0
Fax: 0711-99010-19

Hannover Office
Königstr. 12
30175 Hannover, Germany
Tel: 0511-33402-0
Fax: 0511-33402-34

Benelux Office
Boschdijk 187a
5612 HB Eindhoven,
The Netherlands
Tel: 040-2445845
Fax: 040-2444580

Scandinavia Office
P.O. Box 134
18322 Taeby, Sweden
Tel: 08-6380820
Fax: 08-6380388

NEC Electronics (UK) Limited
Cygnus House, Sunrise Park Way,
Milton Keynes, MK14 6NP, U.K.
Tel: 01908-691-133
Fax: 01908-670-290

NEC Electronics (France) S.A.
9, rue Paul Dautier-BP 187
78142 Velizy-Villacoublay Cédex
France
Tel: 01-30-67-58-00
Fax: 01-30675899

Madrid Office
Juan Esplandiu, 15
28007 Madrid, Spain
Tel: 01-504-2787
Fax: 01-504-2860

NEC Electronics Italiana s.r.l.
Via Fabio Filzi, 25/A,
20124 Milano, Italy
Tel: 02-667541
Fax: 02-66754299

[Asia & Oceania]

NEC Electronics Hong Kong Limited
12/F., Cityplaza 4,
12 Taikoo Wan Road, Hong Kong
Tel: 2886-9318
Fax: 2886-9022/9044

Seoul Branch
10F, ILSONG Bldg., 157-37,
Samsung-Dong, Kangnam-Ku
Seoul, the Republic of Korea
Tel: 02-528-0303
Fax: 02-528-4411

NEC Electronics Taiwan Ltd.
7F, No. 363 Fu Shing North Road
Taipei, Taiwan, R. O. C.
Tel: 02-2719-2377
Fax: 02-2719-5951

NEC Electronics Singapore Pte. Ltd.
101 Thomson Road #04-02/05
United Square, Singapore 307591
Tel: 65-253-8311
Fax: 65-250-3583

G98. 8

Document No. X10679EJGV0SG00(16th edition)
Date Published October 1998 N CP(K)

© NEC Corporation 1991

Printed in Japan